Séminaire de Géométrie Algébrique du Bois-Marie

Étale Cohomology

(SGA $4\frac{1}{2}$)

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Chapter 1

Étale cohomology: the starting points

1 Grothendieck topologies

Grothendieck topologies first appeared as the foundation of Grothendieck's theory of descent (cf. SGA 1 VI, VIII); their use in the corresponding cohomology theories came later. We follow the same path here: by formalizing the classical notions of localization, local property, and patching (1.1, 1.2, 1.3), we obtain the general concept of a Grothendieck topology (1.6); to justify its introduction into algebraic geometry, we prove a faithfully flat descent theorem (1.4), generalizing the classical theorem 90 of Hilbert (1.5).

The reader will find a more complete, but concise, exposition of the formalism in Giraud [19]. M. Artin's notes, "Grothendieck topologies" [1] (Chapters I to III), also remain useful. The 866 pages of Exposés I to V of SGA 4 are valuable when one considers exotic topologies, such as the one giving rise to crystalline cohomology; in order to use the étale topology, which is so close to classical intuition, it is not essential to read them.

1.1 Sieves

Let X be a topological space and let $f : X \to \mathbb{R}$ be a real-valued function on X. The continuity of f is a local property; in other words, if f is continuous on every sufficiently small open subset of X, then f is continuous on all of X. To formalize the notion of a "local property," we introduce some definitions.

A set $\mathcal{U}$ of open subsets of X is called a *sieve* if, for every $U \in \mathcal{U}$ and $V \subset U$, one has $V \in \mathcal{U}$. A sieve is called *covering* if the union of all the open subsets belonging to it is equal to X.

Given a family $\{U_i\}$ of open subsets of X, the sieve generated by $\{U_i\}$ is, by definition, the set of open subsets U of X such that U is contained in one of the U_i.

A property $P(U)$, defined for every open subset U of X, is called *local* if, for every covering sieve $\mathcal{U}$ of every open subset U of X, $P(U)$ is true if and only if $P(V)$ is true for every $V \in \mathcal{U}$. For example, given $f : X \to \mathbb{R}$, the property "f is continuous on U" is local.

1.2 Sheaves

Let us make precise the notion of a function given locally on X.

by P. Deligne, written up by J.F. Boutot

1.2.1 The sieve point of view

Let $\mathcal{U}$ be a sieve of open subsets of X. A function given $\mathcal{U}$-locally on X consists of a function f_U on U for every $U \in \mathcal{U}$, such that if $V \subset U$ then $f_V = f_U|V$.

1.2.2 The Čech point of view

If the sieve $\mathcal{U}$ is generated by a family of open subsets U_i of X, then giving a function $\mathcal{U}$-locally amounts to giving a function f_i on each U_i such that $f_i|U_i \cap U_j = f_j|U_i \cap U_j$.

In other words, if $Z = \coprod U_i$, then giving a function $\mathcal{U}$-locally amounts to giving a function on Z that is constant on the fibres of the natural projection $Z \to X$.

1.2.3

The continuous functions form a sheaf. This means that for every covering sieve $\mathcal{U}$ of an open subset V of X, and every function $\{f_U\}$ given $\mathcal{U}$-locally such that each f_U is continuous on U, there exists a unique continuous function f on V such that $f|U = f_U$ for every $U \in \mathcal{U}$.

1.3 Stacks

Let us now make precise the notion of a vector bundle given locally on X.

1.3.1 The sieve point of view

Let $\mathcal{U}$ be a sieve of open subsets of X. A vector bundle given $\mathcal{U}$-locally on X consists of

a) a vector bundle E_U on each $U \in \mathcal{U}$,

b) if $V \subset U$, an isomorphism $\rho_{U,V} : E_V \xrightarrow{\sim} E_U|V$, satisfying

c) if $W \subset V \subset U$, the diagram

$$\begin{array}{ccc} E_W & \xrightarrow{\rho_{U,W}} & E_U|W \\ & \searrow^{\rho_{V,W}} & \uparrow{\scriptstyle \rho_{U,V}|W} \\ & & E_V|W \end{array}$$

commutes, that is, $\rho_{U,W} = (\rho_{U,V} \text{ restricted to } W) \circ \rho_{V,W}$.

1.3.2 The Čech point of view

If the sieve $\mathcal{U}$ is generated by a family of open subsets U_i of X, then giving a vector bundle $\mathcal{U}$-locally amounts to giving:

a) a vector bundle E_i on each U_i,

b) if $U_{ij} = U_i \cap U_j = U_i \times_X U_j$, an isomorphism $\rho_{ji} : E_i|U_{ij} \xrightarrow{\sim} E_j|U_{ij}$, such that

c) if $U_{ijk} = U_i \times_X U_j \times_X U_k$, the diagram

$$\begin{array}{ccc} E_i|U_{ijk} & \xrightarrow{\rho_{ki}|U_{ijk}} & E_k|U_{ijk} \\ & \searrow^{\rho_{ji}|U_{ijk}} & \uparrow{\scriptstyle \rho_{kj}|U_{ijk}} \\ & & E_j|U_{ijk} \end{array}$$

commutes, that is, $\rho_{ki} = \rho_{kj} \circ \rho_{ji}$ on U_{ijk}.

In other words, if $Z = \coprod U_i$ and $\pi : Z \to X$ is the natural projection, then giving a vector bundle $\mathcal{U}$-locally amounts to giving:

a) a vector bundle E on Z,

b) if x and y are two points of Z such that $\pi(x) = \pi(y)$, an isomorphism $\rho_{yx} : E_x \xrightarrow{\sim} E_y$ between the fibres of E at x and y, depending continuously on (x, y), and such that

c) if x, y, and z are three points of Z such that $\pi(x) = \pi(y) = \pi(z)$, then $\rho_{zx} = \rho_{zy} \circ \rho_{yx}$.

1.3.3

A vector bundle E on X defines a vector bundle $E_{\mathcal{U}}$ given $\mathcal{U}$-locally, namely the system of restrictions E_U of E to the objects of $\mathcal{U}$. The fact that the notion of a vector bundle is local can be expressed as follows: for every covering sieve $\mathcal{U}$ of X, the functor $E \mapsto E_{\mathcal{U}}$ from vector bundles on X to vector bundles given $\mathcal{U}$-locally is an equivalence of categories.

1.3.4

If in 1 one replaces "open subset of X" by "subset of X," one obtains the notion of a sieve of subspaces of X. In this setting there are also patching theorems. For example, let X be a space and let $\mathcal{C}$ be a sieve of subspaces of X generated by a locally finite closed covering of X. Then the functor $E \mapsto E_{\mathcal{C}}$ from vector bundles on X to vector bundles given $\mathcal{C}$-locally is an equivalence of categories.

In algebraic geometry, it is also useful to consider "sieves of spaces over X"; this is what we shall see in the next paragraph.

1.4 Faithfully flat descent

1.4.1

In the setting of schemes, the Zariski topology is not fine enough for the study of nonlinear problems, and one is led to replace the open immersions in the preceding definitions by more general morphisms. From this point of view, descent techniques appear as localization techniques. Thus the following descent statement can be expressed by saying that the properties under consideration are local for the faithfully flat topology (a morphism of schemes is called faithfully flat if it is flat and surjective).

Proposition 1.4.2. *Let A be a ring and B a faithfully flat A-algebra. Then:*

(i) A sequence $\Sigma = (M' \to M \to M'')$ of A-modules is exact if the sequence $\Sigma_{(B)}$ deduced from it by extension of scalars to B is exact.

(ii) An A-module M is of finite type (resp. of finite presentation, flat, locally free of finite rank, invertible (i.e. locally free of rank one)) if the B-module $M_{(B)}$ has the corresponding property.

Proof. (i) Since the functor $M \mapsto M_{(B)}$ is exact (by flatness of B), it suffices to show that if an A-module N is nonzero, then $N_{(B)}$ is nonzero. If N is nonzero, N contains a nonzero cyclic submodule $A/\mathfrak{a}$; hence $N_{(B)}$ contains the cyclic submodule $(A/\mathfrak{a})_{(B)} = B/\mathfrak{a}B$, which is nonzero by surjectivity of the structural morphism $\varphi : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ (if $V(\mathfrak{a})$ is nonempty, then $\varphi^{-1}(V(\mathfrak{a})) = V(\mathfrak{a}B)$ is nonempty).

(ii) For every family (x_i) of elements of $M_{(B)}$, there exists a finitely generated submodule M' of M such that $M'_{(B)}$ contains the x_i. If $M_{(B)}$ is of finite type and the x_i generate $M_{(B)}$, then $M'_{(B)} = M_{(B)}$, and hence $M' = M$ and M is of finite type. □

If $M_{(B)}$ is of finite presentation, the preceding argument yields a surjection $A^n \to M$. If N is the kernel of this surjection, then the B-module $N_{(B)}$ is of finite type, so N is of finite type and M is of finite presentation. The assertion for "flat" follows at once from (i); "locally free of finite rank" means "flat and of finite presentation," and the rank can be tested after extension of scalars to fields.

1.4.3

Let X be a scheme and let $\mathcal{S}$ be a class of X-schemes stable under fibre products over X. A class $\mathcal{U} \subset \mathcal{S}$ is a *sieve* on X (relative to $\mathcal{S}$) if, for every morphism $\varphi : V \to U$ of X-schemes with $U, V \in \mathcal{S}$ and $U \in \mathcal{U}$, one has $V \in \mathcal{U}$. The sieve *generated* by a family $\{U_i\}$ of X-schemes in $\mathcal{S}$ is the class of $V \in \mathcal{S}$ for which there exists a morphism of X-schemes from V to one of the U_i.

1.4.4

Let $\mathcal{U}$ be a sieve on X. A quasi-coherent module given $\mathcal{U}$-locally on X consists of

a) a quasi-coherent module E_U on every $U \in \mathcal{U}$,

b) for every $U \in \mathcal{U}$ and every morphism $\varphi : V \to U$ of X-schemes in $\mathcal{S}$, an isomorphism $\rho_\varphi : E_V \xrightarrow{\sim} \varphi^* E_U$, such that

c) if $\psi : W \to V$ is a morphism of X-schemes in $\mathcal{S}$, the diagram

$$\begin{array}{ccc} E_W & \xrightarrow{\rho_{\varphi\circ\psi}} & \psi^*\varphi^* E_U \\ & \searrow^{\rho_\psi} & \uparrow{\scriptstyle \psi^*\rho_\varphi} \\ & & \psi^* E_V \end{array}$$

commutes, that is, $\rho_{\varphi\circ\psi} = (\psi^*\rho_\varphi) \circ \rho_\psi$.

If E is a quasi-coherent module on X, we write $E_{\mathcal{U}}$ for the module given $\mathcal{U}$-locally whose value is $\varphi_U^* E$ on $\varphi : U \to X$ and such that, for every morphism $\psi : V \to U$, the restriction isomorphism ρ_ψ is the canonical isomorphism $E_V = (\varphi_U \circ \psi)^* E \xrightarrow{\sim} \psi^* \varphi_U^* E = \psi^* E_U$.

Theorem 1.4.5. *Let $\{U_i\} \in \mathcal{S}$ be a finite family of X-schemes flat over X such that X is the union of the images of the U_i, and let $\mathcal{U}$ be the sieve generated by $\{U_i\}$. Then the functor $E \mapsto E_{\mathcal{U}}$ is an equivalence from the category of quasi-coherent modules on X to the category of quasi-coherent modules given $\mathcal{U}$-locally.*

Proof. We treat only the case in which X is affine and $\mathcal{U}$ is generated by an affine X-scheme U faithfully flat over X. The reduction to this case is formal. Put $X = \mathrm{Spec}(A)$ and $U = \mathrm{Spec}(B)$.

If $U \to X$ admits a section, then X belongs to the sieve $\mathcal{U}$ and the assertion is clear. We reduce to this case.

A quasi-coherent module given $\mathcal{U}$-locally defines modules M', M'', and M''' on U, $U \times_X U$, and $U \times_X U \times_X U$, and isomorphisms $\rho : p^*M^\bullet \simeq M^\bullet$ for every projection morphism p between these spaces; this is a *cartesian diagram*

$$M^* : M' \rightrightarrows M'' \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; M'''$$

over

$$U_* : U \;\substack{\leftarrow\\ \leftarrow}\; U \times_X U \;\substack{\leftarrow\\ \leftarrow\\ \leftarrow}\; U \times_X U \times_X U.$$

Conversely, M^* determines the module given $\mathcal{U}$-locally: for $V \in \mathcal{U}$ there exists $\varphi : V \to U$, and we put $M_V = \varphi^* M'$; for $\varphi_1, \varphi_2 : V \to U$ there is a natural identification $\varphi_1^* M' \simeq (\varphi_1 \times \varphi_2)^* M'' \simeq \varphi_2^* M'$, and by using M''' one sees that these identifications are compatible, so the definition is legitimate. In short, giving a module $\mathcal{U}$-locally is equivalent to giving a cartesian diagram M^* over U_*.

In algebraic terms, giving M^* amounts to giving a cartesian diagram of modules

$$M' \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}}\; M'' \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}\\ \xrightarrow{\partial_2}}\; M'''$$

over the diagram of rings

$$B \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}}\; B \otimes_A B \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}\\ \xrightarrow{\partial_2}}\; B \otimes_A B \otimes_A B$$

(more precisely, $\partial_i(bm) = \partial_i(b) \cdot \partial_i(m)$, the usual identities such as $\partial_0\partial_1 = \partial_0\partial_0$ hold, and "cartesian" means that the morphisms $\partial_i : M' \otimes_{B,\partial_i} (B \otimes_A B) \to M''$ and $M'' \otimes_{B\otimes_A B,\partial_i} (B \otimes_A B \otimes_A B) \to M'''$ are isomorphisms).

The functor $E \mapsto E_{\mathcal{U}}$ becomes the functor that associates to an A-module M the diagram

$$M^* = (M \otimes_A B \rightrightarrows M \otimes_A B \otimes_A B \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; M \otimes_A B \otimes_A B \otimes_A B).$$

Its right adjoint is the functor

$$(M' \rightrightarrows M'' \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; M''') \longmapsto \ker(M' \rightrightarrows M''').$$

We must prove that the adjunction maps

$$M \to \ker(M \otimes_A B \rightrightarrows M \otimes_A B \otimes_A B)$$

and

$$\ker(M' \rightrightarrows M'') \otimes_A B \to M'$$

are isomorphisms. By (1.4.2.i), it suffices to prove this after a faithfully flat base change $A \to A'$ (with B becoming $B' = B \otimes_A A'$). Taking $A' = B$ reduces us to the case in which $U \to X$ admits a section. □

1.5 A special case: Hilbert's theorem 90

1.5.1

Let k be a field, let k' be a Galois extension of k, and let $G = \mathrm{Gal}(k'/k)$. Then the homomorphism

$$\begin{aligned} k' \otimes_k k' &\to \oplus_{\sigma \in G} k' \\ x \otimes y &\mapsto \{x \cdot \sigma(y)\}_{\sigma \in G} \end{aligned}$$

is bijective.

It follows that giving a module locally for the sieve on $\mathrm{Spec}(k)$ generated by $\mathrm{Spec}(k')$ is equivalent to giving a k'-vector space equipped with a semilinear action of G, that is:

a) a k'-vector space V',

b) for every $\sigma \in G$, an endomorphism φ_σ of the underlying group of V' such that $\varphi_\sigma(\lambda v) = \sigma(\lambda)\varphi_\sigma(v)$ for all $\lambda \in k'$ and $v \in V'$, satisfying the condition

c) for all $\sigma, \tau \in G$, $\varphi_{\tau\sigma} = \varphi_\tau \circ \varphi_\sigma$.

Let $V = V'^G$ be the group of invariants under this action of G. It is a k-vector space, and by Theorem 1.4.5 one has:

Proposition 1.5.2. *The inclusion of V in V' defines an isomorphism $V \otimes_k k' \xrightarrow{\sim} V'$.*

In particular, if V' has dimension 1 and $v' \in V'$ is nonzero, then φ_σ is determined by the constant $c(\sigma) \in k'^\times$ such that $\varphi_\sigma(v') = c(\sigma)v'$, and condition c) becomes

$$c(\tau\sigma) = c(\tau) \cdot \tau(c(\sigma)).$$

By the proposition, there exists a nonzero invariant vector $v = \mu v'$, $\mu \in k'^\times$. Hence, for every $\sigma \in G$,

$$c(\sigma) = \mu \cdot \sigma(\mu^{-1}).$$

In other words, every 1-cocycle of G with values in $k'^\times$ is a coboundary:

Corollary 1.5.3. *One has* $\mathrm{H}^1(G, k'^\times) = 0$.

1.6 Grothendieck topologies

We now transpose the definitions of the preceding paragraphs to an abstract setting that encompasses both topological spaces and schemes.

1.6.1

Let $\mathcal{S}$ be a category and let U be an object of $\mathcal{S}$. A *sieve* on U is a subset $\mathcal{U}$ of $\mathrm{Ob}(\mathcal{S}/U)$ such that if $\varphi : V \to U$ belongs to $\mathcal{U}$ and $\psi : W \to V$ is a morphism in $\mathcal{S}$, then $\varphi \circ \psi : W \to U$ belongs to $\mathcal{U}$.

If $\{\varphi_i : U_i \to U\}$ is a family of morphisms, the sieve generated by the U_i is, by definition, the set of morphisms $\varphi : V \to U$ that factor through one of the φ_i.

If $\mathcal{U}$ is a sieve on U and $\varphi : V \to U$ is a morphism, the restriction $\mathcal{U}_V$ of $\mathcal{U}$ to V is, by definition, the sieve on V consisting of the morphisms $\psi : W \to V$ such that $\varphi \circ \psi : W \to U$ belongs to $\mathcal{U}$.

1.6.2

A *Grothendieck topology* on $\mathcal{S}$ consists of the choice, for every object U of $\mathcal{S}$, of a set $C(U)$ of sieves on U, called covering sieves, such that the following axioms hold:

a) The sieve generated by the identity of U is covering.

b) If $\mathcal{U}$ is a covering sieve on U and $V \to U$ is a morphism, then the sieve $\mathcal{U}_V$ is covering.

c) A locally covering sieve is covering. In other words, if $\mathcal{U}$ is a covering sieve on U and $\mathcal{U}'$ is a sieve on U such that, for every $V \to U$ belonging to $\mathcal{U}$, the sieve $\mathcal{U}'_V$ is covering, then $\mathcal{U}'$ is covering.

A category equipped with a Grothendieck topology is called a *site*.

1.6.3

Given a site $\mathcal{S}$, a presheaf on $\mathcal{S}$ is a contravariant functor $\mathscr{F}$ from $\mathcal{S}$ to the category of sets. For every object U of $\mathcal{S}$, the elements of $\mathscr{F}(U)$ are called sections of $\mathscr{F}$ over U. For every morphism $V \to U$ and every $s \in \mathscr{F}(U)$, we denote by $s|V$ (s restricted to V) the image of s in $\mathscr{F}(V)$.

If $\mathcal{U}$ is a sieve on U, a section given $\mathcal{U}$-locally consists, for every $V \to U$ belonging to $\mathcal{U}$, of a section $s_V \in \mathscr{F}(V)$ such that, for every morphism $W \to V$, $s_V|W = s_W$. The presheaf $\mathscr{F}$ is called a *sheaf* if, for every object U of $\mathcal{S}$, every covering sieve $\mathcal{U}$ on U, and every section $\{s_V\}$ given $\mathcal{U}$-locally, there exists a unique section $s \in \mathscr{F}(U)$ such that $s|V = s_V$ for every $V \to U$ belonging to $\mathcal{U}$.

Abelian sheaves are defined similarly by replacing the category of sets with the category of abelian groups. One shows that the category of abelian sheaves on $\mathcal{S}$ is an abelian category with enough injectives. A sequence $\mathscr{F} \xrightarrow{f} \mathscr{G} \xrightarrow{g} \mathscr{H}$ of sheaves is exact if, for every object U of $\mathcal{S}$ and every $s \in \mathscr{G}(U)$ such that $g(s) = 0$, there exists locally a t such that $f(t) = s$; i.e. there exists a covering sieve $\mathcal{U}$ on U and, for every $V \in \mathcal{U}$, a section t_V of $\mathscr{F}$ on V such that $f(t_V) = s|V$.

1.6.4 Examples

We have seen two examples above.

a) Let X be a topological space and let $\mathcal{S}$ be the category whose objects are the open subsets of X and whose morphisms are the natural inclusions. The Grothendieck topology on $\mathcal{S}$ corresponding to the usual topology of X is the one for which a sieve $\mathcal{U}$ on an open subset U of X is covering if the union of the open subsets belonging to the sieve equals U. It is clear that the category of sheaves on $\mathcal{S}$ is equivalent to the category of sheaves on X in the usual sense.

b) Let X be a scheme and let $\mathcal{S}$ be the category of schemes over X. The fpqc (faithfully flat quasi-compact) topology on $\mathcal{S}$ is the Grothendieck topology for which a sieve on an X-scheme U is covering if it is generated by a finite family of flat morphisms whose images cover U.

1.6.5 Cohomology

We shall always assume that the category $\mathcal{S}$ has a final object X. The global sections of an abelian sheaf $\mathscr{F}$, denoted by $\Gamma\mathscr{F}$ or $\mathrm{H}^0(X, \mathscr{F})$, form the group $\mathscr{F}(X)$. The functor $\mathscr{F} \mapsto \Gamma\mathscr{F}$ is a left exact functor from the category of abelian sheaves on $\mathcal{S}$ to the category of abelian groups; its derived functors (or satellites) are denoted by $\mathrm{H}^i(X, -)$. These cohomology groups represent the obstructions to passing from the local to the global. By definition, if $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ is an exact sequence of abelian sheaves, then there is a long exact cohomology sequence:

$$0 \longrightarrow \mathrm{H}^0(X, \mathscr{F}) \longrightarrow \mathrm{H}^0(X, \mathscr{G}) \longrightarrow \mathrm{H}^0(X, \mathscr{H}) \longrightarrow \mathrm{H}^1(X, \mathscr{F}) \longrightarrow \cdots$$

$$\ldots \longrightarrow \mathrm{H}^n(X, \mathscr{F}) \longrightarrow \mathrm{H}^n(X, \mathscr{G}) \longrightarrow \mathrm{H}^n(X, \mathscr{H}) \longrightarrow \mathrm{H}^{n+1}(X, \mathscr{F}) \longrightarrow \cdots$$

1.6.6

Given an abelian sheaf $\mathscr{F}$ on $\mathcal{S}$, an *$\mathscr{F}$-torsor* is a sheaf $\mathscr{G}$ equipped with an action $\mathscr{F} \times \mathscr{G} \to \mathscr{G}$ of $\mathscr{F}$ such that locally (after restriction to all the objects of a covering sieve of the final object X), $\mathscr{G}$ with its $\mathscr{F}$-action is isomorphic to $\mathscr{F}$ with the canonical translation action $\mathscr{F} \times \mathscr{F} \to \mathscr{F}$.

One can show that $\mathrm{H}^1(X, \mathscr{F})$ may be interpreted as the set of isomorphism classes of $\mathscr{F}$-torsors.

2 The étale topology

We specialize the definitions of the preceding chapter to the étale topology of a scheme X (2.1, 2.2, 2.3). When X is the spectrum of a field K, the corresponding cohomology agrees with the Galois cohomology of K (2.4).

2.1 The étale topology

We begin by recalling some facts about étale morphisms.

Definition 2.1.1. *Let A be a (commutative) ring. An A-algebra B is called étale if B is an A-algebra of finite presentation and the following equivalent conditions hold:*

a) For every A-algebra C and every square-zero ideal J of C, the canonical map

$$\hom_{A\text{-alg}}(B, C) \to \hom_{A\text{-alg}}(B, C/J)$$

is a bijection.

b) B is a flat A-module and $\Omega_{B/A} = 0$ (where $\Omega_{B/A}$ denotes the module of relative differentials).

c) Let $B = A[X_1, \dots, X_n]/I$ be a presentation of B. Then, for every prime ideal $\mathfrak{p}$ of $A[X_1, \dots, X_n]$ containing I, there exist polynomials $P_1, \dots, P_n \in I$ such that $I_{\mathfrak{p}}$ is generated by the images of $P_1, \dots, P_n$ and $\det(\partial P_i/\partial X_j) \notin \mathfrak{p}$.

(cf. [23, I] or [30, V])

A morphism of schemes $f : X \to S$ is called *étale* if, for every $x \in X$, there exist an affine open neighbourhood $U = \mathrm{Spec}(A)$ of $f(x)$ and an affine open neighbourhood $V = \mathrm{Spec}(B)$ of x in $X \times_S U$ such that B is an étale A-algebra.

2.1.2 Examples

a) If A is a field, an A-algebra B is étale if and only if it is a finite product of separable extensions of A.

b) If X and S are schemes of finite type over $\mathbb{C}$, a morphism $f : X \to S$ is étale if and only if its analytification $f^{\mathrm{an}} : X^{\mathrm{an}} \to S^{\mathrm{an}}$ is a local isomorphism.

2.1.3 Consequences

a) (base change) If $f : X \to S$ is an étale morphism, then so is $f_{S'} : X \times_S S' \to S'$ for every morphism $S' \to S$.

b) If $f : X \to S$ and $g : Y \to S$ are two étale morphisms, then every S-morphism from X to Y is étale.

c) (descent) Let $f : X \to S$ be a morphism. If there exists a faithfully flat morphism $S' \to S$ such that $f_{S'} : X \times_S S' \to S'$ is étale, then f is étale.

2.1.4

Let X be a scheme, and let $\mathcal{S}$ be the category of étale X-schemes. By (2.1.3.c), every morphism in $\mathcal{S}$ is étale. The *étale topology* on $\mathcal{S}$ is the topology for which a sieve on U is covering if it is generated by a finite family of morphisms $\varphi_i : U_i \to U$ such that the union of the images of the φ_i covers U. The *étale site* of X, denoted by X_{et}, is the site defined by $\mathcal{S}$ with the étale topology.

2.2 Examples of sheaves

2.2.1 Constant sheaf

Let C be an abelian group and, for simplicity, suppose that X is noetherian. We denote by $\underline{C}_X$ (or simply C if there is no ambiguity) the sheaf defined by $U \mapsto C^{\pi_0(U)}$, where $\pi_0(U)$ is the (finite) set of connected components of U. The most important case will be $C = \mathbb{Z}/n$. Thus, by definition,

$$\mathrm{H}^0(X, \mathbb{Z}/n) = (\mathbb{Z}/n)^{\pi_0(X)}.$$

Moreover, $\mathrm{H}^1(X, \mathbb{Z}/n)$ is the set of isomorphism classes of $\mathbb{Z}/n$-torsors (1.6.6), in other words, of finite Galois étale coverings of X with group $\mathbb{Z}/n$. In particular, if X is connected and $\pi_1(X)$ is its fundamental group for a chosen base point, then

$$\mathrm{H}^1(X, \mathbb{Z}/n) = \hom(\pi_1(X), \mathbb{Z}/n).$$

2.2.2 Multiplicative group

We denote by $\mathbb{G}_{m,X}$ (or $\mathbb{G}_m$ if there is no ambiguity) the sheaf defined by $U \mapsto \Gamma(U, \mathscr{O}_U^\times)$; it is indeed a sheaf by the faithfully flat descent theorem (1.4.5). By definition,

$$\mathrm{H}^0(X, \mathbb{G}_m) = \mathrm{H}^0(X, \mathscr{O}_X)^\times;$$

in particular, if X is reduced, connected, and proper over an algebraically closed field k, then

$$\mathrm{H}^0(X, \mathbb{G}_m) = k^\times.$$

Proposition 2.2.3. *There is an isomorphism*

$$\mathrm{H}^1(X, \mathbb{G}_m) = \mathrm{Pic}(X),$$

where $\mathrm{Pic}(X)$ *is the group of isomorphism classes of invertible sheaves on* X*.*

Proof. Let $*$ be the functor that associates to an invertible sheaf $\mathscr{L}$ on X the following presheaf $\mathscr{L}^*$ on X_{et}: for $\varphi : U \to X$ étale,

$$\mathscr{L}^*(U) = \mathrm{Isom}_U(\mathscr{O}_U, \varphi^*\mathscr{L}).$$

By (1.4.2.i) and (1.4.5) (full faithfulness), this presheaf is a sheaf; indeed, it is a $\mathbb{G}_m$-torsor. One checks immediately that

a) the functor $*$ is compatible with (étale) localization;

b) it induces an equivalence from the category of trivial invertible sheaves (i.e. isomorphic to $\mathscr{O}_X$) to the category of trivial $\mathbb{G}_m$-torsors: $\mathscr{L}$ is trivial if and only if $\mathscr{L}^*$ is trivial.

Moreover, by (1.4.2.ii) and (1.4.5),

c) the notion of an invertible sheaf is local for the étale topology.

It follows formally from a), b), and c) that $*$ is an equivalence between the category of invertible sheaves on X and that of $\mathbb{G}_m$-torsors on X_{et}; it induces the desired isomorphism. The inverse equivalence is constructed as follows. If $\mathscr{T}$ is a $\mathbb{G}_m$-torsor, there exists a finite étale covering $\{U_i\}$ of X such that the torsors $\mathscr{T}/U_i$ are trivial; $\mathscr{T}$ is then trivial on every V étale over X belonging to the sieve $\mathcal{U} \subset X_{\mathrm{et}}$ generated by $\{U_i\}$. On every $V \in \mathcal{U}$, $\mathscr{T}|V$ corresponds to an invertible sheaf $\mathscr{L}_V$ (by b)), and the $\mathscr{L}_V$ form an invertible sheaf given $\mathcal{U}$-locally, $\mathscr{L}_{\mathcal{U}}$ (by a)). By c), the latter comes from an invertible sheaf $\mathscr{L}(\mathscr{T})$ on X, and $\mathscr{T} \mapsto \mathscr{L}(\mathscr{T})$ is the desired inverse of $*$. □

2.2.4 Roots of unity

For every integer $n > 0$, the sheaf of nth roots of unity, denoted by $\boldsymbol{\mu}_n$, is the kernel of the nth-power map on $\mathbb{G}_m$. If X is a scheme over a separably closed field k and n is invertible in k, the choice of a primitive nth root of unity $\zeta \in k$ defines an isomorphism $i \mapsto \zeta^i$ from $\mathbb{Z}/n$ to $\boldsymbol{\mu}_n$.

The relation between cohomology with coefficients in $\boldsymbol{\mu}_m$ and cohomology with coefficients in $\mathbb{G}_m$ is given by the cohomology exact sequence deduced from the following theorem.

Theorem 2.2.5 (Kummer theory)**.** *If n is invertible on X, the nth-power map on $\mathbb{G}_m$ is an epimorphism of sheaves. Hence there is an exact sequence*

$$0 \to \boldsymbol{\mu}_n \to \mathbb{G}_m \to \mathbb{G}_m \to 0.$$

Proof. Let $U \to X$ be an étale morphism and let $a \in \mathbb{G}_m(U) = \Gamma(U, \mathscr{O}_U^\times)$. Since n is invertible on U, the equation $T^n - a = 0$ is separable; in other words, $U' = \operatorname{Spec}(\mathscr{O}_U[T]/(T^n - a))$ is étale over U. Moreover, $U' \to U$ is surjective and this element has an nth root on U', which proves the result. □

2.3 Fibres, direct images

2.3.1

A *geometric point* of X is a morphism $\bar{x} \to X$, where $\bar{x}$ is the spectrum of a separably closed field $k(\bar{x})$. By abuse of notation, we denote it by $\bar{x}$, leaving the morphism $\bar{x} \to X$ implicit. If x is the image of $\bar{x}$ in X, we say that $\bar{x}$ is centred at x. If $k(\bar{x})$ is an algebraic extension of the residue field $k(x)$, then $\bar{x}$ is called an *algebraic* geometric point of X.

An *étale neighbourhood* of $\bar{x}$ is a commutative diagram

$$\begin{array}{ccc} & & U \\ & \nearrow & \downarrow \\ \bar{x} & \longrightarrow & X, \end{array}$$

where $U \to X$ is étale.

The *strict localization* of X at $\bar{x}$ is the ring $\mathscr{O}_{X,\bar{x}} = \varinjlim \Gamma(U, \mathscr{O}_U)$, where the direct limit is taken over the étale neighbourhoods of $\bar{x}$. It is a strictly henselian local ring whose residue field is the separable closure in $k(\bar{x})$ of the residue field $k(x)$ of X at x. It plays the role of the local ring for the étale topology.

2.3.2

Given a sheaf $\mathscr{F}$ on X_{et}, the *fibre* of $\mathscr{F}$ at $\bar{x}$ is the set (resp. group, ...) $\mathscr{F}_{\bar{x}} = \varinjlim \mathscr{F}(U)$, where the direct limit is again taken over the étale neighbourhoods of $\bar{x}$.

A homomorphism of sheaves $\mathscr{F} \to \mathscr{G}$ is a mono-/epi-/isomorphism if and only if the induced morphisms $\mathscr{F}_{\bar{x}} \to \mathscr{G}_{\bar{x}}$ on fibres have the corresponding property at every geometric point of X. If X is of finite type over an algebraically closed field, it suffices that this hold at the rational points of X.

2.3.3

If $f : X \to Y$ is a morphism of schemes and $\mathscr{F}$ is a sheaf on X_{et}, the *direct image* $f_*\mathscr{F}$ of $\mathscr{F}$ under f is the sheaf on Y_{et} defined by $f_*\mathscr{F}(V) = \mathscr{F}(X \times_Y V)$ for every V étale over Y. The functor $f_* : \operatorname{Sh}(X_{\text{et}}) \to \operatorname{Sh}(Y_{\text{et}})$

is left exact. Its right derived functors $\mathsf{R}^q f_*$ are called the higher direct images. If $\bar{y}$ is a geometric point of Y, then

$$(\mathsf{R}^q f_* \mathscr{F})_{\bar{y}} = \varinjlim \mathrm{H}^q(V \times_Y X, \mathscr{F}),$$

where the direct limit is taken over the étale neighbourhoods V of $\bar{y}$.

Let $\mathscr{O}_{Y,\bar{y}}$ be the strict localization of Y at $\bar{y}$, let $\widetilde{Y} = \mathrm{Spec}(\mathscr{O}_{Y,\bar{y}})$, and let $\widetilde{X} = X \times_Y \widetilde{Y}$. The sheaf $\mathscr{F}$ may be extended to $\widetilde{X}_{\mathrm{et}}$ (this is a special case of the general notion of inverse image) as follows. Let $\widetilde{U}$ be a scheme étale over $\widetilde{X}$. Then there exist an étale neighbourhood V of $\bar{y}$ and a scheme U étale over $X \times_Y V$ such that $\widetilde{U} = U \times_V \widetilde{Y}$; set

$$\mathscr{F}(\widetilde{U}) = \varinjlim \mathscr{F}\left(U \times_V V'\right),$$

where the direct limit is taken over the étale neighbourhoods V' of $\bar{y}$ that dominate V. With this definition,

$$(\mathsf{R}^q f_* \mathscr{F})_{\bar{y}} = \mathrm{H}^q(\widetilde{X}, \mathscr{F}).$$

The functor f_* has a left adjoint f^*, the "inverse image" functor. If $\bar{x}$ is a geometric point of X and $f(\bar{x})$ its image in Y, then $(f^*\mathscr{F})_{\bar{x}} = \mathscr{F}_{f(\bar{x})}$. This formula shows that f^* is exact. Consequently, f_* sends injective sheaves to injective sheaves, and the spectral sequence for the composite functor $\Gamma \circ f_*$ (resp. $g_* f_*$) gives the following theorem.

Theorem 2.3.4 (Leray spectral sequence)**.** *Let $\mathscr{F}$ be an abelian sheaf on X_{et} and let $f : X \to Y$ be a morphism of schemes (resp. let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of schemes). There is a spectral sequence*

$$E_2^{pq} = \mathrm{H}^p(Y, \mathsf{R}^q f_* \mathscr{F}) \Rightarrow \mathrm{H}^{p+q}(X, \mathscr{F})$$

(resp. $E_2^{pq} = \mathsf{R}^p g_* \mathsf{R}^q f_* \mathscr{F} \Rightarrow \mathsf{R}^{p+q}(gf)_* \mathscr{F}$*).*

Corollary 2.3.5. *If $\mathsf{R}^q f_* \mathscr{F} = 0$ for every $q > 0$, then $\mathrm{H}^p(Y, f_* \mathscr{F}) = \mathrm{H}^p(X, \mathscr{F})$ (resp. $\mathsf{R}^p g_*(f_* \mathscr{F}) = \mathsf{R}^p(gf)_* \mathscr{F}$) for every $p \geqslant 0$.*

This applies in particular in the following case:

Proposition 2.3.6. *Let $f : X \to Y$ be a finite morphism (or even, by passage to the limit, an integral morphism), and let $\mathscr{F}$ be an abelian sheaf on X. Then $\mathsf{R}^q f_* \mathscr{F} = 0$ for every $q > 0$.*

Indeed, let $\bar{y}$ be a geometric point of Y, let $\widetilde{Y}$ be the spectrum of the strict localization of Y at $\bar{y}$, and let $\widetilde{X} = X \times_Y \widetilde{Y}$. By the preceding discussion, it suffices to prove that $\mathrm{H}^q(\widetilde{X}, \mathscr{F}) = 0$ for every $q > 0$. But $\widetilde{X}$ is the spectrum of a product of strictly henselian local rings (cf. [30, I]), and the functor $\Gamma(\widetilde{X}, -)$ is exact because every surjective étale $\widetilde{X}$-scheme admits a section. This proves the assertion.

2.4 Galois cohomology

For $X = \mathrm{Spec}(K)$, the spectrum of a field, we shall see that étale cohomology can be identified with Galois cohomology.

2.4.1

We begin with a topological analogy. If K is the function field of an integral affine algebraic variety $Y = \mathrm{Spec}(A)$ over $\mathbb{C}$, then $K = \varinjlim_{f \in A} A[1/f]$.

In other words, $X = \varprojlim U$, where U runs over the open subsets of Y. It is known that there exist arbitrarily small Zariski open subsets that are $K(\pi, 1)$ spaces for the classical topology. It is therefore natural to regard $\mathrm{Spec}(K)$ itself as a $K(\pi, 1)$, where π is the fundamental group (in the algebraic sense) of X, namely the Galois group of $\bar{K}/K$, where $\bar{K}$ is a separable closure of K.

2.4.2

More precisely, let K be a field, let $\bar{K}$ be a separable closure of K, and let $G = \mathrm{Gal}(\bar{K}/K)$ be the topological Galois group. To every finite étale K-algebra A (a finite product of separable extensions of K), associate the finite set $\hom_K(A, \bar{K})$. The Galois group G acts on this set through a discrete (hence finite) quotient. If $A = K[T]/(F)$, this set is identified with the set of roots of F in $\bar{K}$. Galois theory, in the form given to it by Grothendieck, says:

Proposition 2.4.3. *The functor*

$$\left\{ \begin{array}{c} K\text{-algebras} \\ \text{finite étale} \end{array} \right\} \to \left\{ \begin{array}{c} \text{finite sets on which} \\ G \text{ acts continuously} \end{array} \right\}$$

that associates $\hom_K(A, \bar{K})$ *to an étale algebra* A *is an anti-equivalence of categories.*

This yields an analogous description of sheaves for the étale topology on $\mathrm{Spec}(K)$:

Proposition 2.4.4. *The functor*

$$\left\{ \begin{array}{c} \text{étale sheaves} \\ \text{on } \mathrm{Spec}(K) \end{array} \right\} \to \left\{ \begin{array}{c} \text{sets on which} \\ G \text{ acts continuously} \end{array} \right\}$$

that associates to a sheaf $\mathscr{F}$ *its fibre* $\mathscr{F}_{\bar{K}}$ *at the geometric point* $\mathrm{Spec}(\bar{K})$ *is an equivalence of categories.*

The group G is said to act continuously on a set E if the stabilizer of every element of E is an open subgroup of G. The inverse functor is described in the evident way: let A be a finite étale K-algebra, let $U = \mathrm{Spec}(A)$, and let $U(\bar{K}) = \hom_K(A, \bar{K})$ be the corresponding G-set; then $\mathscr{F}(U) = \hom_{G\text{-set}}(U(\bar{K}), \mathscr{F}_{\bar{K}})$.

In particular, if $X = \mathrm{Spec}(K)$, then $\mathscr{F}(X) = \mathscr{F}_{\bar{K}}^G$. Restricting to abelian sheaves and passing to derived functors gives canonical isomorphisms

$$\mathrm{H}^q(X_{\mathrm{et}}, \mathscr{F}) = \mathrm{H}^q(G, \mathscr{F}_{\bar{K}})$$

2.4.5 Examples

a) The constant sheaf $\mathbb{Z}/n$ corresponds to $\mathbb{Z}/n$ with the trivial action of G.

b) The sheaf $\boldsymbol{\mu}_n$ of nth roots of unity corresponds to the group $\boldsymbol{\mu}_n(\bar{K})$ of nth roots of unity in $\bar{K}$, with the natural action of G.

c) The sheaf $\mathbb{G}_m$ corresponds to the group $\bar{K}^\times$ with the natural action of G.

3 The cohomology of curves

For topological spaces, unwindings using the Künneth formula and simplicial decompositions reduce the calculation of cohomology to the interval $I = [0, 1]$, for which $\mathrm{H}^0(I, \mathbb{Z}) = \mathbb{Z}$ and $\mathrm{H}^q(I, \mathbb{Z}) = 0$ for $q > 0$.

In our case, the unwindings lead to more complicated objects, namely curves over an algebraically closed field; in this chapter we calculate their cohomology. The situation is more complicated than in the topological case, because the cohomology groups vanish only for $q > 2$. The essential ingredient in the calculations is the vanishing of the Brauer group of the function field of such a curve (Tsen's theorem, 3.2).

3.1 The Brauer group

We first recall its classical definition.

Definition 3.1.1. *Let K be a field and let A be a finite-dimensional K-algebra. The algebra A is called a central simple algebra over K if the following equivalent conditions hold:*

a) A has no nontrivial two-sided ideal, and its centre is K.

b) There exists a finite Galois extension K'/K such that $A_{K'} = A \otimes_K K'$ is isomorphic to an algebra of square matrices over K'.

c) A is K-isomorphic to an algebra of square matrices over a division algebra with centre K.

Two such algebras are called equivalent if the division algebras associated with them by c) are K-isomorphic. If the algebras have the same dimension, this is equivalent to their being K-isomorphic. Tensor product induces, on passing to the quotient, an abelian group structure on the set of equivalence classes. This group is classically called the *Brauer group* of K and is denoted by $\mathrm{Br}(K)$.

3.1.2

We denote by $\mathrm{Br}(n, K)$ the set of K-isomorphism classes of K-algebras A such that there exists a finite Galois extension K' of K for which $A_{K'}$ is isomorphic to the algebra $M_n(K')$ of $n \times n$ square matrices over K'. By definition, $\mathrm{Br}(K)$ is the union of the subsets $\mathrm{Br}(n, K)$ for $n \in \mathbb{N}$. Let $\bar{K}$ be an algebraic closure of K and let $G = \mathrm{Gal}(\bar{K}/K)$. The set $\mathrm{Br}(n, K)$ is the set of "forms" of $M_n(\bar{K})$, and hence is canonically isomorphic to $\mathrm{H}^1\left(G, \mathrm{Aut}(M_n(\bar{K}))\right)$.

Every automorphism of $M_n(\bar{K})$ is inner. Consequently, $\mathrm{Aut}(M_n(\bar{K}))$ is identified with the projective linear group $\mathrm{PGL}(n, \bar{K})$, and there is a canonical bijection

$$\theta_n : \mathrm{Br}(n, K) \xrightarrow{\sim} \mathrm{H}^1\left(G, \mathrm{PGL}(n, \bar{K})\right).$$

On the other hand, the exact sequence

$$1 \to \bar{K}^\times \to \mathrm{GL}(n, \bar{K}) \to \mathrm{PGL}(n, \bar{K}) \to 1, \tag{3.1.2.1}$$

defines a coboundary operator

$$\Delta_n : \mathrm{H}^1\left(G, \mathrm{PGL}(n, \bar{K})\right) \to \mathrm{H}^2(G, \bar{K}^\times).$$

Composing θ_n and Δ_n gives a map

$$\delta_n : \mathrm{Br}(n, K) \to \mathrm{H}^2(G, \bar{K}^\times).$$

It is readily checked that the maps δ_n are compatible and define a group homomorphism

$$\delta : \mathrm{Br}(K) \to \mathrm{H}^2(G, \bar{K}^\times).$$

Proposition 3.1.3. *The homomorphism $\delta : \mathrm{Br}(K) \to \mathrm{H}^2(G, \bar{K}^\times)$ is bijective.*

This follows from the next two lemmas.

Lemma 3.1.4. *The map $\Delta_n : \mathrm{H}^1\left(G, \mathrm{PGL}(n, \bar{K})\right) \to \mathrm{H}^2(G, \bar{K}^\times)$ is injective.*

By [36], Corollary to Proposition I-44, it suffices to check that whenever the exact sequence (3.1.2.1) is twisted by an element of $\mathrm{H}^1\left(G, \mathrm{PGL}(n, \bar{K})\right)$, the H^1 of the middle group is trivial. This middle group is the group of $\bar{K}$-points of the multiplicative group of a central simple algebra A of rank n^2 over K. To prove that $\mathrm{H}^1(G, A_{\bar{K}}^{\times}) = 0$, interpret $A^{\times}$ as the automorphism group of the free A-module L of rank 1, and H^1 as the set of "forms" of L—A-modules of rank n^2 over K, which are automatically free.

Lemma 3.1.5. *Let $\alpha \in \mathrm{H}^2(G, \bar{K}^{\times})$, let K' be a finite extension of K contained in $\bar{K}$, let $n = [K' : K]$, and let $G' = \mathrm{Gal}(\bar{K}/K')$. If the image of α in $\mathrm{H}^2(G', \bar{K}^{\times})$ is zero, then α belongs to the image of Δ_n.*

First note that

$$\mathrm{H}^2(G', \bar{K}^{\times}) \simeq \mathrm{H}^2\left(G, (\bar{K} \otimes_K K')^{\times}\right).$$

(Geometrically, if $x = \mathrm{Spec}(K)$, $x' = \mathrm{Spec}(K')$, and $\pi : x' \to x$ is the canonical morphism, then $\mathsf{R}^q\pi_*(\mathbb{G}_{m,x'}) = 0$ for $q > 0$ and consequently $\mathrm{H}^q(x', \mathbb{G}_{m,x'}) \simeq \mathrm{H}^q(x, \pi_*\mathbb{G}_{m,x'})$ for $q \geqslant 0$.)

Moreover, the choice of a basis of K' as a vector space over K defines a homomorphism

$$(\bar{K} \otimes_K K')^{\times} \to \mathrm{GL}(n, \bar{K})$$

that sends an element x to the endomorphism of $\bar{K} \otimes_K K'$ given by multiplication by x. There is then a commutative diagram with exact rows:

$$\begin{array}{ccccccccc}
1 & \longrightarrow & \bar{K}^{\times} & \longrightarrow & (\bar{K} \otimes_K K')^{\times} & \longrightarrow & (\bar{K} \otimes_K K')^{\times}/\bar{K}^{\times} & \longrightarrow & 1 \\
 & & \| & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \bar{K}^{\times} & \longrightarrow & \mathrm{GL}(n, \bar{K}) & \longrightarrow & \mathrm{PGL}(n, \bar{K}) & \longrightarrow & 1
\end{array}$$

The lemma follows from the commutative diagram obtained by passing to cohomology:

$$\begin{array}{ccccc}
\mathrm{H}^1\left(G, (\bar{K} \otimes_K K')^{\times}/\bar{K}^{\times}\right) & \longrightarrow & \mathrm{H}^2(G, \bar{K}^{\times}) & \longrightarrow & \mathrm{H}^2\left(G, (\bar{K} \otimes_K K')^{\times}\right) \\
\downarrow & & \| & & \\
\mathrm{H}^1\left(G, \mathrm{PGL}(n, \bar{K})\right) & \xrightarrow{\Delta_n} & \mathrm{H}^2(G, \bar{K}^{\times}). & &
\end{array}$$

Proposition 3.1.6. *Let K be a field, let $\bar{K}$ be an algebraic closure of K, and let $G = \mathrm{Gal}(\bar{K}/K)$. Suppose that $\mathrm{Br}(K') = 0$ for every finite extension K' of K. Then:*

i) $\mathrm{H}^q(G, \bar{K}^{\times}) = 0$ *for every* $q > 0$.

ii) $\mathrm{H}^q(G, \mathscr{F}) = 0$ *for every torsion G-module $\mathscr{F}$ and every $q \geqslant 2$.*

(For the proof, cf. [36].)

3.2 Tsen's theorem

Definition 3.2.1. *A field K is called C_1 if every nonconstant homogeneous polynomial $f(x_1, \ldots, x_n)$ of degree $d < n$ has a nontrivial zero.*

Proposition 3.2.2. *If a field K is C_1, then $\mathrm{Br}(K) = 0$.*

We must show that every division algebra D with centre K and finite over K is equal to K. Let r^2 be the degree of D over K and let $\mathrm{Nrd} : D \to K$ be the reduced norm.

(Locally for the étale topology on K, D is isomorphic—not canonically—to a matrix algebra M_r, and the reduced norm agrees with the determinant map. This is well defined independently of the chosen isomorphism between D and M_r, because every automorphism of M_r is inner and similar matrices have the same determinant. This map, defined locally for the étale topology, descends, by its local uniqueness, to a map $\mathrm{Nrd} : D \to K$.)

The only zero of Nrd is the zero element of D, since for $x \neq 0$ one has $\mathrm{Nrd}(x) \cdot \mathrm{Nrd}(x^{-1}) = 1$. On the other hand, if $\{e_1, \ldots, e_{r^2}\}$ is a basis of D over K and $x = \sum x_i e_i$, then $\mathrm{Nrd}(x)$ is a homogeneous polynomial $\mathrm{Nrd}(x_1, \ldots, x_{r^2})$ of degree r (this is clear locally for the étale topology). Since K is C_1, one has $r^2 \leqslant r$, that is, $r = 1$ and $D = K$.

Theorem 3.2.3 (Tsen)**.** *Let k be an algebraically closed field and let K be an extension of k of transcendence degree 1. Then K is C_1.*

First suppose that $K = k(X)$. Let

$$f(T) = \sum a_{i_1,\ldots i_n} T_1^{i_1} \cdots T_n^{i_n}$$

be a homogeneous polynomial of degree $d < n$ with coefficients in $k(X)$. After multiplying the coefficients by a common denominator, we may suppose that they lie in $k[X]$. Let $\delta = \sup \deg(a_{i_1 \ldots i_n})$. We seek a nontrivial zero in $k[X]$ by the method of undetermined coefficients, writing each T_i $(i = 1, \ldots, n)$ as a polynomial of degree N in X. The equation $f(T) = 0$ then becomes a system of homogeneous equations in the $n \times (N+1)$ coefficients of the polynomials $T_i(X)$, expressing the vanishing of the coefficients of the polynomial in X obtained by replacing T_i with $T_i(X)$. This polynomial has degree at most $\delta + Nd$, so there are $\delta + Nd + 1$ equations in $n \times (N+1)$ variables. Since k is algebraically closed, this system has a nontrivial solution if $n(N+1) > Nd + \delta + 1$, which holds for sufficiently large N when $d < n$.

To prove the theorem in general, it plainly suffices to prove it when K is a finite extension of the purely transcendental extension $k(X)$ of k. Let $f(T) = f(T_1, \ldots, T_n)$ be a homogeneous polynomial of degree $d < n$ with coefficients in K. Let $s = [K : k(X)]$ and let $e_1, \ldots, e_s$ be a basis of K over $k(X)$. Introduce sn new variables U_{ij} such that $T_i = \sum U_{ij} e_j$. For $f(T)$ to have a nontrivial zero in K, it suffices that the polynomial $g(U_{ij}) = N_{K/k(X)}(f(T))$ have a nontrivial zero in $k(X)$. But g is a polynomial of degree sd in sn variables, which proves the result.

Corollary 3.2.4. *Let k be an algebraically closed field and let K be an extension of k of transcendence degree 1. Then the étale cohomology groups $\mathrm{H}^q(\mathrm{Spec}(K), \mathbb{G}_m)$ vanish for every $q > 0$.*

3.3 Cohomology of smooth curves

From now on, unless expressly stated otherwise, all cohomology groups are étale cohomology groups.

Proposition 3.3.1. *Let k be an algebraically closed field and let X be a connected nonsingular projective curve over k. Then*

$$\begin{aligned} \mathrm{H}^0(X, \mathbb{G}_m) &= k^\times, \\ \mathrm{H}^1(X, \mathbb{G}_m) &= \mathrm{Pic}(X), \\ \mathrm{H}^q(X, \mathbb{G}_m) &= 0 \text{ for } q \geqslant 2. \end{aligned}$$

Let η be the generic point of X, let $j : \eta \to X$ be the canonical morphism, and let $\mathbb{G}_{m,\eta}$ be the multiplicative group of the function field $K(X)$. For every closed point x of X, let $i_x : x \to X$ be the

canonical immersion and let $\mathbb{Z}_x$ be the constant sheaf with value $\mathbb{Z}$ on x. Thus $j_*\mathbb{G}_{m,\eta}$ is the sheaf of nonzero meromorphic functions on X and $\oplus_{x\in X} i_{x*}\mathbb{Z}_x$ is the sheaf of divisors; hence there is an exact sequence of sheaves:

$$0 \longrightarrow \mathbb{G}_m \longrightarrow j_*\mathbb{G}_{m,\eta} \xrightarrow{\text{div}} \bigoplus_{x\in X} i_{x*}\mathbb{Z}_x \longrightarrow 0. \tag{3.3.1.1}$$

Lemma 3.3.2. *One has* $\mathsf{R}^q j_*\mathbb{G}_{m,\eta} = 0$ *for every* $q > 0$.

It suffices to show that the fibre of this sheaf at every closed point x of X is zero. If $\tilde{\mathscr{O}}_{X,x}$ is the henselization of X at x and K is the fraction field of $\tilde{\mathscr{O}}_{X,x}$, then

$$\mathrm{Spec}(K) = \eta \times_X \mathrm{Spec}(\tilde{\mathscr{O}}_{X,x}),$$

and hence $(\mathsf{R}^q j_*\mathbb{G}_{m,\eta})_x = \mathrm{H}^q(\mathrm{Spec}(K), \mathbb{G}_m)$.

Now K is an algebraic extension of $k(X)$, hence an extension of k of transcendence degree 1; the lemma follows from (3.2.4).

Lemma 3.3.3. *One has* $\mathrm{H}^q(X, j_*\mathbb{G}_{m,\eta}) = 0$ *for every* $q > 0$.

Indeed, (3.3.2) and the Leray spectral sequence for j give

$$\mathrm{H}^q(X, j_*\mathbb{G}_{m,\eta}) = \mathrm{H}^q(\eta, \mathbb{G}_{m,\eta})$$

for every $q \geqslant 0$, and the right-hand side vanishes for $q > 0$ by (3.2.4).

Lemma 3.3.4. *One has* $\mathrm{H}^q\left(X, \bigoplus_{x\in X} i_{x*}\mathbb{Z}_x\right) = 0$ *for every* $q > 0$.

Indeed, for every closed point x of X one has $\mathsf{R}^q i_{x*}\mathbb{Z}_x = 0$ for $q > 0$, because i_x is a finite morphism (2.3.6), and

$$\mathrm{H}^q(X, i_{x*}\mathbb{Z}_x) = \mathrm{H}^q(x, \mathbb{Z}_x).$$

The right-hand side vanishes for every $q > 0$ because x is the spectrum of an algebraically closed field. (This also shows that the lemma holds more generally for every skyscraper sheaf on X.)

The preceding lemmas and the exact sequence (3.3.1.1) give

$$\mathrm{H}^q(X, \mathbb{G}_m) = 0 \text{ for } q \geqslant 2,$$

and the low-degree exact cohomology sequence

$$1 \to \mathrm{H}^0(X, \mathbb{G}_m) \to \mathrm{H}^0(X, j_*\mathbb{G}_{m,\eta}) \to \mathrm{H}^0\left(X, \bigoplus_{x\in X} i_{x*}\mathbb{Z}_x\right) \to \mathrm{H}^1(X, \mathbb{G}_m) \to 1$$

which is precisely the exact sequence

$$1 \to k^\times \to k(X)^\times \to \mathrm{Div}(X) \to \mathrm{Pic}(X) \to 1.$$

Proposition 3.3.1 implies that the cohomology groups of X with values in $\mathbb{Z}/n$, for n prime to the characteristic of k, have reasonable values:

Corollary 3.3.5. *If X has genus g and n is invertible in k, then $\mathrm{H}^q(X, \mathbb{Z}/n)$ vanishes for $q > 2$ and is free over $\mathbb{Z}/n$ of ranks $1, 2g, 1$ for $q = 0, 1, 2$. Replacing $\mathbb{Z}/n$ by the isomorphic group $\boldsymbol{\mu}_n$, there are canonical isomorphisms*

$$\mathrm{H}^0(X, \boldsymbol{\mu}_n) = \boldsymbol{\mu}_n$$
$$\mathrm{H}^1(X, \boldsymbol{\mu}_n) = \mathrm{Pic}^0(X)_n$$
$$\mathrm{H}^2(X, \boldsymbol{\mu}_n) = \mathbb{Z}/n.$$

Since k is algebraically closed, $\mathbb{Z}/n$ is isomorphic (noncanonically) to $\boldsymbol{\mu}_n$. The Kummer exact sequence

$$0 \to \boldsymbol{\mu}_n \to \mathbb{G}_m \to \mathbb{G}_m \to 0,$$

and Proposition 3.3.1 yield the equalities

$$\mathrm{H}^q(X, \mathbb{Z}/n) = 0 \text{ for } q > 2,$$

and, in low degrees, the exact sequences

$$0 \longrightarrow \mathrm{H}^0(X, \boldsymbol{\mu}_n) \longrightarrow k^\times \xrightarrow{n} k^\times \longrightarrow 0$$

$$0 \longrightarrow \mathrm{H}^1(X, \boldsymbol{\mu}_n) \longrightarrow \mathrm{Pic}(X) \xrightarrow{n} \mathrm{H}^2(X, \boldsymbol{\mu}_n) \longrightarrow 0.$$

Moreover, there is an exact sequence

$$0 \longrightarrow \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}(X) \xrightarrow{\deg} \mathbb{Z} \longrightarrow 0,$$

and $\mathrm{Pic}^0(X)$ is identified with the group of k-rational points of an abelian variety of dimension g, the Jacobian of X. In such a group, multiplication by n is surjective and its kernel is a free $\mathbb{Z}/n\mathbb{Z}$-module of rank $2g$ (because n is invertible in k); this proves the corollary.

A clever unwinding using the "trace method" yields the following proposition as a corollary.

Proposition 3.3.6 ([2, IX 5.7])**.** *Let k be an algebraically closed field, let X be an algebraic curve over k, and let $\mathscr{F}$ be a torsion sheaf on X. Then:*

i) $\mathrm{H}^q(X, \mathscr{F}) = 0$ *for* $q > 2$.

ii) If X is affine, then even $\mathrm{H}^q(X, \mathscr{F}) = 0$ *for* $q > 1$.

For the proof, as well as for an exposition of the "trace method," we refer to [2, IX 5].

3.4 Unwindings

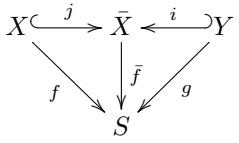
To calculate the cohomology of varieties of dimension > 1, one uses fibrations by curves, thereby reducing to the study of morphisms whose fibres have dimension $\leqslant 1$. This principle has several variants; we mention some of them.

3.4.1

Let A be a k-algebra of finite type, and let $a_1, \ldots, a_n$ be generators of A. If we put $X_0 = \mathrm{Spec}(k)$, $X_i = \mathrm{Spec}(k[a_1, \ldots, a_i])$, and $X_n = \mathrm{Spec}(A)$, the canonical inclusions $k[a_1, \ldots, a_i] \to k[a_1, \ldots, a_i, a_{i+1}]$ define morphisms $X_n \to X_{n-1} \to \cdots \to X_1 \to X_0$ whose fibres have dimension $\leqslant 1$.

3.4.2

For a smooth morphism, one can be more precise. An *elementary fibration* is a morphism of schemes $f : X \to S$ that can be embedded in a commutative diagram



satisfying the following conditions:

i) j is an open immersion, dense in every fibre, and $X = \bar{X} \smallsetminus Y$.

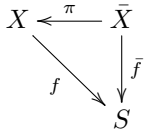
ii) $\bar{f}$ is smooth and projective, with irreducible geometric fibres of dimension 1.

iii) g is an étale covering and no fibre of g is empty.

A *good neighbourhood* relative to S is an S-scheme X for which there exist S-schemes $X = X_n, \ldots, X_0 = S$ and elementary fibrations $f_i : X_i \to X_{i-1}$, $i = 1, \ldots, n$. One can show [2, XI 3.3] that if X is a smooth scheme over an algebraically closed field k, then every rational point of X has an open neighbourhood that is a good neighbourhood (relative to $\mathrm{Spec}(k)$).

3.4.3

A proper morphism $f : X \to S$ can be unwound as follows. By Chow's lemma, there is a commutative diagram



in which π and $\bar{f}$ are projective morphisms, with π in addition an isomorphism over a dense open subset of X. Locally on S, $\bar{X}$ is a closed subscheme of a standard projective space $\mathbb{P}^n_S$.

The latter is unwound by considering the projection $\varphi : \mathbb{P}^n_S \to \mathbb{P}^1_S$ that sends the point with homogeneous coordinates $(x_0, x_1, \ldots, x_n)$ to (x_0, x_1). This is a rational map defined outside the closed subset $Y \simeq \mathbb{P}^{n-2}_S$ of $\mathbb{P}^n_S$ given by the homogeneous equations $x_0 = x_1 = 0$. Let $u : P \to \mathbb{P}^n_S$ be the blow-up with centre Y; the fibres of u have dimension $\leqslant 1$. Moreover, there is a natural morphism $v : P \to \mathbb{P}^1_S$ extending the rational map φ, and v makes P a $\mathbb{P}^1_S$-scheme locally isomorphic to the standard projective space $\mathbb{P}^{n-1}$, which may in turn be projected onto a $\mathbb{P}^1$, and so on.

3.4.4

A smooth projective variety X can be swept out by a *Lefschetz pencil*. The blow-up $\widetilde{X}$ of the intersection of the axis of the pencil with X maps to $\mathbb{P}^1$, and the fibres of this map are the hyperplane sections of X by the hyperplanes of the pencil.

4 The proper base change theorem

4.1 Introduction

This chapter is devoted to the proof and applications of the following theorem.

Theorem 4.1.1. *Let $f : X \to S$ be a proper morphism of schemes and let $\mathscr{F}$ be a torsion abelian sheaf on X. Then, for every $q \geqslant 0$, the fibre of $\mathrm{R}^q f_* \mathscr{F}$ at a geometric point s of S is isomorphic to the cohomology $\mathrm{H}^q(X_s, \mathscr{F})$ of the fibre $X_s = X \otimes_S \mathrm{Spec}\, k(s)$ of f at s.*

For a proper separated continuous map $f : X \to S$ between topological spaces (separated means that the diagonal of $X \times_S X$ is closed) and an abelian sheaf $\mathscr{F}$ on X, the analogous result is well known and elementary. Since f is closed, the $f^{-1}(V)$, with V ranging over the neighbourhoods of s, form a fundamental system of neighbourhoods of X_s, and one checks that $\mathrm{H}^\bullet(X_s, \mathscr{F}) = \varinjlim_U \mathrm{H}^\bullet(U, \mathscr{F})$, where U ranges over the neighbourhoods of X_s. In practice, X_s even has a fundamental system $\mathcal{U}$ of neighbourhoods

U of which it is a deformation retract, and for $\mathscr{F}$ constant one therefore has $\mathrm{H}^\bullet(X_s, \mathscr{F}) = \mathrm{H}^\bullet(U, \mathscr{F})$. Figuratively speaking, the special fibre swallows the general fibre.

For schemes the proof is more delicate, and it is essential to suppose that $\mathscr{F}$ is torsion ([2, XII.2]). In view of the description of the fibres of $\mathsf{R}^q f_* \mathscr{F}$ (2.3.3), Theorem (4.1.1) is essentially equivalent to the following theorem.

Theorem 4.1.2. *Let A be a strictly henselian local ring and let $S = \mathrm{Spec}(A)$. Let $f : X \to S$ be proper, and let X_0 be the closed fibre of f. Then, for every torsion abelian sheaf $\mathscr{F}$ on X and every $q \geqslant 0$, one has $\mathrm{H}^q(X, \mathscr{F}) \xrightarrow{\sim} \mathrm{H}^q(X_0, \mathscr{F})$.*

By passage to the limit, it suffices to prove the theorem when A is the strict henselization of a finitely generated $\mathbb{Z}$-algebra at a prime ideal. We first treat the case $q = 0$ or 1 and $\mathscr{F} = \mathbb{Z}/n$ (4.2). An argument based on constructible sheaves (4.3) shows moreover that it suffices to consider constant $\mathscr{F}$. On the other hand, the unwinding in (3.4.3) allows us to suppose that X_0 is a curve; in that case it remains only to prove the theorem for $q = 2$ (4.4).

Among other applications (4.6), the theorem allows one to define cohomology with proper support (4.5).

4.2 Proof for $q = 0$ or 1 and $\mathscr{F} = \mathbb{Z}/n$

The result for $q = 0$ and $\mathscr{F}$ constant is equivalent to the following proposition (Zariski's connectedness theorem).

Proposition 4.2.1. *Let A be a noetherian henselian local ring and let $S = \mathrm{Spec}(A)$. Let $f : X \to S$ be a proper morphism, and let X_0 be the closed fibre of f. Then the sets of connected components $\pi_0(X)$ and $\pi_0(X_0)$ are in bijection.*

Equivalently, we must show that the sets $\mathrm{Of}(X)$ and $\mathrm{Of}(X_0)$ of open-and-closed subsets are in bijection. The set $\mathrm{Of}(X)$ is known to correspond bijectively to the set of idempotents of $\Gamma(X, \mathscr{O}_X)$, and similarly $\mathrm{Of}(X_0)$ corresponds bijectively to the set of idempotents of $\Gamma(X_0, \mathscr{O}_{X_0})$. Thus we must show that the canonical map

$$\mathrm{Idem}\,\Gamma(X, \mathscr{O}_X) \to \mathrm{Idem}\,\Gamma(X_0, \mathscr{O}_{X_0})$$

is bijective.

Let $\mathfrak{m}$ denote the maximal ideal of A, let $\Gamma(X, \mathscr{O}_X)^\wedge$ denote the completion of $\Gamma(X, \mathscr{O}_X)$ for the $\mathfrak{m}$-adic topology, and, for every integer $n \geqslant 0$, let $X_n = X \otimes_A A/\mathfrak{m}^{n+1}$. By the finiteness theorem for proper morphisms [20, III.3.2], $\Gamma(X, \mathscr{O}_X)$ is a finite A-algebra; since A is henselian, the canonical map

$$\mathrm{Idem}\,\Gamma(X, \mathscr{O}_X) \to \mathrm{Idem}\,\Gamma(X, \mathscr{O}_X)^\wedge$$

is bijective. In particular, the canonical map

$$\mathrm{Idem}\,\Gamma(X, \mathscr{O}_X)^\wedge \to \varprojlim \mathrm{Idem}\,\Gamma(X_n, \mathscr{O}_{X_n})$$

is bijective. But since X_n and X_0 have the same underlying topological space, the canonical map

$$\mathrm{Idem}\,\Gamma(X_n, \mathscr{O}_{X_n}) \to \mathrm{Idem}\,\Gamma(X_0, \mathscr{O}_{X_0})$$

is bijective for every n, which completes the proof.

Since $\mathrm{H}^1(X, \mathbb{Z}/n)$ is in bijection with the set of isomorphism classes of Galois étale coverings of X with group $\mathbb{Z}/n$, the theorem for $q = 1$ and $\mathscr{F} = \mathbb{Z}/n$ follows from the next proposition.

Proposition 4.2.2. *Let A be a noetherian henselian local ring and let $S = \mathrm{Spec}(A)$. Let $f : X \to S$ be a proper morphism, and let X_0 be the closed fibre of f. Then the restriction functor*

$$\mathrm{Rev.\,et}(X) \to \mathrm{Rev.\,et}(X_0)$$

is an equivalence of categories.

(If X_0 is connected and a geometric point of X_0 has been chosen as base point, this says that the canonical map $\pi_1(X_0) \to \pi_1(X)$ on (profinite) fundamental groups is bijective.)

Proposition (4.2.1) shows that this functor is fully faithful. Indeed, if X' and X'' are two étale coverings of X, an X-morphism from X' to X'' is determined by its graph, which is an open-and-closed subset of $X' \times_X X''$.

It remains to show that every étale covering X'_0 of X_0 extends to an étale covering of X. Étale coverings are insensitive to nilpotent elements [23, ch.1]; consequently, X'_0 lifts uniquely to an étale covering X'_n of X_n for every $n \geqslant 0$, in other words, to an étale covering $\mathfrak{X}'$ of the formal scheme $\mathfrak{X}$ obtained by completing X along X_0. By Grothendieck's algebraization theorem for formal coherent sheaves (the existence theorem, [20, III.5]), $\mathfrak{X}'$ is the formal completion of an étale covering $\bar{X}'$ of $\bar{X} = X \otimes_A \hat{A}$.

By passage to the limit, it suffices to prove the proposition when A is the henselization of a finitely generated $\mathbb{Z}$-algebra. We can then apply Artin's approximation theorem to the functor $\mathscr{F} : (A\text{-algebras}) \to (\text{sets})$ that associates to an A-algebra B the set of isomorphism classes of étale coverings of $X \otimes_A B$. Indeed, this functor is locally of finite presentation: if B_i is a filtered direct system of A-algebras and $B = \varinjlim B_i$, then $\mathscr{F}(B) = \varinjlim \mathscr{F}(B_i)$. By Artin's theorem, given an element $\bar{\xi} \in \mathscr{F}(\hat{A})$, here the isomorphism class of $\bar{X}'$, there exists $\xi \in \mathscr{F}(A)$ having the same image as $\bar{\xi}$ in $\mathscr{F}(A/\mathfrak{m})$. In other words, there exists an étale covering X' of X whose restriction to X_0 is isomorphic to X'_0.

4.3 Constructible sheaves

In this paragraph, X is a *noetherian* scheme, and by a sheaf on X we mean an *abelian* sheaf on X_{et}.

Definition 4.3.1. *A sheaf $\mathscr{F}$ on X is called* locally constant constructible *(abbreviated l.c.c.) if it is represented by an étale covering of X.*

Definition 4.3.2. *A sheaf $\mathscr{F}$ on X is called* constructible *if it satisfies the following equivalent conditions:*

(i) There exists a finite surjective family of subschemes X_i of X such that the restriction of $\mathscr{F}$ to X_i is l.c.c.

(ii) There exist a finite family of finite morphisms $p_i : X'_i \to X$, for each i a constant constructible sheaf C_i (i.e. one defined by a finite abelian group) on X'_i, and a monomorphism $\mathscr{F} \to \prod p_{i*}C_i$.

It is readily checked that the category of constructible sheaves on X is an abelian category. Moreover, if $u : \mathscr{F} \to \mathscr{G}$ is a homomorphism of sheaves and $\mathscr{F}$ is constructible, then $\mathrm{im}(u)$ is constructible.

Lemma 4.3.3. *Every torsion sheaf $\mathscr{F}$ is a filtered direct limit of constructible sheaves.*

Indeed, if $j : U \to X$ is an étale scheme of finite type over X, an element $\xi \in \mathscr{F}(U)$ such that $n\xi = 0$ defines a homomorphism of sheaves $j_!\underline{\mathbb{Z}/n} \to \mathscr{F}$ whose image (the smallest subsheaf of $\mathscr{F}$ for which ξ is a local section) is a constructible subsheaf of $\mathscr{F}$. Clearly $\mathscr{F}$ is the direct limit of such subsheaves.

Definition 4.3.4. *Let $\mathcal{C}$ be an abelian category and let T be a functor from $\mathcal{C}$ to the category of abelian groups. The functor T is called* effaceable *in $\mathcal{C}$ if, for every object A of $\mathcal{C}$ and every $\alpha \in T(A)$, there exists a monomorphism $u : A \to M$ in $\mathcal{C}$ such that $T(u)\alpha = 0$.*

Lemma 4.3.5. *The functors* $\mathrm{H}^q(X,-)$ *for* $q>0$ *are effaceable in the category of constructible sheaves on* X.

It suffices to observe that, if $\mathscr{F}$ is a constructible sheaf, there is necessarily an integer $n>0$ such that $\mathscr{F}$ is a sheaf of $\mathbb{Z}/n$-modules. There is then a monomorphism $\mathscr{F}\hookrightarrow\mathscr{G}$, where $\mathscr{G}$ is a sheaf of $\mathbb{Z}/n$-modules and $\mathrm{H}^q(X,\mathscr{G})=0$ for every $q>0$. For example, one can take for $\mathscr{G}$ the Godement resolution $\prod_{x\in X} i_{x*}\mathscr{F}_{\bar{x}}$, where x runs over the points of X and $i_x:\bar{x}\to X$ is a geometric point centred at x. By (4.3.3), $\mathscr{G}$ is a direct limit of constructible sheaves, whence the lemma, because the functors $\mathrm{H}^q(X,-)$ commute with direct limits.

Lemma 4.3.6. *Let* $\varphi^\bullet: T^\bullet\to T'^\bullet$ *be a morphism of cohomological functors defined on an abelian category* $\mathcal{C}$ *with values in the category of abelian groups. Suppose that* T^q *is effaceable for* $q>0$, *and let* $\mathcal{E}$ *be a subset of the objects of* $\mathcal{C}$ *such that every object of* $\mathcal{C}$ *is contained in an object belonging to* $\mathcal{E}$. *Then the following conditions are equivalent:*

(i) $\varphi^q(A)$ *is bijective for every* $q\geqslant 0$ *and every* $A\in\mathrm{Ob}\,\mathcal{C}$.

(ii) $\varphi^0(M)$ *is bijective and* $\varphi^q(M)$ *is surjective for every* $q>0$ *and every* $M\in\mathcal{E}$.

(iii) $\varphi^0(A)$ *is bijective for every* $A\in\mathrm{Ob}\,\mathcal{C}$, *and* T'^q *is effaceable for every* $q>0$.

The proof is by induction on q and presents no difficulty.

Proposition 4.3.7. *Let* X_0 *be a subscheme of* X. *Suppose that, for every* $n>0$ *and every scheme* X' *finite over* X, *the canonical map*

$$\mathrm{H}^q(X',\mathbb{Z}/n)\to\mathrm{H}^q(X'_0,\mathbb{Z}/n),$$

where $X'_0=X'\times_X X_0$, *is bijective for* $q=0$ *and surjective for* $q>0$. *Then, for every torsion sheaf* $\mathscr{F}$ *on* X *and every* $q\geqslant 0$, *the canonical map*

$$\mathrm{H}^q(X,\mathscr{F})\to\mathrm{H}^q(X_0,\mathscr{F})$$

is bijective.

By passage to the limit, it suffices to prove the assertion for constructible $\mathscr{F}$. Apply Lemma (4.3.6) with $\mathcal{C}$ the category of constructible sheaves on X, $T^q=\mathrm{H}^q(X,-)$, $T'^q=\mathrm{H}^q(X_0,-)$, and $\mathcal{E}$ the set of constructible sheaves of the form $\prod p_{i*}C_i$, where $p_i:X'_i\to X$ is finite and C_i is a finite constant sheaf on X'_i.

4.4 End of the proof

By the method of fibration by curves (3.4.3), we reduce to proving the theorem in relative dimension $\leqslant 1$. By the preceding paragraph, it suffices to show the following. If S is the spectrum of a strictly henselian noetherian local ring, $f:X\to S$ is a proper morphism whose closed fibre X_0 has dimension $\leqslant 1$, and n is an integer >0, then the canonical homomorphism

$$\mathrm{H}^q(X,\mathbb{Z}/n)\to\mathrm{H}^q(X_0,\mathbb{Z}/n)$$

is bijective for $q=0$ and surjective for $q>0$.

The cases $q=0$ and 1 were treated above, and $\mathrm{H}^q(X_0,\mathbb{Z}/n)=0$ for $q\geqslant 3$; it therefore suffices to treat $q=2$. We may clearly suppose that n is a power of a prime. If $n=p^r$, where p is the characteristic of the

residue field of S, Artin–Schreier theory shows that $\mathrm{H}^2(X_0, \mathbb{Z}/p^r) = 0$. If $n = \ell^r$, $\ell \neq p$, Kummer theory gives a commutative diagram

$$\begin{array}{ccc} \mathrm{Pic}(X) & \xrightarrow{\alpha} & \mathrm{H}^2(X, \mathbb{Z}/\ell^r) \\ \downarrow & & \downarrow \\ \mathrm{Pic}(X_0) & \xrightarrow{\beta} & \mathrm{H}^2(X_0, \mathbb{Z}/\ell^r) \end{array}$$

in which β is surjective (this was seen in Chapter 3 for a smooth curve over an algebraically closed field, but similar arguments apply to any curve over a separably closed field).

Thus it suffices to prove:

Proposition 4.4.1. *Let S be the spectrum of a henselian noetherian local ring, and let $f : X \to S$ be a proper morphism whose closed fibre X_0 has dimension $\leqslant 1$. Then the canonical restriction map*

$$\mathrm{Pic}(X) \to \mathrm{Pic}(X_0)$$

is surjective. (In fact, it is enough that f be a separated *morphism of finite type.)*

To simplify the proof, we suppose that X is integral, although this is not necessary. Every invertible sheaf on X_0 is associated with a Cartier divisor (because X_0 is a curve, hence quasi-projective), so it suffices to show that the canonical map $\mathrm{Div}(X) \to \mathrm{Div}(X_0)$ is surjective.

Every divisor on X_0 is a linear combination of divisors whose support is concentrated at a single nonisolated closed point of X_0. Let x be such a point, let $t_0 \in \mathscr{O}_{X_0,x}$ be a regular nonunit in $\mathscr{O}_{X_0,x}$, and let D_0 be the divisor concentrated at x with local equation t_0. Let U be an open neighbourhood of x in X on which there is a section $t \in \Gamma(U, \mathscr{O}_U)$ lifting t_0. Let Y be the closed subset of U defined by $t = 0$; after making U sufficiently small, we may suppose that x is the only point of $Y \cap X_0$. Then Y is quasi-finite over S at x. Since S is the spectrum of a henselian local ring, it follows that $Y = Y_1 \amalg Y_2$, where Y_1 is finite over S and Y_2 does not meet X_0. Moreover, since X is separated over S, Y_1 is closed in X.

After replacing U by a smaller open neighbourhood of x, we may suppose that $Y = Y_1$, in other words, that Y is closed in X. We then define a divisor D on X lifting D_0 by setting $D|X \smallsetminus Y = 0$ and $D|U = \mathrm{div}(t)$; this makes sense because t is invertible on $U \smallsetminus Y$.

4.4.2 Remark

When f is proper, one could also give a proof similar to that of Proposition (4.2.2). Indeed, since X_0 is a curve, there is no obstruction to lifting an invertible sheaf on X_0 to the infinitesimal neighbourhoods X_n of X_0, and hence to the formal completion $\mathfrak{X}$ of X along X_0. One then concludes by applying successively Grothendieck's existence theorem and Artin's approximation theorem.

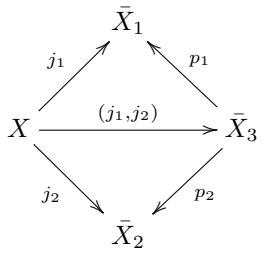
4.5 Cohomology with proper support

Definition 4.5.1. *Let X be a separated scheme of finite type over a field k. By a theorem of Nagata, there exist a scheme $\bar{X}$ proper over k and an open immersion $j : X \to \bar{X}$. For every torsion sheaf $\mathscr{F}$ on X, let $j_!\mathscr{F}$ denote the extension by 0 of $\mathscr{F}$ to $\bar{X}$, and define the groups of* cohomology with proper support *by*

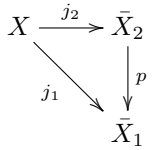
$$\mathrm{H}^q_c(X, \mathscr{F}) = \mathrm{H}^q(\bar{X}, j_!\mathscr{F}).$$

Let us show that this definition is independent of the chosen compactification $j : X \to \bar{X}$. Let $j_1 : X \to \bar{X}_1$ and $j_2 : X \to \bar{X}_2$ be two compactifications. Then X maps into $\bar{X}_1 \times \bar{X}_2$ by $x \mapsto (j_1(x), j_2(x))$,

and the closed image $\bar{X}_3$ of X under this map is a compactification of X. We thus have a commutative diagram



where p_1 and p_2, the restrictions to $\bar{X}_3$ of the natural projections, are proper morphisms. It therefore suffices to treat the case of a commutative diagram



with p proper.

Lemma 4.5.2. *One has $p_*(j_{2!}\mathscr{F}) = j_{1!}\mathscr{F}$ and $\mathsf{R}^q p_*(j_{2!}\mathscr{F}) = 0$ for $q > 0$.*

Let us note at once that the lemma suffices. Using the Leray spectral sequence for p, it gives, for every $q \geqslant 0$,

$$\mathrm{H}^q(\bar{X}_2, j_{2!}\mathscr{F}) = \mathrm{H}^q(\bar{X}_1, j_{1!}\mathscr{F}).$$

To prove the lemma, argue fibre by fibre using the base change theorem (4.1.1) for p. The result is immediate because above a point of X, p is an isomorphism, whereas above a point of $\bar{X}_1 \smallsetminus X$, $j_{2!}\mathscr{F}$ vanishes on the fibre of p.

4.5.3

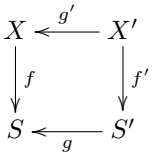
Similarly, if $f : X \to S$ is a separated morphism of finite type between noetherian schemes, there exist a proper morphism $\bar{f} : \bar{X} \to S$ and an open immersion $j : X \to \bar{X}$. The higher direct images with proper support $\mathsf{R}^q f_!$ are defined, for every torsion sheaf $\mathscr{F}$ on X, by

$$\mathsf{R}^q f_!\mathscr{F} = \mathsf{R}^q \bar{f}_*(j_!\mathscr{F}).$$

As before, one checks that this definition is independent of the chosen compactification.

Theorem 4.5.4. *Let $f : X \to S$ be a separated morphism of finite type between noetherian schemes, and let $\mathscr{F}$ be a torsion sheaf on X. Then the fibre of $\mathsf{R}^q f_!\mathscr{F}$ at a geometric point s of S is isomorphic to the cohomology with proper support $\mathrm{H}^q_c(X_s, \mathscr{F})$ of the fibre X_s of f at s.*

This is a simple variant of the proper base change theorem (4.1.1). More generally, if



is a cartesian diagram, there is a canonical isomorphism

$$g^*(\mathsf{R}^q f_!\mathscr{F}) \simeq \mathsf{R}^q f'_!({g'}^*\mathscr{F}). \tag{4.5.4.1}$$

4.6 Applications

Theorem 4.6.1 (Vanishing theorem)**.** *Let $f : X \to S$ be a separated morphism of finite type whose fibres have dimension $\leqslant n$, and let $\mathscr{F}$ be a torsion sheaf on X. Then $\mathsf{R}^q f_!\mathscr{F} = 0$ for $q > 2n$.*

By the base change theorem, we may suppose that S is the spectrum of a separably closed field. If $\dim X = n$, there exists an affine open U of X such that $\dim(X \smallsetminus U) < n$. We then have an exact sequence $0 \to \mathscr{F}_U \to \mathscr{F} \to \mathscr{F}_{X\smallsetminus U} \to 0$, and induction on n reduces us to proving the theorem for $X = U$ affine. The method of fibration by curves (3.4.1) and the base change theorem then reduce us to a curve over a separably closed field, for which the desired result follows from Tsen's theorem (3.3.6).

Theorem 4.6.2 (Finiteness theorem)**.** *Let $f : X \to S$ be a separated morphism of finite type and let $\mathscr{F}$ be a constructible sheaf on X. Then the sheaves $\mathsf{R}^q f_!\mathscr{F}$ are constructible.*

We consider only the case in which $\mathscr{F}$ is annihilated by an integer invertible on X.

To prove the theorem, one reduces to the case in which $\mathscr{F}$ is the constant sheaf $\underline{\mathbb{Z}/n}$ and $f : X \to S$ is proper and smooth with geometrically connected curve fibres of genus g. If n is invertible on X, the sheaves $\mathsf{R}^q f_*\mathscr{F}$ are then locally free of finite rank and vanish for $q > 2$ (4.6.1). Replacing $\mathbb{Z}/n$ by the sheaf $\boldsymbol{\mu}_n$, which is locally isomorphic to it (over S), there are canonical isomorphisms

$$\begin{aligned} &\mathsf{R}^0 f_*\boldsymbol{\mu}_n = \boldsymbol{\mu}_n \\ &\mathsf{R}^1 f_*\boldsymbol{\mu}_n = \underline{\mathrm{Pic}}(X/S)_n \\ &\mathsf{R}^2 f_*\boldsymbol{\mu}_n = \mathbb{Z}/n. \end{aligned} \tag{4.6.2.1}$$

Theorem 4.6.3 (Comparison with classical cohomology)**.** *Let $f : X \to S$ be a separated morphism of schemes of finite type over $\mathbb{C}$, and let $\mathscr{F}$ be a torsion sheaf on X. Denote by a superscript $(-)^{\mathrm{an}}$ the functor passing to the usual topological spaces, and by $\mathsf{R}^q f_!^{\mathrm{an}}$ the derived functors of direct image with proper support under f^{an}. Then*

$$(\mathsf{R}^q f_!\mathscr{F})^{\mathrm{an}} \simeq \mathsf{R}^q f_!^{\mathrm{an}}\mathscr{F}^{\mathrm{an}}.$$

In particular, when $S =$ a point and $\mathscr{F}$ is the constant sheaf $\mathbb{Z}/n$,

$$\mathrm{H}^q_c(X, \mathbb{Z}/n) \simeq \mathrm{H}^q_c(X^{\mathrm{an}}, \mathbb{Z}/n).$$

Unwindings using the base change theorem reduce us to the case in which X is a smooth proper curve, $S =$ a point, and $\mathscr{F} = \mathbb{Z}/n$. The cohomology groups in question then vanish for $q \neq 0, 1, 2$, and we invoke [32]: indeed, if X is proper over $\mathbb{C}$, then $\pi_0(X) = \pi_0(X^{\mathrm{an}})$ and $\pi_1(X) =$ the profinite completion of $\pi_1(X^{\mathrm{an}})$, which proves the assertion for $q = 0, 1$. For $q = 2$, one uses the Kummer exact sequence and the fact that, again by [32], $\mathrm{Pic}(X) = \mathrm{Pic}(X^{\mathrm{an}})$.

Theorem 4.6.4 (Cohomological dimension of affine schemes)**.** *Let X be an affine scheme of finite type over a separably closed field and let $\mathscr{F}$ be a torsion sheaf on X. Then $\mathrm{H}^q(X, \mathscr{F}) = 0$ for $q > \dim(X)$.*

For the very elegant proof, we refer to [2], XIV §2 and 3.

4.6.5 Remark

This theorem is in a sense a substitute for Morse theory. Indeed, consider the classical case in which X is smooth and affine over $\mathbb{C}$, embedded in a standard affine space $\mathbb{C}^N$. Then, for almost every point $p \in \mathbb{C}^N$, the function "distance from p" on X is a Morse function, and the indices of its critical points are smaller than $\dim(X)$. Thus X is obtained by attaching handles of index less than $\dim(X)$, which gives the classical analogue of (4.6.4).

5 Local acyclicity of smooth morphisms

Let X be a complex analytic variety and let $f : X \to D$ be a morphism from X to the disk. Write $[0,t]$ for the closed line segment in D with endpoints 0 and t, and $(0,t]$ for the half-open segment. If f is *smooth*, the inclusion

$$j : f^{-1}((0,t]) \hookrightarrow f^{-1}([0,t])$$

is a homotopy equivalence; the special fibre $X_0 = f^{-1}(0)$ can be pushed into $f^{-1}([0,t])$.

In practice, for sufficiently small t, $f^{-1}((0,t])$ will be a bundle over $(0,t]$, so that the inclusion

$$X_t = f^{-1}(t) \hookrightarrow f^{-1}((0,t])$$

will also be a homotopy equivalence. The *cospecialization morphism* is then the homotopy class of maps

$$\mathrm{cosp} : X_0 \hookrightarrow f^{-1}([0,t]) \xleftarrow{\sim} f^{-1}((0,t]) \xleftarrow{\sim} X_t.$$

Figuratively speaking, for a smooth morphism the general fibre swallows the special fibre.

No longer suppose that f is necessarily smooth (but suppose that $f^{-1}((0,t])$ is a bundle over $(0,t]$). A morphism $\mathrm{cosp}^\bullet$ can still be defined in cohomology whenever $j_*\mathbb{Z} = \mathbb{Z}$ and $\mathsf{R}^q j_*\mathbb{Z} = 0$ for $q > 0$. Under these hypotheses, the Leray spectral sequence for j shows that

$$\mathrm{H}^\bullet\left(f^{-1}([0,t]), \mathbb{Z}\right) \xrightarrow{\sim} \mathrm{H}^\bullet\left(f^{-1}((0,t]), \mathbb{Z}\right)$$

and $\mathrm{cosp}^\bullet$ is the composite morphism

$$\mathrm{cosp}^\bullet : \mathrm{H}^\bullet(X_t, \mathbb{Z}) \xleftarrow{\sim} \mathrm{H}^\bullet\left(f^{-1}((0,t]), \mathbb{Z}\right) \xleftarrow{\sim} \mathrm{H}^\bullet\left(f^{-1}([0,t]), \mathbb{Z}\right) \to \mathrm{H}^\bullet(X_0, \mathbb{Z})$$

The fibre of $\mathsf{R}^q j_*\mathbb{Z}$ at a point $x \in X_0$ is calculated as follows. In an ambient space, take a ball B_ε centred at x with sufficiently small radius ε, and for sufficiently small η put $E = X \cap B_\varepsilon \cap f^{-1}(\eta t)$; this is the *vanishing-cycle variety* at x. One has

$$(\mathsf{R}^q j_*\mathbb{Z})_x \xleftarrow{\sim} \mathrm{H}^q\left(X \cap B_\varepsilon \cap f^{-1}((0,\eta t]), \mathbb{Z}\right) \xrightarrow{\sim} \mathrm{H}^q(E, \mathbb{Z})$$

and the cospecialization morphism is defined in cohomology whenever the vanishing-cycle varieties are acyclic ($\mathrm{H}^0(E,\mathbb{Z}) = \mathbb{Z}$ and $\mathrm{H}^q(E,\mathbb{Z}) = 0$ for $q > 0$), which is expressed by saying that f is *locally acyclic*.

This chapter is devoted to the analogue of this situation for a smooth morphism of schemes in étale cohomology. In this setting, however, it is essential to restrict to torsion coefficients of order *prime to the residue characteristics*. Paragraph 5.1 gives generalities on locally acyclic morphisms and cospecialization maps. In Paragraph 5.2, we prove that a smooth morphism is locally acyclic. In Paragraph 5.3, this result is combined with those of the preceding chapter to obtain two applications: a specialization theorem for cohomology groups (the cohomology of the geometric fibres of a smooth proper morphism is locally constant), and a base change theorem for a smooth morphism.

Throughout what follows, we fix an integer n, and "scheme" means "scheme on which n is invertible." A "geometric point" will always mean an "algebraic geometric point" (2.3.1) $x : \mathrm{Spec}(k) \to X$, with k algebraically closed.

5.1 Locally acyclic morphisms

5.1.1 Notation

Given a scheme S and a geometric point s of S, we denote by $\widetilde{S}^s$ the spectrum of the strict localization of S at s.

Definition 5.1.2. *A geometric point t of S is called a* generization *of s if it is defined by an algebraic closure of the residue field of a point of $\widetilde{S}^s$. We also say that s is a* specialization *of t, and call the* specialization map *the S-morphism $t \to \widetilde{S}^s$.*

Definition 5.1.3. *Let $f : X \to S$ be a morphism of schemes. Let s be a geometric point of S, t a generization of s, x a geometric point of X over s, and $\widetilde{X}_t^x = \widetilde{X}^x \times_{\widetilde{S}^s} t$. Then $\widetilde{X}_t^x$ is called a* vanishing-cycle variety *of f at x.*

The morphism f is called locally acyclic *if the reduced cohomology of every vanishing-cycle variety $\widetilde{X}_t^x$ vanishes:*

$$\hat{\mathrm{H}}^\bullet(\widetilde{X}_t^x, \mathbb{Z}/n) = 0, \tag{5.1.3.1}$$

i.e. $\mathrm{H}^0\left(\widetilde{X}_t^x, \mathbb{Z}/n\right) = \mathbb{Z}/n$ *and* $\mathrm{H}^q(\widetilde{X}_t^x, \mathbb{Z}/n) = 0$ *for* $q > 0$.

Lemma 5.1.4. *Let $f : X \to S$ be a locally acyclic morphism and let $g : S' \to S$ be a quasi-finite morphism (or a projective limit of quasi-finite morphisms). Then the morphism $f' : X' \to S'$ obtained from f by base change is locally acyclic.*

Indeed, every vanishing-cycle variety of f' is a vanishing-cycle variety of f.

Lemma 5.1.5. *Let $f : X \to S$ be a locally acyclic morphism. For every geometric point t of S giving rise to a cartesian diagram*

$$\begin{array}{ccc} X_t & \xrightarrow{\varepsilon'} & X \\ \downarrow & & \downarrow f \\ t & \xrightarrow{\varepsilon} & S \end{array}$$

one has $\varepsilon'_*\mathbb{Z}/n = f^*\varepsilon_*\mathbb{Z}/n$ and $\mathsf{R}^q\varepsilon'_*\mathbb{Z}/n = 0$ for $q > 0$.

Let $\bar{S}$ be the closure of $\varepsilon(t)$, let S' be the normalization of $\bar{S}$ in $k(t)$, and consider the cartesian diagram

$$\begin{array}{ccccc} X_t & \xrightarrow{i'} & X' & \xrightarrow{\alpha'} & X \\ \downarrow & & \downarrow f' & & \downarrow f \\ t & \xrightarrow{i} & S' & \xrightarrow{\alpha} & S \end{array}$$

The local rings of S' are normal with separably closed fraction fields. They are therefore strictly henselian, and the local acyclicity of f' (5.1.4) gives $i'_*\mathbb{Z}/n = f'^*i_*\mathbb{Z}/n$ and $\mathsf{R}^q i'_*\mathbb{Z}/n = 0$ for $q > 0$. Since α is integral, it follows that

$$\mathsf{R}^q\varepsilon'_*\mathbb{Z}/n = \alpha'_*\mathsf{R}^q i'_*\mathbb{Z}/n = \alpha'_* f'^*\mathsf{R}^q i_*\mathbb{Z}/n = f^*\alpha_*\mathsf{R}^q i_*\mathbb{Z}/n = f^*\mathsf{R}^q\varepsilon_*\mathbb{Z}/n,$$

which proves the lemma.

5.1.6

Given a locally acyclic morphism $f : X \to S$ and a specialization map $t \to \widetilde{S}^s$, we shall define canonical homomorphisms, called cospecialization maps,

$$\mathrm{cosp}^\bullet : \mathrm{H}^\bullet(X_t, \mathbb{Z}/n) \to \mathrm{H}^\bullet(X_s, \mathbb{Z}/n),$$

relating the cohomology of the general fibre $X_t = X \times_S t$ to that of the special fibre $X_s = X \times_S s$.

Consider the cartesian diagram

$$\begin{array}{ccccc} X_t & \xrightarrow{\varepsilon'} & \widetilde{X} & \longleftarrow & X_s \\ \downarrow & & \downarrow{\scriptstyle f'} & & \downarrow \\ t & \xrightarrow{\varepsilon} & \widetilde{S}^s & \longleftarrow & s \end{array}$$

obtained from f by base change. By (5.1.4), f' is again locally acyclic. The definition of local acyclicity immediately shows that the restrictions to X_s of the sheaves $\mathsf{R}^q\varepsilon'_*\mathbb{Z}/n$ are $\mathbb{Z}/n$ for $q = 0$ and 0 for $q > 0$. By (5.1.5), we even know that $\mathsf{R}^q\varepsilon'_*\mathbb{Z}/n = 0$ for $q > 0$. We define $\mathrm{cosp}^\bullet$ as the composite map

$$\mathrm{H}^\bullet(X_t, \mathbb{Z}/n) \simeq \mathrm{H}^\bullet(\widetilde{X}, \varepsilon'_*\mathbb{Z}/n) \to \mathrm{H}^\bullet(X_s, \mathbb{Z}/n). \tag{5.1.6.1}$$

Variant Let $\bar{S}$ be the closure of $\varepsilon(t)$ in $\widetilde{S}^s$, let S' be the normalization of $\bar{S}$ in $k(t)$, and let X'/S' be obtained from X/S by base change. Diagram (5.1.6.1) may also be written

$$\mathrm{H}^\bullet(X_t, \mathbb{Z}/n) \simeq \mathrm{H}^\bullet(X', \mathbb{Z}/n) \to \mathrm{H}^\bullet(X_s, \mathbb{Z}/n).$$

Theorem 5.1.7. *Let S be a locally noetherian scheme, s a geometric point of S, and $f : X \to S$ a morphism. Suppose that*

a) f is locally acyclic,

b) for every specialization map $t \to \widetilde{S}^s$ and every $q \geqslant 0$, the cospecialization maps $\mathrm{H}^q(X_t, \mathbb{Z}/n) \to \mathrm{H}^q(X_s, \mathbb{Z}/n)$ are bijective.

Then the canonical homomorphism $(\mathsf{R}^q f_*\mathbb{Z}/n)_s \to \mathrm{H}^q(X_s, \mathbb{Z}/n)$ is bijective for every $q \geqslant 0$.

To prove the theorem, we may clearly suppose that $S = \widetilde{S}^s$. We shall in fact show that, for every sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules $\mathscr{F}$ on S, the canonical homomorphism $\varphi^q(\mathscr{F}) : (\mathsf{R}^q f_* f^*\mathscr{F})_s \to \mathrm{H}^q(X_s, f^*\mathscr{F})$ is bijective.

Every sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules is a filtered direct limit of constructible sheaves of $\mathbb{Z}/n\mathbb{Z}$-modules (4.3.3). Moreover, every constructible sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules embeds in a sheaf of the form $\prod i_{\lambda *}C_\lambda$, where $i_\lambda : t_\lambda \to S$ is a finite family of generizations of s and C_λ is a free $\mathbb{Z}/n\mathbb{Z}$-module of finite rank on t_λ. By the definition of the cospecialization maps, condition b) says that the homomorphisms $\varphi^q(\mathscr{F})$ are bijective when $\mathscr{F}$ has this form.

We conclude by a variant of Lemma (4.3.6):

Lemma 5.1.8. *Let $\mathcal{C}$ be an abelian category in which filtered direct limits exist. Let $\varphi^\bullet : T^\bullet \to T'^\bullet$ be a morphism of cohomological functors commuting with filtered direct limits, defined on $\mathcal{C}$ with values in the category of abelian groups. Suppose that there are two subsets $\mathcal{D}$ and $\mathcal{E}$ of the objects of $\mathcal{C}$ such that:*

a) every object of $\mathcal{C}$ is a filtered direct limit of objects belonging to $\mathcal{D}$,

b) every object belonging to $\mathcal{D}$ is contained in an object belonging to $\mathcal{E}$.

Then the following conditions are equivalent:

(i) $\varphi^q(A)$ *is bijective for every* $q \geqslant 0$ *and every* $A \in \mathrm{Ob}\,\mathcal{C}$.

(ii) $\varphi^q(M)$ *is bijective for every* $q \geqslant 0$ *and every* $M \in \mathcal{E}$.

The proof of the lemma proceeds by passage to the direct limit, induction on q, and repeated application of the five lemma to the diagram of exact cohomology sequences deduced from an exact sequence $0 \to A \to M \to A' \to 0$, with $A \in \mathcal{D}$, $M \in \mathcal{E}$, and $A' \in \mathrm{Ob}\,\mathcal{C}$.

Corollary 5.1.9. *Let* S *be the spectrum of a strictly henselian noetherian local ring and let* $f : X \to S$ *be a locally acyclic morphism. Suppose that for every geometric point* t *of* S *one has* $\mathrm{H}^0(X_t, \mathbb{Z}/n) = \mathbb{Z}/n$ *and* $\mathrm{H}^q(X_t, \mathbb{Z}/n) = 0$ *for* $q > 0$ *(in other words, the geometric fibres of* f *are acyclic). Then* $f_*\mathbb{Z}/n = \mathbb{Z}/n$ *and* $\mathsf{R}^q f_*\mathbb{Z}/n = 0$ *for* $q > 0$.

Corollary 5.1.10. *Let* $f : X \to Y$ *and* $g : Y \to Z$ *be morphisms of locally noetherian schemes. If* f *and* g *are locally acyclic, then so is* $g \circ f$.

We may suppose that X, Y, and Z are strictly local and that f and g are local morphisms. It is then enough to show that, if z is an algebraic geometric point of Z, then $\mathrm{H}^0(X_z, \mathbb{Z}/n) = \mathbb{Z}/n$ and $\mathrm{H}^q(X_z, \mathbb{Z}/n) = 0$ for $q > 0$.

Since g is locally acyclic, one has $\mathrm{H}^0(Y_z, \mathbb{Z}/n) = \mathbb{Z}/n$ and $\mathrm{H}^q(Y_z, \mathbb{Z}/n) = 0$ for $q > 0$. Moreover, the morphism $f_z : X_z \to Y_z$ is locally acyclic (5.1.4), and its geometric fibres are acyclic because they are vanishing-cycle varieties of f. By (5.1.9), $\mathsf{R}^q f_{z*}\mathbb{Z}/n = 0$ for $q > 0$. In addition, $f_{z*}\mathbb{Z}/n$ is constant with fibre $\mathbb{Z}/n$ on Y_z. The result follows from the Leray spectral sequence for f_z.

5.2 Local acyclicity of a smooth morphism

Theorem 5.2.1. *A smooth morphism is locally acyclic.*

Let $f : X \to S$ be a smooth morphism. The assertion is local for the étale topology on X and S, so we may suppose that X is standard affine space of dimension d over S. By passage to the limit, we may suppose that S is noetherian, and the transitivity of local acyclicity (5.1.10) shows that it suffices to treat $d = 1$.

Let s be a geometric point of S and x a geometric point of X centred at a closed point of X_s. We must show that the geometric fibres of $\widetilde{X}^x \to \widetilde{S}^s$ are acyclic. Henceforth put $S = \widetilde{S}^s = \mathrm{Spec}(A)$ and $X = \widetilde{X}^x$. Then $X \simeq \mathrm{Spec}\, A\{T\}$, where $A\{T\}$ is the henselization of $A[T]$ at the point $T = 0$ above s.

If t is a geometric point of S, the fibre X_t is a projective limit of smooth affine curves over t. Thus $\mathrm{H}^q(X_t, \mathbb{Z}/n) = 0$ for $q \geqslant 2$, and it suffices to show that $\mathrm{H}^0(X_t, \mathbb{Z}/n) = \mathbb{Z}/n$ and $\mathrm{H}^1(X_t, \mathbb{Z}/n) = 0$ for n prime to the residue characteristic of S. This follows from the next two propositions.

Proposition 5.2.2. *Let* A *be a strictly henselian local ring,* $S = \mathrm{Spec}(A)$*, and* $X = \mathrm{Spec}\, A\{T\}$*. Then the geometric fibres of* $X \to S$ *are connected.*

By passage to the limit, we may reduce to the case in which A is a strict henselization of a finitely generated $\mathbb{Z}$-algebra.

Let $\bar{t}$ be a geometric point of S lying over t, and let k' be a finite separable extension of $k(t)$ in $k(\bar{t})$. Put $t' = \mathrm{Spec}(k')$ and $X_{t'} = X \times_S \mathrm{Spec}(k')$. We must check that, for all $\bar{t}$ and t', $X_{t'}$ is connected (by which we mean connected and nonempty). Let A' be the normalization of A in k', i.e. the ring of elements of k' integral over the image of A in $k(t)$. One has $A\{T\} \otimes_A A' \xrightarrow{\sim} A'\{T\}$: indeed, the left-hand side is henselian local (because A' is finite over A and local) and is a limit of local étale algebras over $A'\{T\} = A\{T\} \otimes_A A'$. Thus $X_{t'}$ is still the fibre at t' of $X = \mathrm{Spec}\,(A'\{T\})$ over $S' = \mathrm{Spec}(A')$. The local scheme X' is normal, hence integral; its localization $X_{t'}$ is still integral, and a fortiori connected.

Proposition 5.2.3. *Let A be a strictly henselian local ring, $S = \mathrm{Spec}(A)$, and $X = \mathrm{Spec}\,(A\{T\})$. Let $\bar{t}$ be a geometric point of S and $X_{\bar{t}}$ the corresponding geometric fibre. Then every Galois étale covering of $X_{\bar{t}}$ of order prime to the characteristic of the residue field of A is trivial.*

Lemma 5.2.4 (Zariski–Nagata purity theorem in dimension 2)**.** *Let C be a regular local ring of dimension 2 and let C' be a finite normal C-algebra étale over the open complement of the closed point of* $\mathrm{Spec}(C)$*. Then C' is étale over C.*

Indeed, C' is normal of dimension 2, and hence $\mathrm{prof}(C') = 2$. Since $\mathrm{prof}(C') + \dim \mathrm{proj}(C') = \dim(C) = 2$, it follows that C' is free over C. The set of points of C where C' is ramified is then defined by an equation, the discriminant; since it contains no point of height 1, it is empty.

Lemma 5.2.5 (Special case of Abhyankar's lemma)**.** *Let $S = \mathrm{Spec}(V)$ be a trait, let π be a uniformizer, let η be the generic point of S, let X be smooth over S, irreducible, and of relative dimension 1, let $\widetilde{X}_\eta$ be a Galois étale covering of X_η of degree n invertible on S, and let $S_1 = \mathrm{Spec}\left(V[\pi^{1/n}]\right)$. Use a subscript $(-)_1$ to denote base change from S to S_1. Then $\widetilde{X}_{1\eta}$ extends to an étale covering of X_1.*

Let $\widetilde{X}_1$ be the normalization of X_1 in $\widetilde{X}_{1\eta}$. By the structure of the tame inertia groups of the discrete valuation rings obtained by localizing X at the generic points of the special fibre X_s, $\widetilde{X}_1$ is étale over X_1 on the general fibre and at the generic points of the special fibre. By (5.2.4), it is étale everywhere.

5.2.6

Let t denote the point over which $\bar{t}$ lies. We are free to replace A by the normalization of A in a finite separable extension $k(t')$ of $k(t)$ in $k(\bar{t})$ (cf. 5.2.2). This and a preliminary passage to the limit allow us to suppose that

a) A is normal and noetherian, and t is the generic point of S.

b) The étale covering under consideration of $X_{\bar{t}}$ comes from an étale covering of X_t.

c) It comes from an étale covering $\widetilde{X}_U$ of the inverse image X_U of a nonempty open subset U of S (this follows from b): t is the limit of the U).

d) The complement of U has codimension $\geqslant 2$ (at the price of enlarging $k(t)$, by applying (5.2.5) to the discrete valuation rings obtained by localizing S at the points of $S \smallsetminus U$ of codimension 1 in S; there are only finitely many such points).

e) The covering $\widetilde{X}_U$ is trivial over the subscheme $T = 0$. (This may require enlarging $k(t)$ once more.)

Under these conditions, we shall show that the covering $\widetilde{X}_U$ is trivial.

Lemma 5.2.7. *Let A be a normal noetherian strictly henselian local ring, let U be an open subset of* $\mathrm{Spec}(A)$ *whose complement has codimension $\geqslant 2$, let V be its inverse image in $X = \mathrm{Spec}\,(A\{T\})$, and let V' be an étale covering of V. If V' is trivial over $T = 0$, then V' is trivial.*

Let $B = \Gamma(V', \mathscr{O})$. Since V' is the inverse image of V in $\mathrm{Spec}(B)$, it suffices to show that B is finite étale over $A\{T\}$ (and hence split, since $A\{T\}$ is strictly henselian). Let $\widehat{X} = \mathrm{Spec}\, A[\![T]\!]$, and denote by $(-)^\wedge$ base change from X to $\widehat{X}$. The scheme $\widehat{X}$ is faithfully flat over X. Hence $\Gamma(\widehat{V}', \mathscr{O}) = B \otimes_{A\{T\}} A[\![T]\!]$, and it suffices to show that this ring $\widehat{B}$ is finite étale over $A[\![T]\!]$.

Let V_m (resp. V'_m) be the subscheme of $\widehat{V}$ (resp. $\widehat{V}'$) defined by $T^{m+1} = 0$. By hypothesis, V'_0 is a trivial covering of V_0: a sum of n copies of V_0. The same is true of V'_m/V_m, because étale coverings are insensitive to nilpotents. We obtain

$$\varphi : \Gamma(\widehat{V}', \mathscr{O}) \to \varprojlim_m \Gamma(V'_m, \mathscr{O}) = \left(\varprojlim_m \Gamma(V_m, \mathscr{O}) \right)^n .$$

By hypothesis, the complement of U has depth $\geqslant 2$: one has $\Gamma(V_m, \mathscr{O}) = A[T]/(T^{m+1})$, and φ is a homomorphism from $\widehat{B}$ to $A[\![T]\!]^n$. Over U, it provides n distinct sections of $\widehat{V}'/\widehat{V}$; hence $\widehat{V}'$ is trivial, a sum of n copies of $\widehat{V}$. Since the complement of $\widehat{V}$ in $\widehat{X}$ still has codimension $\geqslant 2$ (and therefore depth $\geqslant 2$), it follows that $\widehat{B} = A[\![T]\!]^n$, proving the lemma.

5.3 Applications

Theorem 5.3.1 (Specialization of cohomology groups)**.** *Let $f : X \to S$ be a proper locally acyclic morphism, for example a smooth proper morphism. Then the sheaves $\mathsf{R}^q f_* \mathbb{Z}/n$ are locally constant constructible, and for every specialization map $t \to \widetilde{S}^s$, the cospecialization maps $\mathrm{H}^q(X_t, \mathbb{Z}/n) \to \mathrm{H}^q(X_s, \mathbb{Z}/n)$ are bijective.*

This follows immediately from the definition of the cospecialization maps and the finiteness and proper base change theorems.

Theorem 5.3.2 (Base change by a smooth morphism)**.** *Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

be a cartesian diagram with g smooth. For every sheaf $\mathscr{F}$ on X that is torsion of order prime to the residue characteristics of S, one has

$$g^* \mathsf{R}^q f_* \mathscr{F} \xrightarrow{\sim} \mathsf{R}^q f'_* (g'^* \mathscr{F}).$$

After passing to an open covering of X, we reduce to the case where X is affine, and then, by passage to the limit, to the case where X is of finite type over S. The morphism f then factors as an open immersion $j : X \to \bar{X}$ followed by a proper morphism $\bar{f} : \bar{X} \to S$. The Leray spectral sequence for $\bar{f} \circ j$ and the proper base change theorem show that it suffices to prove the theorem when $X \to S$ is an open immersion.

In this case, if $\mathscr{F}$ has the form $\varepsilon_* C$, where $\varepsilon : t \to X$ is a geometric point of X, the theorem is a corollary of (5.1.5). The general case follows from Lemma (5.1.8).

Corollary 5.3.3. *Let K/k be an extension of separably closed fields, let X be a k-scheme, and let n be an integer prime to the characteristic of k. Then the canonical map $\mathrm{H}^q(X, \mathbb{Z}/n) \to \mathrm{H}^q(X_K, \mathbb{Z}/n)$ is bijective for every $q \geqslant 0$.*

It suffices to observe that $\bar{K}$ is a direct limit of smooth $\bar{k}$-algebras.

Theorem 5.3.4 (Relative purity)**.** *Let*

$$\begin{array}{ccccc} U & \xrightarrow{j} & X & \xleftarrow{i} & Y \\ & \searrow & \downarrow{\scriptstyle f} & \swarrow{\scriptstyle h} & \\ & & S & & \end{array} \tag{5.3.4.1}$$

be a commutative diagram with f smooth of pure relative dimension N, h smooth of pure relative dimension $N-1$, i a closed immersion, and $U = X \smallsetminus Y$. If n is prime to the residue characteristics of S, then

$$\begin{aligned} j_*\mathbb{Z}/n &= \mathbb{Z}/n \\ \mathsf{R}^1 j_*\mathbb{Z}/n &= \mathbb{Z}/n(-1)_Y \\ \mathsf{R}^q j_*\mathbb{Z}/n &= 0 \qquad \textit{for } q \geqslant 2 \end{aligned}$$

In these formulas, $\mathbb{Z}/n(-1)$ denotes the $\mathbb{Z}/n$-dual of $\boldsymbol{\mu}_n$. If t is a local equation for Y, the isomorphism $\mathsf{R}^1 j_*\mathbb{Z}/n \simeq \mathbb{Z}/n(-1)_Y$ is defined by the map $a : \mathbb{Z}/n \to \mathsf{R}^1 j_*\boldsymbol{\mu}_n$ sending 1 to the class of the $\boldsymbol{\mu}_n$-torsor of nth roots of t.

The question is local. This allows us to replace (X, Y) by a locally isomorphic pair, for example

$$\begin{array}{ccccc} \mathbb{A}^1_T & \xrightarrow{j} & \mathbb{P}^1_T & \xleftarrow{i} & T \\ & \searrow{\scriptstyle g} & \downarrow{\scriptstyle f} & \swarrow & \\ & & T & & \end{array}$$

with $T = \mathbb{A}^{N-1}_S$ and $i =$ the section at infinity. Corollary (5.1.9) applies to g and gives $\mathsf{R}^q g_*\mathbb{Z}/n = \mathbb{Z}/n$ for $q = 0$ and 0 for $q > 0$. For f, on the other hand, (4.6.2.1) gives

$$\mathsf{R}^q f_*\mathbb{Z}/n = \begin{cases} \mathbb{Z}/n & q = 0, \\ 0 & q = 1, \\ \mathbb{Z}/n(-1) & q = 2, \\ 0 & q > 2. \end{cases}$$

It is readily checked that $j_*\mathbb{Z}/n = \mathbb{Z}/n$, and that the $\mathsf{R}^q j_*\mathbb{Z}/n = 0$ are concentrated on $i(T)$ for $q > 0$. The Leray spectral sequence $E_2^{pq} = \mathsf{R}^p f_*\mathsf{R}^q j_*\mathbb{Z}/n \Rightarrow \mathsf{R}^{p+q} g_*\mathbb{Z}/n$ therefore reduces to

$$\begin{array}{lllll} i^*\mathsf{R}^q j_*\mathbb{Z}/n & 0 & \cdots & & \\ \\ i^*\mathsf{R}^1 j_*\mathbb{Z}/n & 0 & \cdots & & \\ & \searrow^{d_2} & & & \\ \mathbb{Z}/n & 0 & \mathbb{Z}/n(-1) & 0 & \cdots \end{array}$$

$\mathsf{R}^q j_*\mathbb{Z}/n = 0$ for $q \geqslant 2$, and $\mathsf{R}^1 j_*\mathbb{Z}/n$ is the extension by zero of a locally free sheaf of rank one on T (isomorphic, via d_2, to $\mathbb{Z}/n(-1)$). Since the map a defined above is injective (as may be checked fibre by fibre), it is an isomorphism, which proves (5.3.4).

5.3.5

We refer to [2, XVI.4,5] for proofs of the following applications of the acyclicity theorem (5.2.1).

Theorem 5.3.6. *Let $f : X \to S$ be a morphism of schemes of finite type over $\mathbb{C}$ and let $\mathscr{F}$ be a constructible sheaf on X. Then*

$$(\mathsf{R}^q f_* \mathscr{F})^{\mathrm{an}} \sim \mathsf{R}^q f_*^{\mathrm{an}}(\mathscr{F}^{\mathrm{an}})$$

(cf. 4.6.3: for ordinary cohomology it is necessary to assume that $\mathscr{F}$ is constructible, not merely torsion).

Theorem 5.3.7. *Let $f : X \to S$ be a morphism of schemes of finite type over a field k of characteristic 0, and let $\mathscr{F}$ be a constructible sheaf on X. Then the $\mathsf{R}^q f_* \mathscr{F}$ are also constructible.*

The proof uses resolution of singularities and (5.3.4). It is generalized to a morphism of finite type between excellent schemes of characteristic 0 in [2, XIX.5]. Another proof, independent of resolution, is given in this volume (Finiteness theorems in ℓ-adic cohomology, 1.1) and applies to a morphism of schemes of finite type over a field or over a Dedekind ring.

6 Poincaré duality

6.1 Introduction

Let X be an oriented topological manifold of pure dimension N, and suppose that X admits a finite open covering $\mathcal{U} = (U_i)_{1 \leqslant i \leqslant K}$ such that every nonempty intersection of the open subsets U_i is homeomorphic to a ball. For such a manifold, the Poincaré duality theorem can be presented as follows:

A The cohomology of X is the Čech cohomology associated with the covering $\mathcal{U}$. It is the cohomology of the complex

$$0 \to \mathbb{Z}^{A_0} \to \mathbb{Z}^{A_1} \to \cdots \tag{1}$$

where $A_k = \{(i_0, \ldots, i_k) : i_0 < \cdots < i_k \text{ and } U_{i_0} \cap \cdots \cap U_{i_k} \neq \varnothing\}$.

B For $a \in A_k$, $a = (i_0, \ldots, i_k)$, let $U_a = U_{i_0} \cap \cdots \cap U_{i_k}$ and let j_a be the inclusion of U_a into X. The constant sheaf $\mathbb{Z}$ on X admits the (left) resolution

$$\cdots \longrightarrow \bigoplus_{a \in A_1} j_{a!}\mathbb{Z} \longrightarrow \bigoplus_{a \in A_0} j_{a!}\mathbb{Z} \longrightarrow 0 \tag{2}$$

$$\downarrow$$

$$\mathbb{Z}.$$

The compactly supported cohomology $\mathrm{H}_c^\bullet(X, j_{a!}\mathbb{Z})$ is precisely the cohomology with proper support of the (oriented) ball U_a:

$$\mathrm{H}_c^i(X, j_{a!}\mathbb{Z}) = \begin{cases} 0 & \text{if } i \neq N \\ \mathbb{Z} & \text{if } i = N \end{cases}$$

The hypercohomology spectral sequence for the complex (2) and cohomology with proper support therefore says that $\mathrm{H}_c^i(X)$ is the $(i - N)$th cohomology group of the complex

$$\cdots \to \mathbb{Z}^{A_1} \to \mathbb{Z}^{A_0} \to 0. \tag{3}$$

This complex is dual to the complex (1), whence Poincaré duality.

The essential points of this construction are

a) the existence of a theory of cohomology with proper support;

b) the fact that every point x of a manifold X of pure dimension N has a fundamental system of open neighbourhoods U for which

$$\mathrm{H}^i_c(U) = \begin{cases} 0 & \text{for } i \neq N, \\ \mathbb{Z} & \text{for } i = N. \end{cases} \tag{4}$$

Poincaré duality in étale cohomology can be constructed on this model. For X smooth of pure dimension N over an algebraically closed field k, n invertible on X, and x a closed point of X, the key point is to calculate the projective limit over the étale neighbourhoods U of x:

$$\varprojlim \mathrm{H}^i_c(U, \mathbb{Z}/n) = \begin{cases} 0 & \text{if } i \neq 2N \\ \mathbb{Z}/n & \text{if } i = 2N. \end{cases} \tag{5}$$

Just as for topological manifolds one must first treat directly the case of an open ball (or even simply the interval $(0,1)$), here one must first treat directly the case of curves (6.2). The local acyclicity theorem for smooth morphisms then reduces (5) to this special case (6.3).

The isomorphisms (4) and (5) are not canonical: they depend on the choice of an *orientation* of X. For n invertible on a scheme X, $\boldsymbol{\mu}_n$ is a sheaf of free $\mathbb{Z}/n$-modules of rank one. We denote its Nth tensor power by $\mathbb{Z}/n(N)$ ($N \in \mathbb{Z}$). The intrinsic form of the second line of (5) is

$$\varprojlim \mathrm{H}^{2N}_c(U, \mathbb{Z}/n(N)) = \mathbb{Z}/n, \tag{6}$$

and $\mathbb{Z}/n(N)$ is called the *orientation sheaf* of X. Since the sheaf $\mathbb{Z}/n(N)$ is constant and isomorphic to $\mathbb{Z}/n$, it may be taken outside the limit sign (with N fixed), and one may instead write

$$\varprojlim \mathrm{H}^{2N}_c(U, \mathbb{Z}/n) = \mathbb{Z}/n(-N), \tag{7}$$

and Poincaré duality will take the form of a perfect duality, with values in $\mathbb{Z}/n(-N)$, between $\mathrm{H}^i(X, \mathbb{Z}/n)$ and $\mathrm{H}^{2N-i}_c(X, \mathbb{Z}/n)$.

6.2 The case of curves

6.2.1

Let $\bar{X}$ be a smooth projective curve over an algebraically closed field k, and let n be invertible on $\bar{X}$. The proof of (3.3.5) gives, for connected $\bar{X}$, a canonical isomorphism

$$\mathrm{H}^2(\bar{X}, \boldsymbol{\mu}_n) = \mathrm{Pic}(\bar{X})/n\,\mathrm{Pic}(\bar{X}) \xrightarrow[\sim]{\deg} \mathbb{Z}/n$$

Let D be a reduced divisor of $\bar{X}$ and let $X = \bar{X} \smallsetminus D$:

$$X \overset{j}{\hookrightarrow} \bar{X} \overset{i}{\hookleftarrow} D.$$

The exact sequence $0 \to j_!\boldsymbol{\mu}_n \to \boldsymbol{\mu}_n \to i_*\boldsymbol{\mu}_n \to 0$ and the fact that $\mathrm{H}^i(\bar{X}, i_*\boldsymbol{\mu}_n) = \mathrm{H}^i(D, \boldsymbol{\mu}_n) = 0$ for $i > 0$ give an isomorphism

$$\mathrm{H}^2_c(X, \boldsymbol{\mu}_n) = \mathrm{H}^2(\bar{X}, j_!\boldsymbol{\mu}_n) \xrightarrow{\sim} \mathrm{H}^2(\bar{X}, \boldsymbol{\mu}_n) \xrightarrow{\sim} \mathbb{Z}/n.$$

For disconnected $\bar{X}$, similarly

$$\mathrm{H}^2_c(X, \boldsymbol{\mu}_n) = (\mathbb{Z}/n)^{\pi_0(X)}$$

and the *trace morphism* is defined to be the sum

$$\mathrm{tr} : \mathrm{H}^2_c(X, \boldsymbol{\mu}_n) \simeq (\mathbb{Z}/n)^{\pi_0(X)} \xrightarrow{\Sigma} \mathbb{Z}/n.$$

Theorem 6.2.2. *The pairing* $\mathrm{tr}(a \smile b)$ *identifies each of the two groups* $\mathrm{H}^i(X, \mathbb{Z}/n)$ *and* $\mathrm{H}^{2-i}_c(X, \boldsymbol{\mu}_n)$ *with the dual (with values in* $\mathbb{Z}/n$*) of the other.*

Transcendental proof. Suppose that $\bar{X}$ is a smooth projective curve over a trait S and that $j : X \hookrightarrow \bar{X}$ is the inclusion of the complement of a divisor D étale over S. The cohomology (resp. cohomology with proper support) of the special and generic geometric fibres of X/S are "the same," i.e. the fibres of locally constant sheaves on S. This follows from the analogous facts for $\bar{X}$ and D, by means of the exact sequence $0 \to j_!\mathbb{Z}/n \to \mathbb{Z}/n \to \mathbb{Z}/n_D \to 0$ (for cohomology with proper support) and the formulas $j_*\mathbb{Z}/n = \mathbb{Z}/n$, $\mathsf{R}^1 j_*\mathbb{Z}/n \simeq \mathbb{Z}/n_D(-1)$, and $\mathsf{R}^i j_*\mathbb{Z}/n = 0$ $(i \geqslant 2)$ (for ordinary cohomology) (5.3.4).

This specialization principle reduces (6.2.2) to the case in which k has characteristic 0. By (5.3.3), this case reduces to $k = \mathbb{C}$. Finally, when $k = \mathbb{C}$, the groups $\mathrm{H}^\bullet(X, \mathbb{Z}/n)$ and $\mathrm{H}^\bullet_c(X, \boldsymbol{\mu}_n)$ agree with the groups of the same name calculated for the classical topological space X_{cl}, and, through the isomorphism $\mathbb{Z}/n \to \boldsymbol{\mu}_n : x \mapsto \exp\left(\frac{2\pi i x}{n}\right)$, the trace morphism is identified with "integration over the fundamental class." Thus (6.2.2) follows from Poincaré duality for X_{cl}.

6.2.3 Algebraic proof

For a very economical proof, see (Duality, §2). Here is another, related to the autoduality of the Jacobian.

Return to the notation of (6.2.1). We may suppose—and do suppose—that X is connected. The cases $i = 0$ and $i = 2$ being trivial, we also suppose that $i = 1$. Define ${}_D\mathbb{G}_m$ by the exact sequence $0 \to {}_D\,\mathbb{G}_m \to \mathbb{G}_m \to i_*\mathbb{G}_m \to 0$ (sections of $\mathbb{G}_m$ congruent to 1 mod D). The group $\mathrm{H}^1(\bar{X}, {}_D\,\mathbb{G}_m)$ classifies invertible sheaves on $\bar{X}$ trivialized on D. It is the group of points of $\mathrm{Pic}_D(\bar{X})$, an extension of $\mathbb{Z}$ (the degree) by the group of points of Rosenlicht's generalized Jacobian $\mathrm{Pic}^0_D(\bar{X})$ (corresponding to conductor 1 at each point of D), itself an extension of the abelian variety $\mathrm{Pic}^0(\bar{X})$ by the torus $\mathbb{G}_m^D/(\text{diagonal } \mathbb{G}_m)$.

a) The exact sequence $0 \to j_!\boldsymbol{\mu}_n \to {}_D\,\mathbb{G}_m \xrightarrow{n}{}_D\,\mathbb{G}_m \to 0$ gives an isomorphism

$$\mathrm{H}^1_c(X, \boldsymbol{\mu}_n) = \mathrm{Pic}^0_D(\bar{X})_n. \tag{6.2.3.1}$$

b) The map sending $x \in X(k)$ to the class of the invertible sheaf $\mathscr{O}(x)$ on $\bar{X}$, trivialized by 1 on D, comes from a morphism

$$f : X \to \mathrm{Pic}_D(\bar{X}).$$

For the remainder, fix a base point 0 and put $f_0(x) = f(x) - f(0)$. For every homomorphism $v : \mathrm{Pic}^0_D(\bar{X})_n \to \mathbb{Z}/n$, let $\bar{v} \in \mathrm{H}^1\left(\mathrm{Pic}^0_D(\bar{X}), \mathbb{Z}/n\right)$ be the image under v of the class in $\mathrm{H}^1\left(\mathrm{Pic}^0_D(\bar{X}), \mathrm{Pic}^0_D(\bar{X})_n\right)$ of the torsor defined by the extension

$$0 \longrightarrow \mathrm{Pic}^0_D(\bar{X})_n \longrightarrow \mathrm{Pic}^0_D(\bar{X}) \xrightarrow{n} \mathrm{Pic}^0_D(\bar{X}) \longrightarrow 0$$

Geometric class field theory (as presented in Serre [33]) shows that the map $v \mapsto f_0^*(\bar{v})$

$$\hom\left(\mathrm{Pic}^0_D(\bar{X})_n, \mathbb{Z}/n\right) \to \mathrm{H}^1(X, \mathbb{Z}/n) \tag{6.2.3.2}$$

is an isomorphism. To deduce (6.2.2) from (6.2.3.1) and (6.2.3.2), it remains to know that

$$\mathrm{tr}\left(u \smile f_0^*(\bar{v})\right) = -v(u). \tag{2.3.3}$$

6.3 The general case

Let X be a smooth algebraic variety of pure dimension N over an algebraically closed field k. To state the Poincaré duality theorem, one must first define the trace morphism

$$\mathrm{tr} : \mathrm{H}_c^{2N}(X, \mathbb{Z}/n(N)) \to \mathbb{Z}/n.$$

The definition is a painful unwinding starting from the case of curves [2, XVIII §2]. One then has:

Theorem 6.3.1. *The pairing* $\mathrm{tr}(a \smile b)$ *identifies each of the groups* $\mathrm{H}_c^i(X, \mathbb{Z}/n(N))$ *and* $\mathrm{H}^{2N-i}(X, \mathbb{Z}/n)$ *with the dual of the other.*

Let $x \in X$ be a closed point and let X_x be the strict localization of X at x. For U ranging over the étale neighbourhoods of x, put

$$\mathrm{H}_c^\bullet(X_x, \mathbb{Z}/n) = \varprojlim \mathrm{H}_c^\bullet(U, \mathbb{Z}/n). \tag{6.3.1.1}$$

It would be preferable to consider instead the pro-object "$\varprojlim$" $\mathrm{H}_c^\bullet(U, \mathbb{Z}/n)$, but since the groups involved are finite, the difference is inessential. As we tried to explain in the introduction, (6.3.1) follows from the facts that

$$\begin{gathered} \mathrm{H}_c^i(X_x, \mathbb{Z}/n) = 0 \text{ for } i \neq 2N \text{ and that} \\ \mathrm{tr} : \mathrm{H}_c^{2N}(X_x, \mathbb{Z}/n(N)) \xrightarrow{\sim} \mathbb{Z}/n \text{ is an isomorphism.} \end{gathered} \tag{6.3.1.2}$$

The case $N = 0$ is trivial. If $N > 0$, let Y_y be the strict localization at a closed point of a smooth scheme Y of pure dimension $N - 1$, and let $f : X_x \to Y_y$ be an essentially smooth morphism (of relative dimension one). The proof uses the Leray spectral sequence for cohomology with proper support for f to reduce to the case of curves. Because the "cohomology with proper support" under consideration is defined as limits (6.3.1.1), the existence of such a spectral sequence raises various passage-to-the-limit problems, treated in too much detail in [2, XVIII]. For every geometric point z of Y_y, one has

$$(\mathsf{R}^i f_! \mathbb{Z}/n)_z = \mathrm{H}_c^i(f^{-1}(z), \mathbb{Z}/n).$$

The geometric fibre $f^{-1}(z)$ is a projective limit of smooth curves over an algebraically closed field. It satisfies Poincaré duality. Its ordinary cohomology is given by the local acyclicity theorem for smooth morphisms:

$$\mathrm{H}^i(f^{-1}(z), \mathbb{Z}/n) = \begin{cases} \mathbb{Z}/n & \text{for } i = 0 \\ 0 & \text{for } i > 0. \end{cases}$$

By duality,

$$\mathrm{H}_c^i(f^{-1}(z), \mathbb{Z}/n) = \begin{cases} \mathbb{Z}/n(-1) & \text{for } i = 2 \\ 0 & \text{for } i \neq 2. \end{cases}$$

and the Leray spectral sequence becomes

$$\mathrm{H}_c^i(X_x, \mathbb{Z}/n(N)) = \mathrm{H}_c^{i-2}(Y_y, \mathbb{Z}/n(N-1)).$$

The result follows by induction on N.

6.4 Variants and applications

One can construct in étale cohomology a "duality formalism" (= functors $\mathsf{R}f_*$, $\mathsf{R}f_!$, f^*, and $\mathsf{R}f^!$ satisfying various compatibilities and adjunction formulas) parallel to the one that exists in coherent cohomology. In

this language, the results of the preceding paragraph can be rewritten as follows: if $f : X \to S$ is smooth of pure relative dimension N and $S = \mathrm{Spec}(k)$ with k algebraically closed, then

$$\mathsf{R}f^!\mathbb{Z}/n = \mathbb{Z}/n[2N](N).$$

This result holds without any hypothesis on S. It has the following corollary.

Corollary 6.4.1. *If f is smooth of pure relative dimension N and the sheaves $\mathsf{R}^i f_!\mathbb{Z}/n$ are locally constant, then the sheaves $\mathsf{R}^i f_*\mathbb{Z}/n$ are also locally constant, and*

$$\mathsf{R}^i f_*\mathbb{Z}/n = \underline{\hom}\left(\mathsf{R}^{2N-i} f_!\mathbb{Z}/n(N), \mathbb{Z}/n\right).$$

In particular, under the hypotheses of the corollary, the sheaves $\mathsf{R}^i f_*\mathbb{Z}/n$ are constructible. From this one can show that, if S is of finite type over the spectrum of a field or of a Dedekind ring, then, for every morphism of finite type $f : X \to S$ and every constructible sheaf $\mathscr{F}$ on X, the sheaves $\mathsf{R}^i f_*\mathscr{F}$ are constructible (Finiteness theorems in ℓ-adic cohomology, 1.1).

Chapter 2

Report on the trace formula

In this text, I have attempted to present Grothendieck's cohomological theory of L-functions as directly as possible. I follow very closely some of the lectures given by Grothendieck at the IHES in the spring of 1966. In the spirit of this volume, no appeal will be made to SGA 5—apart from two references to passages in Exposé XI, which is otherwise independent of that seminar.

In §10 (pp. 81–90) of [13], the reader will find a summary of the theory in the crucial case of curves.

1 Some reminders

1.1 Notation

$p, q, \mathbb{F}, \mathbb{F}_q$: p is a prime number, $q = p^f$ a power of p, and $\mathbb{F}$ an algebraic closure of the prime field $\mathbb{F}_p$. For every power p^n of p, denote by $\mathbb{F}_{p^n}$ the subfield with p^n elements of $\mathbb{F}$.

X_0, X: An object over $\mathbb{F}_q$ will often be denoted by a symbol carrying a subscript 0. The same symbol without the 0 then denotes the object obtained from it by extension of scalars to $\mathbb{F}$. For example, if $\mathscr{F}_0$ is a sheaf on a scheme X_0 over $\mathbb{F}_q$, denote by $\mathscr{F}$ its inverse image on $X = X_0 \otimes_{\mathbb{F}_q} \mathbb{F}$.

F: If X_0 is a scheme over $\mathbb{F}_q$, the endomorphism of X_0 that "sends the point with coordinates x to the point with coordinates x^q" is called the *Frobenius morphism* and denoted by F: it is the identity on the underlying topological space and, for x a local section of the structure sheaf, $F^*x = x^q$.

The endomorphism of X obtained from it by extension of scalars is also denoted by F. On the set $X(\mathbb{F}) = X_0(\mathbb{F})$ of rational points of X, F acts as the Frobenius substitution $\varphi \in \mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$ defined by $\varphi(x) = x^q$. In particular, $X^F = X_0(\mathbb{F}_q)$.

In case of ambiguity, we shall write $F_{(q)}$ rather than F.

1.2 Frobenius correspondence

Let X_0 be a scheme over $\mathbb{F}_q$, and let $\mathscr{F}_0$ be a sheaf on X_0. I like to view the *Frobenius correspondence* (an F-endomorphism of $\mathscr{F}_0$) as follows:

$$F^* : F^*\mathscr{F}_0 \xrightarrow{\sim} \mathscr{F}_0 \tag{1.2.1}$$

Regard $\mathscr{F}_0$ as the sheaf of local sections of a space $[\mathscr{F}_0]$ étale over X_0 ($[\mathscr{F}_0]$ is an algebraic space in

the sense of M. Artin). The functoriality of F gives a commutative diagram

$$\begin{array}{ccc} [\mathscr{F}_0] & \xrightarrow{F} & [\mathscr{F}_0] \\ \downarrow & & \downarrow \\ X_0 & \xrightarrow{F} & X_0 \end{array}$$

Because $[\mathscr{F}_0]$ is étale over X_0, this diagram is Cartesian; it gives an isomorphism $[\mathscr{F}_0] \xrightarrow{\sim} F^*[\mathscr{F}_0]$ whose inverse is (1.2.1). We also denote by F^* the correspondences or morphisms deduced from F by extension of scalars or by functoriality, such as $F^* : \mathrm{H}_c^\bullet(X, \mathscr{F}) \to \mathrm{H}_c^\bullet(X, \mathscr{F})$.

For a formal definition, I refer to [21, XI.1,2]. Only the case $q = p$ is considered there, but the general case is treated in the same way.

1.3

The Frobenius correspondence is functorial in $(X_0, \mathscr{F}_0)$. If $f_0 : X_0 \to Y_0$ is a morphism of $\mathbb{F}_q$-schemes, $\mathscr{F}_0$ a sheaf on X_0, $\mathscr{G}_0$ a sheaf on Y_0, and $u \in \hom(f_0^*\mathscr{G}_0, \mathscr{F}_0)$ an f_0-morphism from $\mathscr{G}_0$ to $\mathscr{F}_0$, there is a commutative diagram of spaces $f_0F = Ff_0$, and above it a commutative diagram of sheaves

$$\begin{array}{ccc} X_0 & \xrightarrow{f_0} & Y_0 \\ \downarrow{\scriptstyle F} & & \downarrow{\scriptstyle F} \\ X_0 & \xrightarrow{f_0} & Y_0 \end{array} \qquad \begin{array}{ccc} \mathscr{F}_0 & \xleftarrow{u} & \mathscr{G}_0 \\ \uparrow{\scriptstyle F^*} & & \uparrow{\scriptstyle F^*} \\ \mathscr{F}_0 & \xleftarrow{} & \mathscr{G}_0 \end{array}$$

In particular, the correspondence on $f_{0*}\mathscr{F}_0$ deduced from the Frobenius correspondence of $(X_0, \mathscr{F}_0)$ is the Frobenius correspondence of $(Y_0, f_{0*}\mathscr{F}_0)$.

1.4

The natural morphism $Y \to Y_0$ is a limit of étale morphisms. It follows that for every abelian sheaf $\mathscr{F}_0$ on X_0, the inverse images on Y of the $\mathsf{R}^i f_{0*}\mathscr{F}_0$ are the $\mathsf{R}^i f_*\mathscr{F}$.

On $\mathsf{R}^i f_*\mathscr{F}$, one has

a) the correspondence $F^*\mathsf{R}^i f_*\mathscr{F} \xrightarrow{\sim} \mathsf{R}^i f_*\mathscr{F} \xrightarrow{F^*} \mathsf{R}^i f_*\mathscr{F}$ deduced by functoriality from the Frobenius correspondence;

b) the correspondence $F^*\mathsf{R}^i f_*\mathscr{F} \to \mathsf{R}^i f_*\mathscr{F}$ that is the inverse image on Y of the Frobenius correspondence on $\mathsf{R}^i f_{0*}\mathscr{F}_0$.

It follows from (1.3) that these correspondences coincide. They induce the initial term of a Frobenius action on the entire spectral sequence

$$\mathrm{H}^p(Y, \mathsf{R}^q f_*\mathscr{F}) \to \mathrm{H}^{p+q}(X, \mathscr{F}).$$

Likewise for cohomology with proper support, and for the spectral sequence

$$\mathrm{H}_c^p(Y, \mathsf{R}^q f_*\mathscr{F}) \to \mathrm{H}_c^{p+q}(X, \mathscr{F})$$

1.5 Change of finite field

Let $(X_0, \mathscr{F}_0)$ be over $\mathbb{F}_q$, let n be an integer, and let $(X_1, \mathscr{F}_1)$ over $\mathbb{F}_{q^n}$ be obtained from $(X_0, \mathscr{F}_0)$ by extension of scalars. On the one hand there is the Frobenius correspondence $F_{(q)} : X_0 \to X_0$), $F^*_{(q)} : F^*_{(q)}\mathscr{F}_0 \to \mathscr{F}_0$, and on the other there is $F_{(q^n)} : X_1 \to X_1$, $F^*_{(q^n)} : F^*_{(q^n)}\mathscr{F}_1 \to \mathscr{F}_1$. It is easily verified that $F_{(q^n)}$ is obtained from the n-th power of $F_{(q)}$ by extension of scalars from $\mathbb{F}_q$ to $\mathbb{F}_{q^n}$; after extension of scalars to $\mathbb{F}$, one still has the same identity $F_{(q^n)} = F^n_{(q)}$ between endomorphisms of X and correspondences on X.

If x is a closed point of X_0, with $[k(x), \mathbb{F}_q] = n$, and $\bar{x} \in X_0(\mathbb{F}) = X(\mathbb{F})$ lies over x, then $\bar{x}$ is a fixed point of $F_{(q^n)} = F^n_{(q)}$. Denote by F^*_x the endomorphism of $\mathscr{F}_{\bar{x}}$ induced by $F^*_{(q^n)}$. Up to isomorphism, $(F^*_{\bar{x}}, \mathscr{F}_{\bar{x}})$ does not depend on the choice of $\bar{x}$. The trace, etc. of F^*_x is denoted by notation such as $\mathrm{tr}(F^*_x, \mathscr{F})$.

1.6 L-function

Let X_0 be a scheme of finite type over $\mathbb{F}_q$. Denote by $|X_0|$ the set of its closed points, and for $x \in |X_0|$ set $\deg(x) = [k(x), \mathbb{F}_p]$. If $\mathscr{F}_0$ is a $\mathbb{Q}_\ell$-sheaf on X_0 (for this notion, see below)—or a constructible flat sheaf of modules over a commutative Noetherian ring A—denote by $L(X_0, \mathscr{F}_0)$ the formal series with constant term 1, in $\mathbb{Q}_\ell[\![t]\!]$ or in $A[\![t]\!]$,

$$L(X_0, \mathscr{F}_0) = \prod_{x \in |X_0|} \det\left(1 - F^*_x t^{\deg(x)}, \mathscr{F}\right)^{-1}.$$

1.7 Remark

In definition (1.6), one has $\deg(x) = [k(x), \mathbb{F}_p]$ (absolute degree), and not $\deg(x) = [k(x), \mathbb{F}_q]$ (degree relative to $\mathbb{F}_q$). The effect is that $L(X_0, \mathscr{F}_0)$ depends only on the scheme X_0 and the sheaf $\mathscr{F}_0$, not on q and the structure morphism $X_0 \to \mathrm{Spec}(\mathbb{F}_q)$. The indeterminate t is an avatar of p^{-s}.

1.8 The Galois point of view

This subsection (n°) will not be used in the remainder of the report. For $\mathscr{F}_0$ an abelian sheaf on X_0, the Galois group $\mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$ acts on $\mathbb{F}$, and by transport of structure on $\mathrm{H}^i_c(X, \mathscr{F})$. In particular, the Frobenius substitution φ (1.1) induces an automorphism, also denoted by φ, of $\mathrm{H}^i_c(X, \mathscr{F})$. The Frobenius correspondence, on the other hand, defines an endomorphism F^* of $\mathrm{H}^i_c(X, \mathscr{F})$. One has

$$F^{*-1} = \varphi \qquad (\text{on } \mathrm{H}^i_c(X, \mathscr{F})). \tag{1.8.1}$$

To see this, apply (1.4) to the case where Y_0 is a point: $Y_0 = \mathrm{Spec}(\mathbb{F}_q)$. One finds that $\mathrm{H}^i_c(X, \mathscr{F})$ is the fiber, at the geometric point $\mathrm{Spec}(\mathbb{F})$ of Y_0, of $\mathsf{R}^i f_! \mathscr{F}_0$, and that F^* is the fiber of the Frobenius correspondence of $\mathsf{R}^i f_! \mathscr{F}_0$. Let $[\mathsf{R}^i f_! \mathscr{F}_0]$ be as in (1.2). One has $\mathrm{H}^i_c(X, \mathscr{F}) = [\mathsf{R}^i f_! \mathscr{F}_0](\mathbb{F})$ and

a) φ acts by transport of structure, via its action on $\mathbb{F}$;

b) F^* is the inverse of $F : [\mathsf{R}^i f_! \mathscr{F}_0](\mathbb{F}) \to [\mathsf{R}^i f_! \mathscr{F}_0](\mathbb{F})$.

The conclusion follows from the identity $F = \varphi$ of (1.1).

One advantage of the Galois point of view is that it permits reasoning by transport of structure.

2 $\mathbb{Q}_\ell$-sheaves

The notion of $\mathbb{Q}_\ell$-sheaves is developed in detail in [21, V,VI]. We shall use only the definitions and results below.

Definition 2.1. *Let X be a Noetherian scheme. A $\mathbb{Z}_\ell$-sheaf $\mathscr{F}$ on X is an inverse system of sheaves $\mathscr{F}_n$ ($n \geqslant 0$), with $\mathscr{F}_n$ a constructible sheaf of $\mathbb{Z}/\ell^{n+1}$-modules [2, IX.2], such that the transition morphism $\mathscr{F}_n \to \mathscr{F}_{n-1}$ factors through an isomorphism*

$$\mathscr{F}_n \otimes_{\mathbb{Z}/\ell^{n+1}} \mathbb{Z}/\ell^n \xrightarrow{\sim} \mathscr{F}_{n-1}.$$

The sheaf $\mathscr{F}$ is said to be lisse *if the $\mathscr{F}_n$ are locally constant.*

The terminology of [21] is different: "constructible $\mathbb{Z}_\ell$-sheaf" is used there for "$\mathbb{Z}_\ell$-sheaf," and "constructibly twisted constant" for "lisse."

Likewise below, for $\mathbb{Q}_\ell$-sheaves.

2.2

If k is a separably closed field, a sheaf of $\mathbb{Z}/\ell^n$-modules on $\mathrm{Spec}(k)$ is identified with a $\mathbb{Z}/\ell^n$-module, and the functor $\mathscr{F} \mapsto \varprojlim \mathscr{F}_n$ identifies the $\mathbb{Z}_\ell$-sheaves on $\mathrm{Spec}(k)$ with the $\mathbb{Z}_\ell$-modules of finite type. This justifies the following definition: the *fiber* of the $\mathbb{Z}_\ell$-sheaf $\mathscr{F}$ at the geometric point x of X is the $\mathbb{Z}_\ell$-module $\varprojlim(\mathscr{F}_n)_x$.

2.3

Suppose X is connected, and let x be a geometric point of X. It is known that the "fiber at x" functor is an equivalence between the category of constructible locally constant sheaves of $\mathbb{Z}/\ell^n$-modules and the category of finite-type $\mathbb{Z}/\ell^n$-modules equipped with a continuous action of $\pi_1(X,x)$. Passing to the limit gives the following proposition.

Proposition 2.4. *Under the hypotheses above, the "fiber at x" functor is an equivalence between the category of lisse $\mathbb{Z}_\ell$-sheaves on X and that of finite-type $\mathbb{Z}_\ell$-modules equipped with a continuous action of $\pi_1(X,x)$.*

Proposition 2.5. *Let $\mathscr{F}$ be a $\mathbb{Z}_\ell$-sheaf on a Noetherian scheme X. There is a finite partition of X into locally closed parts X_i such that the $\mathscr{F}|X_i$ are lisse.*

Set $\mathrm{gr}_\ell^n \mathscr{F} = \ell^n \mathscr{F}_m / \ell^{m+1}\mathscr{F}_m$ for $m \geqslant n$, and $\mathrm{gr}_\ell \mathscr{F} = \bigoplus \mathrm{gr}_\ell^n \mathscr{F}$. It is a sheaf of modules over the Noetherian ring $\mathrm{gr}_\ell \mathbb{Z} = \mathbb{Z}/\ell[t]$, the graded ring of $\mathbb{Z}$ for the ℓ-adic filtration. It is a quotient of $\mathrm{gr}_\ell^0 \mathscr{F} \otimes_{\mathbb{Z}/\ell} \mathbb{Z}/\ell[t]$, hence a constructible sheaf of $\mathrm{gr}_\ell \mathbb{Z}$-modules [2, IX.2]; and there is a finite partition of X into locally closed parts X_i such that $\mathrm{gr}_\ell \mathscr{F}|X_i$ is locally constant. Each of the sheaves $\mathrm{gr}_\ell^n \mathscr{F}|X_i$ is then locally constant, as are the $\mathscr{F}_m|X_i$, which are successive extensions of them.

2.6

If $\mathscr{F}$ and $\mathscr{G}$ are two $\mathbb{Z}_\ell$-sheaves on X, set $\hom(\mathscr{F},\mathscr{G}) = \varprojlim(\mathscr{F}_n, \mathscr{G}_n)$; this is a $\mathbb{Z}_\ell$-module. Since $\hom(\mathscr{F}_n,\mathscr{G}_n) = \hom(\mathscr{F}_m,\mathscr{G}_n)$ for $m \geqslant n$, one also has

$$\hom(\mathscr{F},\mathscr{G}) = \varprojlim_n \varinjlim_m \hom(\mathscr{F}_m,\mathscr{G}_n),$$

so that the functor $\mathscr{F} \mapsto \text{“}\varprojlim\text{”}\mathscr{F}_n$ is fully faithful from the category of $\mathbb{Z}_\ell$-sheaves to that of pro-sheaves on X (= pro-objects of the category of sheaves on X). We shall use it to identify $\mathbb{Z}_\ell$-sheaves with certain pro-sheaves (those such that every $\mathscr{F}_n = \mathscr{F}/\ell^{n+1}\mathscr{F}$ is a sheaf, and $\mathscr{F} \xrightarrow{\sim} \text{“}\varprojlim\text{”}\mathscr{F}_n$). It is easily verified that if $\mathscr{F} + \text{“}\varprojlim\text{”}\mathscr{F}_n$, with $\mathscr{F}_n$ a constructible sheaf of $\mathbb{Z}/\ell^n$-modules, then $\mathscr{F}$ is a $\mathbb{Z}_\ell$-sheaf if and only if the inverse systems (in n) $\mathscr{F}_n/\ell^{k+1}\mathscr{F}_n$ are essentially constant: the inverse system of $\mathscr{F}'_k = \varprojlim_n \mathscr{F}_n/\ell^{k+1}\mathscr{F}_n$ then satisfies the conditions of (2.1), and $\mathscr{F} = \text{“}\varprojlim\text{”}\mathscr{F}'_k$.

In what follows, we abandon the notation $\mathscr{F}_n$ of (2.1); if $\mathscr{F}$ is a $\mathbb{Z}_\ell$-sheaf, the sheaf $\mathscr{F}_n$ of (2.1) will be denoted by $\mathscr{F} \otimes \mathbb{Z}/\ell^{n+1}$.

2.7

In the abelian category of pro-sheaves, the subcategory of $\mathbb{Z}_\ell$-sheaves is stable under kernels, cokernels, and extensions. This is clear for cokernels. For kernels, if $f : \mathscr{F} \to \mathscr{G}$ is a morphism of $\mathbb{Z}_\ell$-sheaves, it follows readily from (2.4, 2.5) and Artin–Rees that the inverse system of $\ker(f : \mathscr{F} \otimes \mathbb{Z}/\ell^n \to \mathscr{G} \otimes \mathbb{Z}/\ell^n)$ satisfies the criterion given in (2.6) above. There is even an integer r such that for every n

$$\ker(f) \otimes \mathbb{Z}/\ell^n = \ker(f : \mathscr{F} \otimes \mathbb{Z}/\ell^{n+r} \to \mathscr{G} \otimes \mathbb{Z}/\ell^{n+r}) \otimes \mathbb{Z}/\ell^n.$$

The case of extensions is left to the reader.

2.8

A $\mathbb{Z}_\ell$-sheaf $\mathscr{F}$ is *torsion-free* if $\ker(\ell : \mathscr{F} \to \mathscr{F}) = 0$. Each $\mathscr{F} \otimes \mathbb{Z}/\ell^n$ is then flat over $\mathbb{Z}/\ell^n$ (i.e. its fibers are finite-rank free $\mathbb{Z}/\ell^n$-modules). It follows from (2.4, 2.5) that for every $\mathbb{Z}_\ell$-sheaf $\mathscr{F}$ there is an integer n such that $\mathscr{F}/\ker(\ell^n)$ is torsion-free.

2.9

The abelian category of $\mathbb{Q}_\ell$-*sheaves* on X is the quotient of that of $\mathbb{Z}_\ell$-sheaves by the subcategory of torsion $\mathbb{Z}_\ell$-sheaves. Equivalently, its objects are the $\mathbb{Z}_\ell$-sheaves, and, denoting by $\mathscr{F} \otimes \mathbb{Q}_\ell$ the $\mathbb{Z}_\ell$-sheaf $\mathscr{F}$, viewed as an object of the category of $\mathbb{Q}_\ell$-sheaves, one has

$$\hom(\mathscr{F} \otimes \mathbb{Q}_\ell, \mathscr{G} \otimes \mathbb{Q}_\ell) = \hom(\mathscr{F}, \mathscr{G}) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

The *fiber* of a $\mathbb{Q}_\ell$-sheaf $\mathscr{F} \otimes \mathbb{Q}_\ell$ at a geometric point x of X is the finite-dimensional $\mathbb{Q}_\ell$-vector space $\mathscr{F}_x \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$.

2.10

Let X be a separated scheme of finite type over an algebraically closed field k, and let $\mathscr{F}$ be a constructible $\mathbb{Q}_\ell$-sheaf on X. Let $\mathscr{F}'$ be a constructible $\mathbb{Z}_\ell$-sheaf such that $\mathscr{F} \sim \mathscr{F}' \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$. We shall see in (4.11) that

$$\left(\varprojlim \mathrm{H}^q_c(X, \mathscr{F}' \otimes \mathbb{Z}/\ell^n)\right) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$$

is a finite-dimensional $\mathbb{Q}_\ell$-vector space. It depends only on $\mathscr{F}'$, and is denoted by $\mathrm{H}^q_c(X, \mathscr{F})$. The proof does not use the fact that ℓ is prime to the characteristic, but only that case concerns us.

2.11

In this exposé, all important calculations will be carried out at the level of $\mathbb{Z}/\ell^n$-sheaves, with passage to $\mathbb{Q}_\ell$-sheaves taking place only at the very end. In practice, however, it is often convenient to work systematically with $\mathbb{Q}_\ell$-sheaves. In Exposés V and VI of [21], Jouanolou gives the formalism that makes this possible. One of the essential results (VI 2.2) is that if $f : X \to Y$ is a separated morphism of finite type of Noetherian schemes, and $\mathscr{F}$ a $\mathbb{Z}_\ell$-sheaf on Y, then, for every q, the pro-sheaf "$\varprojlim$"$\mathsf{R}^q f_!(\mathscr{F} \otimes \mathbb{Z}/\ell^n)$ is a $\mathbb{Z}_\ell$-sheaf on Y. He even proves a stronger regularity property (of Artin–Rees type) for the inverse system of the $\mathsf{R}^q f_!(\mathscr{F} \otimes \mathbb{Z}/\ell^n)$.

Jouanolou even works in a more general setting, encompassing the very useful case of E_λ-sheaves, for E_λ a finite extension of $\mathbb{Q}_\ell$. They are defined as the $\mathbb{Q}_\ell$-sheaves were, with $\mathbb{Q}_\ell$ replaced by E_λ, $\mathbb{Z}_\ell$ by the ring $\mathscr{O}_\lambda$ of the valuation λ, and $\mathbb{Z}/\ell^n$ by $\mathscr{O}_\lambda/\pi^n$, where π is a uniformizer. It would amount to the same thing to define an E_λ-sheaf as a $\mathbb{Q}_\ell$-sheaf $\mathscr{F}$ equipped with a homomorphism $E_\lambda \to \mathrm{End}(\mathscr{F})$.

3 Trace formula and L-functions

We use the notation of §1, and ℓ is a prime number $\neq p$.

Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$ ($q = p^f$), and let $\mathscr{F}_0$ be a constructible $\mathbb{Q}_\ell$-sheaf on X_0. Grothendieck's cohomological interpretation of L-functions is the following theorem:

Theorem 3.1. $L(X_0, \mathscr{F}_0) = \prod_i \det\left(1 - F^* t^f, \mathrm{H}^i_c(X, \mathscr{F})\right)^{(-1)^{i+1}}$.

It is obtained as a consequence of the trace formula:

Theorem 3.2. *For every integer n, $\sum_{x \in X^{F^n}} \mathrm{tr}\,(F^{n*}, \mathscr{F}_x) = \sum_i (-1)^i \,\mathrm{tr}\left(F^{n*}, \mathrm{H}^i_c(X, \mathscr{F})\right)$.*

Let us briefly recall the classical method for deducing (3.1) from (3.2). Set $T = t^f$; both sides are formal series in T with constant term 1. Since $\mathbb{Q}_\ell$ has characteristic 0, it suffices to prove that the logarithmic derivatives $T\frac{d}{dT}\log$ of the two sides are equal.

To compute these logarithmic derivatives, we shall use the formula

$$T\frac{d}{dT}\log\det(1 - fT)^{-1} = \sum_{n>0} \mathrm{tr}(f^n)T^n,$$

whose proof (in a slightly more general setting) is recalled below. On the left-hand side, one finds

$$\begin{aligned} T\frac{d}{dT}\log L(X_0, \mathscr{F}_0) &= \sum_{x \in |X_0|} \sum_n [k(x) : \mathbb{F}_q]\,\mathrm{tr}\,(F_x^n, \mathscr{F})\, T^{n \cdot [k(x):\mathbb{F}_q]} \\ &= \sum_{n>0} T^n \sum_{x \in X^{F^n}} \mathrm{tr}\,(F^{n*}, \mathscr{F}_x) \end{aligned}$$

and on the right-hand side

$$\sum_{n>0} T^n \sum (-1)^i \,\mathrm{tr}\left(F^{n*}, \mathrm{H}^i_c(X, \mathscr{F})\right)\ ;$$

the terms are compared one by one.

Proposition 3.3. *Let A be a commutative ring, M a projective A-module of finite type, and f an endomorphism of M. For every invertible formal series Q, set $\frac{d}{dT}\log Q = Q^{-1}\frac{d}{dT}Q$. One has*

$$T\frac{d}{dT}\log\det(1 - fT)^{-1} = \sum_{n>0} \mathrm{tr}(f^n)T^n. \tag{3.3.1}$$

Let M' be an A-module such that $M \oplus M'$ is free of finite rank. Replacing (M, f) by $(M \oplus M', f \oplus 0)$ changes neither side of (3.3), and reduces us to the case where M is free. For $M = A^d$, (3.3.1) is an algebraic identity in the coefficients f_i^j of f. By the principle of extension of algebraic identities, it suffices to verify it for A an algebraically closed field of characteristic 0. Since both sides are additive in M in an extension $0 \to M' \to M \to M'' \to 0$, it even suffices to treat only the case where M has rank 1. If f is multiplication by a, (3.3.1) then reduces to the formula

$$T\frac{d}{dT}\log\left((1-aT)^{-1}\right) = \sum_{n>0} a^n T^n.$$

Corollary 3.4. *Let n be an integer, and let $f^{(n)}$ be the endomorphism $(x_1, \ldots, x_n) \mapsto (f(x_n), x_1, \ldots, x_{n-1})$ of M^n. One has*

$$\det(1 - fT^n) = \det(1 - f^{(n)}T). \tag{3.4.1}$$

Proceeding as in (3.3), we reduce to assuming that A has characteristic 0. It then suffices to verify that the logarithmic derivatives $-T\frac{d}{dT}\log$ of the two sides are equal.

$$-T\frac{d}{dT}\log\det(1-fT^n) = -nT^n\frac{d}{dT^n}\log\det(1-fT^n) = n\sum_{m>0}\operatorname{tr}(f^m)T^{nm},$$

$$-T\frac{d}{dT}\log\det\left(1-f^{(n)}T\right) = \sum_{m>0}\operatorname{tr}\left(f^{(n)m}\right)T^m,$$

and one observes that

$$\operatorname{tr}(f^{(n)m}) = 0 \qquad \text{if } n \nmid m$$
$$\operatorname{tr}(f^{(n)nm}) = n\operatorname{tr}(f^m).$$

3.5 Remark

For $A = \mathbb{Q}_\ell$, (3.4.1) is the special case $X_0 = \operatorname{Spec}(\mathbb{F}_{q^n})$ of (3.1). Using this identity, it is easy to verify directly that the right-hand side of (3.1) is independent of q and of the structure morphism $X_0 \to \operatorname{Spec}(\mathbb{F}_q)$.

3.6 Remark

If $j : X_0 \hookrightarrow \bar{X}_0$ is a compactification of X_0, one has $L(X_0, \mathscr{F}_0) = L(\bar{X}_0, j_!\mathscr{F}_0)$ and $\mathrm{H}^i_c(X, \mathscr{F}) = \mathrm{H}^i(\bar{X}, j_!\mathscr{F})$. This reduces (3.1) to the proper case, and explains the use of cohomology with proper support. For X proper, (3.2) has the form of a Lefschetz trace formula. Note that the local terms $\operatorname{tr}(F^{n*}, \mathscr{F}_x)$ $(x \in X^{F^n})$ depend only on the fiber of $\mathscr{F}$ at x and carry no multiplicity. For X and $\mathscr{F}$ smooth, this corresponds to the fact that the graph of F is transverse to the diagonal (since $dF = 0$).

3.7 Remark

If the formalism of $\mathbb{Q}_\ell$-sheaves is admitted (cf. 2.11; one needs the Leray spectral sequence and the base-change theorem for the higher direct images $\mathsf{R}^q f_!$), it is easy to reduce the proof of (3.1, 3.2) to the case where X_0 is a smooth curve and $\mathscr{F}_0$ is lisse. This is clearly explained in [22, §5] (for 3.1; 3.2 is treated in the same way).

3.8

There are two methods for proving (3.2).

Lefschetz–Verdier If X_0 is proper (cf. 3.6), the general Lefschetz–Verdier trace formula expresses the right-hand side of (3.2) as a sum of local terms, one for each point of X^{F^n}. In the original version of [21], this formula was proved only modulo resolution of singularities. The reader will find an unconditional proof in the definitive version. In the case of curves, to which one can reduce (3.7), all the ingredients of the proof were in fact already available.

To deduce (3.2) from the Lefschetz–Verdier formula, one must be able to compute its local terms. For a curve and the Frobenius endomorphism, this had been done by Artin and Verdier [40].

Nielson–Wecken A method inspired by the work of Nielsen and Wecken makes it possible to reduce (3.2) to a special case proved by Weil; this is what will be explained in the following sections.

4 Reduction to theorems in $\mathbb{Z}/\ell^n$-cohomology

We shall prove (3.2) by passage to the limit from an analogous theorem in $\mathbb{Z}/\ell^n$-cohomology. Its statement presents the following difficulty. Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$, and let $\mathscr{F}_0$ be a flat constructible sheaf of $\mathbb{Z}/\ell^n$-modules. Its fibers are finite-rank free $\mathbb{Z}/\ell^n$-modules, and the left-hand side of (3.2) therefore makes sense (it is an element of $\mathbb{Z}/\ell^n$). The $\mathrm{H}^i_c(X, \mathscr{F})$, on the other hand, will not in general be free, so the right-hand side is not defined. Overcoming this difficulty is one of Grothendieck's essential uses of the theory of derived categories.

4.1

For the definition of the derived category $\mathsf{D}(\mathcal{A})$ of an abelian category $\mathcal{A}$, and that of the subcategories $\mathsf{D}^+(\mathcal{A})$, $\mathsf{D}^-(\mathcal{A})$, $\mathsf{D}^b(\mathcal{A})$, I refer to Verdier's text in this volume. For Λ a ring, write $\mathsf{D}(\Lambda)$ instead of D(category of left Λ-modules); for X a site, write $\mathsf{D}(X, \Lambda)$ instead of D(category of sheaves of left Λ-modules). Likewise for D^-, etc...

The sign R (resp. L) indicates a right-derived (resp. left-derived) functor. For example, $\overset{\mathsf{L}}{\otimes}$ denotes the derived functor of the tensor product: for K in $\mathsf{D}^-(\Lambda^\circ)$ and L in $\mathsf{D}^-(\Lambda)$ (Λ° the opposite ring of Λ), $K \overset{\mathsf{L}}{\otimes} L$ is computed as follows: take quasi-isomorphisms $K' \to K$, $L' \to L$, with K' and L bounded above and one of them having flat components, and take the simple complex $s(K' \otimes_\Lambda L)$ associated with the double complex $K' \otimes_\Lambda L'$.

4.2

Even to treat only the case of $\mathbb{Z}/\ell^n$-cohomology, we shall need the theory of traces of endomorphisms of modules over rings that are not necessarily commutative (finite-group algebras $\mathbb{Z}/\ell^n[G]$). These traces were introduced by Stallings and Hattori [3].

Let Λ be a ring (with identity, but not necessarily commutative). Denote by $\Lambda^\natural$ the quotient of the additive group of Λ by the subgroup generated by the commutators $ab - ba$. If f is an endomorphism of the free left Λ-module Λ^n, with matrix f^j_i, denote by $\mathrm{tr}(f)$ the image of $\sum_i f^i_i$ in $\Lambda^\natural$. If f and g are two homomorphisms $f : \Lambda^n \leftrightarrows \Lambda^m : g$, one immediately verifies that $\mathrm{tr}(fg) = \mathrm{tr}(gf)$.

Let f be an endomorphism of a projective Λ-module of finite type P. Write P as a direct factor of a free module Λ^n: this means choosing a module P' and an isomorphism $\alpha : P \oplus P' \xrightarrow{\sim} \Lambda^n$, or equivalently choosing a homomorphism $a : P \to \Lambda^n$ and a retraction $b : \Lambda^n \to P$ (take $P' = \ker(b)$). Let $f' = \alpha(f \oplus 0)\alpha^{-1} = afb$. Set $\mathrm{tr}(f) = \mathrm{tr}(f')$. This element of $\Lambda^\natural$ depends only on f: if another choice

$c : P \leftrightarrows \Lambda^m : d$ is made, one has $a = a(dc)(ba)$, whence

$$\mathrm{tr}(afb) = \mathrm{tr}(adcbafb) = \mathrm{tr}(cbafbad) = \mathrm{tr}(cfd).$$

If f is an endomorphism of a $\mathbb{Z}/2$-graded projective Λ-module of finite type, with components $f_i^j : P^j \to P^i$, set

$$\mathrm{tr}(f) = \mathrm{tr}(f_0^0) - \mathrm{tr}(f_1^1).$$

If there is a risk of ambiguity, we shall instead write $\mathrm{tr}(f; P)$. If the homomorphisms $f : P \leftrightarrows Q : g$ are homogeneous, reduction to the free case immediately verifies that

$$\mathrm{tr}(fg) = \mathrm{tr}\left((-1)^{\deg f \deg g} gf\right). \tag{4.2.1}$$

If f is an endomorphism of a filtered $\mathbb{Z}/2$-graded Λ-module (P, F), with finite filtration compatible with the grading, and the $\mathrm{gr}_F^i(P)$ are projective of finite type, then P is projective of finite type, and

$$\mathrm{tr}(f; P) = \sum \mathrm{tr}\left(f; \mathrm{gr}_F^i(P)\right) \tag{4.2.2}$$

(split the filtration to reduce to the case of a direct sum).

4.3

A $\mathbb{Z}$-graded module will be regarded as $\mathbb{Z}/2$-graded by the parity of the degree. In particular, if f is an endomorphism of a bounded complex of projective Λ-modules of finite type, set

$$\mathrm{tr}(f) = \sum (-1)^i \mathrm{tr}(f^i). \tag{4.3.1}$$

By (4.2.1), if f is homotopic to zero, $\mathrm{tr}(f) = 0$:

$$f = dH + Hd \to \mathrm{tr}(f) = 0. \tag{4.3.2}$$

Let $\mathsf{K}_{\mathrm{parf}}(\Lambda)$ be the category whose objects are bounded complexes of projective Λ-modules of finite type and whose arrows are morphisms of complexes taken up to homotopy. The functor $\mathsf{K}_{\mathrm{parf}}(\Lambda) \to \mathsf{D}(\Lambda)$ is fully faithful. Denote its essential image by $\mathsf{D}_{\mathrm{parf}}(\Lambda)$. Formulas (4.3.1) and (4.3.2) make it possible to define $\mathrm{tr}(f)$ for f an endomorphism of $K \in \mathrm{Ob}\,\mathsf{D}_{\mathrm{parf}}(\Lambda)$ (and this trace depends only on the isomorphism class of (K, f)).

4.4

Recall that the filtered derived category $\mathsf{DF}(\mathcal{A})$ of an abelian category $\mathcal{A}$ is the category obtained from that of complexes of objects of $\mathcal{A}$ with finite filtration by inverting the filtered quasi-isomorphisms (= morphisms of filtered complexes inducing a quasi-isomorphism on the associated graded). There are functors "underlying complex": $\mathsf{DF}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}) : (K, F) \mapsto K$, and "$p$-th component of the associated graded": $\mathsf{DF}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}) : (K, F) \mapsto \mathrm{gr}_F^p(K)$.

Let $\mathsf{KF}_{\mathrm{parf}}(\Lambda)$ be the category whose objects are bounded filtered complexes (K, F) with finite filtration, with the $\mathrm{gr}_F^p(K^q)$ projective of finite type, and whose arrows are filtration-preserving morphisms of complexes taken up to a filtration-preserving homotopy. The functor $\mathsf{KF}_{\mathrm{parf}}(\Lambda) \to \mathsf{DF}(\Lambda)$ is fully faithful; its essential image $\mathsf{DF}_{\mathrm{parf}}(\Lambda)$ consists of the (K, F) in $\mathsf{DF}(\Lambda)$ such that the $\mathrm{gr}_F^p(K)$ belong to $\mathsf{D}_{\mathrm{parf}}(\Lambda)$ and are almost all zero; if (K, F) belongs to $\mathsf{DF}_{\mathrm{parf}}(\Lambda)$, then K belongs to $\mathsf{D}_{\mathrm{parf}}(\Lambda)$.

By (4.2.2), if f is an endomorphism of $(K, F) \in \mathrm{Ob}\,\mathsf{DF}_{\mathrm{parf}}(\Lambda)$, one has

$$\mathrm{tr}(f, K) = \sum_p \mathrm{tr}(f, \mathrm{gr}_F^p(K)). \tag{4.4.1}$$

4.5

Let Λ be a ring, and let $K \in \mathrm{Ob}\,\mathsf{D}^-(\Lambda)$. We say that K has tor-dimension $\leqslant r$ if, for every right Λ-module N, the hypertors $\mathrm{H}^i(N \overset{\mathsf{L}}{\otimes}_\Lambda K)$ vanish for $i < -r$. It is said to have finite tor-dimension if this condition holds for some r.

The same terminology is used for a complex of sheaves $K \in \mathrm{Ob}\,\mathsf{D}^-(X, \Lambda)$.

Lemma 4.5.1. *Let Λ be a left Noetherian ring, and let $K \in \mathrm{Ob}\,\mathsf{D}^-(\Lambda)$. Then K belongs to $\mathsf{D}_{parf}(\Lambda)$ if and only if it has finite tor-dimension and the $\mathrm{H}^i(K)$ are of finite type.*

After replacing K by an isomorphic complex (in $\mathsf{D}^-(\Lambda)$), we may suppose that K is bounded above and has finite-rank free components (cf. 4.7 below). If K has tor-dimension $\leqslant r$, the $\mathrm{H}^i(K)$ vanish for $i < -r$, $K^{-r}/\operatorname{im}(d)$ admits the free resolution

$$(\cdots \to K^{-r-1} \to K^{-r}) \to K^{-r}/\operatorname{im}(d),$$

and for every right Λ-module N, $\mathrm{H}^{-r-k}(N \overset{\mathsf{L}}{\otimes}_\Lambda K) = \mathrm{H}^{-r-k}(N \otimes_\Lambda K) = \mathrm{Tor}_k(N, K^{-r}/\operatorname{im} d)$ for $k \geqslant 1$.

By hypothesis, these groups vanish; the module $K^{-r}/\operatorname{im} d$ is therefore flat and finitely presented, i.e. projective of finite type. The quotient

$$0 \to K^{-r}/\operatorname{im}(d) \to K^{-r+1} \to K^{-r+2} \to \cdots$$

of K is quasi-isomorphic to K, and this shows that $K \in \mathrm{Ob}\,\mathsf{D}_{\mathrm{parf}}(\Lambda)$.

Proposition–Definition 4.6. *Let X be a Noetherian scheme, and let Λ be a left Noetherian ring. Denote by $\mathsf{D}^b_{\mathrm{ctf}}(X, \Lambda)$ the subcategory of $\mathsf{D}^-(X, \Lambda)$ formed by the complexes $\mathscr{K}$ satisfying the following equivalent conditions:*

i) $\mathscr{K}$ is isomorphic (in $\mathsf{D}^-(X, \Lambda)$) to a bounded complex of flat constructible sheaves of Λ-modules.

ii) $\mathscr{K}$ has finite tor-dimension and the sheaves $\mathrm{H}^i(K)$ are constructible.

The same argument as in (4.5), starting from the following lemma.

Lemma 4.7. *If a complex $\mathscr{K}$ of sheaves of Λ-modules is such that the $\mathrm{H}^i(\mathscr{K})$ are constructible and vanish for large i, there is a quasi-isomorphism $\mathscr{K}' \to \mathscr{K}$, with $\mathscr{K}'$ bounded above and having flat constructible components.*

Construct $\mathscr{K}'$ by proceeding from right to left. For an m such that the $\mathrm{H}^i(\mathscr{K})$ $(i \geqslant m)$ vanish, take $\mathscr{K}'^i = 0$ for $i \geqslant m$. Next suppose that the $\mathscr{K}'^i$ have already been constructed for $i > n$:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & \mathscr{K}^{n-1} & \longrightarrow & \mathscr{K}^n & \longrightarrow & \mathscr{K}^{n+1} & \longrightarrow & \cdots \\ & & & & & & \uparrow & & \\ & & & & & & \mathscr{K}'^{n+1} & \longrightarrow & \cdots, \end{array}$$

with $\mathrm{H}^i(\mathscr{K}') \xrightarrow{\sim} \mathrm{H}^i(\mathscr{K})$ for $i > n+1$, and $\ker(d) \to \mathrm{H}^{n+1}(\mathscr{K})$ surjective in degree $n+1$. Define sheaves $\mathscr{A}$ and $\mathscr{B}$ by the Cartesian diagram

$$\begin{array}{ccccc} \mathscr{K}^n & \longrightarrow & \mathscr{K}^n/\operatorname{im}(d) & \longrightarrow & \ker(d) \\ \uparrow & & \uparrow & & \uparrow \\ \mathscr{A} & \xrightarrow{u} & \mathscr{B} & \longrightarrow & \ker(d). \end{array}$$

Then $\mathscr{B}$ is constructible, u is surjective, and to advance one step it suffices to construct $\mathscr{K}'^n$ flat and constructible and $v : \mathscr{K}'^n \to \mathscr{A}$ such that uv is surjective. The possibility of doing so is guaranteed by the following lemma.

Lemma 4.8. *Let $v : \mathscr{A} \to \mathscr{B}$ be an epimorphism of sheaves of Λ-modules, with $\mathscr{B}$ constructible. There is then $u : \mathscr{K} \to \mathscr{A}$, with $\mathscr{K}$ flat and constructible, such that vu is an epimorphism.*

The sheaf $\mathscr{A}$ is a quotient of a sum of sheaves of the form $\varphi_!\Lambda$, for $\varphi : U \to X$ étale. Since $\mathscr{B}$ is Noetherian ([2, IX.2.10]), a finite sum of these already maps onto $\mathscr{B}$, which proves the lemma.

Theorem 4.9. *Let $f : X \to Y$ be a separated morphism of finite type of Noetherian schemes. If $\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(X, \Lambda)$, then $\mathsf{R}f_!\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(Y, \Lambda)$.*

Recall that if $j : X \hookrightarrow \bar{X} \xrightarrow{\bar{f}} Y$ is a relative compactification of X/Y, then by definition $\mathsf{R}f_! = \mathsf{R}\bar{f}_* \circ j_!$, where $\mathsf{R}\bar{f}_*$ is the derived functor of $\bar{f}_*$. This formula reduces the proof of (4.9) to the case where f is proper. For f proper, f_* has finite cohomological dimension; this makes it possible to define $\mathsf{R}f_*$ on the entire derived category, and also as a functor

$$\mathsf{R}f_* : \mathsf{D}^-(X, \Lambda) \to \mathsf{D}^-(Y, \Lambda).$$

By composition, one likewise defines in general

$$\mathsf{R}f_! : \mathsf{D}^-(X, \Lambda) \to \mathsf{D}^-(Y, \Lambda).$$

For every right Λ-module N, one has (proof given below)

$$N \overset{\mathsf{L}}{\otimes} \mathsf{R}f_!\mathscr{K} \xrightarrow{\sim} \mathsf{R}f_!(N \overset{\mathsf{L}}{\otimes} \mathscr{K}). \tag{4.9.1}$$

Let us deduce (4.9) from (4.9.1).

a) The spectral sequence
$$E_2^{pq} = \mathsf{R}^p f_!\, \mathrm{H}^q(\mathscr{K}) \to \mathrm{H}^{p+q}\, \mathsf{R}f_!\mathscr{K}\,.$$
and the finiteness theorem for $\mathsf{R}f_!$ show that $\mathsf{R}f_!\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}^b(Y, \Lambda)$ and has constructible cohomology. It remains to prove finite tor-dimension (4.6).

b) If $\mathscr{K}$ has tor-dimension $\leqslant -r$, the same is true of $\mathsf{R}f_!\mathscr{K}$: one has $\mathrm{H}^i(N \overset{\mathsf{L}}{\otimes} \mathsf{R}f_!\mathscr{K}) = \mathrm{H}^i(\mathsf{R}f_!(N \overset{\mathsf{L}}{\otimes} \mathscr{K}))$, and in the spectral sequence above for $N \overset{\mathsf{L}}{\otimes} \mathscr{K}$, the E_2^{pq} vanish for $q < r$, hence a fortiori for $p+q < r$.

Let us prove (4.9.1), remaining at the level of complexes.

a) Using the defining formula $\mathsf{R}f_! = \mathsf{R}f_* \circ j_!$, we reduce to assuming that f is proper.

b) Represent $\mathscr{K}$ by a complex bounded above whose components are acyclic for f_*: $\mathsf{R}^p f_* \mathscr{K}^q = 0$ for $p > 0$. One then has $\mathsf{R}f_*K \sim f_*\mathscr{K}$. It is possible to work in this way with complexes bounded above only because f_* has finite cohomological dimension.

c) If L is a free right Λ-module, one has $\mathsf{R}^p f_*(L \otimes \mathscr{K}^q) = L \otimes \mathsf{R}^p f_* \mathscr{K}^q$: this is trivial when Λ is free of finite type, and the general case follows by passage to the limit. The sheaf $L \otimes \mathscr{K}^q$ is acyclic for f_*, and $f_*(L \otimes \mathscr{K}^q) = L \otimes f_*\mathscr{K}^q$.

d) Let $N_\bullet$ be a free resolution of N. One has $N\overset{\mathsf{L}}{\otimes}\mathscr{K} \sim s(N_\bullet \otimes \mathscr{K})$ (the simple complex associated with the double complex $N_\bullet \otimes \mathscr{K}$). By c),

$$\mathsf{R}f_*(N\overset{\mathsf{L}}{\otimes}\mathscr{K}) \sim \mathsf{R}f_*(s(N_\bullet \otimes \mathscr{K})) \sim f_*(s(N_\bullet \otimes \mathscr{K})) \sim s(N_\bullet \otimes f_*\mathscr{K}) \sim N\overset{\mathsf{L}}{\otimes}\mathsf{R}f_*\mathscr{K},$$

which is the asserted formula.

When Y is the spectrum of a separably closed field, the functor $\mathsf{R}f_!$ is identified with a functor denoted by $\mathsf{R}\Gamma_c : \mathsf{D}^b_{\mathrm{ctf}}(X,\Lambda) \to \mathsf{D}_{\mathrm{parf}}(\Lambda)$.

Theorem 4.10. *With the notation of §1, let X_0 be a separated scheme of finite type over $\mathbb{F}_q$, and let Λ be a Noetherian torsion ring annihilated by an integer prime to p. Let $\mathscr{K}_0 \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(X_0,\Lambda)$. With the definition of traces in (4.3), one has for every N*

$$\sum_{x\in X^{F^n}} \mathrm{tr}\,(F^{n*},\mathscr{K}_x) = \mathrm{tr}\,(F^{n*},\mathsf{R}\Gamma_c(X,\mathscr{K}))\,.$$

In the remainder of this §, we shall deduce 3.2 from 4.10, and prove 2.10.

4.11 Proof of 2.10

Let X be a separated scheme of finite type over an algebraically closed field k, and let $\mathscr{G}$ be a $\mathbb{Q}_\ell$-sheaf on X. By 2.8, it can be written in the form $\mathscr{G} = \mathscr{F}\otimes\mathbb{Q}_\ell$, with $\mathscr{F}$ a torsion-free $\mathbb{Z}_\ell$-sheaf. Set

$$K_n = \mathsf{R}\Gamma_c(X, \mathscr{F}\otimes\mathbb{Z}/\ell^{n+1}) \in \mathsf{D}(\mathbb{Z}/\ell^{n+1}).$$

By 4.9, one has $K_n \in \mathrm{Ob}\,\mathsf{D}_{\mathrm{parf}}(\mathbb{Z}/\ell^{n+1})$, and it follows from (4.9.1) that the natural arrow

$$K_n\overset{\mathsf{L}}{\otimes}_{\mathbb{Z}/\ell^{n+1}}\mathbb{Z}/\ell^n \longrightarrow K_{n-1}, \tag{4.11.1}$$

is an isomorphism. The problem of defining this arrow is discussed in 4.12.

We may now invoke [21, XV.3.3] (p. 31, lines 1–2 from the end; this passage is independent of the rest of [21]) to realize each K_n by a bounded complex of finite-rank free $\mathbb{Z}/\ell^{n+1}$-modules, and the arrows (4.11.1) by *isomorphisms of complexes*

$$K_n \otimes \mathbb{Z}/\ell^n \xrightarrow{\sim} K_{n-1}.$$

Let K be the complex $\varprojlim K_n$. It is a bounded complex of free $\mathbb{Z}_\ell$-modules, and $K_n \simeq K\otimes_{\mathbb{Z}_\ell}\mathbb{Z}/\ell^{n+1}$. One has

$$\mathrm{H}^i(K) = \varprojlim \mathrm{H}^i(K_n),$$

(indeed, every inverse system of finite groups satisfies the Mittag–Leffler condition), and $\mathrm{H}^i(K)$ is a $\mathbb{Z}_\ell$-module of finite type. One has

$$\mathrm{H}^i_c(X,\mathscr{G}) = \left(\varprojlim \mathrm{H}^i(K_n)\right)\otimes_{\mathbb{Z}_\ell}\mathbb{Q}_\ell:$$

this is a finite-dimensional $\mathbb{Q}_\ell$-vector space.

4.12 The arrow (4.11.1)

To *define* the arrow (4.11.1), one cannot proceed as in 4.9, i.e. resolve $\mathbb{Z}/\ell^n$ by a complex of free $\mathbb{Z}/\ell^{n+1}$-modules, since one wishes to construct an isomorphism in $\mathsf{D}_{\mathrm{parf}}(\mathbb{Z}/\ell^n)$. The desired arrow is a special case of the following scalar-extension arrow.

(*) Let $\Lambda \to \Lambda'$ be a homomorphism of Noetherian torsion rings. For X a separated scheme of finite type over an algebraically closed k, and $\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}_{\mathrm{ctf}}(X, \Lambda)$, one has an isomorphism

$$\mathsf{R}\Gamma_c(X, \mathscr{K}) \overset{\mathsf{L}}{\otimes}_\Lambda \Lambda' \xrightarrow{\sim} \mathsf{R}\Gamma_c(X, \mathscr{K} \overset{\mathsf{L}}{\otimes}_\Lambda \Lambda').$$

More general arrows are constructed in [2, XVII 4.2.12]. The method is as follows:

a) The definition of $\mathsf{R}\Gamma_c$ and the formula $j_!(\mathscr{K} \otimes_\Lambda \Lambda') = (j_!\mathscr{K}) \otimes_\Lambda \Lambda'$ for an open immersion reduce us to the case where X is proper.

b) For X proper, one must replace $\mathscr{K}$ by a bounded complex whose components are acyclic for the functor Γ, and whose fiber at every geometric point (or merely every closed point) is homotopic to a complex of flat Λ-modules. For such a complex, one has $\Gamma(X, \mathscr{K}) \sim \mathsf{R}\Gamma(X, \mathscr{K})$ and $K \otimes_\Lambda \Lambda' \sim \mathscr{K} \overset{\mathsf{L}}{\otimes}_\Lambda \Lambda'$; the arrow (4.11.1) is defined as the composite

$$\mathsf{R}\Gamma(X, \mathscr{K}) \overset{\mathsf{L}}{\otimes}_\Lambda \Lambda' \to \Gamma(X, \mathscr{K}) \otimes_\Lambda \Lambda' \to \Gamma(X, \mathscr{K} \otimes_\Lambda \Lambda') \xleftarrow{\sim} \mathsf{R}\Gamma(X, \mathscr{K} \overset{\mathsf{L}}{\otimes}_\Lambda \Lambda').$$

c) It remains to show that $\mathscr{K}$ can be replaced by a complex of the desired kind [2, XVII.4.2.10]. First replace $\mathscr{K}$ by a bounded complex with flat components (4.6), then take a truncated canonical flasque resolution of it (this does not modify the homotopy type of the fiber complexes, and one truncates far enough for the acyclicity condition for Γ to hold (i.e. beyond the cohomological dimension of Γ)).

4.13 Proof of (3.2)

To simplify notation, we shall treat only the case $n = 1$ of (3.2). The general case follows by replacing F below by F^n.

Thus let X_0 be separated and of finite type over $\mathbb{F}_q$, and let $\mathscr{G}_0$ be a $\mathbb{Q}_\ell$-sheaf on X_0. One has $\mathscr{G}_0 = \mathscr{F}_0 \otimes \mathbb{Q}_\ell$, for a suitable torsion-free $\mathbb{Z}_\ell$-sheaf $\mathscr{F}_0$. Let $K_n = \mathsf{R}\Gamma_c(X, \mathscr{F} \otimes \mathbb{Z}/\ell^n)$. The Frobenius morphisms induce morphisms

$$F^* : K_n \to K_n \qquad (\text{in } \mathsf{D}_{\mathrm{parf}}(\mathbb{Z}/\ell^{n+1})),$$

which are deduced from one another via the isomorphisms (4.11.1).

As in (4.11), realize the K_n and the isomorphisms (4.11.1) at the level of complexes. The already cited passage [21, XV.3.3] makes it possible to realize the morphisms F^* by endomorphisms of complexes, again denoted by F^*, which reduce to one another. Again denote by F^* their inverse limit and the endomorphisms deduced from it. Since $\mathrm{H}^i_c(X, \mathscr{G}) = \mathrm{H}^i(K) \otimes \mathbb{Q}_\ell = \mathrm{H}^i(K \otimes \mathbb{Q}_\ell)$, one has (with the sign convention of 4.3.1)

$$\mathrm{tr}\,(F^*, \mathrm{H}^\bullet_c(X, \mathscr{G})) = \mathrm{tr}\,(F^*, K^\bullet \otimes \mathbb{Q}_\ell) = \varprojlim_n \mathrm{tr}\,(F^*, K_n^\bullet) = \varprojlim_n \mathrm{tr}\left(F^*, \mathsf{R}\Gamma_c(X, \mathscr{F} \otimes \mathbb{Z}/\ell^{n+1})\right).$$

Apply (4.10) to $\mathscr{F}_0 \otimes \mathbb{Z}/\ell^{n+1}$ (viewed as a complex concentrated in degree 0), for $\Lambda = \mathbb{Z}/\ell^{n+1}$. One finds that

$$\begin{aligned}\operatorname{tr}\left(F^*, \mathsf{R}\Gamma_c(X, \mathscr{F} \otimes \mathbb{Z}/\ell^{n+1})\right) &= \sum_{x \in X^F} \operatorname{tr}\left(F^*, \mathscr{F}_x \otimes \mathbb{Z}/\ell^{n+1}\right) \\ &= \sum_{x \in X^F} \operatorname{tr}\left(F^*, \mathscr{G}_x\right) \mod \ell^{n+1},\end{aligned}$$

and (3.2) follows by passage to the limit.

5 The Nielsen–Wecken method

In this chapter, we prove two special cases of (4.10). The general case will be considered in the next chapter.

Lemma 5.1. *Theorem (4.10) is true for X_0 of dimension 0.*

In dimension 0, formula (4.10) holds for every correspondence, and not only for the iterates of a Frobenius. It is a special case of the following statement (applied to $X(\mathbb{F})$, K, and the F^{*n}).

(*) Let X be a finite set, $F : X \to X$, $\mathscr{K} \in \operatorname{Ob} \mathsf{D}_{\mathrm{parf}}(X, \Lambda)$, and $F^* : F^*\mathscr{K} \to \mathscr{K}$. One has

$$\sum_{X^F} \operatorname{tr}\left(F^*, \mathscr{K}_x\right) = \operatorname{tr}\left(F^*, \Gamma(X, K)\right).$$

One has $\Gamma(X, \mathscr{K}) = \bigoplus_{x \in X} \mathscr{K}_x$, and the formula follows from the fact that for every morphism $u : \bigoplus_{x \in X} \mathscr{K}_x \to \bigoplus_{x \in X} \mathscr{K}_x$, with matrix u^y_x, one has $\operatorname{tr}(u) = \sum_{x \in X} \operatorname{tr}\left(u^x_x\right)$.

Lemma 5.2. *Let Λ be a Noetherian torsion ring annihilated by a power of the prime number $\ell \neq p$, let X_0 be a smooth projective connected curve over $\mathbb{F}_q$, let $j : U_0 \hookrightarrow X_0$ be a dense open subscheme of X_0, and let $\mathscr{G}_0$ be a locally constant sheaf of projective Λ-modules on U_0. One has*

$$\sum_{x \in U^F} \operatorname{tr}\left(F^*, \mathscr{G}_x\right) = \operatorname{tr}\left(F^*, \mathsf{R}\Gamma_c(U, \mathscr{G})\right).$$

More precisely, we shall deduce (5.2) from the following theorem.

Theorem 5.3. *Let X be a nonsingular projective curve over an algebraically closed field k, let f be an endomorphism of X with isolated fixed points, and let $v(f)$ be the number of fixed points of f, each counted with its multiplicity: $v(f)$ is the intersection number of the graph of f with the diagonal of $X \times X$. Then, for ℓ prime to the characteristic exponent of k, one has*

$$v(f) = \sum (-1)^i \operatorname{tr}\left(f^*, \mathrm{H}^i(X, \mathbb{Q}_\ell)\right).$$

This theorem is due to Weil ([41], n° 43, 66, and 68), and Weil's proof is reproduced in Lang [26] (VI §3 combined with VII §2, Th. 3). It can also be deduced from the formalism of the cohomology class associated with a cycle (The cohomology class associated with a cycle, 3.7).

Corollary 5.4. *Let $j : U \hookrightarrow X$ be a dense open subscheme of X, such that $f^{-1}(U) = U$. If the fixed points of f in $X \smallsetminus U$ have multiplicity one, then the number $v_U(f)$ of fixed points of f in U (each counted with its multiplicity) is*

$$v_U(f) = \sum (-1)^i \operatorname{tr}\left(f^*, \mathrm{H}^i_c(U, \mathbb{Q}_\ell)\right).$$

Let F be the finite set $X \smallsetminus U$. The cohomology exact sequence defined by the short exact sequence $0 \to j_!\mathbb{Q}_\ell \to \mathbb{Q}_\ell \to \mathbb{Q}_{\ell F} \to 0$,

$$\xrightarrow{\partial} \mathrm{H}^i_c(U, \mathbb{Q}_\ell) \longrightarrow \mathrm{H}^i(X, \mathbb{Q}_\ell) \longrightarrow \mathrm{H}^i(F, \mathbb{Q}_\ell) \xrightarrow{\partial}$$

gives, with the sign convention of (3.1),

$$\mathrm{tr}\,(f^*, \mathrm{H}^\bullet_c(U, \mathbb{Q}_\ell)) = \mathrm{tr}\,(f^*, \mathrm{H}^\bullet(X, \mathbb{Q}_\ell)) - \mathrm{tr}\,(f^*, \mathrm{H}^\bullet(F, \mathbb{Q}_\ell)) = v(f) - \mathrm{tr}(f^*, \mathbb{Q}_\ell^F).$$

The trace of f^* on $\mathbb{Q}_\ell^F$ is the number $v_F(f)$ of fixed points of f in F; the multiplicity-one hypothesis therefore ensures that

$$\mathrm{tr}\,(f^*, \mathrm{H}^\bullet_c(U, \mathbb{Q}_\ell)) = v(f) - v_F(f) = v_U(f),$$

as promised.

The proof of (5.2) uses some lemmas concerning a special case of noncommutative traces (4.2), which we now develop. *For brevity, we shall say projective for projective of finite type.*

5.5

Let Λ be a ring, and let H be a finite group. The map $\sum a_h \cdot a \mapsto$ class of a_e, from the group algebra $\Lambda[H]$ to $\Lambda^\natural$, vanishes on the commutators $ab - ba$ of $\Lambda[H]$. It therefore factors through

$$\varepsilon : \Lambda[H]^\natural \to \Lambda^\natural.$$

If F is an endomorphism of a projective $\Lambda[H]$-module P, set

$$\mathrm{tr}^H_\Lambda(F) = \varepsilon\, \mathrm{tr}_{\Lambda[H]}(F)$$

where the trace of (4.2) appears on the right-hand side. To avoid ambiguity, we shall sometimes write this trace as $\mathrm{tr}^H_\Lambda(F, P)$.

Proposition 5.6. $\mathrm{tr}_\Lambda(F) = |H|\,\mathrm{tr}^H_\Lambda(F)$.

The formal properties of traces reduce us to treating only the case $P = \Lambda[H]$. The endomorphism F is then right multiplication by an element $\sum a_h \cdot h$ of $\Lambda[H]$, and $\mathrm{tr}_\Lambda(F) = |H| a_e$, $\mathrm{tr}^H_\Lambda(F) = a_e$.

5.7

Let A be a commutative ring, and let Λ be an A-algebra. Multiplication by an element of A passes to the quotient to define an A-module structure on $A^\natural$. If H is a finite group, P a projective $A[H]$-module, and M a $\Lambda[H]$-module projective as a Λ-module, it is known that the Λ-module $P \otimes_A M$, equipped with the diagonal action of H, is a projective $\Lambda[H]$-module. Indeed:

a) It suffices to verify this for $P = A[H]$.

b) The Λ-module $P \otimes_A M$, equipped with the action of H defined by its action on the first factor, is clearly a projective $\Lambda[H]$-module; denote it by $(P \otimes_A M)'$.

c) The map $h \otimes m \mapsto h \otimes h^{-1}m$ induces an isomorphism

$$A[H] \otimes_A M \xrightarrow{\sim} (A[H] \otimes_A M)'. \tag{5.7.1}$$

Proposition 5.8. *With the preceding notation, if u is an endomorphism of P and v an endomorphism of M, then*

$$\mathrm{tr}^H_\Lambda(u \otimes v) = \mathrm{tr}^H_A(u)\,\mathrm{tr}_\Lambda(v).$$

We reduce to assuming that $P = A[H]$; the endomorphism u is then right multiplication m_x by an element $x = \sum a_h h$ of $A[H]$. Isomorphism (5.7.1) takes $u \otimes v$ to $\sum a_h m_h \otimes h^{-1}v$, whose trace is $a_e \cdot \mathrm{tr}_\Lambda(v)$.

5.9

Let G be a group that is an extension of $\mathbb{Z}$ by a finite group H

$$1 \to H \to G \to \mathbb{Z} \to 1,$$

and let G^+ be the inverse image of $\mathbb{N}$. Let Λ be a ring, and let P be a $\Lambda[G^+]$-module projective as a $\Lambda[H]$-module. If $g \in G^+$, denote by Z_g the centralizer of g in H; multiplication by g is an endomorphism of P, viewed as a $\Lambda[Z_g]$-module. Since P is $\Lambda[Z_g]$-projective, a trace $\mathrm{tr}_\Lambda^{Z_g}(g)$ is defined. By 5.6, one has

$$\mathrm{tr}_\Lambda(g) = |Z_g| \cdot \mathrm{tr}_\Lambda^{Z_g}(g). \tag{5.9.1}$$

Suppose that Λ is an algebra over a commutative ring A. For P an $A[G^+]$-module projective as an $A[H]$-module, and M a $\Lambda[G^+]$-module projective as a Λ-module, the $\Lambda[G^+]$-module $P \otimes_A M$ (diagonal action of G^+) is projective as a $\Lambda[H]$-module and, by 5.8, for every $g \in G^+$ one has

$$\mathrm{tr}_\Lambda^{Z_g}(g, P \otimes_A M) = \mathrm{tr}_A^{Z_g}(g, P) \cdot \mathrm{tr}_\Lambda(g, M). \tag{5.9.2}$$

5.10

Let P be a $\Lambda[G^+]$-module projective as a $\Lambda[H]$-module, and let P_H be the coinvariants of H in P. This is a projective Λ-module. The action of $g \in G^+$ passes to the quotient and defines an endomorphism of P_H that depends only on the image of g in $\mathbb{N}$. We denote by F the action of the elements $g \in G^+$ with image 1.

Proposition 5.11. $\mathrm{tr}_\Lambda(F, P_H) = \sum'_{g \mapsto 1} \mathrm{tr}_\Lambda^{Z_g}(g, P)$, *where* $\sum'_{g\mapsto 1}$ *denotes a sum over the* H*-conjugacy classes of elements of* G^+ *with image* 1 *in* $\mathbb{N}$.

After multiplication by $|H|$, this formula can be verified as follows:

$$|H| \cdot \mathrm{tr}_\Lambda(F, P_H) = \mathrm{tr}_\Lambda\left(\sum_{g\mapsto 1} g, P\right) = \sum_{g\mapsto 1}{}' \frac{|H|}{|Z_g|} \mathrm{tr}_\Lambda(g, P) = \sum_{g\mapsto 1}{}' |H| \cdot \mathrm{tr}_\Lambda^{Z_g}(g, P).$$

Choose an element $g \in G^+$ with image 1 in $\mathbb{N}$, and let σ be the automorphism of $\Lambda[H]$ induced by the automorphism $\mathrm{ad}\, g$ of H. Giving a $\Lambda[G^+]$-module amounts to giving a $\Lambda[H]$–module equipped with a σ-linear endomorphism γ: let $g^n h$ act by $\gamma^n h$. In this language, formula 5.11 takes the form: for P a projective $\Lambda[H]$-module and γ a σ-linear endomorphism of P, one has

$$\mathrm{tr}_\Lambda(\gamma, P_H) = \sum_h{}' \mathrm{tr}_\Lambda^{Z_g}(h\gamma, P),$$

where $\sum'$ is a sum over the $\mathrm{ad}\, g$-conjugacy classes in H (orbits of H acting on itself by the maps $x \mapsto hx\, \mathrm{ad}\, g(h)^{-1}$).

In this form, the properties of traces immediately reduce us to the case $P = \Lambda[H]$. The endomorphism γ then has the form $x \mapsto \sigma(x)a$, with $a = \sum a_h h$ in $\Lambda[H]$. It suffices to treat the universal case where the a_h are indeterminates. In this case, $\Lambda = \mathbb{Z}\langle a_h\rangle_{h\in H}$ (noncommutative polynomials), and $\Lambda^\natural$ is torsion-free: one obtains 5.11 by the argument at the beginning and division by $|H|$. One can also calculate directly.

5.11.1 Remark

Everything said above applies equally well to $\mathbb{Z}/2$-graded modules, or to complexes of modules (cf. 4.2, 4.3).

5.12

Retain the notation of 5.2, and let $A = \mathbb{Z}/\ell^n$, with n large enough that Λ is an A-algebra. Let $f : V_0 \to U_0$ be a finite étale connected covering of U_0, Galois with Galois group H, and such that the sheaf $f^*\mathscr{G}_0$ is constant. Denote its constant value $\mathrm{H}^0(V_0, f^*\mathscr{G}_0)$ by M; it is a $\Lambda[H]$-module projective as a Λ-module.

The sheaf f_*A on U_0, equipped with the natural action of H, is a locally free sheaf (of rank one) of $A[H]$-modules. The complex $\mathsf{R}\Gamma_c(U, f_*A)$ is therefore an object of $\mathsf{D}_{\mathrm{parf}}(A[H])$. It is equipped with a Frobenius endomorphism F^*.

For the natural action of H on $f_*f^*\mathscr{G}_0$, the trace morphism $f_*f^*\mathscr{G}_0 \to \mathscr{G}_0$ factors through an isomorphism

$$(f_*f^*\mathscr{G}_0)_H \to \mathscr{G}_0.$$

Moreover, one has $f_*f^*\mathscr{G}_0 = f_*A \otimes_A M$ (with diagonal action of H). The isomorphism

$$\mathscr{G}_0 \xleftarrow{\sim} (f_*A \otimes_A M)_H = (f_*A \otimes_A M) \otimes_{\Lambda[H]} \Lambda$$

derives (4.12) to an isomorphism

$$\mathsf{R}\Gamma_c(U, \mathscr{G}) \sim (\mathsf{R}\Gamma_c(U, f_*A) \otimes_A M) \otimes_{\Lambda[H]} \Lambda.$$

The endomorphism F^* of the left-hand side is deduced from the endomorphism F^* of $\mathsf{R}\Gamma_c(U, f_*A)$. The complex of $\Lambda[H]$-modules $\mathsf{R}\Gamma_c(U, f_*A)$, equipped with F^*, can be viewed as a complex of $\Lambda[G^+]$-modules, for $G^+ = H \times \mathbb{N}$. Formulas 5.10 and (5.9.2), amplified by 5.11.1, therefore give

$$\mathrm{tr}\,(F^*, \mathsf{R}\Gamma_c(U, \mathscr{G})) = \sum_h{}' \mathrm{tr}_A^{Z_h}\,(hF^*, \mathsf{R}\Gamma_c(U, f_*A)) \cdot \mathrm{tr}_\Lambda(h, M)$$

where $\sum'$ is a sum over the conjugacy classes in H. Moreover,

$$|Z_h| \cdot \mathrm{tr}_A^{Z_h}\,(hF^*, \mathsf{R}\Gamma_c(U, f_*A)) = \mathrm{tr}_A\left((Fh^{-1})^*, \mathsf{R}\Gamma_c(V, A)\right).$$

This formula remains true for $A = \mathbb{Z}/\ell^m$, with m increasingly large. Passing to the limit, one finds that $\mathrm{tr}_A^{Z_h}\,(hF^*, \mathsf{R}\Gamma_c(U, f_*A))$ is equal to

$$\frac{1}{|Z_h|}\sum(-1)^i \,\mathrm{tr}\left((Fh^{-1})^*, \mathrm{H}_c^i(V, \mathbb{Q}_\ell)\right)$$

(an element of $\mathbb{Z}_\ell$), and

$$\mathrm{tr}\,(F^*, \mathsf{R}\Gamma_c(U, \mathscr{G})) = \sum_h{}' \frac{1}{|Z_h|}\,\mathrm{tr}\left((Fh^{-1})^*, \mathrm{H}_c^\bullet(V, \mathbb{Q}_\ell)\right) \cdot \mathrm{tr}_\Lambda(h, M). \tag{5.12.1}$$

5.13

It remains to apply 5.4 (note that if $\bar{V}$ is the nonsingular projective completion of V, all the fixed points of Fh^{-1} have multiplicity one). This can be done without any calculation:

A. If $U^F = \varnothing$, $\mathrm{tr}\,(F^*, \mathsf{R}\Gamma_c(U, \mathscr{G})) = 0$.

Indeed, Fh^{-1} has no fixed point on V; 5.4 shows that $\sum(-1)^i \,\mathrm{tr}\left((Fh^{-1})^*, \mathrm{H}_c^i(V, \mathbb{Q}_\ell)\right) = 0$, and one applies (5.12.1).

B. In the general case, let $j : U' = U \smallsetminus U^F \hookrightarrow U$. The filtration $j_!j^*\mathscr{G}_0 \subset \mathscr{G}_0$ of $\mathscr{G}_0$ induces a filtration of $\mathsf{R}\Gamma_c(U,\mathscr{G})$ with successive quotients $\mathsf{R}\Gamma_c(U',\mathscr{G})$ and $\mathsf{R}\Gamma_c(U^F,\mathscr{G})$. By (4.4.1) and 5.1, one has

$$\operatorname{tr}(F^*,\mathsf{R}\Gamma_c(U,\mathscr{G})) = \operatorname{tr}(F^*,\mathsf{R}\Gamma_c(U',\mathscr{G})) + \sum_{x\in U^F} \operatorname{tr}(F^*,\mathscr{G}_x),$$

and, since ${U'}^F = 0$, the first term on the right-hand side is 0.

6 End of the proof of 4.10

Let us prove 4.10 under the additional hypotheses that the ring Λ is annihilated by a power of a prime number $\ell \neq p$, and that $n = 1$. The general case will follow: decompose Λ into its ℓ-primary components (with ℓ running through the prime numbers), and replace $\mathbb{F}_q$ by $\mathbb{F}_{q^n}$ and X_0 by $X_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^n}$.

For X_0 separated and of finite type over $\mathbb{F}_q$, and $\mathscr{K}_0 \in \operatorname{Ob}\mathsf{D}_{\mathrm{ctf}}(X_0,\Lambda)$, set

$$T'(X_0,\mathscr{K}_0) = \sum_{x\in X^F} \operatorname{tr}(F^*,\mathscr{K}_x)$$

and

$$T''(X_0,\mathscr{K}_0) = \operatorname{tr}(F^*,\mathsf{R}\Gamma_c(X,\mathscr{K})).$$

We must prove the identity

$$(1;X_0,\mathscr{K}_0) \qquad T'(X_0,\mathscr{K}_0) = T''(X_0,\mathscr{K}_0).$$

We begin by observing that T' and T'' (denoted generically by T) obey the same formalism:

(A) If $j : U_0 \hookrightarrow X_0$ is an open subscheme of X_0, with closed complement Y_0, one has

$$T(X_0,\mathscr{K}_0) = T(U_0,\mathscr{K}_0|U_0) + T(Y_0,\mathscr{K}_0|Y_0).$$

This is clear for T'. For T'', observe that the filtration $0 \subset j_!j^*\mathscr{K}_0 \subset \mathscr{K}_0$ of $\mathscr{K}_0$ induces a filtration of $\mathsf{R}\Gamma_c(X,\mathscr{K})$ with successive quotients $\mathsf{R}\Gamma_c(U,\mathscr{K})$ and $\mathsf{R}\Gamma_c(Y,\mathscr{K})$, and invoke (4.4.1) (more precisely stated: $\mathscr{K}_0$, equipped with the said filtration, defines an object of $\mathsf{DF}(X,\Lambda)$; applying $\mathsf{R}\Gamma_c$ gives an object of $\mathsf{DF}_{\mathrm{parf}}(\Lambda)$ whose underlying object is $\mathsf{R}\Gamma_c(X,\mathscr{K})$ and whose associated graded is as stated).

(B) If $\mathscr{K}_0$ is a bounded complex with constructible flat components,

$$T(X_0,\mathscr{K}_0) = \sum (-1)^i T(X_0,\mathscr{K}^i).$$

This is clear for T'; for T'', apply the argument above to the stupid filtration of $\mathscr{K}_0$ to obtain

$$T''(X_0,\mathscr{K}_0) = \sum T''(X_0,\mathscr{K}_0^i[-i]) = \sum (-1)^i T''(X_0,\mathscr{K}_0^i).$$

(C) For a morphism $f : X_0 \to Y_0$, one has

$$T''(X_0,\mathscr{K}_0) = T''(Y_0,\mathsf{R}f_!\mathscr{K}_0)$$

and the same identity holds for T' if the fibers of f at the rational points of Y_0 satisfy $\big(1; f^{-1}(y), \mathscr{K}_0|f^{-1}(y)\big)$: use the fact that

$$T'(X_0,\mathscr{K}_0) = \sum_{y\in Y^F} T'\left(f^{-1}(y),\mathscr{K}_0|f^{-1}(y)\right)$$

and

$$\mathsf{R}\Gamma_c(Y,\mathsf{R}f_!\mathscr{K}) = \mathsf{R}\Gamma_c(X,\mathscr{K}).$$

Using this formalism and 4.7, the proof of 4.10 is reduced to the two special cases 5.1 and 5.2.

Chapter 3

L-functions modulo ℓ^n and modulo p

In this exposé, I prove a formula analogous to the main theorem of the report on the trace formula (this volume; cited as "Report" below) for a coefficient ring A that is torsion prime to p, or a field of characteristic p. Essential use will be made of [2, XVII 5.5].

1 Symmetric tensors

1.1

Let A be a commutative ring. For a list of properties of the functors $\Gamma^n : (A\text{-modules}) \to (A\text{-modules})$, I refer to [2, XVII 5.5.1 and 2]. We recall only that for M flat (in fact, only the case where M is projective of finite type concerns us), one has

$$\Gamma^n M = \left(M^{\otimes n}\right)^{S_n} \qquad \text{(symmetric tensors of degree } n\text{)}.$$

1.2

Let X be an S-scheme, let X^n/S be the n-th fiber power of X, and let π be the projection from X^n/S onto $(X^n/S)/S_n = \mathrm{Sym}^n_S(X)$ (assumed to exist). If $\mathscr{G}$ is a sheaf of A-modules on X, the external tensor product of n copies of $\mathscr{G}$, namely $\mathscr{G}^{\boxtimes n}$, is an S_n-equivariant sheaf on X^n/S. We shall have to use the *external* functor Γ^n of [2, XVII 5.5.7 to 9]. We recall only that for $\mathscr{G}$ a sheaf of flat A-modules (in fact, only the constructible flat case concerns us), one has

$$\Gamma^n_{\mathrm{ext}}(\mathscr{G}) = \left(\pi_* \mathscr{G}^{\boxtimes n}\right)^{S_n}.$$

Lemma 1.3. *For $S = \mathrm{Spec}(k)$, with k algebraically closed, X a finite disjoint union of copies of S, and G a flat sheaf of A-modules on X, one has*

$$\Gamma\left(\mathrm{Sym}^n_S(X), \Gamma_{ext}(\mathscr{G})\right) = \Gamma^n \Gamma(X, \mathscr{G})$$

Indeed,

$$\begin{aligned}\Gamma\left(\mathrm{Sym}^n_S(X), \Gamma^n_{\mathrm{ext}}(\mathscr{G})\right) &= \Gamma\left(\mathrm{Sym}^n_S(X), \pi_* \mathscr{G}^{\boxtimes n}\right)^{S_n} \\ &= \Gamma\left(X^n/S, \mathscr{G}^{\boxtimes n}\right)^{S_n} = \left(\Gamma(X, \mathscr{G})^{\otimes n}\right)^{S_n} = \Gamma^n \Gamma(X, \mathscr{G}).\end{aligned}$$

1.4

We shall have to use the derived functors of the nonadditive functors Γ^n and Γ^n_{ext}.

The theory of such derived functors is due to Dold and Puppe [17]. For more complete reminders than those given below, I refer to [2, XVII 5.5.3].

Let $\mathcal{A}$ be an abelian category (or even an additive category in which every idempotent endomorphism is the projection onto a direct factor). Denote by N the normalization functor: N : (cosimplicial complexes of objects of $\mathcal{A}$) $\to$ (differential complexes of objects of $\mathcal{A}$, with K^i for $i < 0$). It is an equivalence of categories. Like its inverse, it takes homotopic morphisms to homotopic morphisms. If Γ is a functor $\mathcal{A} \to \mathcal{B}$, and if K is a complex of objects of $\mathcal{A}$, in nonnegative degrees ($K^i = 0$ for $i < 0$), we set $TK = NTN^{-1}K$.

1.5 Example

If K is concentrated in M in degree 0, TK is concentrated in TM in degree 0.

1.6

If K is a bounded complex, in nonnegative degrees, of projective A-modules of finite type, the complex of A-modules $\Gamma^n K$ has the same properties.

Denote by $\mathsf{D}^{\geqslant 0}_{\text{parf}}(A)$ the subcategory of $\mathsf{D}_{\text{parf}}(A)$ (Report, 4.3) consisting of complexes of tor-dimension $\leqslant 0$. One sees that Γ^n induces a functor

$$\mathsf{L}\Gamma^n : \mathsf{D}^{\geqslant 0}_{\text{parf}}(A) \to \mathsf{D}^{\geqslant 0}_{\text{parf}}(A).$$

For the analogous, but more complicated, definition of $\mathsf{L}\Gamma^n_{\text{ext}}$, I refer to [2, XVII 5.5.14].

1.7

Let K be a bounded complex of projective A-modules of finite type, and let u be an endomorphism of K. We denote by $\det(1 - ut, K)$ the formal series with constant term 1

$$\det(1 - ut, K) = \prod_i \det(1 - ut, K^i)^{(-1)^i}.$$

The verification of the following properties is left to the reader.

1.7.1

Let F be a finite filtration stable under u, such that the $\mathrm{gr}^p_F(K^q)$ are projective of finite type. Then

$$\det(1 - ut, K) = \prod_p \det(1 - ut, \mathrm{gr}^p_F(K)).$$

1.7.2

If K is shifted q places to the left, one has

$$\det(1 - ut, K[q]) = \det(1 - ut, K)^{(-1)^q}.$$

Proposition 1.8. *Let u be an endomorphism of a bounded complex, in nonnegative degrees, of projective A-modules of finite type. One has*

$$\det(1-ut,K)^{-1}=\sum_n \mathrm{tr}(u,\Gamma^n K)t^n. \tag{1.8.1}$$

Denote the right-hand side of (1.8.1) by $D(u,K)$.

Lemma 1.9. *Let u be an endomorphism of a short exact sequence of bounded complexes in degrees $\geqslant 0$ of projective A-modules of finite type*

$$0\to K'\to K\to K''\to 0.$$

One has $D(u,K)=D(u,K')\cdot D(u,K'')$.

If $0\to M'\to M\to M''\to 0$ is a split short exact sequence of A-modules, $\Gamma^n M$ has a natural filtration whose successive quotients are the $\Gamma^i M'\otimes\Gamma^j M''$ $(i+j=n)$. Consequently, $\Gamma^n K$ has a natural filtration whose successive quotients are the complexes $N(N^{-1}\Gamma^i K'\otimes N^{-1}\Gamma^j K'')$ $(i+j=n)$, canonically homotopic to the simple complexes $s(\Gamma^i K'\otimes\Gamma^i K'')$ obtained from the double complexes $\Gamma^i K'\otimes\Gamma^i K''$, and

$$\mathrm{tr}(u,\Gamma^n K)=\sum_{i+j=n}\mathrm{tr}(u,\Gamma^i K')\cdot\mathrm{tr}(u,\Gamma^j K''),$$

whence the lemma. From 1.9, one obtains by induction.

Lemma 1.10. *Let F be a finite filtration of K stable under u, such that the $\mathrm{gr}_F^p(K^q)$ are projective of finite type. Then*

$$D(u,K)=\prod_p D\left(u,\mathrm{gr}_F^p(K)\right).$$

Lemma 1.11. *Let u be an endomorphism of a projective A-module of finite type M, and let $M[-q]$ be the complex concentrated in M in degree q. One has*

$$D\left(u,M[-q]\right)=\Big(\sum\mathrm{tr}(u,\Gamma^n M)t^n\Big)^{(-1)^q}\qquad (q\geqslant 0.)$$

By 1.5, this is true for $q=0$. Proceeding by induction, it remains to show that

$$D\left(u,M[-q]\right)\cdot D\left(u,M[-(q+1)]\right)=1.$$

This formula follows from 1.9, applied to the complex K concentrated in M in degrees q and $q+1$ (with $d=\mathrm{id}$), and to its stupid filtration:

$$(\cdots 0\to M\cdots)\to(\cdots M\to M)\to(\cdots M\to 0\to 0\cdots).$$

Indeed, K is homotopic to 0. In the derived category, one therefore has $\Gamma^n(K)=\Gamma^n(0)$: 0 for $n>0$ and A for $n=0$, and $D(u,K)=1$.

1.12

Let us prove 1.8. By 1.10, applied to the "stupid" filtration of K by the subcomplexes

$$0\to\cdots\to 0\to K^q\to K^{q+1}\to\cdots,$$

one has

$$D(u,K)=\prod_q D\left(u,K^q[-q]\right) =_{(1.10)} \prod_q \left(\sum \operatorname{tr}\left(u,\Gamma^n(K^q)\right)t^n\right)^{(-1)^q}.$$

It remains to prove that for u an endomorphism of a projective module of finite type M, one has

$$\det\left(1-ut,M\right)^{-1}=\sum \operatorname{tr}(u,\Gamma^n M)t^n. \tag{1.12.1}$$

If $u=0$, both sides are equal to 1. By adjoining to M a module N on which $u=0$, and applying 1.9, we reduce to the assumption that M is free. Choose a basis for it. It is then a matter of proving algebraic identities among the coordinates of u. The principle of extension of algebraic identities reduces us to assuming that A is an algebraically closed field. The module M then admits a filtration stable under u with successive quotients of rank 1, and 1.10 reduces us to the case where M has rank 1. Formula (1.12.1) then reduces to the identity

$$\frac{1}{1-at}=\sum a^n t^n.$$

Corollary 1.13. *Let u and u' be endomorphisms of bounded complexes K and K' of projective A-modules of finite type. If, in the derived category, K, equipped with u, is isomorphic to K', equipped with u', then*

$$\det(1-ut,K)=\det(1-u't,K').$$

Applying 1.7.2, we reduce to assuming that K and K' are in nonnegative degrees. One then observes that the right-hand side of 1.8 has the required invariance property.

1.14

This corollary makes it possible to define $\det(1-ut,K)$ for u an endomorphism of $K\in \operatorname{Ob}\mathsf{D}_{\mathrm{parf}}(A)$. This construction is ad hoc; in fact, for v an automorphism of $L\in\operatorname{Ob}\mathsf{D}_{\mathrm{parf}}(\Lambda)$, one can define $\det(v)\in\Lambda^\times$, and $\det(1-ut,K)$ is obtained by taking $v=1-ut$, $\Lambda=A[\![t]\!]$, and $L=K\otimes_A\Lambda$.

2 The theorem

2.1

We retain the notation of (Report, §1). For A a commutative Noetherian torsion ring, X_0 a separated scheme of finite type over $\mathbb{F}_q$, and $\mathscr{K}_0\in\operatorname{Ob}\mathsf{D}_{\mathrm{ctf}}(X_0;A)$, set

$$L(X_0,\mathscr{K}_0)=\prod_{x\in|X_0|}\det\left(1-F_x^* t^{\deg(x)},\mathscr{K}\right)^{-1}$$

(1.7, 1.13, and Report, 1.5, 1.6).

Theorem 2.2. *In each of the following two cases*

a) A is torsion prime to p;

b) A is reduced and of characteristic p

one has

$$L(X_0,\mathscr{K}_0)=\det\left(1-F^*t^f,\mathsf{R}\Gamma_c(X,\mathscr{K})\right)^{-1}. \tag{2.2.1}$$

2.3 Remark

In case a), the method of Report, §3 makes it possible to deduce from the trace formula (Report 4.10) that the logarithmic derivatives of the two sides of (2.2.1) are equal. Since the ring A is a torsion ring, this does not suffice to prove 2.2.

2.4 Remark

If A is a field, one has

$$L(X_0, \mathscr{K}_0) = \prod_i L\left(X_0, \underline{\mathrm{H}}^i(\mathscr{K}_0)\right)^{(-1)^i}$$

and the spectral sequence $E_2^{pq} = \mathrm{H}_c^p(X, \underline{\mathrm{H}}^q(\mathscr{K})) \Rightarrow \mathrm{H}_c^{p+q}(X, \mathscr{K})$ shows that

$$\begin{aligned}\det\left(1 - F^*t, \mathsf{R}\Gamma_c(X, \mathscr{K})\right) &= \prod_n \det\left(1 - F^*t^f, \mathrm{H}_c^n(X, \mathscr{K})\right)^{(-1)^n} \\ &= \prod_{p,q} \det\left(1 - F^*t, \mathrm{H}_c^p(X, \underline{\mathrm{H}}^q(\mathscr{K}))\right)^{(-1)^{p+1}}.\end{aligned}$$

This reduces 2.2 (for A a field) to the following special case.

2.4.1

If $\mathscr{G}_0$ is a constructible sheaf of A-vector spaces, one has

$$L(X_0, \mathscr{G}_0) = \prod_n \det\left(1 - F^*t^f, \mathrm{H}_c^n(X, \mathscr{G})\right)^{(-1)^{n+1}}.$$

2.5 Remark

It is easy to reduce case b) to the case where A is a field (or even a finite field) of characteristic p; cf. 4.2.

2.6 Remark

The special case of b) where $A = \mathbb{Z}/p$, and where X is reduced to the constant sheaf $\mathbb{Z}/p$ in degree 0,

$$Z(X_0, t) = \prod_{x \in |X_0|} \left(1 - t^{\deg(x)}\right)^{-1} \equiv \prod_n \det\left(1 - F^*t^f, \mathrm{H}_c^n(X, \mathbb{Z}/p)\right)^{(-1)^{n+1}} \pmod{p}$$

was proved by N. Katz [16, XXII 3.1]. His method, entirely different from the one followed here, starts from Dwork's p-adic theory of $Z(X_0, t)$.

2.7 Remark

Contrary to what I had rashly asserted to friends, in b) one cannot omit the hypothesis "A reduced." For a counterexample, cf. 4.5.

2.8 Outline of the proof of 2.2

A shift in degrees 1.7.2 reduces us to assuming that $\mathscr{K}_0$ is a bounded complex, in nonnegative degrees, of constructible flat sheaves of A-modules. One then has $\mathsf{R}\Gamma_c(X, \mathscr{K}) \in \mathrm{Ob}\, \mathsf{D}^{\geqslant 0}_{\mathrm{parf}}(A)$ (Report, proof of 4.9, b). We shall use 1.8 to expand both sides of (2.2.1) as power series in $T = t^f$. The equality of the coefficients of T^n will follow from [2, XVII 5.5.21] and a trace formula on $\mathrm{Sym}^n(X)$: in case a), the formula (Report, 4.10), and in case b), a formula to be proved in the next chapter. It is first necessary to reduce to the case where the symmetric powers of X_0 exist (for example, the case where X is affine), using the multiplicativity in X_0 of both sides of (2.2.1): for U_0 an open subscheme of X_0, with closed complement Y_0, one has $L(X_0, \mathscr{K}_0) = L(U_0, \mathscr{K}_0) \cdot L(Y_0, \mathscr{K}_0)$, and the analogous formula for the right-hand side follows from 1.7.1 by the method of (Report §6A).

2.9 The case where X_0 is finite

The multiplicativity in X_0 of both sides of (2.2.1) reduces this case to the case where $X_0 = \mathrm{Spec}(\mathbb{F}_{q^k})$. Returning to the definitions, one sees that (2.2.1) is then equivalent to Report 3.4.

2.10 The left-hand side

By 1.8, one has

$$L(X_0, \mathscr{K}_0) = \prod_{x \in |X_0|} \sum_n \mathrm{tr}\,(F_x^*, \Gamma^n \mathscr{K})\, t^{m \deg(x)}.$$

The coefficient of T^n is therefore the sum, over those formal linear combinations $\sum_{x \in |X_0|} n_x \cdot x$ ($n_x \geqslant 0$, with almost all the n_x zero) of elements of $|X_0|$ such that $\sum n_x [k(x) : \mathbb{F}_q] = n$, of the products

$$\prod_x \mathrm{tr}\,(F_x^*, \Gamma^{n_x} \mathscr{K})\,. \tag{2.10.1}$$

For $x \in |X_0|$, denote by (x) the 0-cycle consisting of the $[k(x) : \mathbb{F}_q]$ points of X above x (an orbit of F). The map $\sum n_x \mapsto \sum n_x \cdot (x)$ identifies the formal linear combinations above, with $\sum n_x [k(x) : \mathbb{F}_q] = n$, with the effective 0-cycles of degree n on X fixed under F, or equivalently with the fixed points of F in $\mathrm{Sym}^n(X)$. The coefficient of T^n in $L(X_0, \mathscr{K}_0)$ thus appears as a sum of products (2.10.1), indexed by $\mathrm{Sym}^n(X)^F$. More precisely,

Lemma 2.11. $L(X_0, \mathscr{K}_0) = \sum_n \sum_{x \in \mathrm{Sym}^n(X)^F} \mathrm{tr}\,(F_x^*, \mathsf{L}\Gamma^n_{\mathrm{ext}} \mathscr{K})\, T^n$.

For Y_0 an increasingly large finite subscheme of X_0, one has

$$L(X_0, \mathscr{K}_0) = \lim L(Y_0, \mathscr{K}_0),$$

i.e., for every m, there is a finite subscheme $Z_0 \subset X_0$ such that if $Y_0 \supset Z_0$, then $L(X_0, \mathscr{K}_0)$ and $L(Y_0, \mathscr{K}_0|Y_0)$ are congruent $\mod T^m$. It suffices for Z_0 to contain the closed points of degree $\leqslant m$. Since the same fact holds for the right-hand side, it suffices to prove 2.11 for X_0 finite. One then has

$$\begin{aligned}
L(X_0, \mathscr{K}_0) &=_{2.9} \det\,(1 - F^* T, \Gamma(X, \mathscr{K}))^{-1} \\
&=_{1.8} \sum_n \mathrm{tr}\,(F^*, \Gamma^n \Gamma(X, \mathscr{K}))\, T^n \\
&=_{1.3} \sum_n \mathrm{tr}\,(F^*, \Gamma(\mathrm{Sym}^n(X), \Gamma^n_{\mathrm{ext}}(\mathscr{K}))\, T^n \\
&= \sum_n \sum_{x \in \mathrm{Sym}^n(X)^F} \mathrm{tr}\,(F_x^*, \Gamma^n_{\mathrm{ext}}(\mathscr{K}))\, T^n
\end{aligned}$$

(by the trace formula in the trivial case of a finite set).

2.12 The right-hand side

By 1.8 and [2, XVII 5.5.12], one has

$$\begin{aligned}\det\left(1-F^*T,\mathsf{R}\Gamma_c(X,\mathscr{K})\right)^{-1} &= \sum_n \operatorname{tr}\left(F^*,\mathsf{L}\Gamma^n\mathsf{R}\Gamma_c(X,\mathscr{K})\right)T^n\\ &= \sum_n \operatorname{tr}\left(F^*,\mathsf{R}\Gamma_c(\operatorname{Sym}^n(X),\Gamma^n_{\text{ext}}\mathscr{K})\right)T^n.\end{aligned}$$

2.13

The theorem is therefore equivalent to the trace formulas

$$\operatorname{tr}\left(F^*,\mathsf{R}\Gamma_c(\operatorname{Sym}^n(X,\Gamma^n_{\text{ext}}\mathscr{K})\right) = \sum_{x\in\operatorname{Sym}^n(X)^F}\operatorname{tr}\left(F^*_x,\Gamma^n_{\text{ext}}\mathscr{K}\right).$$

In case a), they follow from Report 4.10. The analogous formula in case b) will be proved in §4.

3 Artin–Schreier theory

3.1

In what follows, a coherent sheaf on a scheme S will always be regarded as a sheaf on $S_{\text{ét}}$. Let S be a scheme of characteristic $p>0$. If G is a locally constant sheaf of finite-rank $\mathbb{Z}/p$-vector spaces, $\mathscr{G}=G\otimes_{\mathbb{Z}/p}\mathscr{O}$ is a locally free coherent sheaf on S. We shall denote by Φ both the endomorphism $f\mapsto f^p$ of $\mathscr{O}$ and its tensor product with the identity of G: $\phi:\mathscr{G}\to\mathscr{G}$. Artin–Schreier theory [2, IX 3.5] asserts that the sequence

$$0\longrightarrow\mathbb{Z}/p\longrightarrow\mathscr{O}\xrightarrow{\Phi-1}\mathscr{O}\longrightarrow 0$$

is exact. Tensoring with G gives an exact sequence

$$0\longrightarrow G\longrightarrow\mathscr{G}\xrightarrow{\Phi-1}\mathscr{G}\longrightarrow 0$$

3.2

Suppose that S is an open subscheme of a Noetherian scheme $\bar{S}$. Let $j:S\hookrightarrow\bar{S}$ be the inclusion morphism, and let I be a sheaf of ideals defining the closed complement $\bar{S}\smallsetminus S$. We again denote by Φ the extension of Φ to $j_*\mathscr{G}$ by functoriality.

Lemma 3.3. *There exist coherent extensions $\bar{\mathscr{G}}\subset j_*\mathscr{G}$ of $\mathscr{G}$ such that*

$$\Phi(\bar{\mathscr{G}})\subset I\cdot\bar{\mathscr{G}}. \tag{3.3.1}$$

If $\bar{\mathscr{G}}$ is such an extension, the sequence

$$0\longrightarrow j_!G\longrightarrow\bar{\mathscr{G}}\xrightarrow{\Phi-1}\bar{\mathscr{G}}\longrightarrow 0 \tag{3.3.2}$$

is exact.

a) *Existence*: Let $\mathscr{G}' \subset j_*\mathscr{G}$ be a coherent extension of $\mathscr{G}$ to $\bar{S}$. For every n, one has $\Phi(I^n\mathscr{G}') \subset I^{p^n}\Phi(\mathscr{G}')$. Let n be large enough that $I^{n(p-1)-1}\Phi(\mathscr{G}') \subset \mathscr{G}'$, and set $\bar{\mathscr{G}} = I^n\mathscr{G}'$. One has $\Phi(\bar{\mathscr{G}}) \subset I^{pn}\Phi(\mathscr{G}') = I \cdot I^n \cdot I^{n(p-1)-1}\Phi(\mathscr{G}') \subset I \cdot I^n\mathscr{G}' = I\bar{\mathscr{G}}$.

b) *Kernel of* $\Phi - 1$: On S, the assertion follows from 3.1; it remains to verify that a section x of $\ker(\Phi - 1)$, on U étale over $\bar{S}$, vanishes in a neighborhood of the inverse image of $\bar{S} \smallsetminus S$. Indeed, since $x = \Phi(x)$, one has by (3.3.1) that $x \in I\bar{\mathscr{G}}$; moreover, $x \in I^n\bar{\mathscr{G}} \Rightarrow x = \Phi(x) \in \Phi(I^n\bar{\mathscr{G}}) \subset I^{np}\bar{\mathscr{G}}$, and x lies in every $I^n\bar{\mathscr{G}}$, hence is zero near $\bar{S} \smallsetminus S$.

c) *Surjectivity of* $\Phi - 1$: We may suppose $\bar{S}$ affine, and write $\mathscr{G}$ as a quotient of $\mathscr{O}^n$; Φ then lifts to a p-linear map $\widetilde{\Phi} : \mathscr{O}^n \to \mathscr{O}^n$, and it suffices to verify the surjectivity of $\widetilde{\Phi} - 1$:

$$\begin{array}{ccc} \mathscr{O}^n & \xrightarrow{\widetilde{\Phi}-1} & \mathscr{O}^n \\ \downarrow & & \downarrow \\ \mathscr{G} & \xrightarrow{\Phi-1} & \mathscr{G} \end{array}$$

The sheaf $\mathscr{O}^n$ is the sheaf of local sections of the group scheme $\mathbb{G}_a^n$, and $\widetilde{\Phi} - 1$ is induced by a homomorphism from $\mathbb{G}_a^n$ to $\mathbb{G}_a^n$. This homomorphism is étale. To verify that it is surjective, it suffices to check this after every base change $\mathrm{Spec}(k) \to \bar{S}$ (k algebraically closed), and over an algebraically closed k, every étale homomorphism between connected groups of the same dimension is surjective. Finally, an étale surjective morphism of $\bar{S}$-schemes induces an epimorphism between the corresponding sheaves on $\bar{S}_{\mathrm{et}}$.

3.4 Frobenius and Frobenius

The absolute Frobenius morphism $F_{\mathrm{abs}} : S \to S$ is the identity on the underlying topological space of S, and $f \mapsto f^p$ on the structure sheaf. For U/S étale, one has a canonical isomorphism $F_{\mathrm{abs}}^*U = U$: the diagram

$$\begin{array}{ccc} U & \xrightarrow{F_{\mathrm{abs}}} & S \\ \downarrow & & \downarrow \\ S & \xrightarrow{F_{\mathrm{abs}}} & S \end{array}$$

is Cartesian. The endomorphism of the ringed site $(\bar{S}_{\mathrm{et}}, \mathscr{O}) \to (\bar{S}_{\mathrm{et}}, \mathscr{O})$ defined by F_{abs} can therefore be described as follows:

$$F_{\mathrm{abs}}^* \text{ is the identity on } S_{\mathrm{et}}\text{; the homomorphism } F_{\mathrm{abs}}^*\mathscr{O} = \mathscr{O} \to \mathscr{O} \text{ is } \Phi.$$

With this description, for every étale sheaf $\mathscr{G}$ on S, the Frobenius correspondence $F_{\mathrm{abs}}^*\mathscr{G} \to \mathscr{G}$ is the identity.

Giving a p-linear morphism of quasi-coherent sheaves $u : \mathscr{M} \to \mathscr{N}$ on S is equivalent to giving a linear morphism $u' : F_{\mathrm{abs}}^*\mathscr{M} \to \mathscr{N}$. In particular, with the notation of 3.5, the homomorphism $\Phi : \mathscr{G} \to \mathscr{G}$ corresponds to $\Phi' : F_{\mathrm{abs}}^*\mathscr{G} \to \mathscr{G}$.

Lemma 3.5. *With the notation of 3.4, the diagram*

$$\begin{array}{ccc} F_{\mathrm{abs}}^*G & \longrightarrow & G \\ \downarrow & & \downarrow \\ F_{\mathrm{abs}}^*\mathscr{G} & \xrightarrow{\Phi'} & \mathscr{G} \end{array}$$

is commutative.

It suffices to verify this locally. We may therefore suppose G constant, in which case the lemma is evident.

The same theory applies to the powers of F_{abs}.

3.6

Suppose now that $\bar{S}$ is a scheme over $\mathbb{F}_q$ ($q = p^f$), and in accordance with the conventions of Report §1, write $S_0, \bar{S}_0, G_0, \mathscr{G}_0, \bar{\mathscr{G}}_0$ in place of $S, \ldots$. Lemma 3.5, for $F^F_{\mathrm{abs}} : S_0 \to S_0$, gives a commutative diagram

$$\begin{array}{ccc} F^*G_0 & \longrightarrow & G_0 \\ \downarrow & & \downarrow \\ F^*\mathscr{G}_0 & \xrightarrow{(\Phi^f)'} & \mathscr{G}_0 \end{array} \qquad \text{, and then} \qquad \begin{array}{ccc} F^*(j_!G_0) & \longrightarrow & j_!G_0 \\ \downarrow & & \downarrow \\ F^*\bar{\mathscr{G}}_0 & \xrightarrow{(\Phi^f)'} & \mathscr{G}_0 \end{array},$$

and then, by extension of scalars from $\mathbb{F}_q$ to $\mathbb{F}$ and passage to cohomology,

$$\begin{array}{ccc} F^*(j_!G) & \xrightarrow{F_*} & j_!G \\ \downarrow & & \downarrow \\ F^*\bar{\mathscr{G}} & \longrightarrow & \bar{\mathscr{G}} \end{array} \qquad \text{and} \qquad \begin{array}{ccc} \mathrm{H}^\bullet(\bar{S}, j_!G) & \xrightarrow{F^*} & \mathrm{H}^\bullet(\bar{S}, j_!G) \\ \downarrow & & \downarrow \\ \mathrm{H}^\bullet(\bar{S}, \bar{\mathscr{G}}) & \longrightarrow & \mathrm{H}^\bullet(\bar{S}, \bar{\mathscr{G}}) \end{array},$$

where the second row is obtained by extension of scalars from $\mathbb{F}_q$ to $\mathbb{F}$ of the $\mathbb{F}_q$-linear endomorphism Φ^f of $\mathrm{H}^\bullet(\bar{S}_0, \bar{G}_0)$.

Artin–Schreier theory provides a long exact sequence

$$\cdots \longrightarrow \mathrm{H}^i(\bar{S}, j_!G) \longrightarrow \mathrm{H}^i(\bar{S}, \bar{\mathscr{G}}) \xrightarrow{\Phi - 1} \mathrm{H}^i(\bar{S}, \bar{\mathscr{G}}) \longrightarrow \cdots$$

where Φ is the p-linear extension to $\mathrm{H}^\bullet(\bar{S}, \bar{\mathscr{G}}) = \mathrm{H}^\bullet(\bar{S}_0, \bar{\mathscr{G}}_0) \otimes_{\mathbb{F}_q} \mathbb{F}$ of the p-linear endomorphism Φ of $\mathrm{H}^\bullet(\bar{S}_0, \bar{\mathscr{G}}_0)$.

3.7

If $\bar{S}_0$ is proper over $\mathbb{F}_q$, the $\mathrm{H}^i(\bar{S}, \bar{\mathscr{G}})$ are finite-dimensional and $\Phi - 1$ is surjective (cf. 3.6b). Thus

$$\mathrm{H}^i_c(S, G) = \ker\left(\Phi - 1 : \mathrm{H}^i(\bar{S}, \bar{G}) \to \mathrm{H}^i(\bar{S}, \bar{\mathscr{G}})\right),$$

and F^* is induced by the $\mathbb{F}$-linear extension to $\mathrm{H}^i(\bar{S}, \bar{\mathscr{G}}) = \mathrm{H}^i(\bar{S}_0, \bar{\mathscr{G}}_0) \otimes_{\mathbb{F}_q} \mathbb{F}$ of the $\mathbb{F}_q$-linear endomorphism Φ^f of $\mathrm{H}^i(\bar{S}_0 m \bar{\mathscr{G}}_0)$.

Applying Lemma 3.8 below, one deduces the identity

$$\mathrm{tr}\left(F^*, \mathrm{H}^i_c(S, G)\right) = \mathrm{tr}\left(\Phi^f, \mathrm{H}^i(\bar{S}_0, \bar{\mathscr{G}}_0)\right). \tag{3.7.1}$$

Lemma 3.8. *Let Φ be a p-linear endomorphism of a vector space V over $\mathbb{F}_q$: $\phi(\lambda x) = \lambda^p \phi(x)$. Set $F = \Phi^f$ (F is linear), and again denote by Φ and F the respective p-linear extensions of Φ and F to $V \otimes_{\mathbb{F}_q} \mathbb{F}$. Then F stabilizes* $\ker(\Phi - 1) \subset V \otimes_{\mathbb{F}_q} \mathbb{F}$ *and*

$$\mathrm{tr}\left(F, \ker(\Phi - 1)\right) = \mathrm{tr}(F, V) = \mathrm{tr}\left(F, V \otimes_{\mathbb{F}_q} \mathbb{F}\right).$$

On $V \otimes_{\mathbb{F}_q} \mathbb{F}$, one has $F\Phi = \Phi F$: the p-linear endomorphism $F\Phi - \Phi F$ vanishes on V, hence everywhere. Write $V = V' + V''$, with F invertible on V' and nilpotent on V''. The theory of p-linear endomorphisms ensures that

$$\ker(\Phi - 1) \otimes_{\mathbb{F}_p} \mathbb{F} \xrightarrow{\sim} V' \otimes_{\mathbb{F}_q} \mathbb{F}.$$

Since $\mathrm{tr}(F, V'') = 0$, one therefore has

$$\begin{aligned} \mathrm{tr}\left(F, \ker(\Phi - 1)\right) &= \mathrm{tr}\left(F, \ker(\Phi - 1) \otimes_{\mathbb{F}_q} \mathbb{F}\right) = \mathrm{tr}\left(F, V' \otimes_{\mathbb{F}_q} \mathbb{F}\right) \\ &= \mathrm{tr}\left(F, V \otimes_{\mathbb{F}_q} \mathbb{F}\right) = \mathrm{tr}(F, V). \end{aligned}$$

4 Trace formula modulo p

Theorem 4.1. *Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$, let A be a commutative reduced Noetherian ring of characteristic p, and let $\mathscr{K}_0 \in \mathrm{Ob}\, \mathsf{D}_{\mathrm{ctf}}(X_0, A)$. One has*

$$\sum_{x \in X^F} \mathrm{tr}\left(F_x^*, \mathscr{K}_x\right) = \mathrm{tr}\left(F^*, \mathsf{R}\Gamma_c(X, \mathscr{K})\right). \tag{4.1.1}$$

4.2 Reduction to the case where A is a finite field

A standard passage-to-the-limit argument reduces us to the case where A is of finite type over $\mathbb{Z}/p$. For every maximal ideal $\mathfrak{m}$ of A, the image of each side of (4.1.1) in $A/\mathfrak{m}$ is given by the same formula, with $\mathscr{K}_0$ replaced by $\mathscr{K}_0 \overset{\mathsf{L}}{\otimes}_A A/\mathfrak{m} \in \mathrm{Ob}\, \mathsf{D}_{\mathrm{ctf}}(X_0, A/\mathfrak{m})$ (for the right-hand side, cf. Report 4.12). Since the $A/\mathfrak{m}$ are finite, and the intersection of the maximal ideals of A is reduced to 0, one obtains the stated reduction.

4.3 Reduction to the case where $A = \mathbb{Z}/p$

Suppose that A is a finite field, and denote by $T'_A(\mathscr{K}_0)$ and $T''_A(\mathscr{K}_0)$ the two sides of (4.1.1). If, by restriction of scalars, $\mathscr{K}_0$ is regarded as a complex of sheaves of $\mathbb{Z}/p$-modules, one has

$$T'_{\mathbb{Z}/p}(\mathscr{K}_0) = \mathrm{tr}_{A/(\mathbb{Z}/p)}\, T'_A(\mathscr{K}_0)$$

and likewise for T''. If (4.1.1) holds for $A = \mathbb{Z}/p$, one therefore has

$$\mathrm{tr}_{A/(\mathbb{Z}/p)}\, T'_A(\mathscr{K}_0) = \mathrm{tr}_{A/(\mathbb{Z}/p)}\, T''_A(\mathscr{K}_0). \tag{4.3.1}$$

I claim that, for every $\lambda \in A$, one even has

$$\mathrm{tr}_{A/(\mathbb{Z}/p)}\left(\lambda \cdot T'_A(\mathscr{K}_0)\right) = \mathrm{tr}_{A/(\mathbb{Z}/p)}\left(\lambda \cdot T''_A(\mathscr{K}_0)\right), \tag{4.3.2}$$

and hence $T'_A(\mathscr{K}_0) = T''_A(\mathscr{K}_0)$. This is clear for $\lambda = 0$. For λ invertible, denote by $A(\lambda)$ the inverse image on X of the free rank-one sheaf of A-modules on $\mathrm{Spec}(\mathbb{F}_q)$ for which $F^* = \lambda$. Formula (4.3.2) is precisely (4.3.1) for $A(\lambda) \otimes_A \mathscr{K}_0$.

Theorem 4.4. *Let $\bar{S}_0$ be a smooth projective curve, let $j : S_0 \hookrightarrow \bar{S}_0$ be a dense open subscheme of $\bar{S}_0$, and let G_0 be a locally constant sheaf of finite-rank $\mathbb{Z}/p$-vector spaces on S_0. One has*

$$\sum_{x \in S^F} \mathrm{tr}\left(F_x^*, G_x\right) = \sum_i (-1)^i \,\mathrm{tr}\left(F^*, \mathrm{H}^i_c(S, G)\right). \tag{4.4.1}$$

Proof. Apply 3.3 to $(S_0, \bar{S}_0, G_0)$. The coherent extension $\bar{\mathscr{G}_0}$ of $\mathscr{G}_0$ whose existence is guaranteed by 3.3 is locally free because it is torsion-free (we have $\bar{\mathscr{G}_0} \subset j_*\mathscr{G}_0$) and $\bar{S}_0$ is regular of dimension one. The q-linear endomorphism Φ^f of $\bar{\mathscr{G}_0}$ can be interpreted as a morphism

$$u : F^*\bar{\mathscr{G}_0} \to \bar{\mathscr{G}_0}.$$

It factors through $I\bar{G}$, where I defines the closed subset $\bar{S} \smallsetminus S$. Let $v : j_!G_0 \to \bar{\mathscr{G}_0}$ be the unique extension of the natural map $G_0 \to \mathscr{G}_0 = G_0 \otimes_{\mathbb{Z}/p} \mathscr{O}$. The diagram

$$\begin{array}{ccc} F^*j_!G_0 & \xrightarrow{F^*} & j_!G_0 \\ \downarrow{\scriptstyle F^*v} & & \downarrow{\scriptstyle v} \\ F^*\bar{\mathscr{G}_0} & \xrightarrow{u} & \bar{\mathscr{G}_0} \end{array} \tag{4.4.2}$$

Finally, denote by uF^* the composite map

$$\mathrm{H}^\bullet(\bar{S}, \bar{\mathscr{G}}) \xrightarrow{F^*} \mathrm{H}^\bullet(\bar{S}, F^*\bar{\mathscr{G}}) \xrightarrow{u} \mathrm{H}^\bullet(\bar{S}, \bar{\mathscr{G}}),$$

one has by (3.7.1)

$$\mathrm{tr}\left(F^*, \mathrm{H}^i_c(S, G)\right) = \mathrm{tr}\left(uF^*, \mathrm{H}^i(\bar{S}, \bar{\mathscr{G}})\right). \tag{4.4.3}$$

For every closed point $i_x : x \hookrightarrow \bar{S}$ of $\bar{S}$, set $\bar{\mathscr{G}_x} = i_x^*\bar{\mathscr{G}}$. The Lefschetz trace formula in coherent cohomology ("Woods Hole fixed point formula"), applied to F and u, says that

$$\sum_i (-1)^i \,\mathrm{tr}\left(uF^*, \mathrm{H}^i(\bar{S}, \bar{\mathscr{G}})\right) = \sum_{x \in \bar{S}^F} \frac{\mathrm{tr}(u_x, \mathscr{G}_x)}{\det(1 - dF_x)}. \tag{4.4.4}$$

On the right-hand side,

a) since $dF = 0$, the denominators are equal to 1;

b) for $x \in \bar{S}^F \smallsetminus S^F$, the endomorphism u_x of $\bar{\mathscr{G}_x}$ is zero, and therefore so is $\mathrm{tr}(u_x, \bar{\mathscr{G}_x})$;

c) for $x \in S^F$, (4.4.4) gives a contribution

$$\mathrm{tr}(u_x, \bar{\mathscr{G}_x}) = \mathrm{tr}(F_x^*, G_x).$$

Identity (4.4.1) therefore follows from (4.4.3) and (4.4.4).

□

4.5 A counterexample

An example shows that the hypothesis that A be reduced cannot be omitted from 4.1.

Let Y_0 be the elliptic curve over $\mathbb{F}_2$ for which the eigenvalues of Frobenius are $\pm\sqrt{-2}$. It has 3 rational points and is supersingular. The involution $\sigma : x \mapsto -x$ therefore has only one fixed point, the point 0. The quotient of Y_0 by σ is a projective line; the quotient of $Y_0 \smallsetminus 0$ by σ is therefore an affine line $\mathbb{F}_0^1$, and $Y_0 \smallsetminus 0$ is an unramified double covering of $\mathbb{F}_0^1$:

$$\pi : Y_0 \smallsetminus 0 \to \mathbb{A}_0^1 \qquad , \pi = \pi\sigma.$$

The sheaf $\pi_*\mathbb{Z}/2$ is a free module of rank one over the algebra A of the two-element group $\{1, \sigma\}$ over $\mathbb{Z}/2$. This algebra is isomorphic to the algebra of numbers over $\mathbb{Z}/2$.

The sheaf of A-modules $\pi_*\mathbb{Z}/2$ is our example:

a) One has $\mathrm{H}^\bullet_c(\mathbb{A}^1, \pi_*\mathbb{Z}/2) = \mathrm{H}^\bullet_c(Y \smallsetminus 0, \mathbb{Z}/2)$. Since Y_0 is supersingular, one has $\mathrm{H}^i(Y, \mathbb{Z}/2) = 0$ for $i > 0$, and the long exact sequence

$$\cdots \to \mathrm{H}^i_c(Y \smallsetminus 0, \mathbb{Z}/2) \to \mathrm{H}^i(Y, \mathbb{Z}/2) \to \mathrm{H}^i(\{0\}, \mathbb{Z}/2) \to \cdots$$

shows that $\mathrm{H}^\bullet_c(Y \smallsetminus 0, \mathbb{Z}/2) = 0$. Thus $\mathrm{H}^\bullet_c(\mathbb{A}^1, \pi_*\mathbb{Z}/2) = 0$ and

$$\mathrm{tr}_A\left(F^*, \mathsf{R}\Gamma_c(\mathbb{A}^1, \pi_*\mathbb{Z}/2)\right) = 0.$$

b) Since $Y_0 \smallsetminus 0$ has two rational points, at one of the rational points of $\mathbb{A}^0_1$, the Frobenius substitution is the identity, and at the other, it is σ. Thus

$$\sum_{x \in \mathbb{A}^1{}^F} \mathrm{tr}\left(F^*, \pi_*\mathbb{Z}/2\right) = 1 + \sigma,$$

and $1 + \sigma \neq 0$.

The equation of the covering π is $y^2 - y = x^3$.

Chapter 4

The cohomology class associated with a cycle

This exposé is inspired by notes of Grothendieck that formed a preliminary version of [21, IV]. It defines the cohomology class of a cycle X in a separated smooth scheme of finite type over a field, and proves that intersection corresponds to the cup product.

Chapter 1 contains some general formalities. In Chapter 2, the class of a cycle is defined in several situations more general than the one just mentioned. The principal compatibility not considered is that between the direct image of a cycle and the trace morphism in cohomology. In Chapter 3, this formalism is used to deduce the Lefschetz trace formula for an endomorphism with isolated fixed points of a proper smooth scheme over an algebraically closed k—and for a Frobenius endomorphism of a curve.

We adopt the following conventions:

1) "scheme" means a separated Noetherian scheme (this is largely a hypothesis of convenience).

2) In Chapter 2, an integer n is fixed, and n is invertible on all schemes under consideration.

3) In Chapter 3, a prime number ℓ is fixed, and ℓ is invertible on all schemes under consideration. The cohomology used is always ℓ-adic cohomology:

$$\mathrm{H}^\bullet(X) = \mathbb{Q}_\ell \otimes_{\mathbb{Z}_\ell} \varprojlim \mathrm{H}^\bullet(X, \mathbb{Z}/\ell^n).$$

Cohomology classes of cycles, trace morphisms, ... are defined by passage to the limit from the case of finite coefficients $\mathbb{Z}/\ell^n$ (cf. [21, VI]).

1 Cohomology with support and cup products

This section contains reminders from general topology, which the reader is invited to consult only as the need arises.

by A. Grothendieck, written up by P. Deligne

1.1 H^1 and torsors

1.1.1

Let $\mathscr{F}$ be an abelian sheaf on a site X. It is known that $\mathrm{H}^1(X,\mathscr{F})$ classifies $\mathscr{F}$-torsors on X. We normalize (= choose the sign of) the isomorphism (set of isomorphism classes of $\mathscr{F}$-torsors) $\to \mathrm{H}^1(X,\mathscr{F})$ in such a way that for every exact sequence $0 \to \mathscr{F} \xrightarrow{\alpha} \mathscr{G} \xrightarrow{\beta} \mathscr{H} \to 0$ and every $h \in \mathrm{H}^0(X,\mathscr{H})$, the $\mathscr{F}$-torsor $\beta^{-1}(h) \subset \mathscr{G}$, on which $\mathscr{F}$ acts by $(f,x) \mapsto \alpha(f)+x$, has class ∂h.

1.1.2

Let $\mathscr{P}$ be an $\mathscr{F}$-torsor. If (U_i) is an open covering of X, and p_i a section of $\mathscr{P}$ on U_i, one associates to $\mathscr{P}$ the Čech cocycle

$$p_{ij} = p_j - p_i \qquad (p_{ij} \in \mathrm{H}^0(U_i \times U_j, \mathscr{F})).$$

If, according to the usual rule, the map $\check{\mathrm{H}}^\bullet(X,\mathscr{F}) \to \mathrm{H}^\bullet(X,\mathscr{F})$ is defined so as to be a morphism of δ-functors, the image of $(p_{ij}) \in \check{\mathrm{H}}^1(X,\mathscr{F})$ in $\mathrm{H}^1(X,\mathscr{F})$ is the class of $\mathscr{P}$, normalized as in 1.1.1.

1.1.3

The definition of $\mathrm{H}^\bullet(X,\mathscr{F})$ is as follows: for $\mathscr{F}^\bullet$ a resolution of $\mathscr{F}$ with acyclic components, $\mathrm{H}^i(X,\mathscr{F}) = \mathrm{H}^i\,\Gamma(X,\mathscr{F}^\bullet)$. The δ-functor structure is obtained by associating to a short exact sequence of sheaves a short exact sequence of resolutions that remains exact after application of the functor Γ. If $\mathscr{F}^\bullet$ is a resolution of $\mathscr{F}$, the connecting homomorphism ∂ associated with

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}^0 \xrightarrow{d} \ker(d) \longrightarrow 0$$

induces the negative of the defining isomorphism $\mathrm{H}^1(X,\mathscr{F}) = \Gamma\left(X,\ker(d)\right)/d\Gamma(X,\mathscr{F}^0)$.

1.1.4

Let U be an open part of X (a subsheaf of the final sheaf), and let D be the "closed complement." It is known that $\mathrm{H}^1_D(X,\mathscr{F})$ classifies the $\mathscr{F}$-torsors on X trivialized on U. For every short exact sequence $0 \to \mathscr{F} \xrightarrow{\alpha} \mathscr{G} \xrightarrow{\beta} \mathscr{H} \to 0$, and every section with support in D, $h \in \mathrm{H}^0_D(X,\mathscr{H})$, the torsor $\beta^{-1}(h)$, trivialized on U by the section 0, has class ∂h.

The long exact sequence of cohomology with support

$$\cdots \xrightarrow{\partial} \mathrm{H}^i_D(X,\mathscr{F}) \longrightarrow \mathrm{H}^i(X,\mathscr{F}) \longrightarrow \mathrm{H}^i(U,\mathscr{F}) \xrightarrow{\partial} \cdots$$

is defined from the sequence of functors

$$0 \longrightarrow \Gamma_D \longrightarrow \Gamma \longrightarrow \Gamma(U,-) \longrightarrow 0$$

(exact on injective sheaves). For every section $f \in \mathrm{H}^0(U,\mathscr{F})$, $\partial f \in \mathrm{H}^1_D(X,\mathscr{F})$ is the class of the trivial torsor $\mathscr{F}$, trivialized on U by the section f.

1.1.5

Let $j : U \hookrightarrow X$. If $\mathscr{F}$ injects into $j_*j^*\mathscr{F}$, the exact sequence

$$0 \longrightarrow \mathscr{F} \longrightarrow j_*j^*\mathscr{F} \longrightarrow j_*j^*\mathscr{F}/\mathscr{F} \longrightarrow 0$$

gives $\partial : \mathrm{H}^0(j_*j^*\mathscr{F}/\mathscr{F}) = \mathrm{H}^0_D(j_*j^*\mathscr{F}/\mathscr{F}) \to \mathrm{H}^1_D(X, \mathscr{F})$. The composite map

$$\mathrm{H}^0(U, \mathscr{F}) = \mathrm{H}^0(X, j_*j^*\mathscr{F}) \longrightarrow \mathrm{H}^0(X, j_*j^*\mathscr{F}/\mathscr{F}) \xrightarrow{\partial} \mathrm{H}^1_D(X, \mathscr{F})$$

is the negative of the map considered in 1.1.4.

1.1.6

Recall that if $\mathscr{L}$ is an invertible sheaf on a scheme X, the $\mathbb{G}_m$-torsor corresponding to it is the sheaf $\mathrm{Isom}(\mathscr{O}, \mathscr{L})$, on which $\mathbb{G}_m$ acts by $(\lambda, f) \mapsto f \circ (\lambda\cdot) = \lambda f$. Recall also that if D is a Cartier divisor on X, and $j : U \hookrightarrow X$ is the inclusion of the complementary open subscheme, the invertible sheaf $\mathscr{O}(D)$ is the subsheaf of $j_*\mathscr{O}_U$ consisting of local sections s such that sf lies in $\mathscr{O}_X$, where f is a local equation for D.

1.2 Cup products

In this subsection, we develop some observations about cup products in cohomology with support that will be used again in [Duality] to relate Poincaré duality for curves and self-duality of the Jacobian.

1.2.1

Let X be a site, Y a closed part of X, and $\mathscr{F}, \mathscr{G}$ two abelian sheaves on X. For example: X a scheme, Y a closed subscheme, and $\mathscr{F}, \mathscr{G}$ sheaves on X_{et}. Define a product

$$\Gamma_Y(X, \mathscr{F}) \otimes \Gamma(Y, \mathscr{G}) \to \Gamma_Y(X, \mathscr{F} \otimes \mathscr{G}). \tag{1.2.1.1}$$

The product of a section s of $\mathscr{F}$ with support in Y by a section t of $\mathscr{G}$ on Y is obtained as follows: locally on X, t is the restriction to Y of a section t' of $\mathscr{G}$ on X, and one forms the product $s \otimes t'$. It has support in Y and does not depend on the choice of t', which justifies the definition and permits it to be globalized.

Let $i : Y \hookrightarrow X$ be the inclusion morphism. The local analogue of (1.2.1.1) is the product

$$i^!\mathscr{F} \otimes i^*\mathscr{G} \to i^!(\mathscr{F} \otimes \mathscr{G}), \tag{1.2.1.2}$$

from which (1.2.1.1) follows by applying $\Gamma(Y, -)$.

1.2.2

Let us derive these arrows. Let $\mathscr{K}$, $\mathscr{L}$, and $\mathscr{M}$ be in the derived category, with a bilinear map $\mathscr{K} \overset{\mathsf{L}}{\otimes} \mathscr{L} \to \mathscr{M}$. Deriving (1.2.1.1) gives

$$\mathsf{R}\Gamma_Y(X, \mathscr{K}) \overset{\mathsf{L}}{\otimes} \mathsf{R}\Gamma(Y, \mathscr{L}) \to \mathsf{R}\Gamma_Y(X, \mathscr{M}) \tag{1.2.2.1}$$

inducing

$$\mathrm{H}^i_Y(X, \mathscr{K}) \otimes \mathrm{H}^j(Y, \mathscr{L}) \to \mathrm{H}^{i+j}_Y(X, \mathscr{M}) \tag{1.2.2.2}$$

(the cup product). The local arrow (1.2.1.2) gives

$$\mathsf{R}i^!\mathscr{K} \overset{\mathsf{L}}{\otimes} i^*\mathscr{L} \to \mathsf{R}i^!\mathscr{M}, \tag{1.2.2.3}$$

from which (1.2.2.1) follows by applying $\mathsf{R}\Gamma(Y, -)$.

1.2.3

Above, I passed over in silence the conflicts caused by using right-derived functors ($\mathsf{R}\Gamma$) and left-derived functors ($\overset{\mathsf{L}}{\otimes}$) in the same formula.

a) In the next subsection, all the right-derived functors under consideration will have finite cohomological dimension. This makes it possible to work systematically in the derived categories D^-. To have complexes that are both flat and flasque, one uses canonical flasque resolutions as in [2, XVII].

b) For a more general theory, one can no longer interpret bilinear maps from $\mathscr{K}$ and $\mathscr{L}$ to $\mathscr{M}$ as morphisms from $\mathscr{K}\overset{\mathsf{L}}{\otimes}\mathscr{L}$ to $\mathscr{M}$. For example, $\mathsf{R}\Gamma_Y(X,\mathscr{K})\overset{\mathsf{L}}{\otimes}\mathsf{R}\Gamma(Y,\mathscr{L})$ is not defined if both factors lie in D^+. One solution is to work in D^+, define

$$\mathrm{Bil}(\mathscr{K},\mathscr{L};\mathscr{M}) = \varinjlim \hom(\mathscr{K}'\otimes\mathscr{L}',\mathscr{M}'),$$

where the limit is taken over the quasi-isomorphisms $\mathscr{K}' \xrightarrow{\sim} \mathscr{K}$, $\mathscr{L}' \xrightarrow{\sim} \mathscr{L}$, $\mathscr{M} \xrightarrow{\sim} \mathscr{M}'$ ($\mathscr{K}'$, $\mathscr{L}'$, $\mathscr{M}'$ bounded below), where hom means "morphism of complexes, up to homotopy," and systematically use such bilinear maps without ever mentioning $\otimes$.

1.2.4 Second theme

Let U be an open subscheme of X, and let $j : U \hookrightarrow X$ be the inclusion morphism. For $\mathscr{K}$ in the derived category on U, set $\mathsf{R}\Gamma_!(U,\mathscr{K}) = \mathsf{R}\Gamma(X, j_!\mathscr{K})$. For $\mathscr{K}$, $\mathscr{L}$, and $\mathscr{M}$ on U, with a bilinear map $\mathscr{K}\overset{\mathsf{L}}{\otimes}\mathscr{L} \to \mathscr{M}$, we wish to define

$$\mathsf{R}\Gamma(U,\mathscr{K})\overset{\mathsf{L}}{\otimes}\mathsf{R}\Gamma_!(U,\mathscr{L}) \to \mathsf{R}\Gamma_!(U,\mathscr{M}). \tag{1.2.4.1}$$

At the level of sheaves and their global sections, such a product follows from the isomorphism $j_*\mathscr{F}\otimes j_!\mathscr{G} \xleftarrow{\sim} j_!(\mathscr{F}\otimes\mathscr{G})$, but care must be taken because $\mathsf{R}\Gamma_!$ is not in general the derived functor of $\Gamma(X, j_!-)$.

We begin by defining

$$\mathsf{R}j_*\mathscr{K}\overset{\mathsf{L}}{\otimes}j_!\mathscr{L} \xleftarrow{\sim} j_!(\mathscr{K}\overset{\mathsf{L}}{\otimes}\mathscr{L}) \longrightarrow j_!\mathscr{M}.$$

Applying $\mathsf{R}\Gamma(X,-)$ gives

$$\mathsf{R}\Gamma(U,\mathscr{K})\overset{\mathsf{L}}{\otimes}\mathsf{R}\Gamma_!(U,\mathscr{L}) = \mathsf{R}\Gamma(X,\mathsf{R}j_*\mathscr{K})\overset{\mathsf{L}}{\otimes}\mathsf{R}\Gamma(X,j_!\mathscr{L}) \longrightarrow \mathsf{R}\Gamma(X,j_!\mathscr{M}).$$

Here again, the conflict between left and right will be resolved in the next subsection by working in D^-.

1.2.5 Coda

Let $j : U \hookrightarrow X$ be an open part, and let $i : Y \hookrightarrow U$ be a closed part of U. Let $\bar{Y}$ be a closed subset of X such that $\bar{Y}\cap U = Y$ (for example, the complement of $U\smallsetminus Y$). Let $\mathscr{K}$ be on U. Set $\mathsf{R}\Gamma_{Y!}(U,\mathscr{K}) = \mathsf{R}\Gamma_!(Y,\mathsf{R}i^!\mathscr{K})$, where $\mathsf{R}\Gamma_!$ is relative to the inclusion of Y into $\bar{Y}$. For every open subset V of U containing Y, $\mathsf{R}\Gamma_{Y!}(U,\mathscr{K})$ maps to $\mathsf{R}\Gamma_!(V,\mathscr{K})$: if i also denotes the inclusion of $\bar{Y}$ into X, j that of Y into $\bar{Y}$, and k that of V into X, one has $\mathsf{R}i^!\mathscr{K} = j^*\mathsf{R}i^!k_!(k^*\mathscr{K})$, whence a morphism $j_!\mathsf{R}i^!\mathscr{K} \to \mathsf{R}i^!k_!(k^*\mathscr{K})$. Applying $\mathsf{R}\Gamma(\bar{Y},-)$ to it gives $\mathsf{R}\Gamma_{Y!}(U,\mathscr{K}) \to \mathsf{R}\Gamma_{\bar{Y}}(k_!k^*\mathscr{K}) \to \mathsf{R}\Gamma(k_!k^*\mathscr{K}) = \mathsf{R}\Gamma_!(V,\mathscr{K})$.

For $\mathscr{K},\mathscr{L},\mathscr{M}$ on U, with a bilinear map $\mathscr{K}\overset{\mathsf{L}}{\otimes}\mathscr{L} \to \mathscr{M}$, we wish to define

$$\mathrm{H}^n(U,\mathscr{K})\otimes\mathrm{H}^m_!(Y,\mathscr{L}) \to \mathrm{H}^{n+m}_{Y!}(U,\mathscr{M}) \tag{1.2.5.1}$$

(the last group itself mapping to $\mathrm{H}^{n+m}_!(V,\mathscr{M})$). In the derived category, this means defining

$$\mathsf{R}\Gamma_Y(U,\mathscr{K})\overset{\mathsf{L}}{\otimes}\mathsf{R}\Gamma_!(Y,\mathscr{L})\to\mathsf{R}\Gamma_{Y!}(U,\mathscr{M}). \tag{1.2.5.2}$$

We identify $\mathsf{R}\Gamma_Y$ with $\mathsf{R}\Gamma(Y,\mathsf{R}i^!\mathscr{K})$. The desired product is then of type (1.2.4.1), relative to the local product (1.2.2.3) on Y: $\mathsf{R}i^!\mathscr{K}\overset{\mathsf{L}}{\otimes}\mathscr{L}\to\mathsf{R}i^!\mathscr{M}$.

1.2.6

Above, we developed the absolute formalism. There is a parallel relative formalism, with Γ replaced by f_* for f a morphism $X\to S$.

1.3 The Koszul rule

Let A be a commutative ring, and let $(V_i)_{i\in I}$ be a finite family of graded (or $\mathbb{Z}/2$-graded) A-modules. Recall the definition of the graded tensor product $\bigotimes_{i\in I}V_i$, in the sense of the Koszul rule (cf. [2, XVII 1.1]). For each total ordering a on I, we shall define a module $V(a)$. We shall also define a transitive system of isomorphisms $\varphi_{ab}:V(b)\xrightarrow{\sim}V(a)$, and the graded tensor product of the V_i will be the "common value" $\varprojlim V(a)$ of this system of modules. We take

a) $V(a)=\bigotimes_{i\in I}|V_i|$ (the ordinary tensor product of the underlying ungraded modules of the V_i).

b) if the $x_i\in V_i$ are homogeneous, we take $\varphi_{ab}(\otimes x_i)=(-1)^N\otimes x_i$, where N is the sum of the $\deg(x_i)\deg(x_j)$ over the pairs (i,j) such that $i<_a j$ and $i>_b j$.

Example 1

Take $I=\{1,2\}$, and let a be the ordering in which $1<2$, and b the ordering in which $1>2$. Let $v_i\in V_i$ be homogeneous, and denote by $v_1\otimes v_2$ (resp. $v_2\otimes v_1$) the image in the graded tensor product of the product of the v_i in $V(a)$ (resp. $V(b)$). One has

$$v_1\otimes v_2=(-1)^{\deg(v_1)\deg(v_2)}v_2\otimes v_1$$

(the Koszul rule).

Example 2

If all the V_i have degree 1, there is a canonical isomorphism relating the module underlying the graded tensor product and the ordinary tensor product of the modules $|V_i|$ underlying the V_i:

$$\Big|\bigotimes V_i\Big|\simeq\bigotimes|V_i|\otimes_{\mathbb{Z}}\bigwedge^{|I|}\mathbb{Z}^I.$$

Example 3

For A a field, and X_i a finite family of spaces, the Künneth formula is written $\mathrm{H}^\bullet(\prod X_i,A)=\bigotimes\mathrm{H}^\bullet(X_i,A)$.

Let $V^\vee$ be the graded dual of V. The canonical map

$$V^\vee\otimes V=V\otimes V^\vee\to A \tag{1.3.1}$$

is $v'\otimes v\to v'(v)$.

Suppose now that A is a field, and consider only finite-dimensional vector spaces. The canonical isomorphism

$$W \otimes V^\vee \to \hom(V, W) \tag{1.3.2}$$

is $w \otimes v' \mapsto (v \mapsto w \cdot v'(v))$. Via this isomorphism, the composition $\hom(Y,Z) \otimes \hom(X,Y) \to \hom(X,Z)$ is identified with the morphism induced by (1.3.1): $Z \otimes Y^\vee \otimes Y \otimes X^\vee \to Z \otimes X^\vee$.

The trace of $f : V \to V$ (zero for homogeneous f of degree $\neq 0$) is the image of f under

$$\mathrm{tr} : \hom(V,V) \xleftarrow{\sim}_{(1.3.2)} V \otimes V^\vee = V^\vee \otimes V \to_{(1.3.1)} A. \tag{1.3.4}$$

It is easily verified that, for f of degree 0,

$$\mathrm{tr}(f, V) = \sum (-1)^i \,\mathrm{tr}(f, V^i). \tag{1.3.5}$$

Expressing the commutativity of the two morphisms obtained by composing the morphisms (1.3.1) $V^\vee \otimes V \otimes W^\vee \otimes W \to k$, one finds that for homogeneous $f : V \to W$ and $g : W \to V$, one has

$$\mathrm{tr}(fg) = (-1)^{\deg(f)\deg(g)} \,\mathrm{tr}(gf). \tag{1.3.6}$$

2 The cohomology class associated with a cycle

2.1 The class of a divisor

2.1.1

Let D be a Cartier divisor in a scheme X. Away from D, the invertible sheaf $\mathscr{O}(D)$ is trivialized by the section 1. The class $\mathrm{cl}(D)$ of D in $\mathrm{H}^1_D(X, \mathbb{G}_m)$ is the class of the corresponding $\mathbb{G}_m$-torsor trivialized on $X \smallsetminus D$ (1.1.6 and 1.1.4).

Let $\partial : \mathrm{H}^i(X \smallsetminus D, \mathbb{G}_m) \to \mathrm{H}^i_D(X, \mathbb{G}_m)$ be the morphism of 1.1.4. If D admits a global equation f, multiplication by f is an isomorphism from $\mathscr{O}(D)$, trivialized by 1 on $X \smallsetminus D$, to $\mathscr{O}$, trivialized by f on $X \smallsetminus D$. By 1.1.4, one therefore has

$$\mathrm{cl}(D) = \partial f. \tag{2.1.1}$$

For every morphism $u : X' \to X$ such that u^*D is again a Cartier divisor (i.e., $u^{-1}D$ is disjoint from $\mathrm{Ass}(X')$), one has $\mathrm{cl}(u^*D) = u^* \,\mathrm{cl}(D)$. If such functoriality were desired for every morphism u, one would have to consider not Cartier divisors, but more generally invertible sheaves equipped with a section.

Recall that the integer n is henceforth assumed invertible on the schemes under consideration. Let $\partial : \mathrm{H}^i_D(X, \mathbb{G}_m) \to \mathrm{H}^{i+1}_D(X, \boldsymbol{\mu}_n)$ be the boundary map for the Kummer exact sequence $0 \to \boldsymbol{\mu}_n \to \mathbb{G}_m \to \mathbb{G}_m \to 0$.

Definition 2.1.2. *The* class $\mathrm{cl}_n(D)$ *of* D *in* $\mathrm{H}^2_D(X, \boldsymbol{\mu}_n)$ *is* $\partial \,\mathrm{cl}(D)$.

When there is no risk of confusion, the subscript n will be omitted.

2.1.3

The diagram

$$\begin{array}{ccc} \mathrm{H}^0(X \smallsetminus D, \mathbb{G}_m) & \xrightarrow{\partial} & \mathrm{H}^1_D(X, \mathbb{G}_m) \\ \downarrow \partial & & \downarrow \partial \\ \mathrm{H}^1(X \smallsetminus D, \boldsymbol{\mu}_n) & \xrightarrow{\partial} & \mathrm{H}^2_D(X, \boldsymbol{\mu}_n) \end{array}$$

is anticommutative. If D admits a global equation f, $\mathrm{cl}_n(D)$ is therefore the negative of the image under ∂ of the class in $\mathrm{H}^1(X \smallsetminus D, \boldsymbol{\mu}_n)$ of the $\boldsymbol{\mu}_n$-torsor of n-th roots of f.

Proposition 2.1.4. *Let i be the inclusion of D into X. If D and X are regular, the cohomology sheaves with support $\mathsf{R}^p i^! \boldsymbol{\mu}_n$ are zero for $p = 0, 1$, and $\mathsf{R}^2 i^! \boldsymbol{\mu}_n = \underline{\mathbb{Z}/n}$, generated by $\mathrm{cl}_n(D)$.*

It suffices to prove that, for X strictly local and D defined by a regular parameter, one has $\mathrm{H}^p_D(X, \boldsymbol{\mu}_n) = 0$ for $p = 0, 1$ and $\mathrm{H}^2_D(X, \boldsymbol{\mu}_n) = \mathbb{Z}/n$, generated by $\mathrm{cl}(D)$. Denoting reduced cohomology by $\sim$, one has $\widetilde{\mathrm{H}}^{p-1}(X \smallsetminus D, \boldsymbol{\mu}_n) \xrightarrow{\sim} \mathrm{H}^p_D(X, \boldsymbol{\mu}_n)$. The assertion for $p = 0, 1$ says that D does not disconnect X, and for $p = 2$ it follows, via 2.1.3, from Abhyankar's lemma.

This is a partial analogue of the relative theorem (Étale cohomology, 5.3.4). Grothendieck conjectures that the $\mathsf{R}^p i^! \boldsymbol{\mu}_n$ vanish for $p \neq 2$, at least for X excellent (purity conjecture), but this is known only in characteristic 0 ([2, XIX]).

Theorem 2.1.5 (Fundamental compatibility)**.** *Let X be a smooth curve over an algebraically closed field k, let P be a closed point of X, and let* tr *be the composite $\mathrm{H}^2_P(X, \boldsymbol{\mu}_n) \to \mathrm{H}^2_c(X, \boldsymbol{\mu}_n) \xrightarrow{\mathrm{tr}} \mathbb{Z}/n$. One has*

$$\mathrm{tr}\, \mathrm{cl}(P) = 1.$$

Let $\bar{X}$ be the smooth projective curve completing X. The formula says that the invertible sheaf $\mathscr{O}(P)$ on $\bar{X}$ has degree 1.

2.2 Cohomological method

2.2.1

Let X be a (Noetherian) scheme. Recall that a subscheme Y of X is said to be a local complete intersection of codimension c if, locally (on Y), it is defined by a regular sequence of c equations in X. For X the spectrum of a local ring A with maximal ideal $\mathfrak{m}$, and Y with ideal $\mathfrak{a}$, this means that $\mathscr{E}\mathrm{xt}^i(A/\mathfrak{a}, A) = 0$ for $i < \dim(\mathfrak{a}/\mathfrak{m}\mathfrak{a}) = c$, and every sequence of elements of $\mathfrak{a}$ whose image in $\mathfrak{a}/\mathfrak{m}\mathfrak{a}$ is a basis of $\mathfrak{a}/\mathfrak{m}\mathfrak{a}$ is a regular sequence of equations for Y.

2.2.2

Let $i : Y \hookrightarrow X$ be a local complete intersection of codimension c. We propose to define a local fundamental class $\mathrm{cl}(Y)$ that is a global section of the cohomology sheaf with support $\mathsf{R}^{2c} i^! \mathbb{Z}/n(c)$ (recall that $\mathbb{Z}/n(c) = \boldsymbol{\mu}_n^{\otimes c}$).

Locally, Y is the intersection of a sequence of c divisors D_i, and we define $\mathrm{cl}(Y)$ as the cup product of the $\mathrm{cl}(D_i)$. Each $\mathrm{cl}(D_i)$ has support in D_i; their product has support in Y. The fact that this product is locally independent of the choice of the D_i follows from 2.2.3 below and the following invariance properties:

a) compatibility with localization;

b) independence of the ordering of the D_i (the $\mathrm{cl}(D_i)$ have degree 2, which is even, so the cup product is commutative);

c) the product depends only on the "flag" $D_1 \supset D_1 \cap D_2 \supset \cdots \supset Y$.

To prove c), note the existence of a product (a variant of 1.2.1)

$$\mathrm{H}^\bullet_{D_1}(X) \otimes \mathrm{H}^\bullet_{D_1 \cap D_2}(D_1) \otimes \cdots \otimes \mathrm{H}^\bullet_Y(D_1 \cap \cdots \cap D_{c-1}) \to \mathrm{H}^\bullet_Y(X);$$

the product of the $\mathrm{cl}(D_i)$ is also the product of the ($\mathrm{cl}(D_i)$ restricted to $D_1 \cap \cdots \cap D_{i-1}$) = ($\mathrm{cl}(D_1 \cap \cdots \cap D_i)$ in $D_1 \cap \cdots \cap D_{i-1}$).

Lemma 2.2.3. *Let A be a local ring with maximal ideal $\mathfrak{m}$, and let $u = (u_1, \ldots, u_c)$, $v = (v_1, \ldots, v_c)$ be two regular sequences generating the same ideal $\mathfrak{a}$. There is then a sequence w_i $(1 \leqslant i \leqslant N)$ of such sequences joining them, such that w_{i+1} is obtained from w_i by a permutation or by changing only the last element.*

Since permutations are available, it would amount to the same thing to allow one element, rather than the last element, to be changed. By 2.2.1, we are thus reduced to verifying that, in the vector space $\mathfrak{a}/\mathfrak{m}\mathfrak{a}$, one can pass from one basis to another by a sequence of permutations and changes of basis modifying only one vector. Indeed, the linear group is generated by diagonal, elementary, and permutation matrices.

2.2.4

The same methods make it possible to define a local fundamental class for every $Y \subset X$ locally definable by c equations (where fewer equations suffice, the class is zero).

2.2.5

One can pass from such a local fundamental class in $\mathrm{H}^0\left(Y, \mathsf{R}^{2c}i^!\mathbb{Z}/n(c)\right)$ to a global fundamental class in $\mathrm{H}^{2c}_Y(X, \mathbb{Z}/n(c))$ when semipurity results are available:

Proposition 2.2.6. *(i) If $\mathsf{R}^p i^!\mathbb{Z}/n = 0$ for $p < 2c$, then*

$$\mathrm{H}^{2c}_Y(X, \mathbb{Z}/n(c)) \xrightarrow{\sim} \mathrm{H}^0\left(Y, \mathsf{R}^{2c}i^!\mathbb{Z}/n(c)\right).$$

(ii) Let Z be a closed part of Y, with complement V in Y, and let k be the inclusion of Z into X. If $\mathsf{R}^p i^!\mathbb{Z}/n = 0$ for $p \leqslant 2c$, then

$$\mathrm{H}^{2c}_Y(X, \mathbb{Z}/n(c)) \hookrightarrow \mathrm{H}^{2c}_V(X, \mathbb{Z}/n(c)).$$

If $\mathsf{R}^p i^!\mathbb{Z}/n = 0$ for $p \leqslant 2c+1$, this arrow is an isomorphism.

The hypothesis $\mathsf{R}^p i^!\mathbb{Z}/n = 0$ is equivalent to $\mathsf{R}^p i^!\mathbb{Z}/n(c) = 0$. With this understood, (i) follows from the spectral sequence $\mathrm{H}^p(Y, \mathsf{R}^q i^!) \Rightarrow \mathrm{H}^{p+q}_Y(X, -)$. If k_1 is the inclusion of Z into Y, one has $k = ik_1$, whence $\mathsf{R}k^!\mathbb{Z}/n = \mathsf{R}k_1^!\mathsf{R}i^!$, and the long exact cohomology sequence for $Z \subset Y$ gives a long exact sequence

$$\cdots \longrightarrow \mathrm{H}^i\left(Z, \mathsf{R}k^!\mathbb{Z}/n(c)\right) \longrightarrow \mathrm{H}^i_Y(X, \mathbb{Z}/n(c)) \longrightarrow \mathrm{H}^i_V(X, \mathbb{Z}/n(c)) \longrightarrow \cdots$$

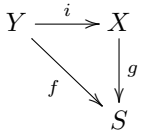
from which (ii) follows.

2.2.7 Amplification

Proposition 2.2.6 remains valid for i any separated morphism of finite type, and $2c$ any integer (positive or negative), provided that $\mathrm{H}^{2c}_Y(X, \mathbb{Z}/n(c))$ is replaced there by $\mathrm{H}^{2c}\left(Y, \mathsf{R}i^!\mathbb{Z}/n(c)\right)$, and likewise for H_V.

The following semipurity results follow from [24, 1.8, 1.10, 1.15]. We recall their proof.

Theorem 2.2.8. *Let there be a morphism of S-schemes of finite type*



Suppose that X is smooth, purely of relative dimension N, and that Y has fiberwise dimension $\leqslant d$. Let $c = N - d$.

(i) One has $\mathsf{R}^p i^!\mathbb{Z}/n = 0$ *for* $p < 2c$*; likewise with* $\mathbb{Z}/n$ *replaced by a sheaf* $g^*\mathscr{F}$*.*

(ii) If, over a dense open subset U *of* S*,* Y *has fiberwise dimension* $< d$*, one has* $\mathsf{R}^p i^!\mathbb{Z}/n = 0$ *for* $p \leqslant 2c$*. If in addition the complement* Z *of* U *does not locally disconnect* S*, one has* $\mathsf{R}^p i^!\mathbb{Z}/n = 0$ *for* $p \leqslant 2c+1$*.*

Since $\mathsf{R}g^!\mathscr{F} = g^*\mathscr{F}(N)[2N]$, the transitivity formula $\mathsf{R}i^!\mathsf{R}g^! = \mathsf{R}f^!$ shows that $\mathsf{R}^{2c+q}i^!(g^*\mathscr{F}) = 0 \Leftrightarrow \mathsf{R}^{-2d+q}f^!(\mathscr{F})$. This reduces us to studying f, i.e. to assuming that $X = S$. Assertion (i) is then [2, XVIII 3.17].

Let Y' be the inverse image of Z in Y:

$$\begin{array}{ccc} Y' & \xrightarrow{v} & Y \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} \\ Z & \xrightarrow{u} & S \end{array}$$

By (i), the $\mathsf{R}^p f^!\mathbb{Z}/n$ have support in Y' for $p \leqslant -2d+1$; and the spectral sequence $\mathsf{R}^a v^!\mathsf{R}^b f^! \Rightarrow \mathsf{R}^{a+b}(fv)^!$ shows that they coincide with the $\mathsf{R}^p(fv)^!\mathbb{Z}/n = \mathsf{R}^p(uf)^!\mathbb{Z}/n$. Applying (i) to Y'/Z and to the spectral sequence $\mathsf{R}^a f^!\mathsf{R}^b u^! \Rightarrow \mathsf{R}^{a+b}(uf)^!$, one finds that (ii) follows from the vanishing of $\mathsf{R}^b u^!\mathbb{Z}/n$ for $b = 0$, or for $b = 0$ and 1, as the case may be.

2.2.9

Grothendieck conjectures the following absolute analogue of 2.2.8 (semipurity conjecture—a consequence of the purity conjecture): for Y of codimension $\geqslant c$ in a regular X, one has $\mathrm{H}^i_Y(X, \mathbb{Z}/n) = 0$ for $i < 2c$, at least if X is excellent.

2.2.10

We are now in a position to define the class of a cycle Y of codimension c in X smooth over a field k. Write $Y = \sum d_i Y_i$, where the Y_i are reduced and irreducible. An open subset U_i of Y_i, whose complement has codimension $> c$, is then a local complete intersection in X. This makes it possible to define the local fundamental class of U_i. By 2.2.6 and 2.2.8, it comes from a unique fundamental class $\mathrm{cl}(Y_i) \in \mathrm{H}^{2c}_{Y_i}(X, \mathbb{Z}/n(c))$, and we set

$$\mathrm{cl}(Y) = \sum d_i \,\mathrm{cl}(Y_i) \in \mathrm{H}^{2c}_{|Y|}(X, \mathbb{Z}/n(c))\,.$$

2.2.11

In complex analytic geometry, a construction of Baum, Fulton, and Mac Pherson [4] makes it possible to define without restriction the cohomology class of a local complete intersection $Y \subset X$. Suppose Y is purely of codimension c, and let $\mathscr{N}$ be the vector bundle on Y that is the normal bundle of Y in X. Its local sections are those of $(\mathscr{I}/\mathscr{I}^2)^\vee$, where $\mathscr{I}$ is the sheaf of ideals of Y. Recall that the sheaf of C^∞ functions on X is defined locally, in terms of local embeddings of X into $\mathbb{C}^n$, as the restriction to X of the quotient of the sheaf of C^∞ functions on $\mathbb{C}^n$ by the ideal generated by the real and imaginary parts of the equations defining X. The bundle $\mathscr{N}$ extends to a complex $C^\infty N$ vector bundle on a neighborhood U of Y in X, and for U sufficiently small, there exist sections f of N whose zero locus is Y and such that, on Y, $df : \mathscr{N} \to N$ is the identity. Two choices of N and f are homotopic on a sufficiently small neighborhood of Y.

The cohomology class $\mathrm{cl}(U)$ of the section 0 of N (denoted $z : U \to N$) is defined: U is a local complete intersection in N, and $\mathsf{R}^i z^! \mathbb{Z} = 0$ for $i \neq 2c$, $\mathsf{R}^{2c} z^! \mathbb{Z} = \mathbb{Z}$. We have $f^* : \mathrm{H}^\bullet_U(N, \mathbb{Z}) \to \mathrm{H}^\bullet_Y(X, \mathbb{Z})$ and set

$$\mathrm{cl}(Y) = f^* \,\mathrm{cl}(U).$$

The same construction works whenever an analytic subspace Y of X has the following normal structure: a locally free sheaf $\mathscr{C}$ of rank c on Y_{red}, and an epimorphism $\mathscr{C} \to \mathscr{I}/\mathscr{I} \cdot \mathscr{I}_{\mathrm{red}}$.

2.3 Homological method

2.3.1

Let $f : Y \to S$ be a flat morphism of finite type with fibers of dimension $\leqslant d$. In [2, XVIII 2.9], we defined a trace morphism

$$\mathrm{tr}_f : \mathsf{R}^{2d} f_! \mathbb{Z}/n(d) \to \mathbb{Z}/n. \tag{2.3.1.1}$$

One has $\mathsf{R}^i f_! \mathbb{Z}/n(d) = 0$ for $i > 2d$, and $\mathsf{R}^i f^! \mathbb{Z}/n = 0$ for $i < -2d$. This, together with the adjunction between $\mathsf{R}f_!$ and $\mathsf{R}f^!$, gives isomorphisms

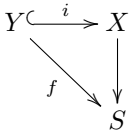
$$\begin{aligned} \hom\left(\mathsf{R}^{2d} f_! \mathbb{Z}/n(d), \mathbb{Z}/n\right) &= \hom\left(\mathsf{R}f_! \mathbb{Z}/n(d), \mathbb{Z}/n[-2d]\right) \\ &= \hom\left(\mathbb{Z}/n, \mathsf{R}f^! \mathbb{Z}/n(-d)[-2d]\right) \\ &= \mathrm{H}^0\left(Y, \mathsf{R}^{-2d} f^! \mathbb{Z}/n\right). \end{aligned} \tag{2.3.1.2}$$

At least when S is a point, the image of tr_f in the last two groups deserves the name fundamental class of Y (in homology).

We again denote by tr_f the image of tr_f in the second group, and the morphisms deduced from it by functoriality. For example, for Y/S proper, the morphism

$$\mathrm{H}^i\left(Y, \mathbb{Z}/n(d)\right) = \mathrm{H}^i\left(S, \mathsf{R}f_! \mathbb{Z}/n(d)\right) \longrightarrow \mathrm{H}^{i-2d}(S, \mathbb{Z}/n). \tag{2.3.1.3}$$

Suppose Y is contained in X, smooth over S and purely of relative dimension N, and set $c = N - d$.



One has $\mathsf{R}f^! = \mathsf{R}i^! \mathsf{R}g^!$, and $\mathsf{R}g^! \mathbb{Z}/n = \mathbb{Z}/n(-N)[-2N]$. The fundamental class of Y is identified with an element of $\mathrm{H}^0\left(Y, \mathsf{R}i^! \mathbb{Z}/n(c)[2c]\right) = \mathrm{H}^{2c}_Y\left(X, \mathbb{Z}/n(c)\right)$, the class $\mathrm{cl}(Y)$ of Y in X. We shall see below that it depends only on $Y \subset X$, not on the projection of X onto S, and that for Y a local complete intersection, it induces the local class of the preceding subsection.

Let us make explicit the isomorphism

$$\mathrm{H}^{2c}_Y\left(X, \mathbb{Z}/n(c)\right) \xrightarrow{\sim} \hom\left(\mathsf{R}^{2d} f_! \mathbb{Z}/n(d), \mathbb{Z}/n\right)$$

that takes $\mathrm{cl}(Y)$ to tr_f: via the isomorphisms

$$\begin{aligned} \mathrm{H}^{2c}_Y\left(X, \mathbb{Z}/n(c)\right) &= \mathrm{H}^{2c}_Y\left(\mathsf{R}i^! \mathbb{Z}/n(c)\right) \\ &= \hom_Y\left(\mathbb{Z}/n(d)[2d], \mathsf{R}i^! \mathbb{Z}/n(N)[2N]\right) \\ &= \hom_X\left(\mathbb{Z}/n(d)[2d]_Y, \mathbb{Z}/n(N)[2N]\right), \end{aligned}$$

it is the top row of

$$\begin{array}{ccccc} \mathrm{H}^{2c}_Y & \longrightarrow & \hom\left(\mathsf{R}f_!\mathbb{Z}/n(d)[2d], \mathsf{R}f_!\mathsf{R}i^!\mathbb{Z}/n(N)[2N]\right) & \xrightarrow{(1)} & \hom\left(\mathsf{R}f_!\mathbb{Z}/n(d)[2d], \mathbb{Z}/n\right) \\ & \searrow^{(3)} & \downarrow{\scriptstyle \mathrm{tr}_i} & & \| \\ & & \hom\left(\mathsf{R}g_!\mathbb{Z}/n(d)[2d]_Y, \mathsf{R}g_!\mathbb{Z}/n(N)[2N]\right) & \xrightarrow{\mathrm{tr}_g} & \hom\left(\mathsf{R}g_!\mathbb{Z}/n(d)[2d]_Y, \mathbb{Z}/n\right), \end{array}$$

where (1) is the adjunction arrow $\mathsf{R}f_!\mathsf{R}f^! \to \mathrm{id}$. The commutativity of (2) expresses the fact that the isomorphism $\mathsf{R}f^! = \mathsf{R}i^!\mathsf{R}g^!$ is defined by adjunction. Finally, the arrow deduced from (3)

$$\mathrm{H}^{2c}_Y\left(X, \mathbb{Z}/n(c)\right) \otimes \mathsf{R}^{2d}f_!\mathbb{Z}/n(d) \to \mathsf{R}^{2N}g_!\mathbb{Z}/n(N)$$

is interpreted as a cup product (cf. 1.2).

Definition 2.3.2. *The class* $\mathrm{cl}(Y) \in \mathrm{H}^{2c}_Y\left(X, \mathbb{Z}/n(c)\right)$ *is characterized by the property that, for every local section* u *of* $\mathsf{R}^{2d}f_!\mathbb{Z}/n(d)$*, one has*

$$\mathrm{tr}_f(u) = \mathrm{tr}_g\left(\mathrm{cl}(Y) \smile u\right).$$

Because $\mathrm{cl}(Y)$ already has an image tr_f in $\hom\left(\mathsf{R}f_!\mathbb{Z}/n(d)[2d], \mathbb{Z}/n\right)$, formula 2.3.2 holds for the arrows deduced from tr_f by functoriality. For example, for X proper over S and $u \in \mathrm{H}^\bullet(Y, \mathbb{Z}/n)$, the formula holds in $\mathrm{H}^\bullet\left(S, \mathbb{Z}/n(-d)\right)$. For $v \in \mathrm{H}^\bullet(X, \mathbb{Z}/n)$, it gives

$$\mathrm{tr}_f(i^*v) = \mathrm{tr}_g\left(\mathrm{cl}(Y) \smile v\right),$$

where the cup product can be computed in $\mathrm{H}^\bullet(X, \mathbb{Z}/n)$.

2.3.3

We shall define the trace morphisms, and hence the class $\mathrm{cl}(Y)$, under more general hypotheses. Consider a diagram

$$\begin{array}{ccc} Y & \hookrightarrow & X \\ & {\scriptstyle f}\searrow & \downarrow{\scriptstyle g} \\ & & S \end{array}$$

with g smooth and purely of relative dimension N, and Y closed in X with fiberwise dimension $\leqslant d$. Equip Y in addition with a "weight" $\mathscr{K}$ of the following kind: $\mathscr{K} \in \mathsf{D}^b_{\mathrm{parf},X}$ is a bounded complex of sheaves of $\mathscr{O}$-modules on X, of finite Tor-dimension over S (or over X, which amounts to the same thing), whose cohomology sheaves are coherent and have support in Y. We propose to define a morphism $\mathrm{tr}_{f,\mathscr{K}} : \mathsf{R}^{2d}f_!\mathbb{Z}/n(d) \to \mathbb{Z}/n$. For Y flat over S and $\mathscr{K} = \mathscr{O}_Y$, this will be the preceding trace morphism. In general, it depends only on the lengths of $\mathscr{K}$ at the generic points y of the irreducible components of Y ($\lg_y(\mathscr{K}) = \sum(-1)^i \lg \mathrm{H}^i(\mathscr{K})_y$). Finally, $\mathrm{cl}(Y, \mathscr{K})$ will denote the class with support in Y such that

$$\mathrm{tr}_g\left(\mathrm{cl}(Y, \mathscr{K}) \smile u\right) = \mathrm{tr}_{f,\mathscr{K}}(u). \tag{2.3.3.1}$$

2.3.4

The construction of $\mathrm{tr}_{f,\mathscr{K}}$ is parallel to that of [2, XVIII.2]; we shall indicate only its main outlines.

A $d = 0$ (f quasi-finite). Let x be a geometric point of Y, s its image in S, $\mathscr{K}_{(x)}$ the inverse image of $\mathscr{K}$ on the strict localization of X at x, $\mathscr{K}_{(x)s} = \mathscr{K}_{(x)} \overset{\mathsf{L}}{\otimes}_{\mathscr{O}^h_{S,s}} k(s)$ its inverse image on the geometric fiber at s, and $n(x) = \sum (-1)^i \dim_{k(s)} \mathrm{H}^i(\mathscr{K}_{(x)s})$. The function $x \mapsto n(x)$ is a weighting of f, and we take the corresponding trace morphism [2, XVII.6.2.5]. Formation of the weighting $n(x)$ and of $\mathrm{tr}_{f,\mathscr{K}}$ is compatible with every base change.

B General case. If $u : X \to \mathbb{A}^d_S$ is such that ui is quasi-finite, set $\mathrm{tr}_{f,\mathscr{K}} = \mathrm{tr}_{\mathbb{A}^d_S} \circ \mathrm{tr}_{ui,\mathscr{K}}$. This trace morphism is clearly compatible with every base change S'/S. To prove that it does not depend on u, we may therefore suppose that S is the spectrum of an algebraically closed field k. In that case, let Y_i be the irreducible components of Y, and let lg_i be the length of $\mathscr{K}$ at the generic point of Y_i. If α_i is the inclusion of $(Y_i)_{\mathrm{red}}$ into Y, one has

$$\mathrm{tr}_{f,\mathscr{K}} = \sum \mathrm{lg}_i \, \mathrm{tr}_{Y_{i,\mathrm{red}/S}} \, \alpha_i^* \tag{2.3.4}$$

(this formula follows from the analogous and easy formula for tr_{ui}).

This defines $\mathrm{tr}_{f,\mathscr{K}}$ locally on Y; one then proceeds as in [2, XVIII.2.9].

Lemma 2.3.5. *If $g = g'g'' : X \xrightarrow{g''} S' \xrightarrow{g'} S$, with g' and g'' smooth and purely of relative dimensions N' and N'', and if Y still has fiberwise codimension $\geqslant c$ over S', then the class $\mathrm{cl}(Y, \mathscr{K})$ is the same whether computed in terms of g or of g''.*

Let $f' = g''i$ and $d' = d - N'$. The formula $\mathrm{tr}_g = \mathrm{tr}_{g'} \mathrm{tr}_{g''}$ ([2, XVIII.2.9] Var 3) ensures the commutativity of

$$\begin{array}{ccc}
 & \mathrm{H}^{2c}_Y(X, \mathbb{Z}/n(c)) & \\
 & \downarrow & \\
\hom_{S'}(\mathsf{R}f_!\mathbb{Z}/n(d')[2d'], \mathsf{R}g''_!\mathbb{Z}/n(N'')[2N'']) & \longrightarrow & \hom_{S'}\left(\mathsf{R}^{2d'} f'_!\mathbb{Z}/n, \mathbb{Z}/n\right) \\
\downarrow{\scriptstyle \mathsf{R}g'_!(N')} & & \downarrow{\scriptstyle \mathrm{tr}_{g'} \circ \mathsf{R}^{2N'} g'_!} \\
\hom_S(\mathsf{R}f_!\mathbb{Z}/n(d)[2d], \mathsf{R}g'_!\mathbb{Z}/n(N)[2N]) & \longrightarrow & \hom_S\left(\mathsf{R}^{2d} f_!\mathbb{Z}/n, \mathbb{Z}/n\right)
\end{array}$$

while the analogous and easily verified formula $\mathrm{tr}_{f,\mathscr{K}} = \mathrm{tr}_{g'} \mathrm{tr}_{f',\mathscr{K}}$ ensures that the image of $\mathrm{tr}_{f',\mathscr{K}}$ is $\mathrm{tr}_{f,\mathscr{K}}$. The assertion follows.

Lemma 2.3.6. *If Y is a divisor in X, the class of 2.3.2 coincides with the class of 2.1.2.*

By semipurity (2.2.6(i) and 2.2.8(i)), the problem is local on Y; using 2.3.5 to replace S by a suitable S', this allows us to suppose that $N = 1$. Using semipurity once more (2.2.6(ii) and 2.2.8(ii)), it suffices to prove 2.3.6 over the generic points of S. An étale localization then reduces us to assuming that S is the spectrum of an algebraically closed field k. Since the classes of 2.3.2 and 2.1.2 are each additive in Y, this reduces us to the case where Y is a closed point on a smooth curve over k. The fact that the characteristic property 2.3.2 is satisfied is then the fundamental compatibility 2.1.5.

Lemma 2.3.7. *The formation of $\mathrm{cl}(Y, \mathscr{K})$ is compatible with every base change S'/S.*

This follows from the same assertion for the trace morphisms.

Theorem 2.3.8. *(i)* $\mathrm{cl}(Y, \mathscr{K})$ *depends only on* $Y \subset X$ *and on the lengths of* $\mathscr{K}$ *at the generic points of* Y*. This class is additive in* $\mathscr{K}$*. For* $\mathscr{K} = \mathscr{O}_Y$ *and* Y *a local complete intersection, it induces the local class of 2.2.2.*

(ii) Let there be a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{u} & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

with X' *smooth over* S'*. If* $u^{-1}(Y)$ *still has fiberwise codimension* $\geqslant c$*, then*

$$\mathrm{cl}\left(u^{-1}(Y), \mathsf{L}u^*\mathscr{K}\right) = u^* \mathrm{cl}(Y, \mathscr{K}).$$

(iii) Let $Y' \subset X$ *have fiberwise codimension* $\geqslant c'$*, let* $\mathscr{K}'$ *be a weight on* Y'*, and suppose that* $Y \cap Y'$ *has fiberwise codimension* $\geqslant c + c'$*. Then*

$$\mathrm{cl}\left(Y \cap Y', \mathscr{K} \overset{\mathsf{L}}{\otimes} \mathscr{K}'\right) = \mathrm{cl}(Y, \mathscr{K}) \smile \mathrm{cl}(Y', \mathscr{K}').$$

(A) Given $Y \subset X/S$ as above, it follows from semipurity (2.2.6, 2.2.8) that to verify that two classes in $\mathrm{H}^{2c}_Y(X, \mathbb{Z}/n(c))$ coincide, it suffices to verify this locally at the generic points of Y, or even after every base change $s \to S$ (s a geometric generic point of S), at the generic points of Y.

(B) Proof of (ii) for $S = S'$ and u a closed immersion By localization (A), we may suppose that X' is the inverse image of the section 0 under a smooth morphism $v : X \to \mathbb{A}^{N'}_S$

$$\begin{array}{ccccc} X' & \xrightarrow{u} & X & & \\ \downarrow & & \downarrow{\scriptstyle v} & & \\ S & \xrightarrow{0} & \mathbb{A}^{N'}_S & \xrightarrow{\pi} & S \end{array}$$

Apply 2.3.5 to $g = \pi v$ and 2.3.7 to the base change $S \to \mathbb{A}^{N'}_S$.

(C) In a product situation: $X = X' \times_S X''$, $Y = Y' \times_S Y''$, $\mathscr{K} = \mathscr{K}' \overset{\mathsf{L}}{\otimes}_S \mathscr{K}''$, one has $\mathrm{cl}(Y, \mathscr{K}) = \mathrm{cl}(Y', \mathscr{K}') \smile \mathrm{cl}(Y'', \mathscr{K}'')$ (external cup product, i.e. $\mathrm{pr}_1^* \mathrm{cl}(Y', \mathscr{K}') \smile \mathrm{pr}_2^* \mathrm{cl}(Y'', \mathscr{K}'')$).

The functorial properties of tr_g [2, XVIII.2.12] imply the commutativity of

$$\begin{array}{ccccc} \mathrm{H}^{2c'}_{X'} \otimes \mathrm{H}^{2c''}_{X''} & \longrightarrow & \hom_S\left(\mathsf{R}f'_!\mathbb{Z}/n(d')[2d'], \mathsf{R}g'_!\mathbb{Z}/n(N')[2N']\right) \otimes \cdots & \longrightarrow & \hom_S\left(\mathsf{R}^{2d'} f'_!\mathbb{Z}/n, \mathbb{Z}/n\right) \otimes \cdots \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{H}^{2c}_X & \longrightarrow & \hom_S\left(\mathsf{R}f_!\mathbb{Z}/n(d)[2d], \mathsf{R}g_!\mathbb{Z}/n(N)[2N]\right) & \longrightarrow & \hom_S\left(\mathsf{R}^{2d} f_!\mathbb{Z}/n, \mathbb{Z}/n\right) \end{array}$$

and one verifies that tr_f is the image of $\mathrm{tr}_{f'} \otimes \mathrm{tr}_{f''}$, whence the assertion.

Proof of (ii) (2.3.7) and a preliminary base change reduce us to assuming that $S = S'$. Factor u through its graph:

$$X' \xrightarrow{(\mathrm{id}, u)} X' \times X \xrightarrow{\mathrm{pr}_2} X.$$

This leads us to treat pr_2 and (id, u) separately: the former is covered by (C) with $Y' = X'$, $\mathscr{K}' = \mathscr{O}_{X'}$ (with $\mathrm{cl}(Y') = 1 \in \mathrm{H}^0(X', \mathbb{Z}/n)$), and the latter by (B).

Proof of (iii) Let $\delta : X \to X \times_S X$ be the diagonal inclusion. We use the formula $\delta^*(Y_1 \times_S Y_2) = Y_1 \cap Y_2$ to reduce to (B) and (C).

Proof of (i) The base change $S_{\mathrm{red}} \to S$ replaces X by X_{red}; this reduces us to assuming X and S reduced. Localization near the generic points of Y then reduces us to the case where Y_{red} is irreducible and the complete intersection of c divisors D_i. The same localization, together with the definition of tr in the quasi-finite case, then shows that $\mathrm{cl}(Y, \mathscr{K}) = \lg \mathrm{cl}(Y_{\mathrm{red}}, \mathscr{O}_{Y_{\mathrm{red}}})$, where lg is the length of $\mathscr{K}$ at the generic point of Y. By (iii), one therefore has

$$\mathrm{cl}(Y, \mathscr{K}) = \lg \prod_i \mathrm{cl}(D_i, \mathscr{O}_{D_i})$$

and the conclusion follows from 2.3.6.

2.3.9 Remark

For S the spectrum of a field, with the class of a cycle defined as in 2.2.10, assertion (iii) says that if two cycles Y' and Y'' intersect with the correct dimension, then

$$\mathrm{cl}(Y' \cap Y'') = \mathrm{cl}(Y') \smile \mathrm{cl}(Y''),$$

provided that the multiplicities of the components of $Y' \cap Y''$ are computed as alternating sums of Tor.

2.3.10 Remark

Let $X/\operatorname{Spec}(k)$, with k algebraically closed. If two cycles of codimension c in X are algebraically equivalent, the existence of the relative theory above implies that the images in $\mathrm{H}^{2c}(X, \mathbb{Z}/n(c))$ of their classes are the fibers, at two points of a suitable connected S, of a section over S of the constant sheaf $\mathrm{H}^{2c}(X, \mathbb{Z}/n(c))$ (the sheaf $\mathsf{R}^{2c} f_* \mathbb{Z}/n(c)$ for $f = \mathrm{pr}_2 : X \times S \to S$): two algebraically equivalent cycles have the same class in $\mathrm{H}^{2c}(X, \mathbb{Z}/n(c))$.

3 Application: the Lefschetz trace formula in the proper smooth case

3.1

Let X and Y be proper smooth algebraic varieties over an algebraically closed field k, purely of dimensions N and M. Via the Künneth isomorphism $\mathrm{H}^\bullet(X \times Y) = \mathrm{H}^\bullet(X) \otimes \mathrm{H}^\bullet(Y)$, the trace morphism $\mathrm{H}^\bullet(X \times Y)(M) \to \mathrm{H}^\bullet(X)$ is simply $\mathrm{H}^\bullet(X) \otimes$ (the trace morphism $\mathrm{H}^\bullet(Y)(M) \to \mathbb{Q}_\ell$). Since the morphism is homogeneous of even degree, there is no sign problem. We denote it by $\int_Y$.

3.2

A class $\eta \in \mathrm{H}^{2q}(X \times Y)(q)$ defines a morphism of degree $2(q - M)$, $\eta_{XY} : \mathrm{H}^\bullet(Y)(M - q) \to \mathrm{H}^\bullet(X)$, by the formula

$$\beta \mapsto \int_Y \eta \cdot \mathrm{pr}_2^* \beta.$$

Proposition 3.3. *Let p and q be two integers, with $p+q = N+M$, and let $\varepsilon \in \mathrm{H}^{2p}(X\times Y)(p)$, $\eta \in \mathrm{H}^{2q}(X\times Y)(q)$. With the trace of $\eta_{XY}\varepsilon_{YX} : \mathrm{H}^\bullet(X) \to \mathrm{H}^\bullet(X)$ understood in the sense of 1.3, one has*

$$\mathrm{tr}_{X\times Y}(\eta\cdot\varepsilon) = \mathrm{tr}\left(\eta_{XY}\varepsilon_{YX}, \mathrm{H}^\bullet(X)\right).$$

To exorcise the signs, it is helpful to retain from the grading of $\mathrm{H}^\bullet$ only the underlying $\mathbb{Z}/2$-grading. To avoid carrying Tate twists along, also fix an isomorphism $\mathbb{Q}_\ell(1) = \mathbb{Q}_\ell$. Let α_Y be the homomorphism $\mathrm{H}^\bullet(Y) \to \mathrm{H}^\bullet(Y)^\vee$ (an isomorphism) that makes the diagram

$$\begin{array}{ccccc} \mathrm{H}^\bullet(Y)\otimes\mathrm{H}^\bullet(Y) & \longrightarrow & \mathrm{H}^\bullet(Y) & \xrightarrow{\mathrm{tr}} & \mathbb{Q}_\ell \\ \downarrow{\scriptstyle \alpha_Y\otimes 1} & & & & \| \\ \mathrm{H}^\bullet(Y)^\vee\otimes\mathrm{H}^\bullet(Y) & & \xrightarrow{(1.3.1)} & & \mathbb{Q}_\ell. \end{array}$$

commute. The map η_{XY} is the image of η under

$$\mathrm{H}^\bullet(X\times Y) = \mathrm{H}^\bullet(X)\otimes\mathrm{H}^\bullet(Y) \xrightarrow{\alpha_Y} \mathrm{H}^\bullet(X)\otimes\mathrm{H}^\bullet(Y)^\vee \overset{(1.3.2)}{=\!=} \hom\left(\mathrm{H}^\bullet(Y), \mathrm{H}^\bullet(X)\right),$$

and likewise for ε. Definition 1.3.4 of the trace then reduces 3.3 to the formula $\mathrm{tr}_{X\times Y} = \mathrm{tr}_X \otimes \mathrm{tr}_Y$.

3.4 Remark

The proof of 3.3 does not use the fact that α_X and α_Y are isomorphisms. It is valid without assuming X and Y smooth. Outside the smooth case, however, it is difficult to find *cohomology* classes to which one would want to apply 3.3.

3.5 Remark

If ε, η are the cohomology classes of algebraic cycles A and B on $X\times Y$, of codimensions p and q, and $A\cap B$ has dimension 0, then $\mathrm{tr}_{X\times Y}(\eta\varepsilon)$ is the intersection multiplicity $A\cdot B$, computed by an alternating sum of Tor (2.3.9).

Proposition 3.6. *For $q = M$ and $\eta = \mathrm{cl}(B)$ the class of a subscheme $B \subset X\times Y$, finite over X, the map η_{XY} is the composite*

$$\eta_{XY} : \mathrm{H}^\bullet(Y) \xrightarrow{\mathrm{pr}_2^*} \mathrm{H}^\bullet(B) \xrightarrow{\mathrm{tr}_{B/X}} \mathrm{H}^\bullet(X).$$

Indeed, by 3.2 and (2.3.3.1) for $\mathscr{K} = \mathscr{O}_B$, one has

$$\eta_{XY}(\beta) = \mathrm{tr}_{X\times Y/X}\left(\mathrm{cl}(B)\cdot \mathrm{pr}_2^*\,\beta\right) = \mathrm{tr}_{B/X}\left(\mathrm{pr}_2^*\,\beta\right).$$

The most important case is that in which B is the graph of a map $f : X \to Y$. In this case, 3.6 shows that $\eta_{XY} = f^*$.

Corollary 3.7. *Let f be an endomorphism of X proper and smooth over an algebraically closed field k. Suppose that the fixed points of f are isolated. Then the trace $\sum(-1)^i \mathrm{tr}(f^*, \mathrm{H}^i(X))$ is the number of fixed points of f, each counted with its multiplicity.*

In 3.5, take $A =$ the graph of the identity and $B =$ the graph of f, and apply 3.3, 3.6.

Corollary 3.8. *Let X be a curve over an algebraically closed k, obtained by extension of scalars from X_0 over $\mathbb{F}_q$, and let f be the Frobenius morphism. Then $\sum(-1)^i \operatorname{tr}\left(f^*, \mathrm{H}^i_c(X)\right)$ is the number of fixed points of f.*

One can replace X by X_{red}, and hence suppose that X is reduced. Let U be the open subset of X on which X is smooth of dimension 1, and let $\bar{U}$ be the smooth projective completion of U. The long exact cohomology sequences for (X, U) and $(\bar{U}, U)$ give

$$\operatorname{tr}\left(f^*, \mathrm{H}^\bullet_c(X)\right) = \operatorname{tr}\left(f^*, \mathrm{H}^\bullet_c(U)\right) + \operatorname{tr}\left(f^*, \mathrm{H}^\bullet_c(X \smallsetminus U)\right)$$

and

$$\operatorname{tr}\left(f^*, \mathrm{H}^\bullet_c(\bar{U})\right) = \operatorname{tr}\left(f^*, \mathrm{H}^\bullet_c(U)\right) + \operatorname{tr}\left(f^*, \mathrm{H}^\bullet_c(\bar{U} \smallsetminus U)\right).$$

The same formulas hold for the numbers of fixed points. Since formula 3.8 is clear for a scheme of dimension 0, this reduces us to proving 3.8 for $\bar{U}$. This case follows from 3.7. The fact that $df = 0$ guarantees that all fixed points have multiplicity one.

3.9 Remark

For X of dimension $\leqslant 1$ over an algebraically closed k, and $f : X \to X$ such that $df = 0$, it is not difficult to verify that 3.8 is still valid.

Chapter 5

Duality

This exposé contains several theorems and compatibility results, all related to Poincaré duality. In Section 1, we give the local biduality theorem in dimension 1 [21, I.5.1], together with some computations of duals. In Section 2, we give a very economical proof of Poincaré duality for curves, taught to me by M. Artin. In Section 3, we prove a compatibility linking two definitions of the pairing giving rise to Poincaré duality for curves: one by cup product, the other by autoduality of the Jacobian. Finally, in Section 4, we prove the compatibility announced in the title.

Throughout the exposé, all schemes are noetherian and separated, and n is an integer invertible on every scheme under consideration.

1 Local biduality in dimension 1

1.1

Let S be a regular scheme, purely of dimension 1. We shall show that the complex with value $\mathbb{Z}/n$ concentrated in degree 0 is dualizing; that is, for $\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}_c^b(S, \mathbb{Z}/n)$ (where $(-)_c$ means constructible), if we put $D\mathscr{K} = \mathsf{R}\mathscr{H}\mathrm{om}(\mathscr{K}, \mathbb{Z}/n)$, then $D\mathscr{K}$ is again constructible with bounded cohomology, and the canonical morphism α from $\mathscr{K}$ to $DD\mathscr{K}$ is an isomorphism.

For $\mathscr{K}$ in D^- and arbitrary $\mathscr{L}$, we have

$$\hom(\mathscr{K} \overset{\mathsf{L}}{\otimes} \mathscr{L}, \mathbb{Z}/n) = \hom(\mathscr{L}, \mathscr{H}\mathrm{om}(\mathscr{K}, \mathbb{Z}/n)).$$

The morphism from $\mathscr{K}$ to $DD\mathscr{K}$ is defined when $D\mathscr{K}$ lies in D^-: if $\beta : D\mathscr{K} \overset{\mathsf{L}}{\otimes} \mathscr{K} \to \mathbb{Z}/n$ is the canonical pairing, it is the morphism defined by the pairing $\mathscr{K} \overset{\mathsf{L}}{\otimes} D\mathscr{K} = D\mathscr{K} \overset{\mathsf{L}}{\otimes} \mathscr{K} \xrightarrow{\beta} \mathbb{Z}/n$.

1.2

Let $f : X \to S$ be a separated morphism of finite type. Put $\mathscr{K}_X = \mathsf{R}f^!\mathbb{Z}/n$. For $\mathscr{K} \in \mathrm{Ob}\,\mathsf{D}^-(X, \mathbb{Z}/n)$, put $D\mathscr{K} = \mathsf{R}\mathscr{H}\mathrm{om}(\mathscr{K}, \mathscr{K}_X)$. The adjunction between $\mathsf{R}f_!$ and $\mathsf{R}f^!$ gives

$$\mathsf{R}f_*\mathscr{K} \overset{\mathsf{L}}{\otimes} \mathsf{R}f_*D\mathscr{K} \longrightarrow \mathsf{R}f_!(\mathscr{K} \overset{\mathsf{L}}{\otimes} D\mathscr{K}) \longrightarrow \mathsf{R}f_!\mathsf{R}f^!\mathbb{Z}/n \longrightarrow \mathbb{Z}/n.$$

The symmetry of this description shows that, when f is proper, the diagram

$$\begin{array}{ccc} \mathsf{R}f_*\mathscr{K} & \longrightarrow & DD\mathsf{R}f_*\mathscr{K} \\ \downarrow & & \downarrow\wr \\ \mathsf{R}f_*DD\mathscr{K} & \xrightarrow{\sim} & D\mathsf{R}f_*D\mathscr{K} \end{array} \tag{1.2.1}$$

is commutative.

If f is the inclusion of a closed point, we know that $\mathsf{R}f^!\mathbb{Z}/n = \mathbb{Z}/n(-1)[-2]$—that is, up to a twist and a shift, $\mathbb{Z}/n$. Over the spectrum of a field this complex is dualizing (Pontryagin duality for $\mathbb{Z}/n$-modules). By (1.2.1), we therefore have $\mathscr{K} \xrightarrow{\sim} DD\mathscr{K}$ for $\mathscr{K}$ of the form $\mathsf{R}f_!\mathscr{L}$, and hence whenever the support of $\underline{\mathrm{H}}^\bullet(\mathscr{K})$ is finite.

Theorem 1.3. *Let $j : U \hookrightarrow S$ be a dense open subscheme of S, and let $\mathscr{F}$ be a locally constant constructible sheaf of $\mathbb{Z}/n$-modules on U. Then $\mathsf{D}j_*\mathscr{F} = j_*D\mathscr{F}$; that is, $\mathscr{H}\mathrm{om}(j_*\mathscr{F}, \mathbb{Z}/n) = j_*\mathscr{H}\mathrm{om}(\mathscr{F}, \mathbb{Z}/n)$ and $\mathscr{E}\mathrm{xt}^i(j_*\mathscr{F}, \mathbb{Z}/n) = 0$ for $i > 0$.*

On U, $\mathscr{E}\mathrm{xt}^i(j_*\mathscr{F}, \mathbb{Z}/n) = 0$ for $i > 0$, since $\mathscr{F}$ is locally constant (and $\mathbb{Z}/n$ is an injective $\mathbb{Z}/n$-module), while for $i = 0$ it is the dual $\mathscr{F}^\vee$ of $\mathscr{F}$. One checks that $\hom(j_*\mathscr{F}, \mathbb{Z}/n) = j_*\mathscr{F}^\vee$, and it remains to verify the vanishing of the $\mathscr{E}\mathrm{xt}^i$ for $i > 0$ at the points of $S \smallsetminus U$. The problem is local at these points. We are thus reduced to assuming that S is a strictly local trait and that U consists of its generic point η. Let $I = \mathrm{Gal}(\bar\eta/\eta)$. The sheaf $\mathscr{F}$ is identified with the Galois module $\mathscr{F}_{\bar\eta}$, and the special fiber of $j_*\mathscr{F}$ with $\mathscr{F}_{\bar\eta}^I$.

Let i be the inclusion of the closed point s, and apply D to the exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & j_!\mathscr{F} & \longrightarrow & j_*\mathscr{F} & \longrightarrow & i_*\mathscr{F}_{\bar\eta}^I & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & j_!\mathscr{F}_{\bar\eta}^I & \longrightarrow & \mathscr{F}_{\bar\eta}^I & \longrightarrow & i_*\mathscr{F}_{\bar\eta}^I & \longrightarrow & 0. \end{array}$$

We obtain a morphism of triangles

$$\begin{array}{ccccc} i_*\left(\mathscr{F}_{\bar\eta}^I\right)^\vee(-1)[-2] & \longrightarrow & Dj_*\mathscr{F} & \longrightarrow & \mathsf{R}j_*\mathscr{F}^\vee \\ \| & & \downarrow & & \downarrow \\ i_*\left(\mathscr{F}_{\bar\eta}^I\right)^\vee(-1)[-2] & \longrightarrow & \left(\mathscr{F}_{\bar\eta}^I\right)^\vee & \longrightarrow & \mathsf{R}j_*\mathscr{F}_{\bar\eta}^I. \end{array}$$

The long exact sequence arising from the first row gives the vanishing of the $\mathscr{E}\mathrm{xt}^i$ for $i > 2$, and, on taking the fiber at s, we find

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \left(\underline{\mathrm{H}}^1(Dj_*\mathscr{F})\right)_s & \longrightarrow & \mathrm{H}^1\left(I, \mathscr{F}_{\bar\eta}^\vee\right) & \xrightarrow{\partial} & \left(\mathscr{F}_{\bar\eta}^I\right)^\vee(-1) & \longrightarrow & \left(\underline{\mathrm{H}}^2 Dj_*\mathscr{F}\right)_s \longrightarrow 0 \\ & & & & \downarrow & & \| & & \\ & & 0 & \longrightarrow & \mathrm{H}^1\left(I, (\mathscr{F}_{\bar\eta}^I)^\vee\right) & \xrightarrow{\partial} & \left(\mathscr{F}_{\bar\eta}^I\right)^\vee(-1) & \longrightarrow & 0. \end{array}$$

Since $\left(\mathscr{F}_{\bar\eta}^I\right)^\vee = \left(\mathscr{F}_{\bar\eta}^\vee\right)_I$, putting $M = \mathscr{F}_{\bar\eta}^\vee$, it remains only to check that

$$\mathrm{H}^1(I, M) \xrightarrow{\sim} \mathrm{H}^1(I, M_I).$$

If p is the residual characteristic exponent, then I is an extension of a group isomorphic to $\widehat{\mathbb{Z}}_{p'} = \varprojlim_{(m,p)=1} \mathbb{Z}/m$ by a p-group P. Since the order of P is prime to the order of M, we have $\mathrm{H}^i(P, M) = 0$ for $i > 0$ and $(M_I)^P = (M^P)_I$. This allows us to replace M by M^P and I by I/P. Finally, there is a functorial isomorphism $\mathrm{H}^1(\widehat{\mathbb{Z}}_{p'}, M) \sim$ (coinvariants of $\widehat{\mathbb{Z}}_{p'}$ in M), which proves the theorem.

Theorem 1.4. *For $\mathscr{K} \in \mathrm{Ob}\, \mathsf{D}_c^b(S, \mathbb{Z}/n)$, one has $\mathscr{K} \xrightarrow{\sim} DD\mathscr{K}$.*

By dévissage, we reduce to the case in which $\mathscr{K}$ is a constructible sheaf $\mathscr{F}$ concentrated in degree 0, and $\mathscr{F}$ either has finite support (1.2) or is of the form $j_*\mathscr{F}_1$ as in 1.3. In the second case, 1.3 reduces us to local biduality for $\mathscr{F}$ locally constant on U.

2 Poincaré duality for curves, after M. Artin

2.1

Let X be a smooth projective curve over an algebraically closed field k. Put $\mathscr{K}_X = \mathbb{Z}/n(1)[2]$ and, for $\mathscr{K} \in \mathrm{Ob}\, \mathsf{D}_c^b(X, \mathbb{Z}/n)$, $D\mathscr{K} = \mathsf{R}\mathscr{H}\mathrm{om}(\mathscr{K}, \mathscr{K}_X)$. For M a $\mathbb{Z}/n$-module, also put $DM = \hom(M, \mathbb{Z}/n)$, and similarly for complexes of modules. The trace morphism $\mathrm{H}^0(X, \mathscr{K}_X) \to \mathbb{Z}/n$, or $\mathsf{R}\Gamma(X, \mathscr{K}_X) \to \mathbb{Z}/n$, defines a pairing $\mathsf{R}\Gamma(X, \mathscr{K}) \overset{\mathsf{L}}{\otimes} \mathsf{R}\Gamma(X, D\mathscr{K}) \to \mathbb{Z}/n$. Poincaré duality between the cohomology and compactly supported cohomology of an open subscheme $j : U \hookrightarrow X$ of the curve X says that, for $\mathscr{K} = j_!\mathbb{Z}/n$, this pairing identifies each factor with the dual of the other.

I give below a proof, communicated to me by M. Artin, that for $\mathscr{K} \in \mathrm{Ob}\, \mathsf{D}_c^b(X, \mathbb{Z}/n)$ this pairing is always perfect; that is, it defines an isomorphism

$$\mathsf{R}\Gamma(X, \mathscr{K}) \xrightarrow{\sim} D\mathsf{R}\Gamma(X, D\mathscr{K}). \tag{2.1.1}$$

For every constructible sheaf $\mathscr{F}$, put

$$'\mathrm{H}^i(X, \mathscr{F}) = \text{the } \mathbb{Z}/n\text{-dual of } \mathrm{H}^{-i}(X, D\mathscr{F}).$$

The assertion that (2.1.1) is an isomorphism is equivalent to the following theorem.

Theorem 2.2. *For a constructible sheaf $\mathscr{F}$ of $\mathbb{Z}/n$-modules, one has*

$$\mathrm{H}^i(X, \mathscr{F}) \xrightarrow{\sim} {}'\mathrm{H}^i(X, \mathscr{F}). \tag{2.2.1}$$

Lemma 2.3. *Let $f : X \to Y$ be a generically étale morphism between smooth curves over k. Then $\mathscr{K}_X = f^*\mathscr{K}_Y = \mathsf{R}f^!\mathscr{K}_Y$, with tr_f as the adjunction morphism $\mathsf{R}f_!\mathsf{R}f^!\mathscr{K}_Y \to \mathscr{K}_Y$.*

We reduce to checking that, for f finite, $f_*\mathsf{R}f^!\mathscr{K}_Y = \mathsf{R}\mathscr{H}\mathrm{om}(f_*\mathbb{Z}/n, \mathscr{K}_Y)$ is $f_*\mathbb{Z}/n$, with tr_f as the adjunction morphism. The vanishing of the $\mathscr{E}\mathrm{xt}^i$ for $i > 0$ follows from 1.3, and the assertion follows.

More explicitly, $\mathsf{R}f^!$ is the derived functor of the functor $f^! : f^!\mathscr{F}(U) = \hom(f_!\mathbb{Z}_U, \mathscr{F})$, and 1.3 implies the vanishing of $\mathsf{R}^i f^!\mathbb{Z}/n$ for $i > 0$.

Corollary 2.4. *Let $f : X' \to X$ be a generically étale morphism of smooth projective curves over k. Then $'\mathrm{H}^i(X, f_*\mathscr{F}) =' \mathrm{H}^i(X', \mathscr{F})$; this isomorphism is functorial, and the diagram*

$$\begin{array}{ccc} \mathrm{H}^i(X, f_*\mathscr{F}) & \longrightarrow & {}'\mathrm{H}^i(X, f_*\mathscr{F}) \\ \| & & \| \\ \mathrm{H}^i(X', \mathscr{F}) & \longrightarrow & {}'\mathrm{H}^i(X', \mathscr{F}) \end{array}$$

is commutative.

The isomorphism is given by 1.2: $Df_*\mathscr{F} = f_*D\mathscr{F}$.

2.5

We prove 2.2. If $\mathscr{F}$ is $\mathbb{Z}/n$ at one point, extended by 0, then $\mathrm{H}^\bullet(D\mathscr{F}) = \mathrm{Ext}^\bullet(\mathscr{F}, \mathscr{K}_X) = \mathbb{Z}/n$ in degree 0, and in the pairing $\mathrm{H}^\bullet(\mathscr{F}) \otimes \mathrm{H}^\bullet(D\mathscr{F}) \to \mathbb{Z}/n$, one has $1 \otimes 1 \mapsto 1$ (Cycle, 2.1.5): the duality is perfect. The case in which $\mathscr{F}$ has finite support is treated in the same way.

For arbitrary constructible $\mathscr{F}$, let $\mathscr{A}$ be the subsheaf of its sections with finite support, and let $\mathscr{G} = \mathscr{F}/\mathscr{A}$. The exact sequence

$$0 \longrightarrow \mathscr{A} \longrightarrow \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow 0 \tag{2.5.1}$$

gives a diagram

$$\begin{array}{ccccccccc} & & & & 0 & \longrightarrow & \mathrm{H}^0(X,\mathscr{A}) & \longrightarrow & \mathrm{H}^0(X,\mathscr{F}) \longrightarrow \mathrm{H}^0(X,\mathscr{G}) \longrightarrow 0 \\ & & & & & & \downarrow\wr & & \downarrow \qquad \downarrow \\ 0 \to {}' \mathrm{H}^{-1}(X,\mathscr{F}) & \to & {}' \mathrm{H}^{-1}(X,\mathscr{G}) & \to & {}' \mathrm{H}^0(X,\mathscr{A}) & \to & {}' \mathrm{H}^0(X,\mathscr{F}) & \to & {}' \mathrm{H}^0(X,\mathscr{G}) \to 0 \end{array} \tag{2.5.2}$$

and compatible isomorphisms $\mathrm{H}^i(X,\mathscr{F}) \xrightarrow{\sim} \mathrm{H}^i(X,\mathscr{G})$, ${}' \mathrm{H}^i(X,\mathscr{F}) \xrightarrow{\sim}{}' \mathrm{H}^i(X,\mathscr{G})$ for $i \neq 0, -1$. If $j : U \to X$ is an open subscheme on which $\mathscr{F}$ is locally constant, there is an exact sequence

$$0 \longrightarrow \mathscr{G} \longrightarrow j_*j^*\mathscr{F} \longrightarrow \mathscr{B} \longrightarrow 0 \tag{2.5.3}$$

with $\mathscr{B}$ of finite support. Since ${}' \mathrm{H}^i(X, j_*j^*\mathscr{F}) = D\,\mathrm{H}^{2-i}(X, j_*(j^*\mathscr{F})^\vee(1))$ by 1.3, one has ${}' \mathrm{H}^i = 0$ for $i < 0$; then (2.5.2) and the long exact sequence defined by (2.5.3) show that ${}' \mathrm{H}^i$ always vanishes for $i < 0$.

If $\mathscr{F} = \mathbb{Z}/n$, the theorem is true for $i = 0$ and 2. This expresses the fact that, for connected X, the trace morphism $\mathrm{H}^2(X, \boldsymbol{\mu}_n) \xrightarrow{\sim} \mathbb{Z}/n$ is an isomorphism. By 2.4, the theorem remains true for $i = 0$ and $\mathscr{F} = f_*\mathbb{Z}/n$.

For arbitrary $\mathscr{F}$ again, with $\mathscr{G}$ as above, there exists $f : X' \to X$ as in 2.4 such that $\mathscr{G}$ embeds into $f_*\mathbb{Z}/n$:

$$0 \longrightarrow \mathscr{G} \longrightarrow f_*\mathbb{Z}/n \longrightarrow \mathscr{Q} \longrightarrow 0.$$

After replacing X' by an X''/X', we may assume that $\mathrm{H}^i(X,\mathscr{F}) \to \mathrm{H}^i(X, f_*\mathbb{Z}/n) = \mathrm{H}^i(X', \mathbb{Z}/n)$ is zero for $i > 0$. Consider then

$$\begin{array}{ccccccccccc} 0 \to \mathrm{H}^0(X,\mathscr{G}) & \to & \mathrm{H}^0(X',\mathbb{Z}/n) & \to & \mathrm{H}^0(X,\mathscr{Q}) & \to & \mathrm{H}^1(X,\mathscr{G}) & \xrightarrow{0} & \mathrm{H}^1(X',\mathbb{Z}/n) & \to & \mathrm{H}^1(X,\mathscr{Q}) \to \cdots \\ \downarrow & & \downarrow\wr & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 \to {}' \mathrm{H}^0(X,\mathscr{G}) & \to & {}' \mathrm{H}^0(X',\mathbb{Z}/n) & \to & {}' \mathrm{H}^0(X,\mathscr{Q}) & \to & {}' \mathrm{H}^1(X,\mathscr{G}) & \to & {}' \mathrm{H}^1(X',\mathbb{Z}/n) & \to & {}' \mathrm{H}^1(X,\mathscr{Q}) \to \cdots \end{array}$$

The isomorphism at $\mathrm{H}^0(X')$ gives $\mathrm{H}^0(X,\mathscr{G}) \hookrightarrow {}' \mathrm{H}^0(X,\mathscr{G})$, and the same injectivity for $\mathscr{F}$ by (2.5.2). Applied to $\mathscr{Q}$, this gives $\mathrm{H}^1(X,\mathscr{G}) \hookrightarrow {}' \mathrm{H}^1(X,\mathscr{G})$. We apply this to $\mathrm{H}^1(X', \mathbb{Z}/n)$. This group has the same order as ${}' \mathrm{H}^1(X', \mathbb{Z}/n) = D(\mathrm{H}^1(X', \mathbb{Z}/n)(1))$, so we obtain an isomorphism, and $\mathrm{H}^1(X,\mathscr{G}) \xrightarrow{\sim}{}' \mathrm{H}^1(X,\mathscr{G})$. The same holds for $\mathscr{F}$. For $i = 2$, we already know that $\mathrm{H}^2(X', \mathbb{Z}/n) \xrightarrow{\sim}{}' \mathrm{H}^2(X', \mathbb{Z}/n)$, and proceed in the same way, then stop for lack of combatants.

3 Poincaré duality for curves

In this section we prove the compatibility (Arcata 2.3.3).

3.1

We use the notation of (Arcata 6.2.3): $\bar{X}$ is a smooth, connected, projective curve over an algebraically closed field k, D is a reduced divisor on $\bar{X}$, and 0 is a point of $X = \bar{X} \smallsetminus D$,

$$X \overset{j}{\hookrightarrow} \bar{X} \overset{i}{\hookleftarrow} D,$$

and $\mathrm{Pic}_D(\bar{X}) = \mathrm{H}^1(\bar{X}, {}_D\mathbb{G}_m)$, where ${}_D\mathbb{G}_m$ is the sheaf of sections of $\mathbb{G}_m$ congruent to 1 mod D. Recall that $\mathrm{H}^1_c(X, \boldsymbol{\mu}_n) = \mathrm{Pic}^0_D(\bar{X})_n$, and that $x \to \mathscr{O}(x)$ defines a canonical map $f : X \to \mathrm{Pic}_D(\bar{X})$. Put $f_0(x) = f(x) - f(0) : X \to \mathrm{Pic}^0_D(\bar{X})$.

Again write j for the inclusion of $X \times X$ in $X \times \bar{X}$. The diagonal Δ of $X \times X$ is closed in $X \times \bar{X}$; it defines a class in $\mathrm{H}^2_\Delta(X \times X, \boldsymbol{\mu}_n) = \mathrm{H}^2_\Delta(X \times \bar{X}, j_!\boldsymbol{\mu}_n)$. Let c denote its image in $\mathrm{H}^2(X \times \bar{X}, j_!\boldsymbol{\mu}_n)$.

The Künneth formula gives

$$\mathrm{H}^\bullet(X \times \bar{X}, j_!\boldsymbol{\mu}_n) = \mathrm{H}^\bullet(X, \mathbb{Z}/n) \otimes \mathrm{H}^\bullet(\bar{X}, j_!\boldsymbol{\mu}_n) = \mathrm{H}^\bullet(X, \mathbb{Z}/n) \otimes \mathrm{H}^\bullet_c(X, \boldsymbol{\mu}_n).$$

We shall compute the $(1,1)$-component of c,

$$c^{1,1} \in \mathrm{H}^1(X, \mathbb{Z}/n) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) = \mathrm{H}^1(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)) = \mathrm{H}^1\left(X, \mathrm{Pic}^0_D(\bar{X})_n\right).$$

Consider the exact sequence

$$0 \longrightarrow \mathrm{Pic}^0_D(\bar{X})_n \longrightarrow \mathrm{Pic}^0_D(\bar{X}) \overset{n}{\longrightarrow} \mathrm{Pic}^0_D(\bar{X}) \longrightarrow 0$$

This makes $\mathrm{Pic}^0_D(\bar{X})$ a torsor under $\mathrm{Pic}^0_D(\bar{X})_n$ over $\mathrm{Pic}^0_D(\bar{X})$. Let u be the class, in $\mathrm{H}^1(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)) = \mathrm{H}^1(X, \mathrm{Pic}^0_D(\bar{X})_n)$, of its inverse image under f_0.

Proposition 3.2. *With the preceding notation, $c^{1,1} = -u$.*

Let e be the difference between the images in $\mathrm{H}^2(X \times \bar{X}, j_!\boldsymbol{\mu}_n)$ of the cohomology classes of Δ and $X \times \{0\}$. Since the latter is of Künneth type $(0,2)$, one has $c^{1,1} = e^{1,1}$. Consider the Leray spectral sequence for $\mathrm{pr}_1 : X \times \bar{X} \to X$ and the sheaf $j_!\boldsymbol{\mu}_n$:

$$E_2^{pq} = \mathrm{H}^p(X, \mathrm{R}^q\mathrm{pr}_{1*} j_!\boldsymbol{\mu}_n) = \mathrm{H}^p(X, \mathrm{H}^q_c(X, \boldsymbol{\mu}_n)) \Rightarrow \mathrm{H}^{p+q}(X \times \bar{X}, j_!\boldsymbol{\mu}_n).$$

It degenerates: it is the tensor product of $\mathrm{H}^\bullet(X)$ with the (trivial) Leray spectral sequence for $\bar{X} \to \mathrm{Spec}(k)$ and the sheaf $j_!\boldsymbol{\mu}_n$. Fiber by fiber, the divisor $\Delta - X \times \{0\}$ has degree 0, so the image of e in E^{02} is zero. We may therefore speak of its image $\bar{e}$ in $E^{1,1}$, and we must show that $\bar{e} = -u$.

Let $\mathscr{G}$ on $X \times \bar{X}$ be the subsheaf of $\mathbb{G}_m$ consisting of local sections whose restriction to the closed subscheme $X \times D$ is 1. There is again an exact sequence

$$0 \longrightarrow j_!\boldsymbol{\mu}_n \longrightarrow \mathscr{G} \longrightarrow \mathscr{G} \longrightarrow 0 \tag{3.2.1}$$

and e is the image under ∂ of the class $e_1 \in \mathrm{H}^1(X \times \bar{X}, \mathscr{G})$ of the invertible sheaf $\mathscr{O}(\Delta - X \times \{0\})$, trivialized by 1 on $X \times D$.

Applying $\mathsf{R}\mathrm{pr}_{1*}$ to the distinguished triangle defined by (3.2.1) gives a distinguished triangle

$$\xrightarrow{\partial} \mathsf{R}\mathrm{pr}_{1*}j_!\boldsymbol{\mu}_n \longrightarrow \mathsf{R}\mathrm{pr}_{1*}\mathscr{G} \longrightarrow \mathsf{R}\mathrm{pr}_{1*}\mathscr{G} \xrightarrow{\partial} \tag{3.2.2}$$

and the diagram

$$\begin{array}{ccc} \mathrm{H}^1(X\times\bar{X},\mathscr{G}) & \xrightarrow{\partial} & \mathrm{H}^2(X\times\bar{X},j_!\boldsymbol{\mu}_n) \\ \| (1) & & \| \\ \mathrm{H}^1(X,\mathsf{R}\mathrm{pr}_{1*}\mathscr{G}) & \xrightarrow{\partial} & \mathrm{H}^2(X,\mathsf{R}\mathrm{pr}_{1*}j_!\boldsymbol{\mu}_n) \end{array}$$

is commutative. Our problem is thus to identify the image in $E^{11}=\mathrm{H}^1(X,\mathrm{H}^1_c(X,\boldsymbol{\mu}_n))$ (the image in E^{02} being zero) of the image under $\partial_{(3.2.2)}:\mathrm{H}^1(X,\mathsf{R}\mathrm{pr}_{1*}\mathscr{G})\to\mathrm{H}^2(X,\mathsf{R}\mathrm{pr}_{1*}j_!\boldsymbol{\mu}_n)$ of the class, still denoted by e_1, corresponding to e_1 under (1).

The sheaf $\mathsf{R}^1\mathrm{pr}_{1*}\mathscr{G}$ is defined by the group scheme over X obtained by pulling back the algebraic group $\mathrm{Pic}_D(\bar{X})$ over $\mathrm{Spec}(k)$, and the image of e_1 in $\mathrm{H}^0(X,\mathsf{R}^1\mathrm{pr}_{1*}\mathscr{G})=\hom(X,\mathrm{Pic}_D(\bar{X}))$ is f_0.

Represent the triangle (3.2.2) by a short exact sequence of complexes of sheaves:

$$0\longrightarrow\mathscr{A}\longrightarrow\mathscr{B}\longrightarrow\mathscr{C}\longrightarrow 0.$$

Let $\mathscr{B}'$ be the subsheaf of

$$\tau_{\leqslant 1}\mathscr{B}:\cdots\mathscr{B}^0\to\ker(d)\to 0\cdots$$

obtained by replacing the degree-one term $\ker(d)$ by the inverse image, under $\ker(d)\to\underline{\mathrm{H}}^1(\mathscr{B})=\mathrm{Pic}_D(\bar{X})$ over X, of $\mathrm{Pic}^0_D(\bar{X})$ over X. Let $\mathscr{A}'=\mathscr{A}\cap\mathscr{B}'$ and let $\mathscr{C}'$ be the image of $\mathscr{B}'$. We have $\mathscr{A}'=\tau_{\leqslant 1}(\mathscr{A})$, and $\mathscr{C}'$ is obtained from $\mathscr{C}$ in the same way as $\mathscr{B}'$ from $\mathscr{B}$. Hence there is a commutative diagram of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathscr{A} & \longrightarrow & \mathscr{B} & \longrightarrow & \mathscr{C} & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & \tau_{\leqslant 1}\mathscr{A} & \longrightarrow & \mathscr{B}' & \longrightarrow & \mathscr{C}' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{H}^1_c(X,\boldsymbol{\mu}_n)[-1] & \longrightarrow & \mathrm{Pic}^0_D(\bar{X})[-1] & \longrightarrow & \mathrm{Pic}^0_D(\bar{X})[-1] & \longrightarrow & 0 \end{array} \tag{3.2.3}$$

where the last row is to be viewed as an exact sequence of complexes of sheaves concentrated in degree 1 on X.

a) $e\in\mathsf{H}^2(X,\mathscr{A})$ comes from $\widetilde{e}\in\mathsf{H}^2(X,\tau_{\leqslant 1}\mathscr{A})$ (because its image in E^{02} is zero). We seek the image $\bar{e}$ of $\widetilde{e}$ in $\mathsf{H}^2(X,\mathrm{H}^1_c(X,\boldsymbol{\mu}_n)[-1])=\mathrm{H}^1(X,\mathrm{H}^1_c(X,\boldsymbol{\mu}_n))$.

b) $e_1\in\mathsf{H}^1(X,\mathscr{C})$ comes from $\widetilde{e}_1\in\mathsf{H}^1(X,\mathscr{C}')$, because its image f_0 in $\mathrm{H}^0(X,\mathrm{Pic}_D(\bar{X}))$ lies in $\mathrm{H}^0(X,\mathrm{Pic}^0_D(\bar{X}))$. We may take $\widetilde{e}=\partial\widetilde{e}_1$.

c) Thus $\bar{e}=\partial f_0$, where ∂ is defined by (3.2.2), and where we use the usual isomorphisms $\mathsf{H}^2(X,\mathscr{F}[-1])=\mathrm{H}^1(X,\mathscr{F})$. These isomorphisms anticommute with ∂, so that $\bar{e}=-\partial f_0$, for ∂ defined by the exact sequence

$$0\longrightarrow\mathrm{H}^1_c(X,\boldsymbol{\mu}_n)\longrightarrow\mathrm{Pic}^0_D(\bar{X})\longrightarrow\mathrm{Pic}^0_D(\bar{X})\longrightarrow 0$$

The conclusion follows from (Cycles, 1.1.1).

3.3

For every homomorphism $\varphi : \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \to \mathbb{Z}/n$, let $\varphi(u) \in \mathrm{H}^1(X, \mathbb{Z}/n)$ be the image of u under $\varphi : \mathrm{H}^1(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)) \to \mathrm{H}^1(X, \mathbb{Z}/n)$. If $x \in \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)$, the cup product $\varphi(u) \smile x$ lies in $\mathrm{H}^2_c(X, \boldsymbol{\mu}_n)$. We shall prove the following.

Proposition 3.4. $\mathrm{tr}(\varphi(u) \smile x) = \varphi(x)$.

This identity follows, after applying φ, from

$$\mathrm{tr}(u \smile x) = x \tag{3.4.1}$$

where tr now denotes the map

$$\mathrm{tr} : \mathrm{H}^2_c(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \otimes \boldsymbol{\mu}_n) \longrightarrow \mathrm{H}^1_c(X, \boldsymbol{\mu}_n),$$

so that $x \mapsto \mathrm{tr}(u \smile x)$ is the composite

$$\begin{aligned} &\mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \xrightarrow{u \otimes x} \mathrm{H}^1(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \xleftarrow{\sim} \mathrm{H}^1(X, \mathbb{Z}/n) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \\ &\xrightarrow{(1)} \mathrm{H}^2_c(X, \boldsymbol{\mu}_n) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \xrightarrow{\mathrm{tr} \otimes \mathrm{id}} \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) \end{aligned} \tag{3.4.2}$$

where (1) is given by $a \otimes b \otimes c \mapsto (a \smile c) \otimes b$.

We express this morphism in terms of the trace morphism $\mathrm{tr} : \mathrm{H}^3_c(X \times X, \boldsymbol{\mu}_n^{\otimes 2}) \to \mathrm{H}^2_c(X, \boldsymbol{\mu}_n)$ relative to the second projection. For $x \in \mathrm{H}^3_c$, $\mathrm{tr}(x)$ is computed as follows: among its Künneth components, retain only the component of type $(2,1)$, and apply $\mathrm{tr} : \mathrm{H}^2_c(X, \boldsymbol{\mu}_n) \to \mathbb{Z}/n$.

Let u' be the image of u under the map

$$\begin{aligned} &\mathrm{H}^1(X, \mathrm{H}^1_c(X, \boldsymbol{\mu}_n)) \xleftarrow{\sim} \mathrm{H}^1(X, \mathbb{Z}/n) \otimes \mathrm{H}^1_c(X, \boldsymbol{\mu}_n) = \mathrm{H}^1(X, \mathbb{Z}/n) \otimes \mathrm{H}^1(\bar{X}, j_!\boldsymbol{\mu}_n) \\ &\longrightarrow \mathrm{H}^2(X \times \bar{X}, \mathbb{Z}/n \boxtimes j_!\boldsymbol{\mu}_n). \end{aligned}$$

In view of the relation between the Künneth formula and cup product, one has

$$\mathrm{tr}(u \smile x) = -\,\mathrm{tr}(u' \smile \mathrm{pr}_1^* x).$$

The second cup product is the one in (Cycle 1.2.4):

$$\mathrm{H}^2(X \times \bar{X}, \mathbb{Z}/n \boxtimes j_!\boldsymbol{\mu}_n) \otimes \mathrm{H}^1(\bar{X} \times \bar{X}, j_!\boldsymbol{\mu}_n \boxtimes \mathbb{Z}/n) \longrightarrow \mathrm{H}^3(\bar{X} \times \bar{X}, j_!\boldsymbol{\mu}_n \boxtimes j_!\boldsymbol{\mu}_n).$$

The minus sign comes from permuting two symbols of degree 1 in (1). Since $c \smile \mathrm{pr}_1^* x$ and $c^{1,1} \smile \mathrm{pr}_1^* x$ have the same $(2,1)$-component in the Künneth decomposition, and since $u' = -c^{1,1}$, one has

$$\mathrm{tr}(u \smile x) = \mathrm{tr}(c \smile \mathrm{pr}_1^* x).$$

The cup product on the right can now be interpreted as the cup product (Cycle 1.2.5) of $\mathrm{cl}(\Delta) \in \mathrm{H}^2_\Delta(X \times X, \boldsymbol{\mu}_n)$ with $\mathrm{pr}_1^* x \in \mathrm{H}^1_c(\Delta, \boldsymbol{\mu}_n)$. On Δ, one has $\mathrm{pr}_1 = \mathrm{pr}_2$, whence

$$\mathrm{tr}(c \smile \mathrm{pr}_1^* x) = \mathrm{tr}(\mathrm{cl}(\Delta) \smile \mathrm{pr}_1^* x) = \mathrm{tr}(\mathrm{cl}(\Delta) \smile \mathrm{pr}_2^* x) = \mathrm{tr}(\mathrm{cl}(\Delta)) \smile x = x.$$

4 Compatibility with SGA 4 XVIII 3.1.10.3

In [2, XVIII], overcome by discouragement, I did not verify the compatibility announced in the title. Although it turns out to be no more than smoke, I feel bound to remedy that omission here. For $f : X \to S$ separated and of finite type, the point was to verify that the adjunction morphism $\mathsf{R}f_!\mathsf{R}f^!\mathscr{L} \to \mathscr{L}$ is compatible with every étale localization $k : V \to S$. Using k and f for several arrows, as below, we must verify commutativity:

$$\begin{array}{ccc} X_V & \xrightarrow{k} & X \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} \\ V & \xrightarrow{k} & S \end{array} \qquad \begin{array}{ccc} k^*\mathsf{R}f_!\mathsf{R}f^!\mathscr{L} & \xrightarrow{k^*(\mathrm{adj})} & k^*\mathscr{L} \\ \downarrow{\scriptstyle \sim} & & \| \\ \mathsf{R}f_!\mathsf{R}f^!k^*\mathscr{L} & \xrightarrow{\mathrm{adj}} & k^*\mathscr{L}. \end{array}$$

At the level of complexes, the required commutativity is written (loc. cit.)

$$\begin{array}{ccc} k^*f_!^\bullet f_\bullet^!\mathscr{L} & \longrightarrow & k^*\mathscr{L} \\ \downarrow{\scriptstyle (1)} & & \| \\ f_!^\bullet f_\bullet^! k^*\mathscr{L} & \longrightarrow & k^*\mathscr{L}. \end{array}$$

It is checked component by component, reducing the problem to a compatibility for $\mathscr{L}$ a sheaf and for each pair of adjoint functors $(f_i^!, f_!^i)$, denoted simply by $f^!$ and $f_!$. The arrow (1) is defined using the isomorphism (2) : $k^*f_! = f_!k^*$, and a morphism (3) : $k^*f^! \to f^!k^*$ deduced from it. In loc. cit., one first introduces a morphism (4) : $k_!f_! \to f_!k_!$, deduced from (2) by the adjunction of $k^* = k^!$ and $k_! : k_!f_! \to k_!f_!k^*k_! \to k_!k^*f_!k_! \to f_!k_!$, and then deduces (3) from (4) by adjunction. Equivalently, one may deduce (3) from (2) by the adjunction of $f_!$ and $f^! : k^* \to f^!f_!k^*f^! \to f^!k^*f_!f^! \to f^!k^*$.

Writing k for k^* and omitting $\mathscr{L}$, the required commutativity is that of the outer boundary of

$$\begin{array}{ccccccc} kf_!f^! & \longrightarrow & f_!kf^! & \longrightarrow & f_!f^!f_!kf^! & \longrightarrow & f_!f^!kf_!f^! \\ \| & & & & & & \| \\ kf_!f^! & & \longleftarrow & & \longleftarrow & & f_!f^!kf_!f^! \\ \downarrow & & & & & & \downarrow \\ k & & \longleftarrow & & \longleftarrow & & f_!f^!k. \end{array}$$

In the first row, we find a homomorphism $kf_!f^! \to f_!f^!kf_!f^!$, defined because $kf_!f^!$ is of the form $f_!X$ (with $X = kf^!$). It admits the adjunction morphism in the second row as a retraction, and the commutativity of the lower square is the functoriality of this adjunction morphism.

Chapter 6

Applications of the trace formula to trigonometric sums

In this exposé, I explain how the trace formula can be used to calculate or study various trigonometric sums and how, together with the Weil conjecture, it can be used to bound them.

The first two sections provide a "user's guide" to these tools. Section 3 presents, in cohomological language, Weil's results on sums in 1 variable. Sections 4 through 6 give a detailed study of Gauss and Jacobi sums – including Weil's old and recent results on Hecke characters defined by Jacobi sums. In Section 7, we study a generalization of Kloosterman sums to several variables. Finally, Section 8 gives some indications of other uses that have been, or can be, made of these methods.

Notations

0.1

We use the notation of Report, Section 1. We will often have to consider a finite extension $\mathbb{F}_{q^n} \subset \mathbb{F}$ of $\mathbb{F}_q$. A subscript 0 will denote an object over $\mathbb{F}_q$, and a subscript 1 an object over $\mathbb{F}_{q^n}$. Replacing a subscript 0 by a subscript 1 (resp. omitting the subscript) means extending scalars to $\mathbb{F}_{q^n}$ (resp. to $\mathbb{F}$).

0.2

We denote by ℓ a prime number $\neq p$. We freely use the language of $\mathbb{Q}_\ell$-sheaves, as well as that of E_λ-sheaves, where E_λ is a finite extension of $\mathbb{Q}_\ell$ (cf. Report, Section 2, especially 2.11).

0.3

Let $H^\bullet$ be a graded vector space. If T is an endomorphism of $H^\bullet$, we put (cf. Cycle, 1.3.5)

$$\mathrm{tr}(T, H^\bullet) = \sum (-1)^i \,\mathrm{tr}(T, H^i).$$

1 Principles

1.1

Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$, let E_λ be a finite extension of $\mathbb{Q}_\ell$, and let $\mathscr{F}_0$ be an E_λ-sheaf on X_0. The trace formula says that

$$\sum_{x\in X^F} \mathrm{tr}(F_x^*, \mathscr{F}) = \sum (-1)^i \,\mathrm{tr}\left(F^*, \mathrm{H}_c^i(X, \mathscr{F})\right). \tag{1.1.1}$$

With the notation 0.3, the right-hand side is simply written $\mathrm{tr}(F^*, \mathrm{H}_c^\bullet(X, \mathscr{F}))$.

This trace formula for E_λ-sheaves can either be deduced from Report 4.10 by passage to the limit (cf. Report 4.11 through 4.13), or be deduced from the trace formula for $\mathbb{Q}_\ell$-sheaves (Report 3.2) by the method of (Functions L mod. ℓ^n, 4.3).

We shall interpret various trigonometric sums as the left-hand side of (1.1.1), for suitable X_0 and $\mathscr{F}_0$.

1.2

Let A be a finite group. An *A-torsor* on a scheme X is a sheaf T on X, equipped with a right action of A, which, locally (for the étale topology) on X, is isomorphic to the constant sheaf A on which A acts by right translation.

If $\tau : A \to B$ is a homomorphism and T is an A-torsor, there exists, up to unique isomorphism, a unique B-torsor $\tau(T)$, equipped with a morphism of sheaves $\tau : T \to \tau(T)$ such that

$$\tau(ta) = \tau(t)\tau(a). \tag{1.2.1}$$

If T is an A-torsor on X and ρ is a linear representation of A, $\rho : A \to \mathrm{GL}(V)$ (where V is a finite-dimensional vector space over E_λ), there exists, up to unique isomorphism, a unique E_λ-sheaf $\mathscr{F}$, smooth of rank $\dim(V)$, equipped with a morphism of sheaves

$$\rho : T \to \underline{\mathrm{Isom}}(V, \mathscr{F}). \tag{1.2.2}$$

such that $\rho(ta) = \rho(t)\rho(a)$. We denote it by $\rho(T)$. For $\tau : A \to B$ and ρ a representation of B, we have a canonical isomorphism $\rho(\tau(T)) = (\rho\tau)(T)$.

In this paragraph (except the appendix), we only consider the case where A is commutative and where $V = E_\lambda$: ρ is a character $A \to E_\lambda^\times$, and $\rho(T)$ is a smooth E_λ-sheaf of rank one. The morphism (1.2.2) is interpreted as an everywhere nonzero morphism

$$\rho : T \to \rho(T) \tag{1.2.3}$$

such that $\rho(ta) = \rho(a)\rho(t)$.

1.3

Let S be a scheme, let G be a commutative group scheme over S, and let G' be an extension of commutative group schemes over S

$$0 \longrightarrow A \longrightarrow G' \xrightarrow{\pi} G \longrightarrow 0.$$

The sheaf T of local sections of π is then an A-torsor on G.

If X is a scheme over S and f, g are two S-morphisms from X to G, the fact that T is defined by an extension gives a canonical isomorphism

$$(f+g)^*T = f^*T + g^*T. \tag{1.3.1}$$

On the left, $f+g$ is the sum of f and g with respect to the group law on G; on the right, $+$ denotes a sum of torsors.

If $f : X \to G$ factors through G', f^*T is trivial (and a factorization gives a trivialization); up to (non-unique) isomorphism, the A-torsor f^*T depends only on the image of f in $\hom_S(X,G)/\pi\hom_S(X,G')$: it is given by the morphism ∂ in the exact sequence

$$\hom(X,G') \longrightarrow \hom(X,G) \xrightarrow{\partial} \mathrm{H}^1(X,A).$$

We will apply these constructions to the exact Kummer sequences $0 \to \boldsymbol{\mu}_n \to \mathbb{G}_m \to \mathbb{G}_m \to 0$, and to Lang torsors.

1.4 The Lang torsor

Let G_0 be a connected commutative algebraic group over $\mathbb{F}_q$ (for the noncommutative case, see the appendix). The group law is written *multiplicatively*. Lang's isogeny

$$\mathfrak{L} : G_0 \to G_0 : x \mapsto Fx \cdot x^{-1}$$

is étale; its image, an open subgroup of G_0, can only be G_0 itself; its kernel is the finite group $G_0(\mathbb{F}_q)$. The *Lang torsor* L is the $G_0(\mathbb{F}_q)$-torsor on G_0 defined by the exact sequence

$$0 \longrightarrow G_0(\mathbb{F}_q) \longrightarrow G_0 \xrightarrow{\mathfrak{L}} G_0 \longrightarrow 0. \tag{1.4.1}$$

Notation We also write L and $\mathfrak{L}$ (or $L_{(q)}$ and $\mathfrak{L}_{(q)}$) for the objects obtained from L and $\mathfrak{L}$ by extending the base field. When necessary, we specify them with a subscript 0 or 1.

Examples For $G_0 = \mathbb{G}_a$ or $\mathbb{G}_m$, the sequences (1.4.1) are written

$$0 \longrightarrow \mathbb{F}_q \longrightarrow \mathbb{G}_a \xrightarrow{x^q - x} \mathbb{G}_a \longrightarrow 0 \tag{1.4.2}$$

$$0 \longrightarrow \boldsymbol{\mu}_{q-1} \longrightarrow \mathbb{G}_m \xrightarrow{x^{q-1}} \mathbb{G}_m \longrightarrow 0 \tag{1.4.3}$$

1.5

Let us compute the endomorphism F^* of the fiber of the Lang torsor at $\gamma \in G^F = G_0(\mathbb{F}_q)$. If $g \in \mathfrak{L}^{-1}(\gamma) \subset G$, we have $Fg = Fg \cdot g^{-1} \cdot g = \gamma g$. Therefore (Report, 1.2)

$$\text{on } L_0(G_0)_\gamma \simeq \mathfrak{L}^{-1}(\gamma),\ F^* \text{ is } g \mapsto g\gamma^{-1}. \tag{1.5.1}$$

1.6

On $\mathbb{F}_{q^n}$, the identity $F_{(q^n)} = F_{(q)}^n$ between endomorphisms of G_1 implies that $\mathfrak{L}_{(q^n)} = \mathfrak{L}_{(q)} \circ \prod_{i=0}^{n-1} F_{(q)}^i$. On $G_0(\mathbb{F}_{q^n})$, $F_{(q)}^i$ acts as the element $x \mapsto x^{q^i}$ of $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$. On $G_0(\mathbb{F}_{q^n})$, $\prod F_{(q)}^i$ is therefore the composite of the norm $G_0(\mathbb{F}_{q^n}) \to G_0(\mathbb{F}_q)$ and the inclusion $G_0(\mathbb{F}_q) \subset G_0(\mathbb{F}_{q^n})$: the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & G_0(\mathbb{F}_{q^n}) & \longrightarrow & G_1 & \xrightarrow{\mathfrak{L}_{(q^n)}} & G_1 & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle N} & & \downarrow{\scriptstyle \prod_{i=0}^{n-1} F_{(q)}^i} & & \| & & \\ 0 & \longrightarrow & G_0(\mathbb{F}_q) & \longrightarrow & G_1 & \xrightarrow{\mathfrak{L}_{(q)}} & G_1 & \longrightarrow & 0 \end{array} \tag{1.6.1}$$

is commutative. In the language of torsors, this gives a canonical isomorphism between $G_0(\mathbb{F}_q)$-torsors on G_1

$$NL_{(q^n)} = L_{(q)}. \tag{1.6.2}$$

Definition 1.7. *Let $f_0 : X_0 \to G_0$ be a morphism and $\chi : G_0(\mathbb{F}_q) \to E_\lambda^\times$ a character. We set $\mathscr{F}(\chi, f_0) = \chi^{-1}(f_0^*(L_0(G_0))) = f_0^*\chi^{-1}(L_0(G_0))$.*

We have the following functoriality properties:

a) $\mathscr{F}(\chi, f_0)$ is bimultiplicative in χ and f_0:

$$\mathscr{F}(\chi, f_0' \cdot f_0'') = \mathscr{F}(\chi, f_0') \otimes \mathscr{F}(\chi, f_0''), \tag{1.7.1}$$

$$\mathscr{F}(\chi'\chi'', f_0) = \mathscr{F}(\chi', f_0) \otimes \mathscr{F}(\chi'', f_0). \tag{1.7.2}$$

This follows from the definition of $\chi(T)$, for T a torsor, combined with (1.3.1) for (1.7.1).

b) For $g_0 : Y_0 \to X_0$ a morphism, we have

$$\mathscr{F}(\chi, f_0 \circ g_0) = g_0^*\mathscr{F}(\chi, f_0). \tag{1.7.3}$$

In particular, $\mathscr{F}(\chi, f_0) = f_0^*\mathscr{F}(\chi)$, where $\mathscr{F}(\chi)$ denotes $\mathscr{F}(\chi, \mathrm{id}_{G_0})$.

c) For $u_0 : G_0 \to H_0$ a morphism, and $\chi : H_0(\mathbb{F}_q) \to E_\lambda^\times$, we have

$$\mathscr{F}(\chi, u_0 f_0) = \mathscr{F}(\chi u_0, f_0). \tag{1.7.4}$$

d) For $G_0 = \prod_{i\in I} G_0^i$, with χ having coordinates χ_i ($i \in I$) and f_0 having coordinates f_0^i, it follows from (1.7.1) through (1.7.4) that

$$\mathscr{F}(\chi, f_0) = \bigotimes_{i\in I} \mathscr{F}(\chi_i, f_0^i). \tag{1.7.5}$$

Note the χ^{-1} in Definition 1.7. It ensures that, for $x \in X^F$, on $\mathscr{F}(\chi, f_0)_x$ one has

$$F_x^* = \chi f_0(x) \tag{1.7.6}$$

(use the fact that the fiber at x of the morphism (1.2.2), for $\rho = \chi^{-1}$, commutes with F_x^*, together with (1.5.1)).

If $\mathscr{F}(\chi, f_0)_1$ is the sheaf obtained from $\mathscr{F}(\chi, f_0)$ by extension from the base field to $\mathbb{F}_{q^n}$, we deduce from (1.6.2) that

$$\mathscr{F}(\chi, f_0)_1 = \mathscr{F}(\chi \circ N, f_1). \tag{1.7.7}$$

1.8 Abuse of notations

(i) If Ξ denotes the composite map $\chi f_0 : X_0(\mathbb{F}_q) \to E_\lambda^\times$, we will sometimes write $\mathscr{F}(\Xi)$ instead of $\mathscr{F}(\chi, f_0)$. By (1.7.1) through (1.7.5), there is little risk of ambiguity. For example:

a) we write $\mathscr{F}(\chi)$ for $\mathscr{F}(\chi, \mathrm{id}_{G_0})$ (notation already used in 1.7b));

b) we write $\mathscr{F}(\chi f_0)$ for $\mathscr{F}(\chi, f_0)$;

c) with the notation of 1.7d), we write $\mathscr{F}(\prod \chi_i f_0^i)$ for $\mathscr{F}(\chi, f_0)$.

With this notation, (1.7.4) shows that the notation $\mathscr{F}(\chi u_0 f_0)$ is unambiguous, (1.7.1) (1.7.2) (1.7.5) express the multiplicativity of $\mathscr{F}(\Xi)$ with respect to Ξ, and (1.7.6) becomes $F_x^* = \Xi(x)$ on $\mathscr{F}(\Xi)_x$.

(ii) If X is a scheme over an extension k of $\mathbb{F}_q$, and f is a morphism of X into $G_0 \otimes_{\mathbb{F}_q} k$, we shall again denote by $\mathscr{F}(\chi, f)$, $\mathscr{F}(\chi f)$, or $\mathscr{F}(\chi)$ (for f an inclusion) the inverse image under f of the E_λ-sheaf obtained from $\chi^{-1}(L_0(G))$ by extension of scalars from $\mathbb{F}_q$ to k. If f_0 is the composite $X \to G_0 \otimes_{\mathbb{F}_q} k \to G_0$, this is again the $\mathscr{F}(\chi, f_0)$ of 1.7. If k is a finite extension of $\mathbb{F}_q$, we have $\mathscr{F}(\chi f) = \mathscr{F}(\chi N_{k/\mathbb{F}_q} f)$.

Applying (1.1.1) to (1.7.6) and (1.7.7), we find:

Theorem 1.9. *Let S_0 be a separated scheme of finite type over $\mathbb{F}_q$, let G_0 be a connected commutative algebraic group over $\mathbb{F}_q$, let $f_0 : S_0 \to G_0$ be a morphism, and let $\chi : G_0(\mathbb{F}_q) \to E_\lambda^\times$. We have*

$$\sum_{s \in S_0(\mathbb{F}_q)} \chi f_0(s) = \mathrm{tr}(F^*, \mathrm{H}_c^\bullet(S, \mathscr{F}(\chi, f_0))) \tag{1.9.1}$$

and, for any integer $n \geqslant 1$,

$$\sum_{s \in S_0(\mathbb{F}_{q^n})} \chi \circ N_{\mathbb{F}_{q^n}/\mathbb{F}_q} f_0(s) = \mathrm{tr}(F^{*n}, \mathrm{H}_c^\bullet(S, \mathscr{F}(\chi, f_0))) \,. \tag{1.9.2}$$

1.9.3 Remark

Take G_0 to be a product. With the notation of 1.7d) and 1.8c), the formula (1.9.2) becomes

$$\sum_{s \in S_0(\mathbb{F}_{q^n})} \prod_i \chi_i \circ N_{\mathbb{F}_{q^n}/\mathbb{F}_q}(f_0^i(s)) = \mathrm{tr}\left(F^{*n}, \mathrm{H}_c^\bullet\left(S, \mathscr{F}\Big(\prod_i \chi_i f_0^i\Big)\right)\right).$$

1.9.4 Remark

Take $G_0 = \{e\}$. Formula (1.9.1) becomes

$$|S_0(\mathbb{F}_q)| = \sum_{s \in S_0(\mathbb{F}_q)} 1 = \mathrm{tr}(F^*, \mathrm{H}_c^\bullet(S, \mathbb{Q}_\ell)).$$

It is rare to be able to compute the right-hand side of (1.9.1) explicitly. Here is an amusing example, with $G_0 = \{e\}$.

1.10 Example

A nonsingular projective quadric of odd dimension over $\mathbb{F}_q$ has the same number of rational points as projective space of the same dimension over $\mathbb{F}_q$.

If, over $\mathbb{C}$, X is a nonsingular quadric hypersurface in the projective space $\mathbb{P}^{2N}$, and Y a hyperplane, we know that for $i \leqslant 2\dim(X) = 2\dim(Y)$, the inclusions $X \hookrightarrow \mathbb{P}^{2N} \hookleftarrow Y$ induce isomorphisms (in ordinary cohomology)

$$\mathrm{H}^i(X, \mathbb{Q}) \xleftarrow{\sim} \mathrm{H}^i(\mathbb{P}^{2N}, \mathbb{Q}) \xrightarrow{\sim} \mathrm{H}^i(Y, \mathbb{Q}).$$

By specialization, it follows that if X_0' is a nonsingular quadric hypersurface in projective space $\mathbb{P}_0^{2N}$ over $\mathbb{F}_q$, and Y_0' a hyperplane, the inclusions $X_0' \hookrightarrow \mathbb{P}_0^{2N} \hookleftarrow Y_0'$ induce isomorphisms

$$\mathrm{H}^i(X', \mathbb{Q}_\ell) \xleftarrow{\sim} \mathrm{H}^i(\mathbb{P}^{2N}, \mathbb{Q}_\ell) \xrightarrow{\sim} \mathrm{H}^i(Y', \mathbb{Q}_\ell).$$

These isomorphisms commute with F^*, and the trace formula applies.

1.11

The trace formula can also be used to understand the classical formulas giving the number of rational points of linear groups over finite fields, and of certain homogeneous spaces (see Section 8).

1.12

There is a dictionary that translates various kinds of classical manipulations of trigonometric sums into cohomological terms. This dictionary will be given in Section 2. Since the cohomological statement is more "geometric," it can sometimes be applied to situations where the classical argument only applies after a extension of the finite base field. Here is an example (a special case of a theorem of Kazhdan).

Theorem 1.13 (Kazhdan)**.** *Let X_0 be a scheme over $\mathbb{F}_q$. Suppose there is an action ρ of $\mathbb{G}_a$ on X, and a morphism $f : X \to Y$ of schemes over $\mathbb{F}$, making X a $\mathbb{G}_a$-torsor over Y. Then the number of rational points of X_0 is divisible by q.*

First suppose that ρ, Y, and f are defined over $\mathbb{F}_q$.

Classical argument The fibers of $f : X_0(\mathbb{F}_q) \to Y_0(\mathbb{F}_q)$ all have q elements: $|X_0(\mathbb{F}_q)| = q \cdot |Y_0(\mathbb{F}_q)|$, hence the divisibility.

Cohomological translation The higher direct image sheaves $\mathsf{R}^i f_! \mathbb{Q}_\ell$ are

$$\mathsf{R}^i f_! \mathbb{Q}_\ell = \begin{cases} \mathbb{Q}_\ell(-1) & \text{if } i = 2 \\ 0 & \text{for } i \neq 2 \end{cases}$$

The Leray spectral sequence for f therefore degenerates, giving an isomorphism

$$\mathrm{H}_c^i(X, \mathbb{Q}_\ell) = \mathrm{H}_c^{i-2}(Y, \mathbb{Q}_\ell)(-1).$$

To understand the effect of a Tate twist on the eigenvalues of Frobenius, it is most convenient to use the Galois point of view (Report 1.8) and note that the Frobenius substitution φ acts on $\mathbb{Q}_\ell(1)$ by multiplication by q. The eigenvalues of F^* on $\mathrm{H}_c^p(X, \mathbb{Q}_\ell)$ are therefore obtained by multiplying by q the

eigenvalues of F^* acting on $\mathrm{H}_c^{i-2}(Y, \mathbb{Q}_\ell)$. We know that the latter are algebraic integers [16, XXI 5.2.2]; hence

$$\begin{gathered}\text{the eigenvalues of } F^* \text{ acting on } \mathrm{H}_c^i(X, \mathbb{Q}_\ell) \text{ are} \\ \text{algebraic integers divisible by } q.\end{gathered} \tag{$*$}$$

Descent Return to the hypotheses of the theorem. For a suitable n, we may suppose that ρ, Y, and f are defined over $\mathbb{F}_{q^n}$. Since the Frobenius morphism relative to $\mathbb{F}_{q^n}$ is the n-th power of the one relative to $\mathbb{F}_q$, statement $(*)$ for the scheme over $\mathbb{F}_{q^n}$ obtained from X_0 by extension of scalars gives:

$$\begin{gathered}\text{the } n\text{-th powers of the eigenvalues of } F^* \text{ acting on} \\ \mathrm{H}_c^i(X, \mathbb{Q}_\ell) \text{ are algebraic integers divisible by } q^n.\end{gathered} \tag{$**$}$$

This assertion implies $(*)$. The number of rational points of X_0, namely $\sum(-1)^i \operatorname{tr}(F^*, \mathrm{H}^i(X, \mathbb{Q}_\ell))$, is therefore an algebraic integer divisible by q. It follows that, as a rational integer, it is divisible by q.

1.14

Formula 1.9.3 controls the dependence on n of the trigonometric sum on the left-hand side. It implies identities between a trigonometric sum and those which are deduced from it by "extension of scalars." The most famous of these identities is that of Hasse–Davenport. Let $\chi : \mathbb{F}_q^\times \to \mathbb{C}^\times$ and $\psi : \mathbb{F}_q \to \mathbb{C}^\times$ be characters, with ψ nontrivial, and define the Gauss sum $\tau(\chi, \psi)$ by

$$\tau(\chi, \psi) = -\sum_{x \in \mathbb{F}_q^\times} \psi(x)\chi^{-1}(x). \tag{1.14.1}$$

Theorem 1.15 (Hasse-Davenport)**.** *We have*

$$\tau\left(\chi \circ N_{\mathbb{F}_{q^n}/\mathbb{F}_q}, \psi \circ \operatorname{tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q}\right) = \tau(\chi, \psi)^n. \tag{1.15.1}$$

Let $E \subset \mathbb{C}$ be a number field containing the values of χ and ψ, and let E_λ be the completion of E at a place of residue characteristic $\ell \neq p$. Identity (1.15.1) is an identity in E. Proving it in $\mathbb{C}$ is equivalent to proving it in E_λ. We shall prove it in E_λ, regarding χ and ψ as taking values in $E_\lambda^\times$.

Apply 1.9.3 with $X_0 = \mathbb{G}_m$ over $\mathbb{F}_q$ and $G_0 = \mathbb{G}_m \times \mathbb{G}_a$. The groups $\mathrm{H}_c^i(\mathbb{G}_m, \mathscr{F}(\chi^{-1}\psi))$ vanish for $i \neq 1$, and H_c^1 has dimension 1 (4.2). By 1.9.3, $\tau(\chi \circ N_{\mathbb{F}_{q^n}/\mathbb{F}_q}, \psi \circ \operatorname{tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q})$ is the unique eigenvalue of $(F^*)^n$ on H_c^1, and (1.15.1) follows.

Similar examples are given in Weil [43, App V].

1.16

An E_λ-sheaf $\mathscr{F}_0$ on X_0 is said to be *pointwise pure of weight* n if for any closed point $x \in |X_0|$, the eigenvalues of F_x^* (Report 1.2) are algebraic numbers all of whose complex conjugates have absolute value $q_x^{n/2}$, where q_x is the number of elements of $k(x)$. The following theorem will be proved in [15].

Theorem 1.17. *If $\mathscr{F}_0$ is pointwise pure of weight n, then for any eigenvalue α of the endomorphism F^* of $\mathrm{H}_c^i(X, \mathscr{F})$, there exists an integer $m \leqslant n + i$ such that the complex conjugates of α are all of absolute value $q^{m/2}$.*

The sheaves considered in 1.9 have weight 0. The theorem therefore gives, for the sum 1.9.1 (or rather, for all its complex conjugates), the bound

$$\left|\sum_{s\in S_0(\mathbb{F}_q)}\prod_i \chi_i(f^i(s))\right| \leqslant \sum_i \dim \mathrm{H}^i_c\left(S, \mathscr{F}\left(\prod \chi_i f^i\right)\right)\cdot q^{i/2}. \tag{1.17.1}$$

1.18 Remarks

a) If $\dim S = n$, for any E_λ-sheaf $\mathscr{F}$ on S, we have

$$\mathrm{H}^i_c(S,\mathscr{F}) = 0 \qquad \text{for } i \notin [0,2n].$$

b) The group $\mathrm{H}^0_c(X,\mathscr{F})$ is the group of global sections of $\mathscr{F}$ on S whose support is proper. If S is connected, not proper and $\mathscr{F}$ is smooth, this group is zero; the same holds if S is connected, $\mathscr{F}$ is smooth of rank one, and is nonconstant.

c) If S is smooth and purely of dimension n, and $\mathscr{F}$ is smooth, Poincaré duality says that the vector spaces $\mathrm{H}^i_c(X,\mathscr{F})$ and $\mathrm{H}^{2n-i}(X,\mathscr{F}^\vee)(n)$ are dual to each other (for the effect of a Tate twist on the eigenvalues of Frobenius, cf. the 2nd part of the proof of 1.13).

d) If $\dim S = n$, Poincaré duality allows us to compute $\mathrm{H}^{2n}_c(S,\mathscr{F})$ as follows: take an open subset U of S_{red} whose complement has dimension $< n$; require this open subset to be smooth and purely of dimension n, with $\mathscr{F}$ smooth on it. We then have $\mathrm{H}^{2n}_c(U,\mathscr{F}) \xrightarrow{\sim} \mathrm{H}^{2n}_c(S,\mathscr{F})$, because in the long exact sequence of cohomology, $\mathrm{H}^i_c(S\smallsetminus U,\mathscr{F}) = 0$ for $i = 2n, 2n-1$. Poincaré duality therefore gives

$$\mathrm{H}^{2n}_c(S,\mathscr{F}) \xrightarrow{\sim} \left(\mathrm{H}^0(U,\mathscr{F}^\vee)(n)\right)^\vee.$$

Let U be connected, and let u be a geometric point of U. The "local system" $\mathscr{F}$ corresponds to a representation of $\pi_1(U,u)$ on $\mathscr{F}_u$, and $\mathrm{H}^0(U,\mathscr{F}^\vee)$ is $(\mathscr{F}^\vee_u)^{\pi_1(U,u)} = ((\mathscr{F}_u)_{\pi_1(U,u)})^\vee$ (dual of coinvariants). We therefore have

$$\mathrm{H}^{2n}_c(S,\mathscr{F}) \simeq (\mathscr{F}_u)_{\pi_1(U,u)}(-n).$$

e) These remarks often allow the calculation of b_0 and b_{2n}. Computing the other b_i can be difficult. If $b_{2n} = 0$ and $b_{2n-1} = 0$, the bound (1.17.1) is a Lang–Weil bound, with, for large q, a gain of $\sqrt{q}$ over the trivial $O(q^n)$ bound. One should bear in mind, however, that (1.17.1) does not formally imply the trivial bound

$$\left|\sum_{s\in S_0(\mathbb{F}_q)}\prod_i \chi_i(f^i(s))\right| \leqslant \left|S_0(\mathbb{F}_q)\right|,$$

so it may be useful to take this into account (cf. the proof of 3.8).

Below I explain a method for proving that $b_i = 0$ for $i \neq n$. When it applies, it gives an estimate of order $O(q^{n/2})$. The constant implicit in O is $b_n = (-1)^n\chi(S,\mathscr{F}(\prod \chi_i f^i))$, which may be difficult to compute.

Proposition 1.19. *Let $\bar{X}$ be a proper scheme over an algebraically closed field k, let $j : X \hookrightarrow \bar{X}$ be an open immersion with domain X, and let $\mathscr{F}$ be an E_λ-sheaf on X. Assume that*

a) $(j_*\mathscr{F})_x = 0$ for $x \in \bar{X}\smallsetminus X$, i.e. $j_!\mathscr{F} \xrightarrow{\sim} j_*\mathscr{F}$;

b) $\mathsf{R}^i j_* \mathscr{F} = 0$ *for* $i > 0$.

Then the maps

$$\mathrm{H}^i_c(X, \mathscr{F}) \to \mathrm{H}^i(X, \mathscr{F})$$

are isomorphisms.

Assumptions a) and b) mean that $j_! \mathscr{F} \xrightarrow{\sim} \mathsf{R} j_* \mathscr{F}$, hence

$$\mathrm{H}^i_c(X, \mathscr{F}) = \mathrm{H}^i(\bar{X}, j_! \mathscr{F}) \xrightarrow{\sim} \mathsf{H}^i(\bar{X}, \mathsf{R} j_* \mathscr{F}) = \mathrm{H}^i(X, \mathscr{F}).$$

In other words, the Leray spectral sequence for j

$$E_2^{pq} = \mathrm{H}^p(\bar{X}, \mathsf{R}^q j_* \mathscr{F}) \Rightarrow \mathrm{H}^{p+q}(X, \mathscr{F})$$

degenerates, by b) into isomorphisms

$$\mathrm{H}^i(\bar{X}, j_* \mathscr{F}) \xrightarrow{\sim} \mathrm{H}^i(X, \mathscr{F}),$$

while, by a)

$$\mathrm{H}^i_c(X, \mathscr{F}) = \mathrm{H}^i(\bar{X}, j_! \mathscr{F}) \xrightarrow{\sim} \mathrm{H}^i(\bar{X}, j_* \mathscr{F}).$$

1.19.1 Example

If X is a curve, or if $\bar{X}$ is smooth, X is the complement of a normal-crossings divisor, and $\mathscr{F}$ is tamely ramified, hypothesis a) of 1.19 (no invariants under local monodromy) implies hypothesis b).

Proposition 1.20. *Let X_0 be a smooth affine scheme, purely of dimension n, over $\mathbb{F}_q$, and let $\mathscr{F}_0$ be a smooth E_λ-sheaf, pointwise pure of weight m, on X_0. We assume that the maps $\mathrm{H}^i_c(X, \mathscr{F}) \to \mathrm{H}^i(X, \mathscr{F})$ are isomorphisms (cf. 1.17). Then, $\mathrm{H}^i_c(X, \mathscr{F}) = 0$ for $i \neq n$, and the complex conjugates of the eigenvalues of F^* on $\mathrm{H}^n_c(X, \mathscr{F})$ are of absolute value $q^{(m+n)/2}$.*

Poincaré duality pairs $\mathrm{H}^i_c(X, \mathscr{F})$ (resp. $\mathrm{H}^i_c(X, \mathscr{F}^\vee)$) with $\mathrm{H}^{2n-i}(X, \mathscr{F}^\vee(n))$ (resp. $\mathrm{H}^{2n-i}(X, \mathscr{F}(n))$). According to [2, XIV 3.2], the ordinary H^i groups vanish for $i > n$. By duality, the H^i_c are zero for $i < n$; since $\mathrm{H}^i_c(X, \mathscr{F}) \xrightarrow{\sim} \mathrm{H}^i(X, \mathscr{F})$, these groups are zero for $i \neq n$. Apply 1.17 to $\mathscr{F}_0$ and to $\mathscr{F}_0^\vee(n)$ (of weight $-m-2n$). We find that the complex conjugates α of the eigenvalues of F^* on $\mathrm{H}^n_c(X, \mathscr{F}) = \mathrm{H}^n_c(X, \mathscr{F}^\vee(n))^\vee$ satisfy

$$|\alpha| \leqslant q^{(m+n)/2} \qquad \text{and}$$
$$|\alpha^{-1}| \leqslant q^{((-m-2n)+n)/2} = q^{-(m+n)/2},$$

hence the assertion.

1.21 Remark

The last argument shows that if X_0 is a smooth separated scheme over $\mathbb{F}_q$, if $\mathscr{F}_0$ is a smooth E_λ-sheaf pointwise pure of weight m on X_0, and if $\mathrm{H}^i_c(X, \mathscr{F}) \hookrightarrow \mathrm{H}^i(X, \mathscr{F})$, then the complex conjugates of the eigenvalues of F^* on $\mathrm{H}^i_c(X, \mathscr{F})$ are of absolute value $q^{(m+i)/2}$.

Appendix. The Lang isogeny in the noncommutative case

1.22

Let G_0 be a connected algebraic group over $\mathbb{F}_q$ and let $\mathfrak{L}$ be the morphism $x \mapsto Fx \cdot x^{-1}$ from G_0 to itself. It is a morphism of G_0-homogeneous spaces if, on the source, G_0 acts by left translation $x \mapsto (y \mapsto xy)$, and on the target by $x \mapsto (y \mapsto Fx \cdot y \cdot x^{-1})$). We have $\mathfrak{L}(xy) = \mathfrak{L}(x)$ for $y \in G_0(\mathbb{F}_q)$, and $\mathfrak{L}$ induces an isomorphism $G_0/G_0(\mathbb{F}_q) \xrightarrow{\sim} G_0$.

As in the commutative case, $\mathfrak{L}$ therefore makes G_0 a $G_0(\mathbb{F}_q)$-torsor on G_0, the *Lang torsor* L. If $\rho : G_0(\mathbb{F}_q) \to \mathrm{GL}(n, E_\lambda)$ is a linear representation of $G_0(\mathbb{F}_q)$, we denote by $\mathscr{F}(\rho)$ the E_λ-sheaf $\rho(L)$ (1.2).

1.23

Let $\gamma \in G^F = G_0(\mathbb{F}_q)$, and compute the endomorphism F^* of $\mathscr{F}(\rho)_\gamma$. Let $g \in G$ be such that $\mathfrak{L}(g) = \gamma$, and choose the frame $\rho(g) \in \mathrm{Isom}(E_\lambda^n, \mathscr{F}(\rho)_\gamma)$. For $e \in E_\lambda^n$, we have

$$F(\rho(g)(e)) = \rho(Fg)(e) = \rho(\gamma g)(e) = \rho(g \cdot g^{-1}\gamma g)(e).$$

We will see that $g^{-1}\gamma g \in G_0(\mathbb{F}_q)$, hence

$$F(\rho(g)(e)) = \rho(g)\rho(g^{-1}\gamma g)(e)$$

and

$$\rho(g)^{-1}(F_\gamma^*)^{-1}\rho(g) = \rho(g^{-1}\gamma g): \tag{1.23.1}$$

$\rho(g)$ identifies the inverse of F^* at γ with the automorphism $\rho(g^{-1}\gamma g)$ of E_λ^n. It remains to understand $g^{-1}\gamma g$.

Lemma 1.24. *(i) If $Fg \cdot g^{-1} \in G_0(\mathbb{F}_q)$, we have $g^{-1}(Fg \cdot g^{-1})g = g^{-1}Fg \in G_0(\mathbb{F}_q)$.*

(ii) Let $\gamma \in G_0(\mathbb{F}_q)$, and g such that $Fg \cdot g^{-1} = \gamma$. The conjugation class of $\gamma' = g^{-1}Fg$ in $G_0(\mathbb{F}_q)$ only depends on that of γ.

(iii) The map induced by $\gamma \mapsto \gamma'$ on the set $G_0(\mathbb{F}_q)^\natural$ of conjugacy classes of $G_0(\mathbb{F}_q)$ is bijective. Its inverse is the corresponding map for the opposite group.

(i) If $\gamma = Fg \cdot g^{-1} \in G_0(\mathbb{F}_q)$, the identities $Fg = \gamma g$ and $F\gamma = \gamma$ give $F(g^{-1}Fg) = (Fg)^{-1}FFg = g^{-1}\gamma^{-1}F(\gamma g) = g^{-1}\gamma^{-1}F\gamma Fg = g^{-1}Fg$, so that $g^{-1}Fg \in G_0(\mathbb{F}_q)$.

(ii) For $\alpha, \gamma \in G_0(\mathbb{F}_q)$, the g such that $Fg \cdot g^{-1} = \gamma$ form a right coset under $G_0(\mathbb{F}_q)$, and if $Fg \cdot g^{-1} = \gamma$, we have $F(\alpha g \alpha^{-1})(\alpha g \alpha^{-1})^{-1} = \alpha\gamma\alpha^{-1}$. As γ ranges over a conjugacy class in $G_0(\mathbb{F}_q)$, the g such that $Fg \cdot g^{-1} = \gamma$ therefore range over a double coset under $G_0(\mathbb{F}_q)$, and $\gamma' = g^{-1}Fg$ ranges over a conjugacy class.

(iii) Finally, the inverse of $Fg \cdot g^{-1} \mapsto g^{-1}Fg$ is $g^{-1}Fg \mapsto Fg \cdot g^{-1}$.

1.25

Formula (1.23.1) can be read succinctly: F^* at γ is $\rho(\gamma')^{-1}$. If γ belongs to the identity component of its centralizer, we can take g in $Z(\gamma)$. We then have $\gamma = \gamma'$. If this condition is not met, γ and γ' need not be conjugate.

1.26 An example

For $G_0 = \mathrm{SL}(2)$, q odd, $\alpha \in \mathbb{F}_q^\times \smallsetminus \mathbb{F}_q^{\times 2}$, and $\bar\gamma$ is the conjugacy class of $-1(1+N)$, with $N^2 = 0$, then $\bar\gamma'$ is the conjugacy class of $-1 \cdot (1+\alpha N)$, so that $\bar\gamma' \neq \bar\gamma$ if $N \neq 0$.

2 Dictionary

Elementary manipulations of trigonometric sums can often be seen, via the trace formula, as counterparts of statements in cohomology. In this section, I list such statements and their counterparts; they carry the same number, with a $*$ appended to the cohomological statement.

2.1

"A sum of algebraic integers is an algebraic integer" has as its cohomological analogue the following theorem, proved in [16, XXI 5.2.2] (see the use of 2.1* in 1.13).

Theorem (2.1*)**.** *Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$ and let $\mathscr{F}_0$ be an E_λ-sheaf on X_0. If, for all $x \in |X_0|$, the eigenvalues of F_x^* on $\mathscr{F}_0$ are algebraic integers, then the eigenvalues of F^* on $\mathrm{H}_c^i(X, \mathscr{F})$ are algebraic integers.*

2.2

Other integrality results are proved in [16, XXI, 5.2.2 et 5.4]: if $\dim(X_0) \leqslant n$, and α is an eigenvalue of F^* on $\mathrm{H}_c^i(X, \mathscr{F})$, we have:

a) under the assumptions of 2.1*, and if $i > n$, then q^{i-n} divides α;

b) if the inverses of the eigenvalues of F_x^* are algebraic integers, then α^{-1} is integral away from p. More precisely, $q^{\inf(n,i)}\alpha^{-1}$ is an algebraic integer.

For the sheaves considered in 1.9, the eigenvalues of F_x^* are roots of unity, so the assumptions of the theorem and of b) above are satisfied.

2.3

Let $f : A \to B$ be a map from one finite set to another, and ε a function on A. We have

$$\sum_{a \in A} \varepsilon(a) = \sum_{b \in B} \sum_{f(a)=b} \varepsilon(a). \tag{2.3.1}$$

2.3*

Let $f : X \to Y$ be a morphism of separated schemes of finite type over an algebraically closed field k, and let $\mathscr{F}$ be an E_λ-sheaf on X. The Leray spectral sequence for f in cohomology with compact support is

$$E_2^{pq} = \mathrm{H}_c^p(Y, \mathsf{R}^q f_! \mathscr{F}) \Rightarrow \mathrm{H}_c^{p+q}(X, \mathscr{F}). \tag{2.3.1*}$$

Let $f_0 : X_0 \to Y_0$ be a morphism of separated schemes of finite type over $\mathbb{F}_q$ and let $\mathscr{F}_0$ be an E_λ-sheaf on X_0. Between (2.3.1) and (2.3.1*) there is the following compatibility.

a) The sheaf $\mathsf{R}^q f_!\mathscr{F}$ is obtained by extending scalars from $\mathbb{F}_q$ to $\mathbb{F}$ in the sheaf $\mathsf{R}^q f_{0!}\mathscr{F}_0$ on Y_0. As such, it is equipped with a Frobenius correspondence $F^* : F^*\mathsf{R}^q f_!\mathscr{F} \to \mathsf{R}^q f_!\mathscr{F}$. It is obtained by applying the functoriality of $\mathsf{R}f_!$ to the Frobenius correspondence of $\mathscr{F}$: namely the composite

$$F^*\mathsf{R}^q f_!\mathscr{F} \xrightarrow{\sim} \mathsf{R}^q f_! F^*\mathscr{F} \xrightarrow{\sim} \mathsf{R}^q f_!\mathscr{F}.$$

b) The trace formula for $\mathscr{F}_0$ reads

$$\sum_{x\in X_0(\mathbb{F}_q)} \operatorname{tr}(F_x^*, \mathscr{F}_0) = \operatorname{tr}(F^*, \mathrm{H}_c^\bullet(X, \mathscr{F})); \tag{1}$$

while that for $\mathsf{R}^q f_!\mathscr{F}$ reads

$$\sum_{y\in Y_0(\mathbb{F}_q)} \operatorname{tr}(F_y^*, \mathsf{R}^q f_!\mathscr{F}) = \operatorname{tr}(F^*, \mathrm{H}_c^\bullet(Y, \mathsf{R}^q f_!\mathscr{F})). \tag{2_q}$$

We have, for $y \in Y(\mathbb{F})$, $(\mathsf{R}^q f_!\mathscr{F})_y = \mathrm{H}_c^q(f^{-1}(y), \mathscr{F})$, and if $y \in Y_0(\mathbb{F}_q)$,

$$\operatorname{tr}(F_y^*, \mathsf{R}^q f_!\mathscr{F}) = \operatorname{tr}(F^*, \mathrm{H}_c^q(f^{-1}(y), \mathscr{F})),$$

hence by the trace formula on the fiber

$$\sum_{\substack{x\in X_0(\mathbb{F}_q)\\ f(x)=y}} \operatorname{tr}(F_x^*, \mathscr{F}_0) = \sum (-1)^q \operatorname{tr}(F_y^*, \mathsf{R}^q f_!\mathscr{F}).$$

Taking the alternating sum of the identities (2_q), we therefore find

$$\sum_{y\in Y_0(\mathbb{F}_q)} \sum_{\substack{x\in X_0(\mathbb{F}_q)\\ f(x)=y}} \operatorname{tr}(F_x^*, \mathscr{F}_0) = \operatorname{tr}(F^*, \mathrm{H}_c^\bullet(Y, \mathsf{R}^\bullet f_!\mathscr{F})). \tag{2}$$

Equality of the left-hand sides of (1) and (2) follows from (2.3.1); equality of the right-hand sides follows from (2.3.1*) (taking into account that F^* is an endomorphism of the entire spectral sequence).

2.4

Let $(A_i)_{i\in I}$ be a finite family of finite sets, $A = \prod_{i\in I} A_i$, ε_i a function on A_i, and ε the function $a \mapsto \prod \varepsilon_i(a_i)$ on A. We have

$$\sum_{a\in A} \varepsilon(a) = \prod_{i\in I} \sum_{a_i\in A_i} \varepsilon_i(a_i). \tag{2.4.1}$$

2.4*

Let $(X_i)_{i\in I}$ be a finite family of separated schemes of finite type over an algebraically closed field k, let $X = \prod_{i\in I} X_i$, let $\mathscr{F}_i$ be an E_λ-sheaf on X_i and $\mathscr{F}$ the external tensor product $\boxtimes_{i\in I} \mathscr{F}_i = \bigotimes_{i\in I} \mathrm{pr}_i^*(\mathscr{F}_i)$ of the $\mathscr{F}_i$. The Künneth formula gives

$$\mathrm{H}_c^\bullet(X, \mathscr{F}) = \bigotimes_{i\in I} \mathrm{H}_c^\bullet(X_i, \mathscr{F}_i). \tag{2.4.1*}$$

For the isomorphism (2.4.1*) to be canonical and independent of the choice of a total ordering of I, the tensor product on the right must be understood as a graded tensor product, with the Koszul sign rule (Cycle, 1.3).

2.5

Let A be a finite set, let B be a subset of A, and let ε be a function on A. We have

$$\sum_{a\in A}\varepsilon(a)=\sum_{a\in A\smallsetminus B}\varepsilon(a)+\sum_{a\in B}\varepsilon(a). \tag{2.5.1}$$

2.5*

Let X be a separated scheme of finite type over an algebraically closed field k, let Y be a closed subscheme, and let $\mathscr{F}$ be an E_λ-sheaf on X. We have a long exact cohomology sequence

$$\cdots\xrightarrow{\partial}\mathrm{H}^i_c(X\smallsetminus Y,\mathscr{F})\longrightarrow\mathrm{H}^i_c(X,\mathscr{F})\longrightarrow\mathrm{H}^i_c(Y,\mathscr{F})\xrightarrow{\partial}\cdots \tag{2.5.1*}$$

More generally, given a finite filtration of X by closed subsets X_p ($p\in\mathbb{Z}$; assume that $X_p\supset X_{p+1}$, that $X_p=X$ for p sufficiently small, and that $X_p=\varnothing$ for p sufficiently large), there is a spectral sequence

$$E_1^{pq}=\mathrm{H}^{p+q}_c(X_p\smallsetminus X_{p+1},\mathscr{F})\Rightarrow\mathrm{H}^{p+q}_c(X,\mathscr{F}). \tag{2.5.2*}$$

2.6

Let A be a finite set, $(A_i)_{i\in I}$ a finite covering of A and ε a function on A. For $J\subset I$, denote by A_J the intersection of A_j ($j\in J$). We have

$$\sum_{J\subset I}(-1)^{|J|}\sum_{a\in A_J}\varepsilon(a)=0. \tag{2.6.1}$$

2.6*

Let X be a separated scheme of finite type over an algebraically closed field k, $(X_i)_{i\in I}$ a closed (resp. open) finite covering of X by subschemes, and $\mathscr{F}$ an E_λ-sheaf on X. For $J\subset I$, let X_J be the intersection of the X_j ($j\in J$). We have the respective Leray spectral sequences

$$E_1^{pq}=\bigoplus_{|J|=p+1>0}\mathrm{H}^q_c(X_J,\mathscr{F})\otimes_{\mathbb{Z}}\textstyle\bigwedge^{|J|}\mathbb{Z}^J\Rightarrow\mathrm{H}^{p+q}_c(X,\mathscr{F}) \tag{2.6.1*}$$

$$E_1^{pq}=\bigoplus_{|J|=1-p>0}\mathrm{H}^q_c(X_J,\mathscr{F})\otimes_{\mathbb{Z}}\textstyle\bigwedge^{|J|}\mathbb{Z}^J\Rightarrow\mathrm{H}^{p+q}_c(X,\mathscr{F}) \tag{2.6.2*}$$

If $J=\varnothing$ were not excluded, there would also be spectral sequences converging to 0. The factors $\bigwedge^{|J|}\mathbb{Z}^J$ are included to avoid choosing a total order on I.

2.7

Let A be a finite commutative group, and $\chi:A\to E_\lambda^\times$ a nontrivial character. We have

$$\sum_{a\in A}\chi(a)=0. \tag{2.7.1}$$

We will prove the cohomological analogue of (2.7.1) by transposing the following proof: if $x \in A$, $a \mapsto xa$ is a permutation of A, and

$$\sum_{a\in A} \chi(a) = \sum_{a\in A} \chi(xa) = \chi(x) \sum_{a\in A} \chi(a) \qquad \text{, whence}$$
$$(\chi(x) - 1) \sum_{a\in A} \chi(a) = 0,$$

and by hypothesis there exists x such that $\chi(x) - 1 \neq 0$.

Theorem (2.7*)**.** *Let G_0 be a connected commutative algebraic group over $\mathbb{F}_q$ and $\chi : G_0(\mathbb{F}_q) \to E_\lambda^\times$ a nontrivial character. We have*

$$\mathrm{H}_c^\bullet(G, \mathscr{F}(\chi)) = 0. \tag{2.7.1*}$$

For x a rational point of G, let t_x be the translation $t_x(g) = xg$ of G. The formula $\mathfrak{L}t_x = t_{\mathfrak{L}(x)}\mathfrak{L}$ expresses that $(t_x, t_{\mathfrak{L}(x)})$ is an automorphism of the diagram $G \xrightarrow{\mathfrak{L}} G$ (G equipped with the torsor of Lang). Let $\rho(g)$ be the automorphism of $(G, \mathscr{F}(\chi))$ induced by it. For $g \in G_0(\mathbb{F}_q)$, i.e. for $\mathfrak{L}(g) = e$, it is the identity on G, and multiplication by $\chi(g)^{-1}$ on $\mathscr{F}(\chi)$.

Let $\rho_H(g)$ denote the automorphism of $\mathrm{H}_c^\bullet(G, \mathscr{F}(\chi))$ induced by $\rho(g)$. A homotopy argument (Lemma 2.8 below) shows that $\rho_H(g) = \rho_H(e)$, and hence is the identity. On the other hand, for $g \in G_0(\mathbb{F}_q)$, $\rho_H(g)$ is multiplication by $\chi^{-1}(g)$: on $\mathrm{H}^\bullet(G, \mathscr{F}(\chi))$, multiplication by $(\chi^{-1}(g) - 1)$ is zero. Taking g such that $\chi(g) \neq 1$, we obtain 2.7.

Lemma 2.8. *Let X and Y be two schemes over an algebraically closed field k, with X separated and of finite type and Y connected. Let $\mathscr{F}$ be a sheaf on X and let (ρ, ε) be a family of endomorphisms of $(X, \mathscr{F})$ parametrized by Y:*

$$\rho : Y \times_k X \to Y \times_k X \qquad \textit{is a } Y\textit{-morphism, and}$$
$$\varepsilon : \rho^* \mathrm{pr}_2^* \mathscr{F} \to \mathrm{pr}_2^* \mathscr{F} \qquad \textit{a morphism of sheaves.}$$

Assume that ρ is proper. For $y \in Y(k)$, let $\rho_H(y)^*$ denote the endomorphism of $\mathrm{H}_c^\bullet(X, \mathscr{F})$ induced by $\rho_y : X \to X$ and $\varepsilon_y : \rho_y^* \mathscr{F} \to \mathscr{F}$. Then, $\rho_H(y)^*$ is independent of y.

Indeed, $\mathsf{R}^p \mathrm{pr}_{1!} \mathrm{pr}_2^* \mathscr{F}$ is the constant sheaf on Y with fiber $\mathrm{H}_c^p(X, \mathscr{F})$, and $\rho_H(y)^*$ is the fiber at y of the endomorphism

$$\mathsf{R}^p \mathrm{pr}_{1!} \mathrm{pr}_2^* \mathscr{F} \xrightarrow{\rho^*} \mathsf{R}^p \mathrm{pr}_{1!} \rho^* \mathrm{pr}_2^* \mathscr{F} \xrightarrow{\varepsilon} \mathsf{R}^p \mathrm{pr}_{1!} \mathrm{pr}_2^* \mathscr{F}$$

of this sheaf.

2.9 Remark

Let G_0 be a connected algebraic group over $\mathbb{F}_q$ and $\rho : G_0(\mathbb{F}_q) \to \mathrm{GL}(V)$ a linear representation such that $V^{G_0(\mathbb{F}_q)} = 0$. Then again $\mathrm{H}_c^\bullet(G, \rho(L)) = 0$.

3 Sums in one variable

3.1

Weil was the first to apply "cohomological" methods to the study of trigonometric sums in one variable; since he had proved the analogue of the Riemann hypothesis for Artin's L functions over function fields,

these methods gave him excellent estimates (of order O(square root of the number of terms)) and an understanding of how these sums behave under "extension of the base field."

The substance of this section is a presentation, in cohomological language, of his results.

3.2

Cohomological methods here lead us to compute the groups $\mathrm{H}^i_c(X, \mathscr{F})$, where $\mathscr{F}_0$ is an E_λ-sheaf on a curve X_0 over $\mathbb{F}_q$. These groups vanish for $i \neq 0, 1, 2$, and for $i = 0, 2$, they have a simple interpretation (1.8 a,b,c). Furthermore, the Euler-Poincaré characteristic $\chi_c(\mathscr{F}) = \sum(-1)^i \dim \mathrm{H}^i_c(X, \mathscr{F})$ (which is also equal to $\chi(\mathscr{F}) = \sum(-1)^i \dim \mathrm{H}^i(X, \mathscr{F})$) can be computed in terms of X, the rank of $\mathscr{F}$ at the generic points of X, and the ramification of $\mathscr{F}$. The essential result is the following.

Let $\bar{X}$ be a smooth connected projective curve of genus g over an algebraically closed field k, let X be the dense open subset of $\bar{X}$ complementary to a finite set S of points, and let $\mathscr{F}$ be a smooth E_λ-sheaf on X. Let $\chi(X) = 2 - 2g - \#S$ be the Euler–Poincaré characteristic of X, and let $\mathrm{rg}(\mathscr{F})$ be the rank of $\mathscr{F}$. For each point $s \in S$, we define an integer $\mathrm{Sw}_s(\mathscr{F})$, the Swan conductor, measuring the wild ramification of $\mathscr{F}$, and

$$\chi_c(\mathscr{F}) = \mathrm{rg}(\mathscr{F}) \cdot \chi(X) - \sum_{s \in S} \mathrm{Sw}_s(\mathscr{F}) \tag{3.2.1}$$

[29].

3.3

Let $\bar{X}$, $j : X \hookrightarrow \bar{X}$ and $\mathscr{F}$ as in 3.2. The Poincaré duality theorem takes the particularly convenient form: $\mathrm{H}^i(\bar{X}, j_*\mathscr{F})$ and $\mathrm{H}^{2-i}(\bar{X}, j_*(\mathscr{F}^\vee(1)))$ are dual to each other (Duality, 1.3).

3.4

Take $k = \mathbb{F}$, and suppose that X, $\bar{X}$, and $\mathscr{F}$ arise from X_0, $\bar{X}_0$, and $\mathscr{F}_0$ over $\mathbb{F}_q$. If $\mathscr{F}_0$ becomes trivial on a finite covering of X_0, Weil proved that the complex conjugates of the eigenvalues of F^* on $\mathrm{H}^i(\bar{X}, j_*\mathscr{F})$ have absolute value $q^{i/2}$. Such sheaves $\mathscr{F}_0$ correspond to Artin L-functions over the function field of $\bar{X}_0$.

3.5

Under the same hypotheses, or more generally if E_λ is the completion at a place λ of a number field E and if $\mathscr{F}_0$ belongs to an infinite compatible system of v-adic representations (v a place of E), we can compute

$$\prod_i \det(-F^*, \mathrm{H}^i(X, \mathscr{F}))^{(-1)^i}$$

from *local* information on $\mathscr{F}_0$. This is explained in [13]. For trivial $\mathscr{F}_0$ on a finite covering of X_0, it is the expression of the constant in the functional equation of an Artin L-function as a product of local constants.

3.5 Example

Let ψ be the character $\exp\left(\frac{2\pi i}{p} \mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}\right)$ of $\mathbb{F}_q$; we have $\psi(a^p - a) = 1$. Let X_0 be a smooth projective geometrically irreducible curve of genus g over $\mathbb{F}_q$, and let f be a rational function on X_0, i.e. a morphism

$f : X_0 \to \mathbb{P}^1$, not identically equal to ∞. We are interested in the sum

$$S'_f = \sum_{\substack{x \in X_0(\mathbb{F}_q) \\ f(x) \neq \infty}} \psi(f(x)).$$

This sum is zero for f of the form $g^p - g$. This suggests modifying the sum S'_f as follows:

a) for any closed point x of X_0, set $v_x(f) =$ the order of the pole of f at x if $f(x) = \infty$, and $v_x(f) = 0$ otherwise; also set $v_x^*(f) = \inf v_x(f + g^p - g)$ (the infimum being over g).

b) If $v_x^*(f) = 0$, $x \in X_0(\mathbb{F}_q)$, and $f + g^p - g$ is regular at x, put $\psi(f(x)) = \psi((f + g^p - g)(x))$. Finally set

$$S_f = \sum_{x \in X_0(\mathbb{F}_q)}{}' \psi(f(x)), \tag{3.5.1}$$

where $(-)'$ indicates that the sum runs over those x for which $v_x^*(f) = 0$.

We have $S_f = S_{f+g^p-g}$. We shall verify that if f is not of the form $g^p - g + c^{te}$, then

$$|S_f| \leqslant \left(2g - 2 + \sum_{v_x^*(f) \neq 0} [k(x) : \mathbb{F}_q](1 + v_x^*(f))\right) q^{1/2}. \tag{3.5.2}$$

We begin by passing from complex coefficients to ℓ-adic coefficients, replacing ψ by a character of the form $\psi_0 \circ \mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}$, with $\psi_0 : \mathbb{F}_p \hookrightarrow E_\lambda^\times$. For $j : U_0 \hookrightarrow X_0$ the inclusion of the open where $f \neq \infty$, we then prove that

$$S_f = \sum (-1)^i \,\mathrm{tr}(F^*, \mathrm{H}^i(X, j_* \mathscr{F}(\psi f))). \tag{3.5.3}$$

The trace formula reduces this statement to the following (for $x \in X_0(\mathbb{F}_q)$):

a) If $v_x^*(f) = 0$, then F_x^* on $j_* \mathscr{F}(\psi f)$ is $\psi(f(x))$;

b) If $v_x^*(f) \neq 0$, then $j_* \mathscr{F}(\psi f)$ is ramified at x: $(j_* \mathscr{F}(\psi f))_{\bar{x}} = 0$.

Formula a) follows from (1.7.6) if $v_x(f) = 0$, and the general case reduces to this one by replacing f by $f + g^p - g$: up to isomorphism, this does not change $\mathscr{F}(\psi f)_0$ on the open where f and g are regular (1.3).

Formula b) follows from

$$\mathrm{Sw}_x \, \mathscr{F}(f, \psi) = v_x^*(f). \tag{3.5.4}$$

After reduction to the case $v_x(f) = v_x^*(f)$ (hence $p \nmid v_x(f)$) and extension of scalars to $\mathbb{F}$, this formula can be found in [34, 4.4].

Finally, we check that the sheaf $\mathscr{F}(\psi f)$ on U is nonconstant if f is not of the form $g^p - g + c^{te}$: since ψ_0 is injective, the triviality of $\mathscr{F}(\psi f) = \mathscr{F}(\psi_0 f)$ (1.8.ii) is equivalent to that of the $\mathbb{F}_p$-torsor on U defined by $T^p - T - f = 0$. If it were trivial, the torsor on U_0 defined by the same equation would be the inverse image of an $\mathbb{F}_q$-torsor on $\mathrm{Spec}(\mathbb{F}_q)$, with equation $T^p - T - \lambda = 0$, and the $\mathbb{F}_p$-torsor equation $T^p - T - (f - \lambda) = 0$ on U_0 would be trivial and f would therefore be of the form $g^p - g + \lambda$. The $\mathrm{H}^i(\bar{X}, j_* \mathscr{F}(\psi f))$ therefore vanish for $i \neq 1$, and formula (3.5.3) reduces to

$$S_f = -\,\mathrm{tr}(F^*, \mathrm{H}^1(\bar{X}, j_* \mathscr{F}(\psi f))) \tag{3.5.5}$$

and, by (3.2.1) and (3.5.4), the dimension of H^1 is $2g - 2 + \sum_{v_x^*(f) > 0} [k(x) : \mathbb{F}_q](1 + v_x^*(f))$; moreover, $-S_f$ is the sum of this many eigenvalues of F^*, each having complex absolute value $q^{1/2}$.

3.6

Suppose that, for an automorphism σ of X_0, we have

$$f(\sigma x) = -f(x).$$

The sum S_f is then real. If σ is an involution, Poincaré duality allows us to say a little more: set $\mathscr{F}_0 = j_*\mathscr{F}(\psi f)$, $\mathscr{G}_0 = j_*\mathscr{F}(\psi(-f))$, we have

a) $\mathscr{F}_0$ and $\mathscr{G}_0$ are in duality,

b) $\sigma^*\mathscr{F}_0 \xrightarrow{\sim} \mathscr{G}_0$ and $\sigma^*\mathscr{G}_0 \xrightarrow{\sim} \mathscr{F}_0$, for natural isomorphisms such that the composite $\mathscr{F}_0 = (\sigma^2)^*\mathscr{F}_0 \xrightarrow{\sim} \sigma^*\mathscr{G}_0 \xrightarrow{\sim} \mathscr{F}_0$ is the identity.

c) the pairing $\mathscr{F}_0 \otimes \mathscr{G}_0 \to E_\lambda$ satisfies $\sigma^*(f \cdot g) = \sigma^*(f) \cdot \sigma^*(g)$.

Passing to cohomology, we find that $\mathrm{H}^1(X, \mathscr{F})$ and $\mathrm{H}^1(X, \mathscr{G})$ are in perfect duality with values in $E_\lambda(1)$, and that the bilinear form $\alpha \cdot \sigma^*\beta$ on $\mathrm{H}^1(X, \mathscr{F})$ is *alternating*: σ^* acts trivially on $E_\lambda(1)$, and

$$\alpha \cdot \sigma^*\beta = \sigma^*(\alpha \cdot \sigma^*\beta) = \sigma^*\alpha \cdot \beta = -\beta \cdot \sigma^*\alpha.$$

We conclude that $\mathrm{H}^1(X, \mathscr{F})$ is of even dimension (we also check easily that each nonzero $v_x^*(f)$ is odd) and that the eigenvalues of F^* on $\mathrm{H}^1(X, \mathscr{F})$ occur in pairs α and q/α.

3.7 Example

This applies to the Kloosterman sums $\sum_{x \in \mathbb{F}_q^\times} \psi\left(x + \frac{a}{x}\right)$ (take $X_0 = \mathbb{P}^1$, $f = x + \frac{a}{x}$, $\sigma x = -x$; the poles of f are 0 and ∞, and at each of them $v_x^*(f) = 1$; the dimension of H^1 is $-2 + 2 + 2 = 2$). We therefore have (for $a \neq 0$)

$$\sum_{\substack{xy=a \\ x,y \in \mathbb{F}_{q^n}}} \exp\left(\frac{2\pi i}{p} \operatorname{tr}_{\mathbb{F}_{q^n}/\mathbb{F}_p}(x+y)\right) = -(\alpha^n + \bar{\alpha}^n) \qquad , \alpha\bar{\alpha} = q.$$

This result is due to L. Carlitz [9].

We now give an application of a Lang–Weil method.

Proposition 3.8. *Let $P \in \mathbb{F}_q[X_1, \ldots, X_n]$ be a polynomial in n variables, of degree d, which is not of the form $Q^p - Q + C^{te}$. Again setting $\psi(x) = \exp\left(\frac{2\pi i}{p} \operatorname{tr}_{\mathbb{F}_q/\mathbb{F}_p}(x)\right)$, we have*

$$\left|\sum_{x_i \in \mathbb{F}_q} \psi P(x_1, \ldots, x_n)\right| \leqslant (d+1)q^{n-\frac{1}{2}}.$$

For the left-hand side we have the trivial estimate q^n. It is therefore enough to prove 3.8 when $d - 1 < q^{1/2}$; assume therefore that $d < q + 1$, and let P_d be the homogeneous part of degree d of P. Recall:

Lemma 3.9. *A hypersurface of degree d in $\mathbb{P}^r$ ($r \geqslant 1$) cannot contain all rational points over $\mathbb{F}_q$ unless $d \geqslant q + 1$.*

We argue by induction: if there exists a rational hyperplane not entirely contained in the hypersurface (this is not the case for $r = 1$), we apply the induction hypothesis to the intersection of the hypersurface with this hyperplane. Otherwise, the degree d is $\geqslant$ the number of rational hyperplanes, $\geqslant q + 1$.

Let us now distinguish two cases.

Case 1: $p \nmid d$**.** Applying the lemma to P_d, we see that, after making a linear change of variables if necessary, we may assume that $P_d(1,0,\dots,0) \neq 0$, i.e. that the coefficient of X_1^d in P is nonzero. A polynomial in 1 variable

$$S(X) = \sum_{i=0}^{d} a_i x^i,$$

with $a_d \neq 0$ ($p \nmid d$) is never of the form $Q^p - Q + C^{te}$, and the (3.5.2) estimate is reduced to

$$\left|\sum \psi(S(x))\right| \leqslant (d-1)q^{1/2}.$$

Applying this estimate to the partial sums obtained by varying only x_1, we find

$$\left|\sum \psi P(x_1,\dots,x_n)\right| = \left|\sum_{x_2,\dots,x_n}\sum_{x_1} \psi P(x_1,\dots,x_n)\right| \leqslant q^{n-1}(d-1)q^{1/2}$$

as promised.

Case 2: $p \mid d$ **($d > 0$).** If P_d is a p-th power, replacing P by $P - (P_d - P_d^{1/p})$ does not change the sum under consideration and lowers the degree: this case is disposed of by induction on d. Otherwise, the differential of P_d is not identically zero; applying 3.9, one can assume, after a linear change of variables if necessary, that it is nonzero at the point $(1,0,\dots,0)$. If we write

$$P_d = \sum_{i=0}^{d} X_1^{d-i} S_i(X_2,\dots,X_n),$$

this means that the linear form S_1 is not identically zero.

The form S_0 is constant, and by replacing P with $P - (S_0 X_1^d - S_0^{1/p} X_1^{d/p})$, we can assume that it is zero. Finally, denote by $-\lambda$ the coefficient of X_1^{d-1} in P. For fixed $x_2,\dots,x_n$, we have

$$P(X_1,x_2,\dots,x_n) = (S_1(x_2,\dots,x_n) - \lambda)X_1^{d-1} + \text{terms of lower degree in } X_1$$

and if $S_1(x_2,\dots,x_n) \neq \lambda$, we therefore have

$$\left|\sum_{x_1} \psi P(x_1,\dots,x_n)\right| \leqslant (d-2)q^{1/2}.$$

In total,

$$\begin{aligned}
\left|\sum \psi P(x_1,\dots,x_n)\right| &\leqslant \left|\sum_{S_1=\lambda} \psi P(x_1,\dots,x_n)\right| + \sum_{S_1(X_2,\dots,X_n)\neq\lambda} \left|\sum_{x_1} \psi P(x_1,\dots,x_n)\right| \\
&\leqslant q^{n-1} + (q^{n-1} - q^{n-2})(d-2)q^{1/2} \\
&< (d-2)q^{n-1/2} + q^{n-1} \\
&< (d-1)q^{n-1/2}.
\end{aligned}$$

Another result of this nature is given by R.A. Smith [37].

4 Gauss and Jacobi sums

4.1

Let k be a finite field of characteristic p, let χ be a character of $k^\times$, and let ψ be a nontrivial character of the additive group of k. We take the following as the definition of Gauss sums:

$$\tau(\chi,\psi) = -\sum_{x\in k^\times} \psi(x)\chi^{-1}(x) \tag{4.1.1}$$

(note the sign). Classically, χ and ψ are complex-valued. We take them to have values in $E_\lambda^\times$ (see the proof of 1.15). Regard $-\tau(\chi,\psi)$ as a sum over the rational points of the scheme $\mathbb{G}_m$ over k. By Section 1, we have $-\tau(\chi,\psi) = \mathrm{tr}(F^*, \mathrm{H}_c^*(\mathbb{G}_m, \mathscr{F}(\psi\chi^{-1}))$. Moreover,

Proposition 4.2. *The cohomology of $\mathbb{G}_m$ with coefficients in $\mathscr{F}(\psi\chi^{-1})$ satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq 1$, *and* $\dim \mathrm{H}_c^1 = 1$.

(ii) F^*, *acting on* H_c^1, *is multiplication by* $\tau(\chi,\psi)$.

(iii) If χ *is nontrivial, we have* $\mathrm{H}_c^\bullet \xrightarrow{\sim} \mathrm{H}^\bullet$.

Proof. For any n prime to p, denote by K_n the $\boldsymbol{\mu}_n$-torsor on $\mathbb{G}_m$ defined by the exact Kümmer sequence $0 \to \boldsymbol{\mu}_n \to \mathbb{G}_m \xrightarrow{x^n} \mathbb{G}_m \to 0$. If k has q elements, then over k one has $\boldsymbol{\mu}_{q-1} = k^\times$, and the Lang torsor on $\mathbb{G}_m/k$ is K_{q-1}. Hence $\mathscr{F}(\chi^{-1}) = \chi(K_{q-1})$. Assertion 4.2(ii) follows from (i,iii), which in turn follow from the following geometric statement, where $\mathscr{F}(\psi)$ denotes the sheaf on $\mathbb{G}_m/\mathbb{F}$ obtained by extending scalars from k to $\mathbb{F}$ in $\mathscr{F}(\psi)$ on $\mathbb{G}_m/k$ (1.8.b). □

Proposition 4.3. *Let* $\chi : \boldsymbol{\mu}_n \to E_\lambda^\times$. *The cohomology of* $\mathbb{G}_m$ *with coefficients in* $\mathscr{F}(\psi)\otimes\chi(K_n)$ *satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq 1$, *and* $\dim \mathrm{H}_c^1 = 1$.

(ii) If χ *is nontrivial, we have* $\mathrm{H}_c^\bullet \xrightarrow{\sim} \mathrm{H}^\bullet$.

Proof. The sheaf $\mathscr{F}(\psi)$ is the restriction to $\mathbb{G}_m$ of a sheaf locally constant on $\mathbb{G}_a$, wildly ramified at infinity, with Swan conductor 1. The sheaf $\chi(K_n)$ is constant if $\chi = 1$; if $\chi \neq 1$, it is tamely ramified at 0 and ∞.

The sheaf $\mathscr{F}(\psi)\otimes\chi(K_n)$ is therefore ramified at ∞; applying 1.18(b,c), we find that $\mathrm{H}_c^i(\mathbb{G}_m, \mathscr{F}(\psi)\otimes\chi(K_n)) = 0$ for $i \neq 1$. If $\chi \neq 1$, it is ramified at 0 and ∞, and (ii) follows from 1.19.a.

The Swan conductors are 0 at 0 and 1 at ∞. By (3.2.1), the Euler–Poincaré characteristic is therefore -1, which completes the proof. □

4.4 Remark

If χ is nontrivial, 4.2(iii) and Poincaré duality show that $\mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\psi\chi^{-1}))$ and $\mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\psi^{-1}\chi))$ are in duality (duality with values in $E_\lambda(-1)$). We therefore have $\tau(\chi,\psi)\cdot\tau(\chi^{-1},\psi^{-1}) = q$, i.e. $|\tau(\chi,\psi)| = q^{1/2}$.

4.5

If k is an extension of degree N of $\mathbb{F}_q$, we can also regard (4.1.1) as a sum in N variables over $\mathbb{F}_q$. More generally, let k be an étale algebra over $\mathbb{F}_q$, of degree N over $\mathbb{F}_q$. It is a product of fields k_i of degrees N_i, and we set

$$\varepsilon(k) = (-1)^{\sum(N_i+1)}. \tag{4.5.1}$$

This is the sign of the permutation of $S = \hom_{\mathbb{F}_q}(k, \mathbb{F})$ induced by the Frobenius substitution $\varphi \in \mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$.

Let $\psi : \mathbb{F}_q \to E_\lambda^\times$ be a nontrivial character, and let $\chi : k^\times \to E_\lambda^\times$. We set

$$\tau_{\mathbb{F}_q}(\chi, \psi) = (-1)^N \sum_{x \in k^\times} \psi \operatorname{tr}_{k/\mathbb{F}_q}(x) \cdot \chi^{-1}(x). \tag{4.5.2}$$

If χ has coordinates $\chi_i : k_i^\times \to E_\lambda^\times$, we have the elementary identity

$$\tau_{\mathbb{F}_q}(\chi, \psi) = \varepsilon(k) \prod \tau(\chi_i, \psi \circ \operatorname{tr}_{k_i/\mathbb{F}_q}). \tag{4.5.3}$$

After some preliminaries, we give in 4.10 a cohomological interpretation of the sums (4.5.2). In 4.12, we will interpret the Hasse–Davenport identity as a twisted form (cf. 1.12) of the following special case of (4.5.3): for $k = \mathbb{F}_q^N$, and χ a character of $\mathbb{F}_q^\times$, we have $\tau_{\mathbb{F}_q}(\chi \circ N_{k/\mathbb{F}_q}, \psi) = \tau(\chi, \psi)^N$.

4.6

Recall that for M a finitely generated projective module over a ring A, the functor $\mathrm{Spec}(B) \mapsto M \otimes_A B$ from affine schemes over $\mathrm{Spec}(A)$ to Ens is represented by the affine scheme $\mathbb{V}(M^\vee)$ (EGA notation), the spectrum of $\mathrm{Sym}_A^\bullet(M^\vee)$. If $M = A^n$, this is the standard affine space of dimension n.

Suppose that M is a unital A-algebra, and let $V = \mathbb{V}(M^\vee)$. By definition, for any B equipped with an A-algebra structure, the set $V(B)$ of points of V with coordinates in B is $M \otimes_A B$. Since the functor $B \mapsto M \otimes_A B$ takes values in unital rings, V is a unital ring scheme over $\mathrm{Spec}(A)$. The norm and trace morphisms $M \otimes_A B \to B$ are functorial in B; they therefore correspond to morphisms of schemes N and T from V to $\mathbb{G}_a$. We shall consider the following schemes obtained from V:

a) V^* is the open subscheme of invertible elements of V (a group scheme under $\cdot$);

b) W is the hyperplane of equation $T = 0$ and $W^* = W \cap V^*$;

c) P is the hyperplane at infinity $V \smallsetminus \{0\}/\mathbb{G}_m$ of the affine space V over $\mathrm{Spec}(A)$. If Q is the hyperplane at infinity of W, $W^*/\mathbb{G}_m$ and $V^*/\mathbb{G}_m$ are open subsets of the projective spaces Q and P over $\mathrm{Spec}(A)$.

$$\begin{array}{ccc} W^* & \hookrightarrow & V^* \\ \downarrow{\scriptstyle\pi} & & \downarrow{\scriptstyle\pi} \\ W^*/\mathbb{G}_m & \hookrightarrow & V^*/\mathbb{G}_m \end{array} \qquad \subset \qquad \begin{array}{ccc} W \smallsetminus \{0\} & \hookrightarrow & V \smallsetminus \{0\} \\ \downarrow{\scriptstyle\pi} & & \downarrow{\scriptstyle\pi} \\ Q & \hookrightarrow & P. \end{array} \tag{4.6.1}$$

If $M = A^I$, I a finite set, then $V \simeq \mathbb{G}_a^I$, $V^* = \mathbb{G}_m^I$, $V^*/\mathbb{G}_m$ is the torus $\mathbb{G}_m^I/(\mathbb{G}_m \text{ diagonal})$, N and T are given by $\prod x_i$ and $\sum x_i$, and Q is therefore the projective hyperplane with equation $\sum x_i = 0$ in P.

The case of interest is that in which M is finite étale over A. The previous description then applies locally to $\mathrm{Spec}(A)$ (for the étale topology) and we can write $V^* = \mathbb{G}_m^I$, for I a locally constant sheaf of finite sets on $\mathrm{Spec}(A)$. If, for example, A is a field with algebraic closure $\bar{A}$, and M is a separable A-algebra, we put $I = \hom_A(M, \bar{A})$, and over $\bar{A}$ we have $V \sim \mathbb{G}_a^I$. Via this isomorphism, N and T are given by $\prod x_i$ and $\sum x_i$.

4.7 The Kummer torsor

Let S be a scheme, let n be an integer invertible on S, and let G be a commutative group scheme with connected fibers (written multiplicatively) over S. The sequence

$$0 \longrightarrow G_n \longrightarrow G \xrightarrow{x^n} G \longrightarrow 0$$

is exact. It defines a G_n-torsor $K_n(G)$ (or simply K_n) on G. For $\chi : G_n \to E_\lambda^\times$ a homomorphism from the étale S-sheaf G_n to the constant sheaf $E_\lambda^\times$, we denote by $\mathscr{K}_n(\chi)$ the E_λ-sheaf $\chi^{-1}(K_n)$.

The K_n torsors form an inverse system of torsors under the projective system of groups G_n (transition morphisms $x \mapsto x^d : G_{nd} \to G_n$). This expresses the commutativity of the diagrams

$$\begin{array}{ccccccccc} 0 & \longrightarrow & G_{nd} & \longrightarrow & G & \xrightarrow{x^{nd}} & G & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle x^d} & & \downarrow{\scriptstyle x^d} & & \| & & \\ 0 & \longrightarrow & G_n & \longrightarrow & G & \xrightarrow{x^n} & G & \longrightarrow & 0. \end{array}$$

We therefore have $\mathscr{K}_n(\chi) = \mathscr{K}_{nd}(\chi \circ x^d)$.

4.8

Apply the construction of 4.6 with $A = \mathbb{F}_q$ and with $M = k$ an étale algebra over $\mathbb{F}_q$. In accordance with our general conventions, we denote by a subscript 0 the $\mathbb{F}_q$-schemes denoted $V, V^*, \ldots$ in 4.6. The same letters without a subscript denote the schemes over $\mathbb{F}$ obtained by extension of scalars.

We put $I = \hom(k, \mathbb{F})$; then $V \sim \mathbb{G}_a^I$ and T is given by $(x_i) \mapsto \sum x_i$.

$$\mathscr{F}(\psi \circ \mathrm{tr}_{k/\mathbb{F}_q}) = T^* \mathscr{F}(\psi) = \bigotimes_i \mathrm{pr}_i^* \mathscr{F}(\psi). \tag{4.8.1}$$

Since $V^* \sim \mathbb{G}_m^I$, a character χ of $V_n^* \sim \boldsymbol{\mu}_n^I$, with values in $E_\lambda^\times$, amounts to a family $(\chi_i)_{i \in I}$ of characters of $\boldsymbol{\mu}_n$ indexed by I. It is defined over $\mathbb{F}_q$ if, for all $\sigma \in \mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$, we have $\chi_{\sigma_i} = \chi_i \circ \sigma^{-1}$. It then defines a E_λ-sheaf $\mathscr{K}_n((\chi_i)_{i \in I})$ on V_0^*. If the product of the χ_i is trivial, χ factors through a character of $(V_0^*/\mathbb{G}_m)_n$ and $\mathscr{K}_n((\chi_i)_{i \in I})$ is the inverse image of a sheaf, denoted in the same way, on $V_0^*/\mathbb{G}_m$. This construction is related to the Lang torsor as follows.

Lemma 4.9. *(i) For n sufficiently divisible, we have a commutative diagram*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & (V_0^*)_n & \longrightarrow & V_0^* & \xrightarrow{x^n} & V_0^* & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \tau} & & \downarrow{\scriptstyle \tau} & & \| & & \\ 0 & \longrightarrow & k^\times & \longrightarrow & V_0^* & \xrightarrow{\mathfrak{L}} & V_0^* & \longrightarrow & 0; \end{array}$$

if χ is a character $\chi : k^\times \to E_\lambda^\times$, we have $\mathscr{F}(\chi) = \mathscr{K}_n(\chi \circ \tau)$.

(ii) For sufficiently divisible n, τ identifies $k^\times$ with the coinvariants of $\mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$ acting on $V_n^* \sim \boldsymbol{\mu}_n^I$.

(iii) For k a field, $N = [k : \mathbb{F}_q]$, $\omega \in I$ an embedding of k in $\mathbb{F}$, and $n = q^N - 1$, the index component ω of $\chi \circ \tau$ is $\chi \circ \omega^{-1}$.

(iv) If χ is nontrivial on each factor of k, the $(\chi \circ \tau)_i$ are all nontrivial.

It is enough to prove the lemma when k is a field of degree N over $\mathbb{F}_q$. In this case, n is "sufficiently divisible" if $q^N - 1 \mid n$. Choose an embedding ω of k into $\mathbb{F}$; then I is identified with $\mathbb{Z}/N$: the element $i \in \mathbb{Z}/N$ corresponds to $\omega_i = \omega^{q^i}$. Via the isomorphism $V_0^*(\mathbb{F}) = \mathbb{F}^{\times I}$, we have

a) $x \in k^\times$ corresponds to $(\omega_i(x)) \in \mathbb{F}^{\times I}$;

b) $F((x_i)_{i\in\mathbb{Z}/N}) = (x_{i-1}^q)_{i\in\mathbb{Z}/N}$.

A character $\chi = (\chi_i)$ of $V_n^* \sim \boldsymbol{\mu}_n^I$ is defined over $\mathbb{F}_q$ if $\chi_{-i}(x) = \chi_0(x^{q^i})$ $(i \in \mathbb{Z})$. If $q^N - 1 \mid n$, there are $q^N - 1$ such characters: χ_0 factors through $\boldsymbol{\mu}_{q^N-1}$ and determines the χ_i. To prove (ii), it is therefore sufficient to verify (i) and (iii) (or its corollary (iv)) which ensures that $\chi \mapsto \chi \circ \tau$ is injective.

Let us prove (i) and (iii) for $n = q^N - 1$. Write the group of endomorphisms of $V^* \sim \mathbb{G}_m^I$ additively; in particular, denote by n the $x \mapsto x^n$ operator. If α is the cyclic permutation operator $(x_i) \mapsto (x_{i-1})$, we have $q^N - 1 = (q\alpha)^N - 1 = (q\alpha - 1)((q\alpha)^{N-1} + \cdots + 1)$. This determines τ: we have $\tau((x_i)) = \prod_{0\leqslant j<N} x_{i-j}^{q^j}$, and the induced map from $(V_0^*)_n$ to $k^\times$ is [as in a) above] $(x_i) \mapsto \prod x_{-i}^{q^i}$; hence (iii) follows.

Proposition 4.10. *The cohomology of V^* with coefficients in $\mathscr{F}(\chi^{-1} \cdot \psi \operatorname{tr}_{k/\mathbb{F}_q})$ satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq N$, *and* $\dim \mathrm{H}_c^N = 1$.

(ii) On H_c^N, F^* *is multiplication by* $\tau_{\mathbb{F}_q}(\chi, \psi)$.

(iii) If χ *is nontrivial on each factor of* k, *we have* $\mathrm{H}_c^\bullet \xrightarrow{\sim} \mathrm{H}^\bullet$.

Applying 4.9(i), over $\mathbb{F}$, if $\chi \circ \tau = (\chi_i)_{i\in I}$, we have $\mathscr{F}(\chi) = \mathscr{K}_n((\chi_i)_{i\in I})$. Points (i) and (iii) therefore follow from the following more geometric statement (for (iii), apply 4.9(iv)), and (ii) follows from the trace formula:

Proposition 4.11. *Let $(\chi_i)_{i\in I}$ be a family of characters of $\boldsymbol{\mu}_n(\mathbb{F})$. The cohomology of $V^* \simeq \mathbb{G}_m^I$ with coefficients in $\mathscr{K}_n((\chi_i)_{i\in I}) \otimes \mathscr{F}(\psi \operatorname{tr}_{k/\mathbb{F}_q})$ satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq N$, *and* $\dim \mathrm{H}_c^N = 1$.

(ii) If the χ_i are all nontrivial, we have $\mathrm{H}_c^\bullet \xrightarrow{\sim} \mathrm{H}^\bullet$.

We have $V^* \sim \mathbb{G}_m^I$, $\mathscr{K}((\chi_i)) = \bigotimes_i \mathrm{pr}_i^* \chi_i(K_n(\mathbb{G}_m))$ and $\mathscr{F}(\psi \circ \operatorname{tr}_{k/\mathbb{F}_q}) = T^*\mathscr{F}(\psi) = \bigotimes_i \mathrm{pr}_i^* \mathscr{F}(\psi)$. The Künneth formula then applies, and 4.11 follows from 4.3.

4.12

From 2.4* and (Cycle, 1.3 Example 2), it also follows that the group of permutations σ of I such that $\chi_i = \chi_{\sigma i}$ $(i \in I)$ acts on H_c^N by multiplication by the sign $\varepsilon(\sigma)$. We deduce a second proof of the identity of Hasse-Davenport.

If χ is a character of $\mathbb{F}_q^\times$, the fact that, over $\mathbb{F}$, N is given by $(x_i) \mapsto \prod x_i$ gives $\mathscr{F}(\chi \circ N) = N^*\mathscr{F}(\chi) = \bigotimes_i \mathrm{pr}_i^* \mathscr{F}(\chi)$ and the Künneth formula gives

$$\mathrm{H}_c^\bullet(V^*, \mathscr{F}(\chi^{-1} \circ N, \psi \circ \mathrm{tr})) \sim \mathrm{H}_c^\bullet(\mathbb{G}_m, \mathscr{F}(\chi^{-1}\psi))^{\otimes I},$$

where the tensor product on the right-hand side is taken in the sense of 2.4*. Since $\mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\psi\chi^{-1}))$ is of dimension 1, its tensor power $\otimes I$, in the ordinary sense, depends only on the cardinality n of I. Applying (Cycle, 1.3 Example 2), we obtain a canonical isomorphism

$$\mathrm{H}_c^N\left(V^*, \mathscr{F}(\chi^{-1} \circ N) \otimes T^*\mathscr{F}(\psi)\right) \sim \mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\psi\chi^{-1})) \otimes_{\mathbb{Z}} \bigwedge^N \mathbb{Z}^I. \tag{4.12.1}$$

To compute the action of F^*, it is most convenient to adopt the Galois point of view and observe that, since the isomorphism (4.12.1) is canonical, it is compatible with the transport-of-structure action of $\mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$. If $\varepsilon(k)$ is the sign of the permutation φ of I, we find that

$$\begin{aligned}\tau_{\mathbb{F}_q}(\chi\circ N_{k/\mathbb{F}_q},\psi) &= \mathrm{tr}\left(\varphi^{-1}, \mathrm{H}_c^N\left(V^*, N^*\mathscr{F}(\chi^{-1})\otimes T^*\mathscr{F}(\psi)\right)\right)\\ &= \varepsilon(k)\,\mathrm{tr}\left(\varphi^{-1}, \mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\chi^{-1}\psi))\right)^N\\ &= \varepsilon(k)\tau(\chi,\psi)^N,\end{aligned}$$

that is,

$$\tau_{\mathbb{F}_q}\left(\chi\circ N_{k/\mathbb{F}_q},\psi\right) = \varepsilon(k)\tau(\chi,\psi)^N. \tag{4.12.2}$$

If k is a field, φ is a cyclic permutation of I, $\varepsilon(k) = (-1)^{N+1}$, and we obtain the Hasse–Davenport identity:

$$\tau\left(\chi\circ N_{k/\mathbb{F}_q}, \psi\circ \mathrm{tr}_{k/\mathbb{F}_q}\right) = (-1)^{N+1}\tau_{\mathbb{F}_q}\left(\chi\circ N_{k/\mathbb{F}_q},\psi\right) = \tau(\chi,\psi)^N.$$

Lemma 4.13. *For $\chi\circ\tau = (\chi_i)_{i\in I}$ as in 4.9(i), the following conditions are equivalent*

(i) $\chi|\mathbb{F}_q^\times$ is trivial.

(ii) The product of χ_i is trivial.

(iii) $\mathscr{F}(\chi) = \mathscr{K}_n((\chi_i))$ is the inverse image of a unique sheaf on $V^*/\mathbb{G}_m$.

(iv) The inverse image of $\mathscr{F}(\chi)$ on $\mathbb{G}_m$ (embedded in V^* by the structural morphism $\mathbb{F}_q\to k$) is trivial.

(i)⇒(ii). We regard χ as a character of $V^*/\mathbb{G}_m(\mathbb{F}_q) = k^\times/\mathbb{F}_q^\times$, and we take $\mathscr{F}(\chi)$ on $V^*/\mathbb{G}_m$.
(ii)⇒(iii). Likewise, we regard (χ_i) as a character of $(V^*/\mathbb{G}_m)_n$. The uniqueness in (iii) follows from the fact that V^* is a fibration with connected fibers over $V^*/\mathbb{G}_m$, and (iii)⇒(iv) is trivial.
(iv)⇒(i), (ii). This inverse image is $\mathscr{F}(\chi|\mathbb{F}_q^\times)$ and $\mathscr{K}_n(\prod\chi_i)$.

4.14

Let k be an étale algebra over $\mathbb{F}_q$, of dimension $N+1$, and let χ be a nontrivial character of $k^\times$, trivial on $\mathbb{F}_q^\times$. The Jacobi sum $J(\chi)$ is defined by

$$J(\chi) = (-1)^{N-1}\sum_{\substack{x\in k^\times/\mathbb{F}_q^\times\\ \mathrm{tr}(x)=0}}\chi^{-1}(x). \tag{4.14.1}$$

We have the following identity between Gauss sums and Jacobi sums.

Proposition 4.15. *For χ as above and ψ a nontrivial additive character of $\mathbb{F}_q$, we have*

$$qJ(\chi) = \tau_{\mathbb{F}_q}(\chi,\psi).$$

In the special case $k=\mathbb{F}_q^n$, formula (4.5.3) rewrites this as

$$qJ(\chi) = \prod_i \tau(\chi_i,\psi), \tag{4.15.1}$$

or equivalently as

$$\chi_0(-1)J(\chi) = (-1)^{N-1} \sum_{\substack{x_1,\dots,x_N \in \mathbb{F}_q^\times \\ \sum x_i = 1}} \prod_{i=1}^{N} \chi_i^{-1}(x_i) = \tau(\chi_0^{-1},\psi)^{-1} \prod_{i=1}^{N} \tau(\chi_i,\psi)$$

$$\chi_0^{-1} = \prod_{i=1}^{N} \chi_i$$

Proof of 4.15.

$$\tau_{\mathbb{F}_q}(\chi,\psi) = (-1)^{N+1} \sum_{x\in k^\times} \chi(x)^{-1}\psi\,\mathrm{tr}(x) = (-1)^{N+1} \sum_{x\in k^\times/\mathbb{F}_q^\times} \chi(x)^{-1} \sum_{\lambda\in\mathbb{F}_q^\times} \psi(\lambda\,\mathrm{tr}\,x).$$

The sum $\sum_{\lambda\in\mathbb{F}_q^\times} \psi(\lambda\,\mathrm{tr}\,x)$ equals $q-1$ if $\mathrm{tr}(x)=0$, and -1 if $\mathrm{tr}(x)\neq 0$. Therefore,

$$\tau_{\mathbb{F}_q}(\chi,\psi) = (-1)^{N+1} \left(q \sum_{\substack{x\in k^\times/\mathbb{F}_q^\times \\ \mathrm{tr}(x)=0}} \chi(x)^{-1} - \sum_{x\in k^\times/\mathbb{F}_q^\times} \chi(x)^{-1} \right).$$

The second term on the right-hand side is zero, since it is the sum of the values of a character, and 4.15 follows. □

Proposition 4.15 therefore has the following cohomological counterpart:

Proposition 4.16. *The cohomology of $W^*/\mathbb{G}_m$ (4.6, 4.8) with coefficients in $\mathscr{F}(\chi^{-1})$ satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq N-1$, *and* $\dim \mathrm{H}_c^{N-1} = 1$.

(ii) On H_c^{N-1}, F^* *is multiplication by* $J(\chi)$.

(iii) H_c^{N-1} *is canonically isomorphic to* $\mathrm{H}_c^{N+1}\left(V^*, \mathscr{F}(\chi^{-1}, \psi\circ \mathrm{tr}_{k/\mathbb{F}_q})\right)(1)$.

Assertion (ii) follows from (i) and the trace formula; (i) and (iii) follow from 4.11 and the following more geometric statement.

Proposition 4.17. *Let $(\chi_i)_{i\in I}$ be a family of characters of $\boldsymbol{\mu}_n(\mathbb{F})$ that are not all trivial and whose product is* 1. *Then* $\mathrm{H}_c^{i-1}\left(W^*/\mathbb{G}_m, \mathscr{K}_n((\chi_i)_{i\in I})\right)$ *is canonically isomorphic to* $\mathrm{H}_c^{i+1}\left(V^*, \mathscr{K}_n((\chi_i)_{i\in I}) \otimes \mathscr{F}(\psi\,\mathrm{tr}_{k/\mathbb{F}_q})\right)(1)$.

The first line of 4.15, namely the projection formula, gives (π as in (4.6.1)).

$$\mathsf{R}^i\pi_!\left(\mathscr{K}_n((\chi_i)_{i\in I}) \otimes \mathscr{F}(\psi\,\mathrm{tr}_{k/\mathbb{F}_q})\right) = \mathscr{K}_n((\chi_i)_{i\in I}) \otimes \mathsf{R}^i\pi_!\mathscr{F}(\psi\,\mathrm{tr}_{k/\mathbb{F}_q}) \tag{4.17.1}$$

(on $V^*/\mathbb{G}_m$). Let us compute the sheaves $\mathsf{R}^i\pi_!\mathscr{F}(\psi\,\mathrm{tr}_{k/\mathbb{F}_q}) = \mathsf{R}^i\pi_! T^*\mathscr{F}(\psi)$. Let $v_0 : \widetilde{V}_0 \to V_0$ be the blow-up of V_0 at $\{0\}$. In the diagram

$$\begin{array}{ccccc} V_0 \smallsetminus \{0\} & \hookrightarrow & \widetilde{V}_0 & \xrightarrow{v_0} & V_0 \\ & \searrow^{\pi} & \downarrow{\scriptstyle \bar{\pi}} & & \\ & & P_0 & & \end{array} \tag{4.17.2}$$

π is a fibration whose fibers are punctured affine lines, $\widetilde{V}_0$ is the corresponding line bundle, and $Z_0 = v_0^{-1}(0)$ is its section 0. The $T^*\mathscr{F}(\psi)$ sheaf on $V_0 \smallsetminus \{0\}$ extends to $(Tv_0)^*\mathscr{F}(\psi)$ on $\widetilde{V}_0$, and the restriction of this sheaf to Z_0 is the constant sheaf E_λ, because $T_0 v_0$ vanishes on Z_0. The long exact sequence for cohomology with compact supports is therefore

$$\longrightarrow \mathsf{R}^i\pi_!T^*\mathscr{F}(\psi) \longrightarrow \mathsf{R}^i\bar{\pi}_!(Tv_0)^*\mathscr{F}(\psi) \longrightarrow (E_\lambda \text{ for } i=0,\ 0 \text{ otherwise}) \longrightarrow \tag{4.17.3}$$

The inverse image $\bar{\pi}^{-1}(Q_0)$ is the blow-up $\widetilde{W}_0$ of W_0 at 0; $T_0 v_0$ vanishes on $\widetilde{W}_0$; hence $(Tv_0)^*\mathscr{F}(\psi) = E_\lambda$ on $\bar{\pi}^{-1}(Q_0)$ and, since $\bar{\pi}$ is an affine-line fibration,

$$\mathsf{R}^i\bar{\pi}_!(Tv_0)^*\mathscr{F}(\psi)|Q = \begin{cases} 0 & \text{for } i \neq 2 \\ E_\lambda(-1) & \text{for } i = 2. \end{cases} \tag{4.17.4}$$

If $x \in P$, $x \notin Q$, the line $D = \pi^{-1}(x)$ is mapped isomorphically onto $\mathbb{G}_a$ by Tv_0; therefore (by 2.7*)

$$\mathrm{H}_c^\bullet(D, (Tv_0)^*\mathscr{F}(\psi)) = \mathrm{H}_c^\bullet(\mathbb{G}_a, \mathscr{F}(\psi)) = 0 \qquad \text{and}$$
$$\mathsf{R}^i\bar{\pi}_!(Tv_0)^*\mathscr{F}(\psi) = \begin{cases} 0 & \text{for } i \neq 2 \\ E_\lambda(-1)_Q & \text{for } i = 2 \end{cases} \tag{4.17.5}$$

Combining (4.17.3) and (4.17.5), we finally find

$$\mathsf{R}^i\pi_!T^*\mathscr{F}(\psi) = \begin{cases} 0 & \text{for } i \neq 1, 2 \\ E_\lambda & \text{for } i = 1 \\ E_\lambda(-1)_Q & \text{for } i = 2. \end{cases} \tag{4.17.6}$$

4.18

Let us compute the compactly supported cohomology of the sheaf $\mathscr{K}_n((\chi_i)) \otimes T^*\mathscr{F}(\psi)$ on V^* using the Leray spectral sequence of $\pi : V^* \to V^*/\mathbb{G}_m$. Applying (4.17.1) and (4.17.6), we obtain as initial terms $E_2^{p,1} = \mathrm{H}_c^p(V^*/\mathbb{G}_m, \mathscr{K}_n(\chi_i)) = 0$ (because $(\chi_i) \neq 1$) and $E_2^{p,2} = \mathrm{H}_c^p(W^*/\mathbb{G}_m, \mathscr{K}(\chi_i))(-1)$.

The spectral sequence reduces to an isomorphism, and 4.17 follows.

4.19

The sheaves $\mathscr{K}_n((\chi_i)_{i\in I})$ make sense over any field, or even over any base scheme. This allows us to generalize 4.16(i). With the notation of 4.6, suppose that M is finite étale of constant rank N over A, and let n be invertible in A. Let a be the projection of $W^*/\mathbb{G}_m$ to $\mathrm{Spec}(A)$. Let also χ be a character (morphism of sheaves) $\chi : (V^*/\mathbb{G}_m)_n \to E_\lambda^\times$, and $\mathscr{K}_n(\chi)$ the sheaf corresponding. Locally for the étale topology, we can regard χ as a family of characters $(\chi_i)_{i\in I}$ of $\boldsymbol{\mu}_n$, whose product is trivial.

Proposition 4.20. *(i) If χ is everywhere nontrivial, we have $\mathsf{R}^i a_!(\mathscr{K}_n(\chi)) = 0$ for $i \neq N-2$, and $\mathsf{R}^{N-2}a_!(\mathscr{K}_n(\chi))$ is smooth of rank 1.*

(ii) An automorphism σ of M which preserves χ acts on this $\mathsf{R}^{N-2}a_!$ by multiplication by the sign $\varepsilon(\sigma)$ of σ, seen as a permutation of I.

(iii) If the χ_i are all nontrivial, we have $\mathsf{R}a_! \xrightarrow{\sim} \mathsf{R}a_*$.

Proof. The scheme $W^*/\mathbb{G}_m$ is the complement of a relative normal-crossings divisor in the scheme Q, which is proper and smooth over $\mathrm{Spec}(A)$, and $\mathscr{K}_n(\chi)$ is locally constant on $W^*/\mathbb{G}_m$ and is tamely ramified along the boundary. Consequently, the $\mathsf{R}^i a_!$ and $\mathsf{R}^i a_*$ are smooth sheaves, compatible with every change of base. A standard argument then reduces us to the case in which A is a finite field, and (i) follows from 4.17 and 4.11(i). To prove (ii), use the compatibility of the action of σ with the isomorphism 4.17, together with 4.12. For (iii), note that if the χ_i are all nontrivial, then $\mathscr{K}_n(\chi)$ on $W^*/\mathbb{G}_m \subset Q$ is ramified along every divisor at infinity, and apply 1.19.1. □

5 Hecke characters

5.1

Let F be a number field (of finite degree over $\mathbb{Q}$) and let k be a field of characteristic 0. Here are several equivalent descriptions of an *algebraic* homomorphism $\alpha : F^\times \to k^\times$.

(i) Let $\{e_i\}$ be a basis of F over $\mathbb{Q}$. The homomorphism α is algebraic if it is given by a formula $\alpha(\sum x^i e_i) = A(x^i)$, $A \in k(X^i)$.

This means that α agrees on $F^\times$ with a rational map defined over k from $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m)$ to $\mathbb{G}_m$. By the Zariski density of $F^\times$, and because a rational homomorphism is defined everywhere, such a map is a homomorphism of group schemes:

(ii) α is induced by a homomorphism of group schemes over k

$$\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m) \otimes_{\mathbb{Q}} k \to \mathbb{G}_m.$$

If k is a number field, the adjunction property of restriction of scalars R shows that this is equivalent to

(iii) (k a number field) α is induced by a homomorphism of group schemes over $\mathbb{Q}$: $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m) \to \mathrm{R}_{k/\mathbb{Q}}(\mathbb{G}_m)$.

Let $\bar{k}$ be an algebraic closure of k, and let $I = \hom(F, \bar{k})$. Over $\bar{k}$, the character group of $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m)$ is $\mathbb{Z}^I$, with basis the embeddings of F into $\bar{k}$. The characters defined over k are those invariant under $\mathrm{Gal}(\bar{k}/k)$; they can be described either as those sending $F^\times$ into $k^\times \subset \bar{k}$, or in terms of orbits of $\mathrm{Gal}(\bar{k}/k)$ on I, which in turn correspond to the factors of $F \otimes k$.

(iv) α is of the form $\alpha = \prod_{\omega \in I} \omega^{n_\omega}$. The allowed families of exponents (n_ω) are those such that $n_\omega = n_{\omega'}$ for ω and ω' in the same orbit of $\mathrm{Gal}(\bar{k}/k)$. Equivalently, these are those such that $\prod_\omega \omega(x)^{n_\omega} \in k$ for every $x \in F$.

(v) Write $F \otimes k = \prod_{j \in J} F_j$, where the F_j are fields. Then α can be written $\alpha = \prod N_{F_j/k}^{m_j}$.

In the special case where k contains a normal closure of F, these expressions become: $\alpha = \prod \omega^{n_\omega}$, where ω runs over the $[F : \mathbb{Q}]$ embeddings of F into k.

5.2

Let k be a number field, and let α be an algebraic homomorphism from $F^\times$ to $k^\times$. There is then a unique homomorphism, also denoted α, of the group $I(F)$ of fractional ideals of F to that of k, such that $\alpha((x)) = (\alpha(x))$. Uniqueness follows from the fact that every ideal a has a power that is principal, and that $I(k)$ is torsion-free. To prove existence, we use for example 5.1(v): we have $\alpha(a) = \prod N_{F_j/k}^{m_j}((a))$.

5.3

Let us recall the definition of algebraic Hecke characters, also called Weil Hecke characters (or grössencharaktere) of type A_0. Let F be a number field, let $\mathfrak{m}$ be an ideal of F (i.e., of the ring of integers of F), let $I_{\mathfrak{m}}$ be the group of fractional ideals of F prime to $\mathfrak{m}$, and let k be a field of characteristic 0. A homomorphism $\chi : I_{\mathfrak{m}} \to k^\times$ is an *algebraic Hecke character* (of conductor $\leqslant \mathfrak{m}$) if there exists an algebraic homomorphism $\chi_{\mathrm{alg}} : F^\times \to k^\times$ satisfying

$$\text{For } x \in F^\times, \text{ prime to } \mathfrak{m}, \text{ totally positive, and } \equiv 1 \pmod{\mathfrak{m}}, \text{ one has } \chi((x)) = \chi_{\mathrm{alg}}(x). \qquad (*)$$

By the density of the set of x in $(*)$, χ_{alg} is entirely determined by χ. This is the *algebraic part* of χ. If $\chi((x)) = \chi_{\mathrm{alg}}(x)$ for x totally positive and $\equiv 1 \pmod{\mathfrak{m}'}$, χ extends to a Hecke character of conductor $\leqslant \inf(\mathfrak{m}, \mathfrak{m}')$: $I_{\mathfrak{m}+\mathfrak{m}'} \to k^\times$. We identify Hecke characters that coincide on their common domain of definition, and we call *conductor* of χ the smallest $\mathfrak{m}'$ such that χ is of conductor $\leqslant \mathfrak{m}'$.

5.4 Remark

If an algebraic Hecke character χ takes its values in a subfield k' of k, it is already a Hecke character with values in k': it suffices to observe that $\chi_{\mathrm{alg}} : \mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m) \to \mathbb{G}_m$ is already defined over k', and this follows from the Zariski density in $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m)$ of the set of totally positive $x \equiv 1 \pmod{\mathfrak{m}}$. We can always take for k' a subfield of k of finite degree over $\mathbb{Q}$.

5.5

If ε is a totally positive unit $\equiv 1 \pmod{\mathfrak{m}}$, we have $\chi_{\mathrm{alg}}(\varepsilon) = \chi((\varepsilon)) = 1$. The homomorphism χ_{alg} therefore factors through the quotient $T_{\mathfrak{m}}$ of $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m)$ by the Zariski closure of the group $E_{\mathfrak{m}} \subset F^\times$ of totally positive units $\equiv 1 \pmod{\mathfrak{m}}$.

According to Serre [35, II.3], if k is an algebraic closure of $\mathbb{Q}$, and $\mathfrak{m}$ is large enough, the characters $\prod \omega^{n_\omega}$ of $\mathrm{R}_{F/\mathbb{Q}}(\mathbb{G}_m)$ of the form χ_{alg} are characterized as follows: there must exist an integer N, the *weight* of χ (or χ_{alg}), such that for any element σ of $\mathrm{Gal}(k/\mathbb{Q})$ conjugate to complex conjugation, we have $n_\omega + n_{\sigma\omega} = N$. If F_1 is the largest subfield of F which is a quadratic extension that is totally imaginary over a totally real field F_1', this amounts to saying that χ_{alg} is of the form $\chi_1 \circ N_{F/F_1}$, and that $\chi_1 | F_1' = (N_{F_1'/\mathbb{Q}})^N$.

If χ is of weight N, for any ideal $\mathfrak{a}$ of F, $\chi(\mathfrak{a})$ is a algebraic number all of whose conjugates have absolute value $N(\mathfrak{a})^{N/2}$ [35, II.3 prop.2].

5.6

For a fixed number field k, additional conditions are imposed on χ_{alg}. For any ideal $\mathfrak{a}$ of F prime to the conductor, we have in fact

$$\chi_{\mathrm{alg}}(\mathfrak{a}) = (\chi(\mathfrak{a})), \qquad (5.6.1)$$

so that $\chi_{\mathrm{alg}}(\mathfrak{a})$ is principal. Since the group of ideal classes of k is torsion, it is enough to prove the n-th power of (5.6.1), for a suitable $n \neq 0$; this allows us to replace $\mathfrak{a}$ by $\mathfrak{a}^n$, therefore to assume $\mathfrak{a} = (x)$, with x totally positive $\equiv 1 \pmod{\mathfrak{m}}$. All that remains is to use the definitions.

Together with 5.5, this shows that χ_{alg} determines the valuations of $\chi(\mathfrak{a})$ at all places of k.

5.7

Serre's group $S_{\mathfrak{m}}$ can be characterized as the group of multiplicative type whose group of characters (on any field) is the group of algebraic Hecke characters of conductor $\leqslant \mathfrak{m}$ (see [35, II 2.1 et 2.2]). Its relation with ℓ-adic representations is explained in [35, II 2.3].

Theorem 5.8 ([35])**.** *Let χ be an algebraic Hecke character of F with values in E_λ, for E_λ a finite extension of $\mathbb{Q}_\ell$. There is then a unique homomorphism $\chi_\lambda : \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to E_\lambda^\times$, such that*

(i) χ_λ is unramified outside the conductor $\mathfrak{f}$ of χ and outside ℓ.

(ii) For $\mathfrak{p}$ a prime ideal of F prime to $\mathfrak{f}$ and to ℓ, and $F_\mathfrak{p} \in \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}}$ the geometric Frobenius at $\mathfrak{p}$, we have

$$\chi_\lambda(F_\mathfrak{p}) = \chi(\mathfrak{p}).$$

5.9

At the end of this section, we give a criterion for a λ-adic representation to arise from a Hecke character, and in the next section we apply it to Jacobi sums. We thereby recover, in a somewhat generalized form, Weil's results [42, 44], with the second part of Weil's proof replaced by a ℓ-adic cohomology argument.

Theorem 5.10. *Let F and k be two number fields, let λ be a place of k of residue characteristic ℓ, and let $\chi_\lambda : \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to k_\lambda^\times$ be a homomorphism, unramified outside ℓ and a finite set S of places. We assume that there exists a set T of places (containing S and the places above ℓ), of density 0, and an algebraic homomorphism $\chi_0 : F^\times \to k^\times$ such that*

(i) For $\bar{k}$ an algebraic closure of k, $\chi_0 : F^\times \to k^\times \hookrightarrow \bar{k}^\times$ is of the type considered in 5.5, of weight N;

(ii) For $\mathfrak{p}$ prime $\notin T$, $\chi_\lambda(F_\mathfrak{p})$ lies in $k^\times$, $(\chi_\lambda(F_\mathfrak{p})) = \chi_0(\mathfrak{p})$, and all complex conjugates of $\chi_\lambda(F_\mathfrak{p})$ have absolute value $(N\mathfrak{p})^{N/2}$.

Then χ_λ is defined (5.8) by an algebraic Hecke character of F with values in k, whose algebraic part is χ_0.

Let χ' be an algebraic Hecke character with values in a finite Galois extension k' of k, of algebraic part χ_0 (5.5). After enlarging T if necessary, we may suppose that the conductor of χ' is supported in T. Let λ' be a place of k' above λ, and let $\chi_{\lambda'}$ be the composite $\mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to k_\lambda^\times \to {k'}_{\lambda'}^\times$, and $\chi'_{\lambda'} : \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to {k'}_{\lambda'}^\times$ defined by χ' (5.8). Set $\varepsilon = \chi_{\lambda'} \cdot {\chi'}_{\lambda'}^{-1}$.

Hypothesis (ii), together with 5.5, 5.6, and 5.8, ensures that for $\mathfrak{p} \notin T$, we have $\varepsilon(F_\mathfrak{p}) \in {k'}^\times$, and that in every completion of k' this number has absolute value 1. It follows that $\varepsilon(F_\mathfrak{p})$ belongs to the finite group $\boldsymbol{\mu}$ of roots of unity in k'. By the Chebotarev density theorem and the density-zero hypothesis on T, the $F_\mathfrak{p}$ ($\mathfrak{p} \notin T$) are dense in $\mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}}$, so that $\varepsilon(\sigma) \in \boldsymbol{\mu}$ for all $\sigma \in \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}}$: the character ε is a finite order character $\mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to \boldsymbol{\mu}$. By class field theory, it corresponds to a finite-order character of the idèle class group of F, i.e. to an algebraic Hecke character with trivial algebraic part. After correcting χ' by this character, we are reduced to the case in which $\chi_{\lambda'} = \chi'_{\lambda'}$, and it remains to show that χ', a priori taking values in k', in fact takes values in k. If $\sigma \in \mathrm{Gal}(k'/k)$, χ' and ${\chi'}^\sigma$ coincide on every prime ideal $\mathfrak{p} \notin T$, since $\chi'(\mathfrak{p}) = \chi'_{\lambda'}(F_\mathfrak{p}) = \chi_{\lambda'}(F_\mathfrak{p}) \in k^\times$. The characters $\chi'_{\lambda'}$, and ${\chi'_{\lambda'}}^\sigma$, coincide on a dense subset of $\mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}}$ and hence are equal. Thus $\chi' = {\chi'}^\sigma$, so χ' takes values in k.

6 Hecke characters defined by Jacobi sums

The idea of this section is as follows. Consider a Gauss sum $\tau_{\mathbb{F}_q}(\chi,\psi)$. Both this sum and the corresponding sheaf $\mathscr{F}(\chi^{-1}\psi)$ are tied in two ways to the characteristic p and to the base field $\mathbb{F}_q$:

a) because the additive character ψ and the sheaf (ψ) occur in it;

b) because $\mathscr{F}(\chi)$ is defined from the exact Lang sequence

$$0 \longrightarrow k^\times \longrightarrow V^* \longrightarrow V^* \longrightarrow 0.$$

These present two difficulties if we wish to lift the corresponding cohomological constructions to a number field F and find ℓ-adic representations of $\mathrm{Gal}(\bar{F}/F)$ whose Frobenius eigenvalues are Gauss sums.

If χ is trivial on $\mathbb{F}_q^\times$, the sum $\tau_{\mathbb{F}_q}(\chi,\psi)$ is independent of ψ and is q times a Jacobi sum. The latter is related to the cohomology of $W^*/\mathbb{G}_m$, with coefficients in a sheaf obtained from χ: thus ψ, and with it difficulty a), disappear. To resolve b), it suffices to describe the sheaves in question by means of exact Kümmer sequences (which make sense in every characteristic). This is possible by 4.9.

Once the ℓ-adic representations have been found, we use 5.10 and Stickelberger's theorem to show that they come from algebraic Hecke characters.

6.1

If E is a field with algebraic closure $\bar{E}$, and λ is a character, of order dividing n, of $\mu_d(\bar{E})$, we denote by ${}_n\lambda$ the character of $\mu_n(\bar{E})$ such that λ and ${}_n\lambda$ induce the same character of $\widehat{\mathbb{Z}}(1)_{\bar{E}} = \varinjlim \mu_n(\bar{E})$. The same notation is used when λ is a character of $\widehat{\mathbb{Z}}(1)_{\bar{E}}$.

Let F and k be two number fields, $\bar{F}$ an algebraic closure of F, I a finite set equipped with an action of $\mathrm{Gal}(\bar{F}/F)$ and $\lambda = (\lambda_i)_{i\in I}$ a family of characters $\lambda_i : \widehat{\mathbb{Z}}(1)_{\bar{F}} \to k^\times$. It is assumed that $\lambda_i \neq 1$ $(i \in I)$, that the product of λ_i is trivial and that the family λ is defined over F, i.e. that for every $\sigma \in \mathrm{Gal}(\bar{F}/F)$, we have $\lambda_{\sigma(i)} = \lambda_i \circ \sigma^{-1}$.

The Galois set I is the set of homomorphisms into $\bar{F}$ of an algebra E separable over F. If C is the set of orbits of $\mathrm{Gal}(\bar{F}/F)$ in I, we have a decomposition of E as a product of fields $E = \prod_{\alpha\in C} E_\alpha$, where α is identified with the set of embeddings of E_α in $\bar{F}$. If the characters λ_i, for $i \in \alpha$, are of order d_α, the field E_α contains the d_α-th roots of 1 and there exists $\lambda_\alpha : \mu_{d_\alpha}(E_\alpha) \hookrightarrow k^\times$ such that, if $i \in \alpha$ corresponds to the embedding σ_i of E_α in $\bar{F}$, then ${}_{d_\alpha}\lambda_i = \lambda_\alpha \sigma_i^{-1}$.

6.2

Let $n > 0$ be a multiple of the d_α, let Σ be the set of places of F at which E/F ramifies or which divide n, let μ be a place of k of residue characteristic ℓ, and let A be the ring of elements of F integral outside Σ and ℓ. The integral closure M of A is finite étale of rank $N = |I|$ over A. With the notation of 4.6, 4.8, ${}_n\lambda_i$ define

$${}_n\lambda : (V^*/\mathbb{G}_m)_n \to k^\times \subset k_\mu^\times.$$

Let a be the projection of $W^*/\mathbb{G}_m$ onto $\mathrm{Spec}(A)$. By 4.20, the sheaf $\mathsf{R}^{N-2}a_!\mathscr{K}_n({}_n\lambda)$ is smooth of rank one; it defines an ℓ-adic representation

$$j_\mu[\lambda] : \mathrm{Gal}(\bar{F}/F)^{\mathrm{ab}} \to k_\mu^\times$$

(or simply j_μ) unramified outside Σ and ℓ.

6.3

Let us compute $j_\mu(F_\mathfrak{p})$. By 4.20, it suffices to do so after reduction modulo $\mathfrak{p}$. Put $f = A/\mathfrak{p}$, $e = M/\mathfrak{p}M$, and let $\bar{f}$ be the algebraic closure of f determined by a place of $\bar{F}$ above $\mathfrak{p}$. We still have $I = \hom_f(e, \bar{f})$. After reduction, ${}_n\lambda$ still has a description as in 6.1:

a) Let D be the set of orbits of $\mathrm{Gal}(\bar{f}/f)$ in I. The decomposition of e as a product of fields has the form $e = \prod_{\beta\in D} e_\beta$, where σ_i $(i \in \beta)$ are identified with the embeddings of e_β into $\bar{f}$. For $\beta \in D$, denote by $\alpha(\beta)$ the element of C containing β, and put $d_\beta = d_{\alpha(\beta)}$. Let

$$\lambda_\beta : \mu_{d_\beta}(e_\beta) \xleftarrow{\sim} \mu_{d_\beta}(E_{\alpha(\beta)}) \xrightarrow{\lambda_{\alpha(\beta)}} k^\times.$$

b) If $i \in \beta$ corresponds to the embedding σ_i of e_β into $\bar{f}$, the reduction ${}_{d_\beta}\bar{\lambda}_i : \mu_{d_\beta}(\bar{f}) \to k^\times$ of ${}_{d_\beta}\lambda_i$ is $\lambda_\beta \circ \sigma_i^{-1}$.

Let q_β be the number of elements of e_β, $\chi_\beta = {}_{q_\beta-1}\lambda_\beta$, and $\chi : e^\times \to k^\times$ the character with coordinates χ_β. On the affine space over $A/\mathfrak{p}$ defined by e (4.6), or rather on the torus $e_\beta^\times$, we have isomorphisms $\mathscr{K}_n({}_n\bar{\lambda}_i)_{i\in\beta}) = \mathscr{K}_{d_\beta}((\lambda_\beta \circ \sigma_i^{-1})_{i\in\beta}) = \mathscr{K}_{q_\beta-1}((\chi_\beta \circ \sigma_i^{-1})_{i\in\beta}) = \mathscr{F}(\chi_\beta)$ (4.9). On $e^\times$, we therefore have $\mathscr{K}_n({}_n\bar{\lambda}_i)_{i\in I}) = \mathscr{F}(\chi)$, and this isomorphism descends to the reduction modulo $\mathfrak{p}$ of $V^*/\mathbb{G}_m$. It follows that

Proposition 6.4. *With the notations of 6.2 and 6.3, $j_\mu(F_\mathfrak{p})$ is the Jacobi sum $J(\chi)$.*

Let us rewrite the formulas linking $(\lambda_i)_{i\in I}$ to $(\chi_\beta)_{\beta\in D}$: for $i \in \beta \subset \alpha$,

$$\begin{aligned} {}_{d_\alpha}\lambda_i &= \lambda_\alpha \circ \sigma_i^{-1} && \lambda_\alpha : \mu_{d_\alpha}(E_\alpha) \to k^\times \\ \lambda_\beta &= \text{reduction of } \lambda_\alpha && : \mu_{d_\alpha}(e_\beta) \to k^\times \\ \chi_\beta &= {}_{q_\beta-1}\lambda_\beta : x \mapsto \lambda_\beta(x^{(q_\beta-1)/d_\alpha}) && : e_\beta^\times \to k^\times \end{aligned}$$

Theorem 6.5. *The representation j_μ is defined by an algebraic Hecke character of F with values in $k^\times$.*

We shall verify this using 5.10. Here we follow Weil's proof [42, 44], using Stickelberger's theorem, and I shall give only a few indications.

6.6

Given $j : \mathrm{Gal}(\bar{F}/F) \to k_\lambda^\times$ and a finite extension F'/F, one verifies using 5.10 that j is defined by a Hecke character if and only if $j|\,\mathrm{Gal}(\bar{F}/F')$ is. This remark allows us, in 6.5, to reduce to the case where Galois acts trivially on I and where F contains the n-th roots of unity. We may then reduce to the case $F = \mathbb{Q}(\sqrt[n]{1})$ and take $k = F$. In this case, each ${}_n\lambda_i$ is defined by $a_i \in \mathbb{Z}/n$: it is $x \mapsto x^{a_i} : \mu_n \to k^\times$. We have $\sum a_i = 0$, and $a_i \neq 0$.

6.7

In a number field, the places of degree greater than 1 always form a set of density 0. By applying 5.10, we may therefore disregard them and use Stickelberger only for a prime field. Let p be prime, let $0 < a < p-1$, and let $\widetilde{x} : \mathbb{F}_p^\times \to \mathbb{Q}_p^\times$ be the "multiplicative lift" character; put $g = -\sum_{x\in\mathbb{F}_p^\times} \widetilde{x}^{-a}\zeta^x \in \mathbb{Q}_p(\zeta)$, for ζ a p-th root of 1. Again denote by x the integer in $[0, p-1]$ lifting x, and expand $\zeta^x = (1+\pi)^x$ by the binomial

formula. We find $g = -\sum_{i=0}^{p-1} A_i \pi^i$ with $A_i \in \mathbb{Z}_p$, whose reduction $\bmod p$ is the sum $\sum_{x\in\mathbb{F}_p^\times} x^{-a}\binom{x}{i}$. For b not divisible by $p-1$, we have $\sum_{x\in\mathbb{F}_p^\times} x^b = 0$. If $i < a$, $\binom{x}{i}$ is a polynomial in x of degree $i < a$, whence $A_i \equiv 0 \pmod p$. Likewise, $A_a \equiv \frac{p-1}{a!} \pmod p$, and $g \sim \frac{\pi^a}{a!}$: the valuation of g is given by $v(g)/v(p) = \frac{a}{p-1}$.

6.8

Let σ_i ($i \in (\mathbb{Z}/n)^\times$) be the element of $\mathrm{Gal}(\mathbb{Q}(\sqrt[n]{1})/\mathbb{Q})$ such that $\sigma_i(\zeta) = \zeta^i$ for $\zeta^n = 1$, and write the group of algebraic homomorphisms $\mathbb{Q}(\sqrt[n]{1})^\times \to \mathbb{Q}(\sqrt[n]{1})^\times$ additively. For λ as in 6.6, defined by a family $(a_i)_{i\in I}$ of nonzero integers $\bmod n$ whose sum is zero, 5.10, (4.15.1) and 6.7 show that $j_\mu[\lambda]$ is defined by an algebraic Hecke character $j[a]$ and that, if N denotes the norm, its algebraic part is given by

$$(Nj[a])_{\mathrm{alg}} = \sum_{i\in(\mathbb{Z}/n)^\times} \sum_{j\in I} \left\langle \frac{ia_j}{n} \right\rangle \sigma_i^{-1} \tag{6.8.1}$$

where $\langle\cdot\rangle$ is the fractional part.

6.9

The most interesting case is that in which $F \subset \mathbb{Q}(\sqrt[n]{1})$ and the family of ${}_n\lambda_i$ is constructed as follows:

a) Take a finite family $(A_j)_{j\in J}$ of orbits of $\mathrm{Gal}(\mathbb{Q}(\sqrt[n]{1})/F) = H \subset (\mathbb{Z}/n)^\times$ in $\mathbb{Z}/n$. Assume that $A_j \neq \{0\}$ and that $\sum_{j\in J}\sum_{a\in A_j} a = 0$.

b) Take $I = \sqcup_{j\in J} A_j$; Galois acts on each A_j.

c) Take $\bar{F} \supset \mathbb{Q}(\sqrt[n]{1})$, $k = \mathbb{Q}(\sqrt[n]{1})$ and, if $i \in I$ corresponds to $a \in A_j \subset (\mathbb{Z}/n)^\times$, ${}_n\lambda_i(x) = x^a$.

Using 4.20, one verifies that a general j_μ (relative to F_1) is obtained from a j_μ as above (for $F \subset \mathbb{Q}(\sqrt[n]{1})$, with F_1 an extension of F) by restriction to $\mathrm{Gal}(\bar{F}/F_1)$ and multiplication by a character of order two.

For $a \in \mathrm{Gal}(k/\mathbb{Q}) = (\mathbb{Z}/n)^\times$, a transforms the family of A_j into the family of aA_j. If $a \in H$, the family of A_j is therefore preserved, and the $j_\mu(F_\mathfrak{p})$ are invariant under H:

Proposition 6.10. *Under the assumptions of 6.9, j_μ is defined by a character that is an algebraic Hecke character of F with values in F.*

Let us again take $F = \mathbb{Q}(\sqrt[n]{1})$ and consider the Hecke characters $j[a]$ defined by a family $(a_i)_{i\in I}$, $i \in \mathbb{Z}/n$, $a_i \neq 0$, $\sum a_i = 0$. The following theorem was obtained independently by Vishik.

Theorem 6.11. *Every algebraic Hecke character of $\mathbb{Q}(\sqrt[n]{1})$ has a power in the group generated by the $j[a]$ and the norm.*

Since we work "up to torsion," only the algebraic part of the Hecke characters under consideration matters. Applying (6.8.1) to $j[a]$, with $a =$ an element of $\mathbb{Z}/n$ repeated n times, we are reduced to the following lemma.

Lemma 6.12. *Write $\sum x_j\sigma_j$ for the elements of the group algebra $\mathbb{Q}[(\mathbb{Z}/n)^\times]$, and let X be the subgroup consisting of those for which $x_j + x_{-j} = C\underline{te}$. Then X is generated over $\mathbb{Q}$ by $N = \sum\sigma_j$ and by the elements $g_a = \sum\left\langle\frac{ia}{n}\right\rangle\sigma_i^{-1}$ ($a \in \mathbb{Z}/n$, $a \neq 0$).*

Both X and $Y = \langle N, (g_a)\rangle$ are ideals of the group algebra, and it suffices to check that they are annihilated by the same complex characters χ of $(\mathbb{Z}/n)^\times$. This amounts to saying that a odd character ($\chi(j) = -\chi(-j)$) never annihilates all the g_a. If χ is primitive, we have

$$\chi(g_1) = \sum \frac{i}{n}\chi^{-1}(i) = -L(\chi, 0) \neq 0.$$

In the general case, if χ comes from a primitive character of $(\mathbb{Z}/d)^\times$, $\chi(g_{n/d})$ reduces to an analogous sum over $(\mathbb{Z}/d)^\times$, hence the lemma.

7 Generalized Kloosterman sums

7.1

Let $\mathbb{F}_q$ be a finite field with q elements, and let ψ be a nontrivial additive character. In this section, we undertake a cohomological study of generalized Kloosterman sums

$$\mathrm{K}_{n,a} = \sum_{x_1\cdots x_n = a} \psi(x_1 + \cdots + x_n) \qquad (n \geqslant 1). \tag{7.1.1}$$

For $a = 0$, this sum is elementary (see 7.7):

$$\mathrm{K}_{n,0} = (-1)^{n-1}. \tag{7.1.2}$$

The interesting case is $a \neq 0$. The x_i then belong to $\mathbb{F}_q^\times$. In this case we shall prove the bound

$$|\mathrm{K}_{n,a}| \leqslant nq^{\frac{n-1}{2}}. \tag{7.1.3}$$

Our essential tools will be the cohomological counterparts of the following identities:

$$\mathrm{K}_{n,a} = \sum_{x_1} \psi(x_1) \sum_{x_2\cdots x_n = a/x_1} \psi(x_2 + \cdots + x_n) \tag{7.1.4}$$

$$= \sum_{x\in\mathbb{F}_q^\times} \psi(x)\mathrm{K}_{n-1,a/x} \qquad (a \neq 0, n \geqslant 2)$$

$$\sum_a \mathrm{K}_{n,a} = \sum_a \sum_{x_1\cdots x_n = a} \psi\left(\sum x_i\right) = \sum_{x_1,\ldots,x_n} \psi\left(\sum x_i\right) = 0 \tag{7.1.5}$$

For χ a nontrivial character of $\mathbb{F}_q^\times$,

$$\sum \chi(a)\mathrm{K}_{n,a} = \sum_{x_1,\ldots,x_n} \chi\left(\prod x_i\right) \cdot \psi\left(\sum x_i\right) = (-\tau(\chi^{-1}, \psi))^n \tag{7.1.6}$$

(Gauss sum).

7.2

Let k be an étale algebra of degree n over $\mathbb{F}_q$, and let $\varepsilon(k)$ be as in (4.5.1). Put

$$\mathrm{K}_{k,a} = \sum_{N_{k/\mathbb{F}_q}(x) = a} \psi\,\mathrm{tr}(x).$$

We likewise have

$$\mathrm{K}_{k,0} = (-1)^{n-1} \cdot \varepsilon(k) \tag{7.2.1}$$

$$\sum_a \mathrm{K}_{k,a} = 0 \tag{7.2.2}$$

$$\sum_a \chi(a)\mathrm{K}_{k,a} = (-1)^n \tau_{\mathbb{F}_q}\left(\chi^{-1} \circ N_{k/\mathbb{F}_q}, \psi\right). \tag{7.2.3}$$

By Fourier inversion on $\mathbb{F}_q^\times$, we deduce an expression of $\mathrm{K}_{k,a}$ in terms of Gauss sums:

$$\mathrm{K}_{k,a} = \frac{1}{q-1}\sum_\chi \chi(a) \sum_{b\in\mathbb{F}_q^\times} \chi^{-1}(b)\mathrm{K}_{k,b}$$

$$(-1)^n\mathrm{K}_{k,a} = \frac{1}{q-1}\left(\varepsilon(k) + \sum_{\chi\neq 1} \chi(a)\tau_{\mathbb{F}_q}\left(\chi \circ N_{k/\mathbb{F}_q}, \psi\right)\right). \tag{7.2.4}$$

The Hasse–Davenport identity now makes it possible to compare $\mathrm{K}_{k,a}$ with $\mathrm{K}_{n,a} = \mathrm{K}_{\mathbb{F}_{q^n},a}$:

$$\mathrm{K}_{k,a} = \varepsilon(k)\mathrm{K}_{n,a}. \tag{7.2.5}$$

I know of no cohomological interpretation of (7.2.4), but I shall give one for (7.2.5). Let us return to the sums $\mathrm{K}_{n,a}$.

7.3

Let $\mathbb{F}$ be an algebraic closure of $\mathbb{F}_q$ and, for $a \in \mathbb{F}$, let V_a^{n-1}, or simply V_a, be the hypersurface in $\mathbb{A}^n$ with equation $x_1 \cdots x_n = a$. As usual, we shall regard ψ as taking ℓ-adic rather than complex values. With the notation of 1.8(ii), (7.1.3) follows from

Theorem 7.4. *The cohomology of V_a^{n-1} with coefficients in $\mathscr{F}(\psi(\sum x_i))$ satisfies*

(i) $\mathrm{H}_c^i = 0$ *for* $i \neq n-1$;

(ii) $\mathrm{H}_c^\bullet \xrightarrow{\sim} \mathrm{H}^\bullet$;

(iii) for $a \neq 0$, $\dim \mathrm{H}_c^{n-1} = n$;

(iv) for $a = 0$, H_c^{n-1} *is canonically isomorphic to* E_λ.

7.5

As usual, this theorem also controls the dependence of $\mathrm{K}_{n,a}$ on q: for each $a \in \mathbb{F}_q^\times$, there are n Frobenius eigenvalues $\alpha_1, \ldots, \alpha_n$, of complex absolute value $q^{\frac{n-1}{2}}$, such that

$$\sum_{\substack{x_1\cdots x_n = a \\ x_i \in \mathbb{F}_{q^m}}} \psi \circ \mathrm{tr}\left(\sum x_i\right) = (-1)^{n-1}\left(\alpha_1^m + \cdots + \alpha_n^m\right).$$

For even n, $x \mapsto -x$ is an involution of V_a, which transforms $\mathscr{F}(\psi)$ into its dual. Arguing as in 3.6, we deduce a canonical nondegenerate alternating form on $\mathrm{H}_c^{n-1}(V_a, \mathscr{F}(\psi))$, with values in $\mathbb{Q}_\ell(-(n-1))$, and hence the following.

Corollary 7.6. *With the notations of 7.5, if n is even, the α_i group together into $\frac{n}{2}$ pairs of roots $\alpha, \bar{\alpha}$ whose product is q^{n-1}.*

7.7

Let us verify the case $a = 0$ of 7.5. The hypersurface V_0 is the union of the coordinate hyperplanes $x_i = 0$. Let us compute the Leray spectral sequence (2.6.1*) for this covering of V_0, for cohomology with or without supports and with coefficients in $\mathscr{F}(\psi)$. By 2.7*, all the initial terms vanish except the cohomology of the intersection $\{0\}$ of all the coordinate hyperplanes. What remains is an isomorphism $\mathrm{H}^0_c \xrightarrow{\sim} \mathrm{H}^0 = E_\lambda$ which justifies 7.4.

We shall prove 7.4 and 7.8 below by simultaneous induction on n, using the spectral sequence of which (7.1.4) is the counterpart. This requires control, as a varies, of $\mathrm{K}_{n,a}$.

Let $\pi : \mathbb{A}^n \to \mathbb{A}$ be the map "product of the coordinates," and let σ be the sum of the coordinates. The hypersurfaces V_a are the fibers of π.

Theorem 7.8. *(i) The sheaf $\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)$ is smooth of rank n on $\mathbb{A}^1 \smallsetminus \{0\}$.*

(ii) Its extension by 0 to $\mathbb{P}^1$ is the direct image of its restriction to $\mathbb{A}^1 \smallsetminus \{0\}$.

(iii) At 0, the monodromy is unipotent, with a single Jordan block.

(iv) At ∞, wild inertia acts without fixed points $\neq 0$, and the Swan conductor is 1.

(v) We have $\mathsf{R}^i\pi_!\mathscr{F}(\psi\sigma) \xrightarrow{\sim} \mathsf{R}^i\pi_*\mathscr{F}(\psi\sigma)$ (zero for $i \neq n-1$).

Proof. that 7.4 (for a given value of n) implies 7.8 (for the same value of n).

The fiber of $\mathsf{R}^i\pi_!\mathscr{F}(\psi\sigma)$ at a is $\mathrm{H}^i_c(V_a, \mathscr{F}(\psi\sigma))$, and on the dense open subset of $\mathbb{A}^1$ where the $\mathsf{R}^i\pi_!\mathscr{F}(\psi\sigma)$ are locally constant, the fiber of $\mathsf{R}^i\pi_*\mathscr{F}(\psi\sigma)$ is $\mathrm{H}^i(V_a, \mathscr{F}(\psi\sigma))$ (Finiteness Theorem 2.1). The hypothesis 7.4(n) therefore gives us the following.

Lemma 7.9. *(i) $\mathsf{R}^i\pi_!\mathscr{F}(\psi\sigma) = 0$ for $i \neq n-1$.*

(ii) For $i = n-1$, the fiber of this sheaf at any point $a \neq 0$ has constant rank n. At 0, it has rank 1.

(iii) On a dense open U, we have $\mathsf{R}^i\pi_!\mathscr{F}(\psi\sigma) \xrightarrow{\sim} \mathsf{R}^i\pi_*\mathscr{F}(\psi\sigma)$.

7.10

We shall now use the Leray spectral sequence of π for cohomology with or without compact supports, with coefficients in $\mathscr{F}(\psi\sigma)$. The abutment is known: $\mathrm{H}^\bullet_c(\mathbb{A}^n, \mathscr{F}(\psi\sigma)) = \mathrm{H}^\bullet(\mathbb{A}^n, \mathscr{F}(\psi\sigma)) = 0$ (2.7*). For cohomology with compact supports, 7.9(i) ensures that $E_2^{pq} = 0$ for $q \neq n-1$. The terms E_2 are therefore all zero:

$$\mathrm{H}^p_c(\mathbb{A}^1, \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)) = 0. \tag{7.10.1}$$

For $p = 0$, this means that $\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)$ has no sections with point support. In view of the constancy of rank in 7.9(ii), we obtain

$$\begin{array}{l}\text{The sheaf } \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma) \text{ on } \mathbb{A}^1 \text{ is smooth on } \mathbb{A}^1 \smallsetminus \{0\}, \text{ and is} \\ \text{a subsheaf of the direct image } \mathscr{G} \text{ of its restriction to } \mathbb{A}^1 \smallsetminus \{0\}.\end{array} \tag{7.10.2}$$

The long exact cohomology sequence of

$$0 \longrightarrow \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma) \longrightarrow \mathscr{G} \longrightarrow \mathscr{Q} \longrightarrow 0.$$

where the support of $\mathscr{Q}$ is concentrated at 0, gives

$$\mathrm{H}^0_c(\mathbb{A}^1, \mathscr{G}) \longrightarrow \mathscr{Q}_0 \longrightarrow \mathrm{H}^1_c(\mathbb{A}^1, \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)).$$

Since the outer terms vanish, $\mathscr{Q} = 0$ and

$$\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma) \text{ is the direct image of its restriction to } \mathbb{A}^1 \smallsetminus \{0\}. \tag{7.10.3}$$

Let j be the inclusion of $\mathbb{A}^1$ into $\mathbb{P}^1$. The "derived-category version" of the Leray spectral sequences is

$$\mathsf{H}^\bullet(\mathbb{P}^1, j_!\mathsf{R}\pi_!\mathscr{F}(\psi\sigma)) = \mathsf{H}^\bullet_c(\mathbb{A}^1, \mathsf{R}\pi_!\mathscr{F}(\psi\sigma)) = \mathrm{H}^\bullet_c(\mathbb{A}^n, \mathscr{F}(\psi\sigma)) = 0$$
$$\mathsf{H}^\bullet(\mathbb{P}^1, \mathsf{R}j_*\mathsf{R}\pi_*\mathscr{F}(\psi\sigma)) = \mathsf{H}^\bullet(\mathbb{A}^1, \mathsf{R}\pi_*\mathscr{F}(\psi\sigma)) = \mathrm{H}^\bullet(\mathbb{A}^n, \mathscr{F}(\psi\sigma)) = 0.$$

Let Δ be the mapping cylinder of $j_!\mathsf{R}\pi_!\mathscr{F}(\psi\sigma) \to \mathsf{R}j_*\mathsf{R}\pi_*\mathscr{F}(\psi\sigma)$. According to 7.9(iii), the cohomology sheaves of this complex of sheaves have finite support. On the other hand, the long exact sequence in cohomology of the triangle $(j_!\mathsf{R}\pi_!\mathscr{F}(\psi\sigma), \mathsf{R}j_*\mathsf{R}\pi_*\mathscr{F}(\psi\sigma), \Delta)$ shows that $\mathsf{H}^\bullet(\mathbb{P}^1, \Delta) = 0$. We have $\mathsf{H}^\bullet(\mathbb{P}^1, \Delta) = \mathrm{H}^0(\mathbb{P}^1, \mathscr{H}^\bullet(\Delta))$, and finally $\Delta = 0$:

$$j_!\mathsf{R}\pi_!\mathscr{F}(\psi\sigma) \xrightarrow{\sim} \mathsf{R}j_*\mathsf{R}\pi_*\mathscr{F}(\psi\sigma).$$

In particular, $\mathsf{R}\pi_!\mathscr{F}(\psi\sigma) \xrightarrow{\sim} \mathsf{R}\pi_*\mathscr{F}(\psi\sigma)$ and, since these complexes of sheaves have only one nonzero cohomology sheaf,

$$j_!\mathsf{R}\pi_!\mathscr{F}(\psi\sigma) \xrightarrow{\sim} j_*\mathsf{R}\pi_!\mathscr{F}(\psi\sigma). \tag{7.10.4}$$

This completes the proof of (i)(ii)(v).

7.11

Let Sw_0 and Sw_∞ be the Swan conductors at 0 and at ∞ of $\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)$. We have $\chi_c(\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)) = 0$ (7.10.1) and the formula of Euler–Poincaré (3.2.1) reduces to

$$\mathrm{Sw}_0 + \mathrm{Sw}_\infty = 1: \tag{7.11.1}$$

one of these conductors is 0 (tame ramification), and the other is 1.

7.12

Let $\mathscr{G}$ be a sheaf of rank 1 on $\mathbb{G}_m$, of Kummer type 4.7 ($\mathscr{G} = \mathscr{K}_n(\chi)$) and nonconstant ($\chi \neq 1$). The sheaf $\mathscr{F}(\psi\sigma) \otimes \pi^*\mathscr{G}$ on $V^* = \pi^{-1}(\mathbb{G}_m) = \mathbb{A}^n \smallsetminus \pi^{-1}(0)$ is then of the type encountered in the study of Gauss sums, and 4.11(ii) gives us

$$\mathrm{H}^\bullet_c(V^*, \pi^*\mathscr{G} \otimes \mathscr{F}(\psi\sigma)) \xrightarrow{\sim} \mathrm{H}^\bullet(V^*, \pi^*\mathscr{G} \otimes \mathscr{F}(\psi\sigma)). \tag{7.12.1}$$

The Leray spectral sequence of π transforms this statement into

$$\mathrm{H}^\bullet_c(\mathbb{G}_m, \mathscr{G} \otimes \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)) \xrightarrow{\sim} \mathrm{H}^\bullet(\mathbb{G}_m, \mathscr{G} \otimes \mathsf{R}^{n-1}\pi_*\mathscr{F}(\psi\sigma)). \tag{7.12.2}$$

Let i be the inclusion of $\mathbb{G}_m$ into $\mathbb{P}^1$, and let Δ be the mapping cylinder of $\delta : i_!(\mathscr{G} \otimes \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)) \to \mathsf{R}i_*(\mathscr{G} \otimes \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma))$ (where $\mathsf{R}^{n-1}\pi_! = \mathsf{R}^{n-1}\pi_*$). Reasoning as in 7.10, we deduce from (7.12.2) that $\Delta = 0$. In particular

$$\text{At 0 and at } \infty, \text{ inertia acts without fixed points} \neq 0 \text{ on } \mathscr{G} \otimes \mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma). \tag{7.12.3}$$

At ∞, this holds even if $\mathscr{G}$ is trivial (7.10.4), so wild inertia acts without a fixed point. In particular, the ramification is wild: by 7.11, we have $\mathrm{Sw}_\infty = 1$, and the ramification at 0 is tame.

At 0, (7.12.3) forces the ramification – which is tame – of $\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)$ to be unipotent. By (7.10.3) and 7.9(ii), there is a single Jordan block. This completes the proof of 7.8(n).

7.13

For $n = 1$, 7.4 is trivial. To complete the proof of 7.4 and 7.8, we still have to prove that 7.8, for a given value of n, implies 7.4 for $n + 1$. The case $a = 0$ having been treated (7.7), we assume that $a \neq 0$. We shall write $x_0, \ldots, x_n$ for the coordinates of $\mathbb{A}^{n+1}$. Let $g = x_0 : V_a \to \mathbb{G}_m$, τ the involution $x \mapsto ax^{-1}$ of $\mathbb{G}_m$, and let π, as above, be the product of the coordinates: $\mathbb{A}^{n+1} \to \mathbb{A}^1$, with σ their sum. We shall denote by $\mathscr{F}(\psi\sigma)$ the sheaf $\mathscr{F}(\psi(\sum x_i))$ both on $\mathbb{A}^{n+1}$ and on $\mathbb{A}^n$.

7.14

Let us write down the Leray spectral sequence for g. Since the sheaf $\mathscr{F}(\psi\sigma)$ on $\mathbb{A}^{n+1}$ is the external tensor product of the sheaves $\mathscr{F}(\psi)$ on $\mathbb{A}^1$ and $\mathscr{F}(\psi\sigma)$ on $\mathbb{A}^n$, we find (cf. (7.1.4)).

$$'E_2^{pq} = \mathrm{H}_c^p(\mathbb{G}_m, \mathscr{F}(\psi) \otimes \tau^* \mathsf{R}^q\pi_!\mathscr{F}(\psi\sigma)) \Rightarrow \mathrm{H}_c^{p+q}(V_a, \mathscr{F}(\psi\sigma))$$
$$''E_2^{pq} = \mathrm{H}^p(\mathbb{G}_m, \mathscr{F}(\psi) \otimes \tau^* \mathsf{R}^q\pi_*\mathscr{F}(\psi\sigma)) \Rightarrow \mathrm{H}^{p+q}(V_a, \mathscr{F}(\psi\sigma)).$$

By hypothesis, $\mathsf{R}^q\pi_!\mathscr{F}(\psi\sigma) \xrightarrow{\sim} \mathsf{R}^q\pi_*\mathscr{F}(\psi\sigma)$, and this sheaf is zero for $q \neq n-1$. The sheaf $\mathscr{F}(\psi)$ on $\mathbb{G}_m$ is wildly ramified at ∞ and unramified at 0, while $\tau^*\mathsf{R}^q\pi_!\mathscr{F}(\psi\sigma)$ is wildly ramified at 0 (without invariants under wild inertia, and with Swan conductor 1) and tamely ramified at ∞. Their tensor product is therefore

a) wildly ramified at 0 and ∞, without invariants under wild inertia;

b) of Swan conductor 1 at 0 and n (the rank of $\mathsf{R}^q\pi_!\mathscr{F}(\psi\sigma)$) at ∞.

We have ${}'E_2^{pq} \xrightarrow{\sim} {}''E_2^{pq}$, and ${}'E_2^{pq} = 0$ except for $p = 1$ and $q = n-1$. The Euler-Poincaré formula finally gives

$$\dim {}'E_2^{1,n-1} = 1 + n;$$

this completes the proof. □

7.15 Remark

The isomorphism obtained in 7.14

$$\mathrm{H}_c^n(V_a^n, \mathscr{F}(\psi\sigma)) = \mathrm{H}_c^1(\mathbb{G}_m, \mathscr{F}(\psi) \otimes \tau^*\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)) \tag{7.15.1}$$

allows us, with the notation of 7.5, to calculate the product $\det(F)$ of the Frobenius eigenvalues α_i. The formalism of [13] applies, since $\mathscr{F}(\psi)$ and $\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)$ belong to infinite compatible systems of ℓ-adic representations. This reduces us to local problems, which can be simplified by noting that (cf. [13, 9.5])

a) the global constants for $\mathscr{F}(\psi)$ and $\tau^*\mathsf{R}^{n-1}\pi_!\mathscr{F}(\psi\sigma)$ are easy to calculated easily, since the cohomology of these sheaves is elementary.

b) at every point, for either of these sheaves, the corresponding ℓ-adic representation of the decomposition group has unramified semisimplification.

The result, for the sum with n variables, is that

$$\det\left(F, \mathrm{H}_c^{n-1}(V_a^{n-1}, \mathscr{F}(\psi))\right) = q^{\frac{n(n-1)}{2}}. \tag{7.15.2}$$

7.16 Remark

For $p = 2$, the sheaf (ψ) is orthogonal. For $a \neq 0$ and n odd, we deduce from 7.4(ii) and Poincaré duality a symmetric bilinear form on $\mathrm{H}_c^{n-1}(V_a^{n-1}, \mathscr{F}(\psi))$ with values in $\bar{\mathbb{Q}}_\ell(1-n)$. With respect to this form, F is an orthogonal similitude of multiplier q^{n-1}, and $q^{\frac{1-n}{2}}F$ belongs to the special orthogonal group, by (7.15.2). One of the eigenvalues of F is therefore $q^{\frac{n-1}{2}}$, and the others occur in pairs of roots α, $\bar{\alpha}$ whose product is q^{n-1}.

For $n = 3$, L. Carlitz [8] obtained a more precise result, equivalent to the following proposition.

Proposition 7.17. *For $p = 2$, and $\psi(x) = (-1)^{\mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_2}(x)}$, we have an isomorphism*

$$\mathrm{H}_c^2(V_a^2, \mathscr{F}(\psi\sigma)) = \mathrm{Sym}^2 \mathrm{H}_c^1(V_a^1, \mathscr{F}(\psi\sigma)).$$

The second tensor power of $\mathrm{H}_c^1(V_a^1, \mathscr{F}(\psi\sigma))$ is $\mathrm{H}_c^2(V_a^1 \times V_a^1, \mathscr{F}(\psi\sigma) \boxtimes \mathscr{F}(\psi\sigma))$ (Künneth), with the symmetry $x \otimes y \mapsto y \otimes x$ represented geometrically by $-\tau^*$, where τ is the automorphism $(x, y) \mapsto (y, x)$ of $V_a \times V_a$. If X' is the quotient of $V_a^1 \times V_a^1$ by $\{\mathrm{id}, \tau\}$ and π is the projection of $V_a^1 \times V_a^1$ onto X', the group $\mathrm{Sym}^2 \mathrm{H}_c^1(V_a^1, \mathscr{F}(\psi\sigma))$ is therefore identified with the cohomology of X' with coefficients in the part anti-invariant under τ of $\pi_*(\mathscr{F}(\psi\sigma) \boxtimes \mathscr{F}(\psi\sigma))$. Denote by s the function on X' such that $s\pi = \sigma\mathrm{pr}_1 + \sigma\mathrm{pr}_2$. We have $\mathscr{F}(\psi\sigma) \boxtimes \mathscr{F}(\psi\sigma) = \mathscr{F}(\psi(\sigma\mathrm{pr}_1 + \sigma\mathrm{pr}_2)) = \mathscr{F}(\psi s\pi) = \pi^*\mathscr{F}(\psi s)$ (an isomorphism compatible with τ). On the image $\pi(\Delta)$ of the diagonal, the desired sheaf of anti-invariants is therefore zero, while on the complement $X = X' \smallsetminus \pi(\Delta)$ it is the tensor product of $\mathscr{F}(\psi s)$ by $\varepsilon(P)$, where ε is the unique character of order 2 of $\{1, \tau\}$, and P the $\{1, \tau\}$-torsor on X given by the double covering induced by $V_a^1 \times V_a^1$. We have

$$\mathrm{Sym}^2 \mathrm{H}_c^1(V_a^1, \mathscr{F}(\psi\sigma)) = \mathrm{H}_c^2(X, \mathscr{F}(\psi s) \otimes \varepsilon(P)). \tag{7.17.1}$$

Let us compute X, s and $\varepsilon(P)$. On $V_a^1 \times V_a^1$, put $x_1 = x_1 \circ \mathrm{pr}_1$, $x_2 = x_2 \circ \mathrm{pr}_2$, $x_1' = x_1 \circ \mathrm{pr}_2$, $x_2' = x_2 \circ \mathrm{pr}_1$. We have $x_1x_2 = x_1'x_2' = a$. Identify functions on X' with functions on X invariant under τ. Let $s_1 = x_1 + x_1'$, $s_2 = x_2 + x_2'$, $p_1 = x_1x_1'$, $p_2 = x_2x_2'$. We have $s_1p_2 = as_2$, $s_2p_1 = as_1$ and X is defined in X' by $s_1 \neq 0$, $s_2 \neq 0$. On X, we therefore have $p_1 = as_1s_2^{-1}$, and (s_1, s_2) identifies X with $\mathbb{G}_m \times \mathbb{G}_m$. On X, the covering P has equation $T^2 - s_1T + as_1s_2^{-1}$ (take T to be the coordinate x_1). If $T = s_1t$, this is rewritten $t^2 - t + as_1^{-1}s_2^{-1}$. We therefore have $\varepsilon(P) = \mathscr{F}(\psi(as_1^{-1}s_2^{-1}))$. Since $s = s_1 + s_2$, we have $\mathscr{F}(\psi s) \otimes \varepsilon(P) = \mathscr{F}(\psi(s_1 + s_2 + as_1^{-1}s_2^{-1}))$ and the right-hand side of (7.17.1) is identified with the left-hand side of 7.17.

Remark $\mathrm{H}_c^1(V_a^1, \mathscr{F}(\psi\sigma))$ is also the first cohomology group of the elliptic curve obtained by completing the affine curve with equations

$$y^2 - y = x_1 + x_2$$
$$x_1x_2 = a;$$

its modular invariant is $j = a^{-2}$.

7.18 Remark

For χ a multiplicative character of $\mathbb{F}_q^\times$, and $a \neq 0$, the method of 7.14 applies to the study of the sum

$$S = \sum_{x_0 \cdots x_n = a} \chi(x_0)\psi(x_0 + \cdots + x_n)$$

(replace the sheaf $\mathscr{F}(\psi)$ on $\mathbb{G}_m$ by $\mathscr{F}(\chi \cdot \psi)$). We again find a cohomology group of dimension $n + 1$, and $|S| \leqslant (n + 1)q^{n/2}$. Such sums were considered by Salie and Mordell [28].

One may hope that our methods make it possible to study the sums

$$S_{k,\chi,a} = \sum_{N(x)=a} \chi(x)\psi\,\mathrm{tr}(x)$$

(k an étale algebra over $\mathbb{F}_q$, $a \in \mathbb{F}_q^\times$, χ a character of $k^\times$). The analog of 7.4 (for $a \neq 0$) should be true.

7.19

The symmetric group S_n acts on $\mathbb{A}^n$, preserving V_a and the map σ, and hence also the sheaf $(\psi\sigma)$, the inverse image under σ of the sheaf (ψ) on $\mathbb{G}_a$. It therefore acts on $\mathrm{H}_c^{n-1}(V_a, \mathscr{F}(\psi\sigma))$. Formula (7.2.5) admits the following cohomological interpretation.

Proposition 7.20. *The action of S_n on $\mathrm{H}_c^{n-1}(V_a, \mathscr{F}(\psi\sigma))$ is multiplication by the sign character.*

It suffices to show that a transposition τ acts by multiplication by -1. Since $\tau^2 = 1$, the only possible eigenvalues of τ are ± 1. Let V_a/τ be the quotient of V_a for $\{\mathrm{id}, \tau\}$. The map σ, and hence the sheaf $(\psi\sigma)$, descend to the quotient, and the subspace of $\mathrm{H}_c^{n-1}(V_a, \mathscr{F}(\psi\sigma))$ fixed by τ is $\mathrm{H}_c^{n-1}(V_a/\tau, \mathscr{F}(\psi\sigma))$. We must prove that this cohomology is zero.

Suppose $a \neq 0$. For $\tau = (1,2)$ the quotient V_a/τ is identified with $\mathbb{G}_a \times \mathbb{G}_m^{n-2}$, the quotient map being $(x_1, x_2, x_3, \ldots, x_n) \mapsto (x_1 + x_2, x_3, \ldots, x_n)$. Indeed, x_1 and x_2 are determined up to order by $x_1 + x_2$ and $x_1 x_2 = a/x_3 \cdots x_n$. In the coordinates $(t, x_3, \ldots, x_n)$, the sheaf takes the form $\mathscr{F}(\psi(t + x_3 + \cdots + x_n)) = \mathscr{F}(\psi(t)) \otimes \mathscr{F}(\psi(x_3 + \cdots + x_n))$. The Künneth formula and $\mathrm{H}_c^\bullet(\mathbb{G}_a, \mathscr{F}(\psi)) = 0$ therefore give the required vanishing.

For $a = 0$, we can deduce 7.20 from calculation 7.7.

7.21

The sums $\mathrm{K}_{n,a}$ in this section were studied by S. Sperber, in his thesis, and in [38] by methods p-adic methods inspired by Dwork and by Bombieri's article [6]. For $p \neq 2$, these methods determine the dependence on q in 7.5, the functional equation $\alpha_i = q^{n-1}\beta_i^{-1}$ between the Frobenius eigenvalues for the sums $\mathrm{K}_{n,a}$ and $\mathrm{K}_{n,b}$ if $b = (-1)^n a$ (see 7.6), and the value 7.15 of the product of the roots. For $p \geqslant n+3$, they also determine the p-adic valuations of the roots. The result is very beautiful: if we arrange the roots α_i in a suitable order, with $0 \leqslant i < n$, $\alpha_i q^{-i}$ are units. The restrictions on p are probably not essential.

8 Other applications

8.1

The identity 1.9.4 can sometimes be used to calculate the number of rational points of an algebraic variety over $\mathbb{F}_q$. In most cases where an exact result has been obtained, cohomology is expressed in terms of algebraic cycles; this is the case for rational surfaces, quadrics, smooth intersections of two quadrics of odd dimension, flag varieties.... This is not, however, always the case; in Lusztig's beautiful article [27], algebraic cycles do not appear.

8.2

For reductive groups, cohomology can be calculated by lifting to characteristic 0: if B is a Borel subgroup of G, G is a B-torsor over G/B; the cohomology of the smooth projective variety G/B is invariant under

specialization, and we use the Leray spectral sequence of $G \to G/B$ to prove the same result for G. It can be expressed conveniently in terms of *the* maximal torus T of G and the action *of the* Weyl group W on T (for the definition *of the* maximal torus, cf. [7, VIII §2 Rem.2] or [10, p.105]). It is the exterior algebra on its primitive part, which agrees up to a shift with the quotient $I(\mathrm{H}^\bullet(BG, \mathbb{Q}_\ell))$ of the cohomology of BG formed by the "indecomposable elements," and $\mathrm{H}^\bullet(BG, \mathbb{Q}_\ell) = \mathrm{H}^\bullet(BT, \mathbb{Q}_\ell)^W$. If X is the group of characters of T, we have altogether

$$\mathrm{H}^\bullet(G, \mathbb{Q}_\ell) = \bigwedge \left(I \left(\mathrm{Sym}^\bullet(X \otimes \mathbb{Q}_\ell(-1))^W \right) [-1] \right).$$

If G is defined over a finite field $\mathbb{F}_q$, $\mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$ acts on X and this isomorphism is Galois-compatible. Passing to cohomology with compact supports, we recover the classical formulas for the number of rational points of G. If $F^* : X \to X$ is defined by the Frobenius morphism $F : T \to T$, and $\deg_G(F)$ is the degree $q^{\dim G}$ of $F : G \to G$, we have

$$|G_0(\mathbb{F}_q)| = \deg_G(F) \cdot \det \left(1 - {F^*}^{-1}, I(\mathrm{Sym}^\bullet(X \otimes \mathbb{Q})^W) \right). \tag{8.2.1}$$

The same formula applies to the Ree and Suzuki groups; these groups are of the form G^F (reductive G, F^2 a Frobenius), and it suffices to verify that the Lefschetz trace formula holds for F acting on G. If B is a Borel subgroup stable by F, we verify it directly for F acting on B; for the action on G/B, which is proper and smooth, we can invoke the general theorems.

8.3

The trace formula was used by Deligne–Lusztig [10], Kazhdan [25] and Springer [39] in the study of complex representations of finite groups $G_0(\mathbb{F}_q)$, for G_0 reductive over $\mathbb{F}_q$. The work of Kazhdan and Springer contains admirable examples of how the trace formula permits "analytic continuation" from a split situation to a nonsplit situation (see 1.13).

Chapter 7

Finiteness theorems in ℓ-adic cohomology

1 Statement of the theorems

Throughout this exposé, S will be a Noetherian scheme and A a left Noetherian torsion ring annihilated by an integer invertible on S. Our main result is the following.

Theorem 1.1. *Suppose that S is regular of dimension 0 or 1. Let $f : X \to Y$ be a morphism of S-schemes of finite type, and let $\mathscr{F}$ be a constructible sheaf of left A-modules on X. Then the sheaves $\mathsf{R}^i f_* \mathscr{F}$ are constructible.*

The proof will be given in Section 2 and in 3.10.

1.2 Remark

In characteristic 0, this result is less general than [2, XIX paragraphe 5], which proves the conclusion of the theorem for every morphism of finite type of excellent schemes of characteristic 0. The proof of [2, XIX] uses, on the one hand, resolution of singularities and, on the other, the assumption that the schemes have equal characteristic (in order to deduce the "purity theorem" from resolution).

1.3 Remark

We shall say that a complex $K \in \mathrm{Ob}\,\mathsf{D}(X, A)$ is *constructible* if its cohomology sheaves are constructible, and shall use a subscript c for the subcategory of $\mathsf{D}(X, A)$ (or D^+, D^-, D^b) consisting of constructible complexes. In the language of derived categories, which we shall use freely, 1.1 says that $\mathsf{R}f_* : \mathsf{D}^+(X, A) \to \mathsf{D}^+(Y, A)$ sends D^+_c into D^+_c. More explicitly:

a) For K concentrated in a sheaf $\mathscr{F}$ in degree 0, one has $\mathscr{H}^i \mathsf{R}f_* K = \mathsf{R}^i f_* \mathscr{F}$; the derived statement therefore implies the theorem.

b) Conversely, one invokes the spectral sequence

$$E_1^{pq} = \mathsf{R}^q f_* \mathscr{H}^p(K) \Rightarrow \mathscr{H}^{p+q} \mathsf{R}f_* K.$$

Speaking the language of derived categories has the advantage of replacing, by the simple transitivity formula $\mathsf{R}(fg)_* = \mathsf{R}f_*\mathsf{R}g_*$, the Leray spectral sequence $E_2^{pq} = \mathsf{R}^p f_* \mathsf{R}^q g_* \Rightarrow \mathsf{R}^{p+q}(fg)_*$.

Under the hypotheses of 1.1, it follows from [2] that $\mathsf{R}f_*$ has finite cohomological dimension. This makes it possible above to replace D^+ by D, D^-, or D^b.

1.4 Ideas of the proof

a) Poincaré duality treats the case where X is smooth, $\mathscr{F}$ is locally constant, and $Y = S$. The dimension hypothesis is seen naïvely when calculating the $\mathsf{R}\mathscr{H}\mathrm{om}$ on S.

b) Factor f as gj: g proper and j an open immersion. One has $\mathsf{R}f_* = \mathsf{R}g_*\mathsf{R}j_*$, and the finiteness theorem for proper morphisms controls $\mathsf{R}g_*$. By dévissage, one reduces to assuming that X is smooth, $\mathscr{F}$ is locally constant, f is an open immersion, and Y is proper over S.

c) Naively, an induction on $\dim X$ (with a change of S) makes it possible to suppose that the theorem is true outside a part of Y finite over S. Denoting the structure morphisms by $a : X \to S$ and $b : Y \to S$, one then shows that if the theorem were false for $\mathscr{F}$ and $\mathsf{R}f_*$, it would also be false for $\mathscr{F}$ and $\mathsf{R}b_*\mathsf{R}f_* = \mathsf{R}a_*$: a contradiction.

Corollary 1.5. *Under the hypotheses of the theorem, the categories* $\mathsf{D}_c(X, A)$ *and* $\mathsf{D}_c(Y, A)$ *are taken into one another by the* 4 *operations* $\mathsf{R}f_*$, $\mathsf{R}f_!$, f^*, $\mathsf{R}f^!$.

The case of $\mathsf{R}f_*$ is treated above, that of $\mathsf{R}f_!$ in [2, XVII 5.3], and that of f^* is clear. It remains to consider $\mathsf{R}f^!$. The problem is local, which makes it possible to suppose that f factors as $X \xrightarrow{i} Z \xrightarrow{g} Y$ (i a closed immersion and g smooth, purely of relative dimension n). Poincaré duality $\mathsf{R}g^!K \xrightarrow{\sim} g^*K(n)[2n]$ and the transitivity $\mathsf{R}f^! = \mathsf{R}i^!\mathsf{R}g^!$ reduce us to proving 1.5 for i. For every $L \in \mathsf{D}(Z, A)$, if j is the inclusion into Z of $U = Z \smallsetminus X$, then $i_*\mathsf{R}i^!L$ is the mapping cylinder of $L \to \mathsf{R}j_*j^*L$, and 1.5 for i follows from 1.1 for j.

Corollary 1.6. *Let X be as in the theorem, and suppose that A is commutative. Then, if $\mathscr{F}$ and $\mathscr{G}$ are constructible sheaves of A-modules on X, the $\mathscr{E}\mathrm{xt}^i_A(\mathscr{F}, \mathscr{G})$ are constructible:* $\mathsf{R}\mathscr{H}\mathrm{om}$ *sends* $\mathsf{D}^-_c(X, A) \times \mathsf{D}^+_c(X, A)$ *into* $\mathsf{D}^+_c(X, A)$.

By dévissage of $\mathscr{F}$, we reduce to assuming that $\mathscr{F}$ has the form $j_!\mathscr{F}_1$ ($j : Y \hookrightarrow X$ a locally closed immersion and $\mathscr{F}_1$ locally constant on Y). One then has

$$\mathsf{R}\mathscr{H}\mathrm{om}(j_!\mathscr{F}_1, \mathscr{G}) = \mathsf{R}j_*\,\mathscr{H}\mathrm{om}(\mathscr{F}_1, \mathsf{R}j^!\mathscr{G}):$$

by 1.3, we are reduced to the case where $\mathscr{F}$ is locally constant—or even constant, since the problem is local. For $\mathscr{F}$ locally constant and constructible, one has $\mathscr{H}\mathrm{om}(\mathscr{F}, \mathscr{G})_x = \mathrm{hom}(\mathscr{F}_x, \mathscr{G}_x)$, and likewise for $\mathsf{R}\mathscr{H}\mathrm{om}$. For $\mathscr{F}$ constant, the $\mathscr{E}\mathrm{xt}^i$ can therefore be calculated using a projective resolution of finite type of its constant value, and the $\mathscr{E}\mathrm{xt}^i$ are constructible if $\mathscr{G}$ is—whence the corollary.

1.7 Remark

Since $\mathsf{R}f_*$ has finite cohomological dimension, it takes complexes of finite tor-dimension (resp. $\leqslant d$) to complexes of finite tor-dimension (resp. $\leqslant d$) [2, XVII 5.2.11]. [2, XVII 5.2.10] and the proof of 1.5 then show that finite tor-dimension is stable under the 4 operations $\mathsf{R}f_*$, $\mathsf{R}f_!$, f^*, $\mathsf{R}f^!$. A variant of the argument of 1.6 (apply dévissage to K along a partition of X so that the cohomology sheaves become

locally constant, then localize so as to replace K by a finite complex of locally free sheaves of finite type) then shows that, for A commutative, $\mathsf{R}\mathscr{H}\mathrm{om}$ induces

$$\mathsf{R}\mathscr{H}\mathrm{om} : \mathsf{D}^b_{\mathrm{ctf}}(X, A) \times \mathsf{D}^b_{\mathrm{tf}}(X, A) \to \mathsf{D}^+_{\mathrm{tf}}(X, A)$$

(t.f.: finite tor-dimension).

1.8

Under the hypotheses of the theorem, the same method allows us to prove a local biduality theorem $K \xrightarrow{\sim} DDK$ (Section 4). We also prove a finiteness theorem for vanishing-cycle sheaves. For arbitrary S, one still obtains a "generic" theorem:

Theorem 1.9. *Let $f : X \to Y$ be a morphism of S-schemes of finite type, and let $\mathscr{F}$ be a constructible sheaf of A-modules on X. There is a dense open subset U of S such that*

(i) Above U, the $\mathsf{R}^i f_* \mathscr{F}$ are constructible, and all but finitely many of them vanish.

(ii) Formation of the $\mathsf{R}^i f_* \mathscr{F}$ is compatible with every base change $S' \to U \subset S$.

For S the spectrum of a field, one has $U = S$.

If S is the spectrum of a field, a passage-to-the-limit argument gives compatibility with S-base changes for the $\mathsf{R}^i f_*$ for every quasi-compact quasi-separated morphism of S-schemes and every sheaf $\mathscr{F}$.

Corollary 1.10. *Let X be a scheme of finite type over a separably closed k, and let $\mathscr{F}$ be a constructible sheaf of A-modules. Then the $\mathrm{H}^i(X, \mathscr{F})$ are of finite type.*

This is the special case $S = \mathrm{Spec}(k)$, $Y = S$.

Corollary 1.11. *Let X and Y be two schemes of finite type over a separably closed k, and let $K \in \mathrm{Ob}\, \mathsf{D}^-(X, A^\circ)$, $L \in \mathrm{Ob}\, \mathsf{D}^-(Y, A)$ be complexes of sheaves of right and left modules. The Künneth arrow*

$$\mathsf{R}\Gamma(X, K) \overset{\mathsf{L}}{\otimes} \mathsf{R}\Gamma(Y, L) \to \mathsf{R}\Gamma(X \times Y, \mathrm{pr}_1^* K \overset{\mathsf{L}}{\otimes} \mathrm{pr}_2^* L)$$

is an isomorphism.

Proceed as in [2, XVII 5.4.3]:

$$\begin{array}{ccc} X \times Y & \xrightarrow{\mathrm{pr}_1} & X \\ \downarrow{\scriptstyle \mathrm{pr}_2} & & \downarrow{\scriptstyle a} \\ Y & \xrightarrow{b} & \mathrm{Spec}(k) \end{array}$$

base change by b shows that $\mathsf{R}\mathrm{pr}_{2,*}\mathrm{pr}_1^* K$ is $b^* \mathsf{R}\Gamma(X, K)$. One then has (cf. [2, XVII 5.2.11] and the proof)

$$\begin{aligned} \mathsf{R}\Gamma(X \times Y, \mathrm{pr}_1^* K \overset{\mathsf{L}}{\otimes} \mathrm{pr}_2^* L) &= \mathsf{R}\Gamma(Y, \mathsf{R}\mathrm{pr}_{2,*}(\mathrm{pr}_1^* K \overset{\mathsf{L}}{\otimes} \mathrm{pr}_2^* L)) \\ &= \mathsf{R}\Gamma(Y, (\mathsf{R}\mathrm{pr}_{2,*}\mathrm{pr}_1^* K) \overset{\mathsf{L}}{\otimes} L) \\ &= \mathsf{R}\Gamma(Y, b^* \mathsf{R}\Gamma(X, K) \overset{\mathsf{L}}{\otimes} L) \\ &= \mathsf{R}\Gamma(X, K) \overset{\mathsf{L}}{\otimes} \mathsf{R}\Gamma(Y, L) \end{aligned}$$

1.12

Finally, as a local counterpart of 1.9(ii), we shall prove a generic local acyclicity theorem (2.13).

2 Generic theorems

In this section, we prove 1.9 and the generic local acyclicity theorem 2.13. To prove 1.9, we immediately reduce to the case where S is integral. Let η be its generic point.

2.1

We begin by proving 1.9 under the following additional assumptions: X is smooth over S, purely of relative dimension n, $A = \mathbb{Z}/m$, $\mathscr{F}$ is locally constant, and $Y = S$. Let $\mathscr{F}' = \mathscr{H}\mathrm{om}(\mathscr{F}, \mathbb{Z}/m)$. We shall show that if the $\mathsf{R}^i f_! \mathscr{F}'$ are locally constant, the conclusions of 1.9 hold with $U = S$. Recall that if $\mathscr{L}$ is a constructible locally constant sheaf, the $\mathscr{E}\mathrm{xt}^i(\mathscr{L}, \mathscr{G})$ may be computed fiber by fiber, and are constructible if $\mathscr{G}$ is. Recall also that $\mathbb{Z}/m$ is an injective $\mathbb{Z}/m$-module and is the dualizing module, so that $\mathscr{F} = \mathsf{R}\mathscr{H}\mathrm{om}(\mathscr{F}', \mathbb{Z}/m)$. Poincaré duality

$$\mathsf{R}f_* \, \mathsf{R}\mathscr{H}\mathrm{om}(K, \mathsf{R}f^! L) = \mathsf{R}\mathscr{H}\mathrm{om}(\mathsf{R}f_! K, L),$$

for $K = \mathscr{F}'$, $L = \mathbb{Z}/m$, $\mathsf{R}f^! L = \mathbb{Z}/m[2n](n)$, therefore gives

$$\mathsf{R}^{2n-i} f_* \mathscr{F} = \mathscr{H}\mathrm{om}(\mathsf{R}^i f_! \mathscr{F}', \mathbb{Z}/m)(-n).$$

The sheaf on the right is locally constant and its formation commutes with every base change, which proves the assertion.

2.2

Let us prove 1.9 under the additional assumptions that X is smooth over S, $\mathscr{F}$ is locally constant, and $Y = S$.

a) By decomposing X into connected components, we reduce to the case where it is purely of some relative dimension n over S.

b) By decomposing A as a product, we reduce to the case where $\ell^m A = 0$, with ℓ a prime invertible on S; we then filter $\mathscr{F}$ by the $\ell^k \mathscr{F}$ and use the corresponding spectral sequence to reduce to the case $\ell \mathscr{F} = 0$. We may then replace A by $A/\ell A$ and assume that $\ell A = 0$.

c) After replacing S by a suitable U, we may assume that there exists a finite étale Galois covering X_1/X, with Galois group G, such that the inverse image $\mathscr{F}_1$ of $\mathscr{F}$ on X_1 is a constant sheaf, with constant value F. Writing f_1 for the projection of X_1 onto S, we then have

$$\mathsf{R}^i f_{1*} \mathscr{F}_1 = (\mathsf{R}^i f_{1*} \mathbb{Z}/\ell) \otimes_{\mathbb{Z}/\ell} F.$$

We also have the Hochschild–Serre spectral sequence (or the Leray spectral sequence for the covering X_1/X)

$$\underline{\mathrm{H}}^p(G, \mathsf{R}^q f_{1*} \mathscr{F}_1) \Rightarrow \mathsf{R}^{p+q} f_* \mathscr{F}.$$

By 2.1, after shrinking U if necessary, we may assume that the $\mathsf{R}^i f_{1*} \mathbb{Z}/\ell$ are locally constant and that their formation commutes with every base change. The same property then holds for the $\mathsf{R}^i f_* \mathscr{F}$. Finally, if $\bar{\eta}$ is a geometric point localized at the generic point η of S, almost all the $\mathsf{R}^i f_* \mathscr{F}$ vanish, because the $(\mathsf{R}^i f_* \mathscr{F})_{\bar{\eta}} = \mathrm{H}^i(X_{\bar{\eta}}, \mathscr{F})$ do.

2.3

We prove by induction on n that

$$\begin{array}{c}\text{The conclusions of 1.9 hold when } \dim X_\eta \leqslant n \\ \text{and } f \text{ is an open immersion with dense image.}\end{array} \qquad (*_n)$$

For $n = 0$, after shrinking S if necessary, we have $X = Y$: $(*_0)$ is clear. Assume $(*_{n-1})$, and prove $(*_n)$. In $(*_{n-1})$, "open immersion" may be replaced by "immersion," as is seen by factoring it as an open immersion followed by a closed immersion.

Lemma 2.4. *After shrinking S if necessary, the conclusions of 1.9 hold over $Y' \subset Y$, where the complement Y_1 of Y' is finite over S.*

The assertion is local on Y, which we may assume affine: $Y \subset \mathbb{A}^n_S$. The induction hypothesis $(*_{n-1})$ applies to

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow & \downarrow{\scriptstyle \mathrm{pr}_i} \\ & & \mathbb{A}^1_S \end{array}$$

Thus, for each i, there exists a dense open subset U_i of $\mathbb{A}^1_S$ such that the conclusions of 1.9 hold over $\mathrm{pr}_i^{-1}(U_i)$; they hold over the union of the $\mathrm{pr}_i^{-1}(U_i)$, and 2.4 follows.

2.6

Let us prove $(*_n)$ when X is smooth over S and $\mathscr{F}$ is locally constant on X. Since the problem is local on Y, we may first assume Y affine and then projective (replacing Y by its closure in a projective space). Let $i : Y_1 \hookrightarrow Y$ and $j : Y' \to Y$ be as in 2.4.

$$\begin{array}{ccccc} X & \hookrightarrow & Y & \xleftarrow{i} & Y_1 \\ & {\scriptstyle a}\searrow & \downarrow{\scriptstyle b} & \swarrow{\scriptstyle b_1} & \\ & & S & & \end{array}$$

After shrinking S if necessary, we know that $j^*\mathsf{R}f_*\mathscr{F}$ is constructible and that its formation commutes with every base change in S; we also know that $\mathsf{R}a_*\mathscr{F} = \mathsf{R}b_*\mathsf{R}f_*\mathscr{F}$ is constructible and that its formation commutes with every base change.

Apply $\mathsf{R}b_*$ to the triangle defined by the exact sequence

$$0 \longrightarrow j_!j^*\mathsf{R}f_*\mathscr{F} \longrightarrow \mathsf{R}f_*\mathscr{F} \longrightarrow i_!i^*\mathsf{R}f_*\mathscr{F} \longrightarrow 0: \qquad (1)$$

we obtain a triangle

$$\longrightarrow \mathsf{R}b_*j_!j^*\mathsf{R}f_*\mathscr{F} \longrightarrow \mathsf{R}a_*\mathscr{F} \longrightarrow b_{1*}i^*\mathsf{R}f_*\mathscr{F} \longrightarrow \qquad (2)$$

in which the first two terms are constructible and their formation commutes with every base change in S (for the first, by the finiteness theorem for the proper morphism b). The same therefore holds for the third. Since b_1 is finite, it follows that $i^*\mathsf{R}f_*\mathscr{F}$ is constructible and its formation commutes with every base change in S; the same holds for $\mathsf{R}f_*\mathscr{F}$ by (1).

2.7

Let us prove $(*_n)$ in general. We first reduce to the case where X contains a dense open subset V smooth over S. When S is the spectrum of a perfect field, it is enough to replace X by X_{red} (and Y by Y_{red}). In general, we must shrink S, make a finite, radicial, surjective base change $S' \to S$, and replace X and Y by X_{red} and Y_{red}. Since the étale topology is insensitive to finite, radicial, surjective morphisms, this is harmless. After shrinking V, we may assume that $\mathscr{F}$ is locally constant on V.

$$V \overset{j}{\hookrightarrow} X \overset{f}{\hookrightarrow} Y.$$

Define Δ by the triangle

$$\longrightarrow \mathscr{F} \longrightarrow \mathsf{R}j_*j^*\mathscr{F} \longrightarrow \Delta \longrightarrow . \tag{1}$$

The cohomology sheaves of Δ are supported on $X \smallsetminus V$, and $\dim(X \smallsetminus V)_\eta < n$. The induction hypothesis therefore allows us to assume that $\mathsf{R}f_*\Delta$ is constructible. Applying $\mathsf{R}f_*$ to the triangle (1), we obtain a triangle

$$\longrightarrow \mathsf{R}f_*\mathscr{F} \longrightarrow \mathsf{R}(fj)_*j^*\mathscr{F} \longrightarrow \mathsf{R}f_*\Delta \longrightarrow$$

in which two of the vertices are constructible and their formation commutes with every base change in S. The same therefore holds for the third.

2.8

Let us prove 1.9. The problem is local on Y, which we may assume affine. Taking an affine covering of X and invoking the Leray spectral sequence, we reduce to the case where X is affine as well. This ensures that we can factor f as an open immersion followed by a proper morphism: $f = gj$, hence $\mathsf{R}f_* = \mathsf{R}g_*\mathsf{R}j_*$. The open immersion is covered by $(*_n)$, and the proper morphism by the finiteness theorem.

The following corollaries are proved as in Section 1.

Corollary 2.9. *Under the hypotheses of the theorem, for K in $\mathsf{D}^b_c(X,A)$ or $\mathsf{D}^b_c(Y,A)$ respectively, there exists a nonempty open subset U of S over which $\mathsf{R}f_*K$, $\mathsf{R}f_!K$, f^*K, and $\mathsf{R}f^!K$ belong to $\mathsf{D}^b_c(Y,A)$ or $\mathsf{D}^b_c(X,A)$, and their formation commutes with every base change $S' \to U \subset S$.*

Corollary 2.10. *Let X be as in the theorem, and suppose that A is commutative. Then, if $\mathscr{F}$ and $\mathscr{G}$ are constructible sheaves of A-modules on X, over a dense open subset U of S, the $\mathscr{E}\mathrm{xt}^i_A(\mathscr{F},\mathscr{G})$ are again constructible and their formation commutes with every base change $S' \to U \subset S$.*

2.11

If x is a geometric point of a scheme X, we denote by X_x the strict henselization of X at x. For $f : X \to S$ and t a geometric point of S, we denote by X_t the geometric fiber of X at t. Finally, for x a geometric point of X and t a geometric point of $S_{f(x)}$, $(X_x)_t$ is the fiber at t of $X_x \to S_{f(x)}$.

Definition 2.12. *Let $f : X \to S$ and $K \in \mathrm{Ob}\,\mathsf{D}^+(X,A)$. We say that f is* locally acyclic at x relative to *K if, for every geometric point t of $S_{f(x)}$, we have $K_x = \mathsf{R}\Gamma(X_x,K) \xrightarrow{\sim} \mathsf{R}\Gamma(X_{x,t},K)$. We say that f is* locally acyclic relative to *K if this holds for every x, and* universally locally acyclic relative to *K if it remains true after every base change $S' \to S$.*

Theorem 2.13. *Let $f : X \to S$ be a morphism of finite type and let $\mathscr{F}$ be constructible on X. There exists a dense open subset U of S over which f is universally locally acyclic relative to $\mathscr{F}$.*

We shall use the following result, proved in the appendix.

Lemma 2.14. *Let $X \xrightarrow{f} S_1 \xrightarrow{g} S_2$. If g is smooth and f is universally locally acyclic relative to K, then so is gf.*

We reduce to the case where S is integral, with generic point η, and argue by induction on $\dim X_\eta$. We first deduce from the induction hypothesis that

$$\begin{array}{c}\text{After shrinking } S\text{, there exists } T \subset X\text{, finite over } S\text{, such that } f \\ \text{is universally locally acyclic outside } T.\end{array} \tag{A}$$

Since this question is local, we localize on X and factor f as $X \xrightarrow{u} \mathbb{A}^1_S \to S$. The induction hypothesis applies to u, and we conclude by 2.14 and the usual arguments.

To prove the theorem, we may assume that X is proper, since the problem is local. We shrink S so that the $\mathsf{R}^i f_* \mathscr{F}$ are locally constant and (A) applies, and use the following lemma.

Lemma 2.15. *Let $f : X \to S$ be proper and let $\mathscr{F}$ be constructible. Suppose that the $\mathsf{R}^i f_* \mathscr{F}$ are locally constant and that, outside $T \subset X$ finite over S, f is locally acyclic relative to $\mathscr{F}$. Then f is locally acyclic relative to $\mathscr{F}$.*

Let s be a geometric point of S, let S_s be the strict localization of S at s, and let t be a geometric point of S_s. Let $X_{(s)}$ be the inverse image of X on S_s, let $\bar{j} : X_t \to X_{(s)}$, and let $i : X_s \hookrightarrow X_{(s)}$. Again denoting by $\mathscr{F}$ the inverse image of $\mathscr{F}$ on $X_{(s)}$ or X_t, we must prove that $i^*\mathscr{F} \xrightarrow{\sim} i^*\mathsf{R}\bar{j}_*\mathscr{F}$. Let Δ be the mapping cylinder of this map. Its cohomology sheaves are supported on T_s. To prove that $\Delta = 0$, it therefore suffices to prove that $\mathsf{R}\Gamma(X_s, \Delta) = 0$. This is the mapping cylinder of $(\mathsf{R}f_*\mathscr{F})_s = \mathsf{R}\Gamma(X_s, \mathscr{F}) \to \mathsf{R}\Gamma(X_s, i^*\mathsf{R}\bar{j}_*\mathscr{F}) = \mathsf{R}\Gamma(X_{(s)}, \mathsf{R}\bar{j}_*\mathscr{F}) = \mathsf{R}\Gamma(X_t, \mathscr{F}) = (\mathsf{R}f_*\mathscr{F})_t$. By hypothesis this specialization morphism is an isomorphism, and $\Delta = 0$.

Corollary 2.16. *When S is the spectrum of a field, every S-scheme X is universally locally acyclic relative to K, for every $K \in \mathrm{Ob}\,\mathsf{D}^+(X, A)$.*

This follows from 2.13 by passage to the limit (which is possible because the U of 2.13 is always equal to S.

3 Proof of 1.1 and constructibility of the vanishing-cycle sheaves

3.1

Let S be a strictly local trait (the spectrum of a henselian discrete valuation ring with separably closed residue field). Denote its closed and generic points by s and η, and let $\bar{\eta}$ be a geometric point localized at η (a separable closure of $k(\eta)$). Let X be over S and let $\mathscr{F}$ be a sheaf on X_η. Recall the definition of the vanishing-cycle sheaves $\mathsf{R}^i\Psi_\eta(\mathscr{F})$ (sheaves on X_s, endowed with an action of $\mathrm{Gal}(\bar{\eta}/\eta)$). Let $i : X_s \hookrightarrow X$, $j : X_\eta \hookrightarrow X$, and let $\bar{j}$ be the composite $X_{\bar{\eta}} \to X_\eta \hookrightarrow X$; then

$$\mathsf{R}^i\Psi_\eta(\mathscr{F}) = i^*\mathsf{R}^i\bar{j}_*(\bar{j}^*\mathscr{F}).$$

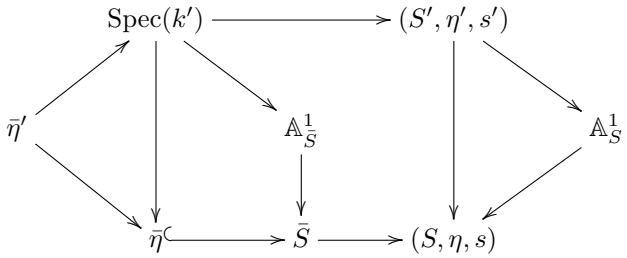
Here is the main theorem of this subsection. It improves [16, XIII 2.3.1, 2.4.2].

Theorem 3.2. *Suppose that X is of finite type over S and that $\mathscr{F}$ is constructible. Then the vanishing-cycle sheaves $\mathsf{R}^i\Psi_\eta(\mathscr{F})$ are constructible.*

We argue by induction on the dimension n of X_η. We may moreover assume—and do assume—that X_η is dense in X (replacing X by $\bar{X}_\eta$ does not change the $\mathsf{R}^i\Psi$, which are supported on $\bar{X}_\eta$).

Lemma 3.3. *If 3.2 holds when $X_\eta < n$, and if $\dim X_\eta = n$, then there exist constructible subsheaves $\mathscr{G}_i$ of the $\mathsf{R}^i\Psi_\eta(\mathscr{F})$ such that the supports of the local sections of $\mathsf{R}^i\Psi_\eta(\mathscr{F})/\mathscr{G}_i$ are finite.*

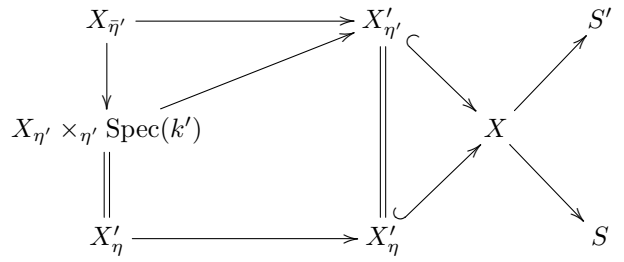
Let s' be a geometric generic point of the affine line over s, and let S' be the strict localization at s' of the affine line $\mathbb{A}^1_S$ over S. It is again a strictly local trait, and uniformizers for S are uniformizers on S'.



Let $k' = k(\bar{\eta}) \otimes_{k(\eta)} k(\eta')$. If $\bar{S}$ is the normalization of S in $\bar{\eta}$, then k' is the fraction field of the strict localization of $\mathbb{A}^1_{\bar{S}}$ at s'; it is therefore a field, and $\mathrm{Gal}(\bar{\eta}/\eta) = \mathrm{Gal}(k'/\eta')$. Let $\bar{\eta}'$ be the spectrum of an algebraic closure of k'. The Galois group $P = \mathrm{Gal}(\bar{\eta}'/k')$ is a pro-p group, where p is the characteristic exponent of $k(s)$.

Lemma 3.4. *For every scheme X' over S', and every sheaf $\mathscr{F}$ on $X'_\eta = X'_{\eta'}$, the vanishing-cycle sheaves for X'/S' and for X'/S are related by $\mathsf{R}^i\Psi_\eta(\mathscr{F}) = \mathsf{R}^i\Psi_{\eta'}(\mathscr{F})^P$.*

The following diagram compares the morphisms used in the definition of $\mathsf{R}\Psi$ for X'/S' and X'/S:



and the lemma follows from the Hochschild–Serre spectral sequence, since P is a pro-p group and p is invertible in A.

Let us prove 3.3. The question is local; this allows us to assume that X is affine, $X \subset \mathbb{A}^n_S$. Let f be one of the projections $X \subset \mathbb{A}^n_S \to \mathbb{A}^1_S$. Let X' be the "localization" $X_{\mathbb{A}^1_S} S'$ of X, and let $\mathscr{F}'$ be the inverse image of $\mathscr{F}$ on X'.

$$\begin{array}{ccccc} \mathscr{F}' \text{ on } X' & \longrightarrow & S' & & \\ \downarrow{\scriptstyle\lambda} & & \downarrow{\scriptstyle\lambda} & & \\ \mathscr{F} \text{ on } X & \xrightarrow{f} & \mathbb{A}^1_S & \longrightarrow & S. \end{array}$$

On $X'_s = X'_{s'}$, we have

$$\lambda^*\mathsf{R}^i\Psi_\eta(\mathscr{F}) = \mathsf{R}^i\Psi_\eta(\mathscr{F}') = \mathsf{R}^i\Psi_{\eta'}(\mathscr{F}')^P:$$

this sheaf is constructible because the induction hypothesis applies to X'/S', and we use the following lemma, applied to $X_s \subset \mathbb{A}^n_S$ and to the vanishing-cycle sheaves.

Lemma 3.5. *Let $\mathbb{A}^n$ be affine n-space over a field k, let $X \subset \mathbb{A}^n$, let $\mathscr{F}$ be a sheaf of A-modules on X, and let $\bar\eta$ be a geometric generic point of $\mathbb{A}^1$. Let $X_{\bar\eta,i}$ be the geometric generic fiber of $X \subset \mathbb{A}^n \xrightarrow{\mathrm{pr}_i} \mathbb{A}^1$, and let $\mathscr{F}_{\bar\eta,i}$ be the restriction of $\mathscr{F}$ to $X_{\bar\eta,i}$. If the $\mathscr{F}_{\bar\eta,i}$ are constructible, there exists a constructible $\mathscr{F}' \subset \mathscr{F}$ such that the local sections of $\mathscr{F}/\mathscr{F}'$ have finite support.*

By passage to the limit, for each i

a) There exists an étale neighborhood U of $\bar\eta$ and a constructible sheaf $\mathscr{H}$ on $X_{U,i} = X \times_{\mathbb{A}^1,\mathrm{pr}_i} U$ whose restriction $\mathscr{F}_{\bar\eta,i}$ to $X_{\bar\eta,i}$ is the given one.

b) For a suitable U, the isomorphism $\mathscr{H}_{\bar\eta,i} \xrightarrow{\sim} \mathscr{F}_{\bar\eta,i}$ comes from $u : \mathscr{H} \to \mathscr{F}_{U,i}$, where $\mathscr{F}_{U,i}$ is the inverse image of $\mathscr{F}$ on $X_{U,i}$. Let $\varphi : X_{U,i} \to X$. The morphism u defines $v_i : \varphi_!\mathscr{H} \to \mathscr{F}$, with image $\mathscr{F}'_i$. We have $(\mathscr{F}/\mathscr{F}'_i)_{\bar\eta,i} = 0$, and for $\mathscr{F}'$ we take the sum of the $\mathscr{F}'_i$.

3.6

Let us prove 3.2 in dimension n. We may assume that X is affine (because the problem is local on X_s), and then projective over S (by taking the closure of X in a projective embedding). We then have a spectral sequence [16, I 2.2.3] (the Leray spectral sequence for $\bar j$, taking into account the base-change theorem for the proper morphism $X \to S$).

$$E_2^{pq} = \mathrm{H}^p(X_s, \mathsf{R}^q\Psi_\eta(\mathscr{F})) \Rightarrow \mathrm{H}^{p+q}(X_{\bar\eta}, \mathscr{F}). \tag{1}$$

Let $\mathscr{G}^q \subset \mathsf{R}^q\Psi_\eta(\mathscr{F})$ be as in 3.3, and let $\mathscr{H}^q$ be the quotient sheaf. We have $\mathrm{H}^i(X_s, \mathscr{H}^q) = 0$ for $i \neq 0$, and for $\mathscr{H}^q$, hence $\mathsf{R}^q\Psi_\eta(\mathscr{F})$, to be constructible, it suffices that $\mathrm{H}^0(X_s, \mathscr{H}^q)$ be of finite type.

Compute (1) modulo modules of finite type (i.e. in the quotient of the category of A-modules by the thick subcategory of A-modules of finite type). The long cohomology exact sequence deduced from $0 \to \mathscr{G}^q \to \mathsf{R}^q\Psi_\eta(\mathscr{F}) \to \mathscr{H}^q \to 0$, together with 1.10 applied to $\mathscr{G}^q$, gives $E_2^{pq} \sim \mathrm{H}^p(X_s, \mathscr{H}^q)$. In particular, $E_2^{pq} \sim 0$ for $p \neq 0$, $E_2^{0q} \sim \mathrm{H}^q(X_{\bar\eta}, \mathscr{F}) \sim 0$, and this completes the proof.

Parallel arguments give the following result, which improves [16, XIII 2.1.12, 2.4.2].

Proposition 3.7. *The formation of the vanishing-cycle sheaves commutes with changes of traits.*

Let $g : (S', \eta', s') \to (S, \eta, s)$ be a (surjective) morphism of strictly local traits, let X/S, let $\mathscr{F}$ be on X_η, of torsion prime to the residual characteristic, and let $(X', \mathscr{F}')$ be their inverse images on S'. Again denoting by g the morphisms parallel to g, such as $X'_{s'} \to X_s$, we must prove that

$$g^*\mathsf{R}\Psi_\eta(\mathscr{F}) \xrightarrow{\sim} \mathsf{R}\Psi_{\eta'}(\mathscr{F}'). \tag{3.7.1}$$

This morphism is defined by the commutative diagram

$$\begin{array}{ccccc} X'_{s'} & \xrightarrow{i'} & X' & \xleftarrow{\bar j'} & X'_{\eta'} \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle g} \\ X_s & \xrightarrow{i} & X & \xleftarrow{\bar j} & X_{\bar\eta} \end{array}$$

A passage to the limit reduces us to the case where X is of finite type over S, and we argue by induction on $\dim X_\eta$. Since the question is local, we may assume that X is affine, and then projective. Assume (3.7.1) when $\dim X_\eta < n$. When $\dim X_\eta = n$, it then follows from the following two assertions, where Δ is the mapping cylinder of (3.7.1).

(A) The support of the local sections of the cohomology sheaves of Δ is finite.

(B) $\mathsf{R}\Gamma(X_s, \Delta) = 0$, hence $\Delta = 0$, by (A).

3.8

The proof of (A) parallels that of 3.3. We localize on X and consider maps $X \to \mathbb{A}^1_S$; denoting by $S(x)$ the strict henselization of $\mathbb{A}^1_S$ at the generic point of the special fiber, we apply the induction hypothesis to $X_1/S(x)$ ($X_1 = X \times_{\mathbb{A}^1} S(x)$) and to $S'(x)$. Taking invariants under a pro-p group, we find that Δ vanishes on the geometric generic fiber of $X_{s'} \to \mathbb{A}^1_s$. Since this holds locally on X and for every projection, we obtain (A).

3.9

For (B), the properness of X over S ensures that

$$\mathsf{R}\Gamma(X'_{s'}, \mathsf{R}\Psi_{\eta'}(\mathscr{F}')) = \mathsf{R}\Gamma(X'_{\bar{\eta}}, \mathscr{F}')$$

and

$$\mathsf{R}\Gamma(X_s, \mathsf{R}\Psi_{\eta}(\mathscr{F})) = \mathsf{R}\Gamma(X_{\bar{\eta}}, \mathscr{F}).$$

We have $\mathsf{R}\Gamma(X_s, \mathsf{R}\Psi_\eta(\mathscr{F})) \xrightarrow{\sim} \mathsf{R}\Gamma(X'_{s'}, g^*\mathsf{R}\Psi_\eta(\mathscr{F}))$ (invariance under change of the separably closed base field); thus $\mathsf{R}\Gamma(X'_{s'}, \Delta)$ is the mapping cylinder of $\mathsf{R}\Gamma(X_{\bar{\eta}}, \mathscr{F}) \to \mathsf{R}\Gamma(X'_{\bar{\eta}}, \mathscr{F}')$, an isomorphism by the same invariance theorem.

3.10

Let us prove 1.1. We reduce to the case where S is connected, hence integral. If $\dim S = 0$ (S is the spectrum of a field), apply 1.9, 1.10. If S has dimension 1, Theorem 1.9 ensures that the $\mathsf{R}^i f_* \mathscr{F}$ are constructible over the complement of a finite set T of points of S. It remains to see that their restrictions to the Y_t ($t \in T$) are constructible. It suffices to see this after localization at t: we may assume that S is a strictly local trait. The essential case is the following.

Lemma 3.11. *Let X be of finite type over S, let $j : X_\eta \hookrightarrow X$, $i : X_s \hookrightarrow X$, and let $\mathscr{F}$ be constructible on X_η. Then $i^*\mathsf{R}j_*\mathscr{F}$ is constructible.*

Let $I = \mathrm{Gal}(\bar{\eta}/\eta)$ be the inertia group. It is known to be an extension of $\widehat{\mathbb{Z}}_{p'}(1)$ by a pro-p group P (p being the characteristic exponent of the residue field). The Hochschild–Serre spectral sequence gives

$$E_2^{pq} = \mathscr{H}^p(I, \mathsf{R}^q\Psi_\eta(\mathscr{F})) \Rightarrow i^*\mathsf{R}^{p+q}j_*\mathscr{F}.$$

Let $R^q_t = \mathsf{R}^q\Psi_\eta(\mathscr{F})^P$ (t for tame). If σ is a generator of $\widehat{\mathbb{Z}}_{p'}(1)$, then $E_2^{0q} = \ker(\sigma - 1, R^q_t)$, $E_2^{1q} = \mathrm{coker}(\sigma - 1, R^q_t)$, and $E_2^{pq} = 0$ for $p \neq 0, 1$. These sheaves are constructible because the vanishing-cycle sheaves are, and 3.11 follows.

3.12

Let us treat the general case, with S still a strictly henselian trait. If $X = X_\eta$, then f is the composite $X \to Y_\eta \to Y$, and we apply 1.9 and 3.11. In the general case, let $j : X_\eta \hookrightarrow X$ and let Δ be the mapping cylinder of $\mathscr{F} \to \mathsf{R}j_*j^*\mathscr{F}$. Its cohomology sheaves are supported on X_s; by 1.9 applied to $X_s \to Y_s$, $\mathsf{R}f_*\Delta$ is therefore constructible. The complex $\mathsf{R}f_*\mathsf{R}j_*j^*\mathscr{F} = \mathsf{R}(fj)_*j^*\mathscr{F}$ is also constructible, and we conclude from the triangle

$$\longrightarrow \mathsf{R}f_*\mathscr{F} \longrightarrow \mathsf{R}f_*\mathsf{R}j_*j^*\mathscr{F} \longrightarrow \mathsf{R}f_*\Delta \longrightarrow .$$

4 Local biduality

4.1

Let S be regular of dimension 0 or 1, and put $K_S = A$ (the constant sheaf with value A). For $K \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(S, A)$, put $DK = \mathsf{R}\mathscr{H}\mathrm{om}(K, K_S) \in \mathrm{Ob}\,\mathsf{D}^+(S, A^\circ)$ (a complex of sheaves of right A-modules). We still have $DK \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(S, A^\circ)$, and an explicit calculation, possible because S has dimension 1, shows that $K \xrightarrow{\sim} DDK$ (for $A = \mathbb{Z}/n$, Duality 1.4).

4.2

We assume that A is commutative, an assumption that is probably unnecessary. For $a : X \to S$ of finite type over S, put $K_X = \mathsf{R}a^! K_S$. For $K \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(X, A)$, put $DK = \mathsf{R}\mathscr{H}\mathrm{om}(K, K_X) \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(X, A)$. Our main result is the following.

Theorem 4.3. *We have $K \xrightarrow{\sim} DDK$ ($K \in \mathrm{Ob}\,\mathsf{D}^b_{\mathrm{ctf}}(X, A)$).*

Lemma 4.4. *If a is proper, then $\mathsf{R}a_* K \xrightarrow{\sim} \mathsf{R}a_* DDK$.*

Poincaré duality

$$\mathsf{R}a_! \,\mathsf{R}\mathscr{H}\mathrm{om}(K, \mathsf{R}a^! K_S) = \mathsf{R}\mathscr{H}\mathrm{om}(\mathsf{R}a_! K, K_S)$$

gives us an isomorphism $\mathsf{R}a_* D = D\mathsf{R}a_! = D\mathsf{R}a_*$. The lemma follows from the commutativity of the diagram (Duality, 1.2)

$$\begin{array}{ccc} \mathsf{R}a_* K & \longrightarrow & \mathsf{R}a_* DDK \\ \downarrow \wr & & \downarrow \wr \\ DD\mathsf{R}a_* K & \xrightarrow{\sim} & D\mathsf{R}a_* DK. \end{array}$$

4.5

By localization, it is enough to treat the cases where S is the spectrum of a field or a trait. When S is the spectrum of a field, we argue by induction on $\dim X$. Since the problem is local, we may assume that X is proper. Moreover, the induction hypothesis, by the usual arguments, ensures that the mapping cylinder Δ of $K \to DDK$ has skyscraper cohomology sheaves. The lemma then ensures that $\mathsf{R}a_* \Delta = 0$, hence that $\Delta = 0$.

4.6

When S is a trait, we already know that 4.3 holds on the generic fiber (4.5). We may also assume that X is proper over S, and we argue by induction on $\dim X_s$, with X proper over S. The mapping cylinder Δ again has skyscraper cohomology sheaves on X_s, and we conclude as above.

4.7

Suppose that $A = \mathbb{Z}/m$. We then have a variant of the preceding theory in which K is arbitrary in $\mathsf{D}^b_c(X, A)$. The complex $\mathsf{R}a^! \mathbb{Z}/m$ has finite injective dimension [2, XVIII 3.1.7]. This ensures that DK still belongs to $\mathsf{D}^b_c(X, A)$. To prove local biduality over S, we use the fact that $\mathbb{Z}/m$ is the dualizing $\mathbb{Z}/m$-module. We then proceed as above.

5 Appendix, by L. Illusie

In this appendix, which takes up certain parts of the late [21, II], and a letter from P. Deligne to the author, we prove 2.14, as well as various generalizations and variants of the specialization theorems [2, XVI 2.1] and (Étale cohomology 5.1.7).

The schemes and morphisms considered will be assumed quasi-compact and quasi-separated. If X is a scheme and x a geometric point of X, we denote by $X(x)$ the strict localization of X at x.

Fix a base scheme S and a sheaf of rings A on S (not necessarily commutative or noetherian). If X is an S-scheme, we write $\mathsf{D}(X)$ for $\mathsf{D}(X, A_X)$.

5.1 Cohomological properness

Proposition 5.1.1. *Let $f : X \to Y$ be an S-morphism and let $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$. The following conditions are equivalent:*

(i) The formation of $\mathsf{R}f_*E$ commutes with every finite base change $S' \to S$.

(ii) The formation of $\mathsf{R}f_*E$ commutes with every quasi-finite base change (or inverse limit of quasi-finite morphisms) $S' \to S$.

(iii) For every specialization arrow $i : t \to S(s)$ (Étale cohomology, 5.1.2), if $\bar{S}$ denotes the normalization in $k(\bar{t})$ of the integral scheme that is the closure of $i(t)$ in $S(s)$ (by [20, IV, 6.15.5, 18.6.11, 18.8.16], $\bar{S}$ is therefore local and integral, with closed point radicial over s), if $\bar{f} : \bar{X} \to \bar{Y}$ is the morphism deduced from f by the base change $\bar{S} \to S$, and if $f_s : X_s \to Y_s$ is the fiber of f at s, then the base-change arrow

$$(\mathsf{R}\bar{f}_*(E|\bar{X}))|Y_s \to \mathsf{R}f_{s*}(E|X_s)$$

is an isomorphism.

The equivalence of (i) and (ii) follows from Zariski's Main Theorem, and it is clear that (ii) implies (iii). Let us prove that (iii) implies (ii). For every $S' \to S$ that is an inverse limit of quasi-finite morphisms, and every geometric point s of S', denote by $C(S', s)$ the cone of the base-change arrow

$$(\mathsf{R}f'_*(E|X'))|Y_s \to \mathsf{R}f_{s*}(E|X_s),$$

where $f' : X' \to Y'$ is deduced from f by the base change $S' \to S$, and $f_s : X_s \to Y_s$ is the fiber of f at s. We must prove that, for every (S', s) as above, $C(S', s)$ is acyclic. We shall prove by induction on N that $\mathrm{H}^i\, C(S', s) = 0$ for $i \leqslant N$. By the Main Theorem, passage to the limit, and elimination of radicial morphisms, we may assume that S and S' are strictly local, that S' is integral with closed point s, and that $S' \to S$ is finite. Let t be a geometric generic point of S', and let $\bar{S}$ be the normalization of S' in $k(t)$. Let $\bar{S}_\bullet$ be the simplicial scheme that is the coskeleton of $\bar{S} \to S'$ ($\bar{S}_0 = \bar{S}, \ldots \bar{S}_n = (\bar{S}/S')^{n+1}, \ldots$), and let $\bar{X}_\bullet = X' \times_{S'} \bar{S}_\bullet$. Let $g_\bullet : \bar{X}_\bullet \to X'$ be the canonical projection. Put $E|X' = E'$. The bicomplex $g_{\bullet *}g_\bullet^* E'$, whose columns are the $g_{n*}g_n^* E'$, is a resolution of E' [2, VIII 8]. We may therefore represent $C(S', s)$ by a bicomplex whose columns identify with the $C(\bar{S}_n, s)$. By (iii), $C(\bar{S}_0, s) = C(\bar{S}, s)$ is acyclic. On the other hand, for $n > 0$, we have $\mathrm{H}^i\, C(\bar{S}_n, s) = 0$ for $i \leqslant N$ by the induction hypothesis. It follows that $\mathrm{H}^i\, C(S', s) = 0$ for $i \leqslant N + 1$, which completes the proof.

5.1.2 Remark

Let (f, E) satisfy condition (i) of 5.1.1.

a) If $E \in \mathrm{Ob}\,\mathsf{D}^b(X)$ and f_* has finite cohomological dimension, then for every $F \in \mathrm{Ob}\,\mathsf{D}^-(S, A^\circ)$, the canonical arrow

$$\mathsf{R}f_*(E) \overset{\mathsf{L}}{\otimes}_A q^*F \to \mathsf{R}f_*(E \overset{\mathsf{L}}{\otimes}_A p^*F), \qquad (*)$$

where $p : X \to S$ and $q : Y \to S$ are the projections, is an isomorphism. The same conclusion holds with $E \in \mathrm{Ob}\,\mathsf{D}^-(X)$.

Indeed, by dévissage, it is enough to verify the assertion for F of the form $i_!A_{S'}$, with $i : S' \to S$ quasi-finite. By the Main Theorem, we reduce to the case where i is finite, and the conclusion is then equivalent to the commutation of $\mathsf{R}f_*(E)$ with base change by i.

b) Suppose that A is noetherian and constructible. Then, for every constructible $F \in \mathrm{Ob}\,\mathsf{D}^b(S, A^\circ)$ of finite tor-dimension, the arrow $(*)$ above is an isomorphism.

Indeed, by dévissage, we reduce to the case where F is of the form $i_!M$, with $i : S' \to S$ locally closed, A locally constant on S', and M constructible, of finite tor-dimension, and with locally constant cohomology on S'. After base change from S to S', we are thus reduced to the case where F is perfect [5, I], hence finally, by localization and dévissage, to the case $F = A$, for which the conclusion is immediate.

5.1.3

Let $f : X \to Y$ be an S-morphism and let $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$. We say that (f, E) is *cohomologically proper* rel. to S if the formation of $\mathsf{R}f_*E$ commutes with every base change $S' \to S$. Equivalently, after every base change $S' \to S$, the formation of $\mathsf{R}f'_*E'$ (where $f' = f \times_S S'$ and $E' =$ the inverse image of E on $X \times_S S'$) commutes with every finite base change $S'' \to S'$ (cf. [2, XII 6.1]). Here are some examples of cohomological properness (for a study of the analogous notion for sheaves of sets or of noncommutative groups, the reader may consult [23, XIII]).

1. Suppose that A is torsion and f is proper. Then, for every $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$, (f, E) is cohomologically proper rel. to S (proper base-change theorem, [2, XII 5.1]).

2. Suppose that S is finite, that X and Y are of finite type over S, and that A is annihilated by an integer invertible on S. Then, for every $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$, (f, E) is cohomologically proper rel. to S (1.9).

3. Suppose that A is constant, noetherian, and annihilated by an integer n invertible on S, and that $f : X = Y \smallsetminus D \to Y$ is the inclusion of the complement of a divisor D on Y with normal crossings relative to S [23, XIII 2.1]. Let E be a sheaf of A-modules on X satisfying one of the following conditions:

   (i) E is constructible locally constant and tamely ramified along D;

   (ii) E is, locally for the étale topology on Y and on S, the inverse image of a sheaf of A-modules on S.

   Then (f, E) is cohomologically proper rel. to S, and $\mathsf{R}f_*E$ is constructible (compare [23, XIII 2.4]).

Let us sketch the proof in case (i). The question is local near a point y of Y. We may assume that D is a sum of smooth divisors, $D = \sum_{i=1}^m D_i$. By the relative Abhyankar lemma [23, XIII 5.5], near y we may trivialize E by a finite covering $\widetilde{Y} \to Y$ of the form $\widetilde{Y} = Y[T_1, \ldots, T_r]/(T_1^{n_1} - t_1, \ldots, T_r^{n_r} - t_r)$, where the t_i are local equations of the smooth divisors passing through y, and the n_i are integers prime to the char. of $k(y)$. The inverse image $\widetilde{D}$ of D in $\widetilde{Y}$ still has normal crossings rel. to S, and if $g : \widetilde{X} = \widetilde{Y} \smallsetminus \widetilde{D} \to X$ is

the projection, g^*E is constant. Since E injects into g_*g^*E and the quotient is tamely ramified, an easy dévissage reduces us to the case where E is constant. Denote the projections by $p : X \to S$, $q : Y \to S$, and, for $1 \leqslant i \leqslant m$, let $f_i : Y \smallsetminus D_i \to Y$ be the canonical inclusion. From the relative purity theorem [2, XVI 3.7], one easily deduces by induction on m that, for every locally constant A-module M on S, the canonical arrow

$$\mathsf{R}f_*(\mathbb{Z}/n)\overset{\mathsf{L}}{\otimes}q^*M \to \mathsf{R}f_*(p^*M) \tag{5.1.3.1}$$

is an isomorphism, and that on the other hand we have:

$$\begin{aligned} f_{i*}(\mathbb{Z}/n) &= \mathbb{Z}/n, \\ \mathsf{R}^1 f_{i*}(\mathbb{Z}/n) &= (\mathbb{Z}/n)(-1)_{D_i}, \end{aligned} \tag{5.1.3.2}$$

where a canonical isomorphism is given by the fundamental class $\mathrm{cl}(D_i)$ (Cycle 2.1.4),

$$\begin{aligned} \mathsf{R}^q f_{i*}(\mathbb{Z}/n) &= 0 \text{ for } q > 1, \\ \mathsf{R}f_*(\mathbb{Z}/n) &= \bigotimes_i^{\mathsf{L}} \mathsf{R}f_{i*}(\mathbb{Z}/n) \text{ , i.e.:} \\ \mathsf{R}^1 f_*(\mathbb{Z}/n) &= \bigoplus_{i=1}^{m} (\mathbb{Z}/n)(-1)_{D_i}, \\ \textstyle\bigwedge^{\bullet} \mathsf{R}^1 f_*(\mathbb{Z}/n) &\xrightarrow{\sim} \mathsf{R}^{\bullet} f_*(\mathbb{Z}/n). \end{aligned}$$

The conclusion of 3, in case (i), therefore follows from (5.1.3.1) and (5.1.3.2). In case (ii), by passage to the limit we reduce to the case where $E = p^*M$, with M constructible, and then, by dévissage and base change (using (i)), to the already treated case where M is constant.

It follows from the preceding proof that (5.1.3.1) is an isomorphism for every $M \in \mathrm{Ob}\,\mathsf{D}^+(S)$.

One may compare the formulas (5.1.3.2) with the analogous formulas in integral cohomology when $S = \mathrm{Spec}(\mathbb{C})$ (see, for example, [12, 3.1]), which follow from the fact that, locally for the classical topology, the complement of D has the homotopy type of a torus.

By passage to the limit, one deduces from (5.1.3.2) similar formulas in ℓ-adic cohomology, i.e. with $\mathbb{Z}/n$ replaced by $\mathbb{Z}_\ell$ (ℓ a prime invertible on S). Suppose that $q : Y \to S$ is proper and smooth. It then follows from the Weil conjectures [14, 15] that the Leray spectral sequence

$$E_2^{ij} = \mathsf{R}^i q_* \mathsf{R}^j f_* \mathbb{Z}_\ell \Rightarrow \mathsf{R}^{\bullet} p_* \mathbb{Z}_\ell$$

degenerates at E_3 modulo torsion [11, §6]. The same phenomenon occurs in integral cohomology when $S = \mathrm{Spec}(\mathbb{C})$; see (loc. cit.) and [12, 3.2.13].

5.2 Specialization and cospecialization

Let $f : X \to S$ and $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$.

5.2.1

Suppose that the formation of $\mathsf{R}f_*E$ commutes with every finite base change. For every specialization $u : t \to S(s)$ of geometric points of S, we then define an arrow, called the *specialization arrow*,

$$\mathrm{sp}(u) : \mathsf{R}\Gamma(X_s, E|X_s) \to \mathsf{R}\Gamma(X_t, E|X_t), \tag{5.2.1.1}$$

as follows. Let, as in 5.1.1(iii), $\bar{S}$ be the normalization in $k(t)$ of the closure of $u(t)$ in $S(s)$, and let $\bar{f} : \bar{X} \to \bar{S}$ be the morphism deduced from f by the base change $\bar{S} \to S$. By the immediate implication (i)⇒(iii) of 5.1.1, the base-change arrow

$$\mathsf{R}\Gamma(\bar{X}, E|\bar{X}) = \mathsf{R}\bar{f}_*(E|\bar{X})_s \to \mathsf{R}\Gamma(X_s, E|X_s) \tag{5.2.1.2}$$

is an isomorphism. We define (5.2.1.1) as the composite of the inverse of (5.2.1.2) with the restriction arrow

$$\mathsf{R}\Gamma(\bar{X}, E|\bar{X}) \to \mathsf{R}\Gamma(X_t, E|X_t). \tag{5.2.1.3}$$

5.2.2

Suppose that f is locally acyclic rel. to E (2.12). For every specialization $u : t \to S(s)$ of geometric points of S, we then define an arrow, called the *cospecialization arrow*,

$$\mathrm{cosp}(u) : \mathsf{R}\Gamma(X_t, E|X_t) \to \mathsf{R}\Gamma(X_s, E|X_s), \tag{5.2.2.1}$$

as follows. Let $g : X_t \to X$ and $g' : \bar{X} \to X$ be the canonical arrows. Viewed vis-à-vis these two maps, the canonical arrow

$$\mathsf{R}g'_*(E|\bar{X}) \to \mathsf{R}g_*(E|X_t) \tag{5.2.2.2}$$

is an isomorphism by the local-acyclicity hypothesis. Indeed, we may assume that S is strictly local and integral, with closed point s, and that t is a geometric generic point. Compute the fiber of (5.2.2.2) at a geometric point x of X, with image y in S. After replacing S by the strict localization of S at y, we may assume that $y = s$. Since $\bar{S} \to S$ is an inverse limit of finite morphisms and is radicial at s, we find that $\mathsf{R}g'_*(E|\bar{X})_x = E_x$ and that the fiber of (5.2.2.2) at x identifies with the restriction $E_x = \mathsf{R}\Gamma(X(x), E) \to \mathsf{R}\Gamma(X(x)_t, E|X(x)_t)$, which is an isomorphism by hypothesis. Applying $\mathsf{R}\Gamma(X, -)$ to (5.2.2.2), we deduce that the restriction arrow (5.2.1.3) is an isomorphism. We define (5.2.2.1) as the composite of the inverse of (5.2.1.3) with the base-change arrow (5.2.1.2).

5.2.3

If $s'' \xrightarrow{u'} s' \xrightarrow{u} s$ are specialization arrows of geometric points of S, one verifies that

$$\mathrm{sp}(u')\,\mathrm{sp}(u) = \mathrm{sp}(uu'),$$
$$\mathrm{cosp}(u)\,\mathrm{cosp}(u') = \mathrm{cosp}(uu'),$$

whenever both sides are defined.

5.2.4

Suppose that the formation of $\mathsf{R}f_*E$ commutes with every finite base change and that f is locally acyclic rel. to E. It follows from the definitions that, for every specialization $u : t \to S(s)$ of geometric points of S, the arrows $\mathrm{sp}(u)$ and $\mathrm{cosp}(u)$ are isomorphisms, inverse to one another.

The preceding hypotheses hold in particular when A is noetherian and annihilated by an integer n invertible on S, when f is proper and smooth, and when E has locally constant cohomology (cohomological properness of proper morphisms (5.1.3), and local acyclicity of smooth morphisms [2, XV]). This recovers the "specialization theorem" [2, XVI 2.1].

On the other hand, 5.1.1 has the following consequence:

Proposition 5.2.5. *Suppose that f is locally acyclic relative to E and that, for every specialization $u : t \to S(s)$, the cospecialization arrow $\mathrm{cosp}(u)$ is an isomorphism. Then the formation of $\mathsf{R}f_*E$ commutes with every finite base change.*

In view of 5.2.4, 5.2.5 generalizes (Étale cohomology 5.1.7).

Corollary 5.2.6. *Suppose that f is locally acyclic rel. to E. Let x be a geometric point of X, with image s in S, and denote by $f_{(x)} : X(x) \to S(s)$ the morphism induced by f. Then the formation of $\mathsf{R}f_{(x)_*}(E|X(x))$ commutes with every finite base change $S' \to S(s)$.*

By 5.2.5, it is enough to verify that the cospecialization arrows for $(f_{(x)}, E|X(x))$ are isomorphisms. By the transitivity of the cospecialization arrows (5.2.3), it is enough to show that if $u : t \to S(s)$ is a specialization, then $\mathrm{cosp}(u) : \mathsf{R}\Gamma(X(x)_t, E|X(x)_t) \to \mathsf{R}\Gamma(X(x)_s, E|X(x)_s)$ is an isomorphism. But since s is closed in $S(s)$, $X(x)_s$ is strictly local with closed point x, $\mathsf{R}\Gamma(X(x)_s, E|X(x)_s) = E_x$, and $\mathrm{cosp}(u)$ is an isomorphism, inverse to the restriction isomorphism $E_x \to \mathsf{R}\Gamma(X(x)_t, E|X(x)_t)$.

Corollary 5.2.7. *Let $f : X \to Y$ be an S-morphism, let $p : X \to S$, $q : Y \to S$ be the projections, and let $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$. Suppose that A is locally constant, that f is locally acyclic rel. to E, and that q is locally acyclic at every geometric point y of Y rel. to every sheaf locally constant in a neighborhood of y. Then p is locally acyclic relative to E.*

Let x be a geometric point of X, with images y and s in Y and S, let $t \to S(s)$ be a specialization, and let $f_{(x)} : X(x) \to Y(y)$ be the morphism induced by f. We must show that the restriction

$$E_x \to \mathsf{R}\Gamma(X(x)_t, E|X(x)_t) \qquad (*)$$

is an isomorphism. We may assume that A is constant. By 5.2.4 and 5.2.5, for every specialization $z \to Y(y)$, the specialization arrow

$$E_x = \mathsf{R}f_{(x)_*}(E|X(x))_y \to (\mathsf{R}f_{(x)_*}(E|X(x)))_z = \mathsf{R}\Gamma(X(x)_z, E|X(x)_z)$$

is an isomorphism; in other words, $\mathsf{R}f_{(x)_*}(E|X(x))$ is constant with value E_x. The acyclicity hypothesis on q therefore plays the decisive rôle and ensures that the restriction

$$E_x \to \mathsf{R}\Gamma(Y(y)_t, (E_x)|Y(y)_t) = \mathsf{R}\Gamma\left(Y(y)_t, \mathsf{R}f_{(x)_*}(E|X(x))|Y(y)_t\right)$$

is an isomorphism. But this arrow identifies with $(*)$, which proves the assertion.

The hypothesis of 5.2.7 on q holds in particular when q is smooth and A is annihilated by an integer n invertible on S [2, XV]. On the other hand, if the hypotheses of 5.2.7 hold after every base change $S' \to S$, then it follows that p is universally locally acyclic rel. to E. These remarks imply, in particular, the statement of 2.14.

Corollary 5.2.8. *Let $f : X \to S$ and $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$. Suppose that f is universally locally acyclic rel. to E, that the formation of $\mathsf{R}f_*E$ commutes with every finite base change, and that, for every morphism $s' \to s$, where s is a geometric point of S and s' is the spectrum of a separably closed field, the base-change arrow $\mathsf{R}\Gamma(X_s, E|X_s) \to \mathsf{R}\Gamma(X_{s'}, E|X_{s'})$ is an isomorphism. Then (f, E) is cohomologically proper (5.1.3).*

Let $f' : X' \to S'$ be the morphism deduced from f by a base change $S' \to S$; we must show that the formation of $\mathsf{R}f'_*(E|X')$ commutes with every finite base change. By 5.2.5, it is enough to show that, for every specialization $u' : t' \to S'(s')$, the cospecialization arrow $\mathrm{cosp}(u')$: $\mathsf{R}\Gamma(X'_{t'}, E|X'_{t'}) \to \mathsf{R}\Gamma(X'_{s'}, E|X'_{s'})$

is an isomorphism. Let s (resp. t) be the geometric point that is the image of s' (resp. t') in S, and let $u : t \to S(s)$ be the specialization defined by u'. One readily sees that there is a commutative square

$$\begin{array}{ccc} \mathsf{R}\Gamma(X_t, E|X_t) & \xrightarrow{\mathrm{cosp}(u)} & \mathsf{R}\Gamma(X_s, E|X_s) \\ \downarrow & & \downarrow \\ \mathsf{R}\Gamma(X'_{t'}, E|X'_{t'}) & \xrightarrow{\mathrm{cosp}(u')} & \mathsf{R}\Gamma(X'_{s'}, E|X'_{s'}) \end{array}$$

in which the vertical arrows are the base-change arrows. By hypothesis, these are isomorphisms. On the other hand, (5.2.4) $\mathrm{cosp}(u)$ is an isomorphism, hence $\mathrm{cosp}(u')$ is an isomorphism, which completes the proof.

Here is an example of an application of 5.2.8. Suppose that A is annihilated by an integer n invertible on S, and that f is locally of finite type and universally locally acyclic. Let x be a geometric point of X, with image s in S, and let $f_{(x)} : X(x) \to S(s)$ be the morphism induced by f. It follows from 5.2.6 and 1.9 (by passage to the limit) that the hypotheses of 5.2.8 are satisfied by the pair $(f_{(x)}, E|X(x))$, which is therefore cohomologically proper.

5.2.9

Suppose that A is constant. Let $f : X \to S$, and let $E \in \mathrm{Ob}\,\mathsf{D}^+(X)$ be a complex of finite tor-dimension. We say that f is *strongly locally acyclic* rel. to E if, for every geometric point x of X, with image s in S, every specialization $t \to S(s)$, and every A°-module M, the restriction arrow

$$E_x \overset{\mathsf{L}}{\otimes}_A M \to \mathsf{R}\Gamma(X(x)_t, (E|X(x)_t) \overset{\mathsf{L}}{\otimes}_A M)$$

is an isomorphism. I do not know whether "locally acyclic" implies "strongly locally acyclic."

Proposition 5.2.10. *Under the hypotheses of 5.2.9, suppose that A is noetherian and torsion, that S is noetherian, and that f is strongly locally acyclic rel. to E. Then, for every diagram with cartesian squares*

$$\begin{array}{ccccc} X & \xleftarrow{j} & X' & \xleftarrow{j'} & X'' \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f''} \\ S & \xleftarrow{i} & S' & \xleftarrow{i'} & S'', \end{array}$$

with i quasi-finite and i' an open immersion, and every $F \in \mathrm{Ob}\,\mathsf{D}^+(S'', A^\circ)$, the canonical arrow

$$j^* E \overset{\mathsf{L}}{\otimes} f'^* \mathsf{R}i'_* F \to \mathsf{R}j'_*((jj')^* E \overset{\mathsf{L}}{\otimes} f''^* F) \qquad (*)$$

is an isomorphism.

Compare this statement with [2, XV 1.17].

Let us prove 5.2.10. We readily reduce to the case where $S = S'$ is strictly local, and to F constructible and concentrated in degree 0. We then resolve F on the right by finite sums of sheaves of the form $u_* M$, where $u : t \to S$ is a geometric point and M is an A°-module. The hypothesis, vis-à-vis this resolution, implies that $(*)$ is an isomorphism for $F = u_* M$, and the result follows.

Chapter 8

Derived categories

1 Triangulated categories: definitions and examples

1.1 Definition of triangulated categories

1.1.0

Let $\mathcal{A}$ be an additive category and let T be an additive automorphism of $\mathcal{A}$. For two objects X and Y of $\mathcal{A}$, we set

$$\hom^i(X,Y) = \hom_{\mathcal{A}}(X, T^i(Y)) \qquad i \in \mathbb{Z}$$

Let X, Y, and Z be three objects of $\mathcal{A}$, let $\alpha \in \hom^i(X,Y)$, and let $\beta \in \hom^j(Y,Z)$. We define the composite $\beta \circ \alpha$ belonging to $\hom^{i+j}(X,Z)$ by

$$\beta \circ \alpha = T^i(\beta) \circ \alpha.$$

One checks that this gives a new category whose groups of morphisms between objects are graded. By an abuse of language, this new category will be called the category $\mathcal{A}$ graded by the translation functor T.

Let $\mathcal{A}$ be a category graded by a translation functor T. Whenever a morphism is named or denoted without its degree being specified, it will always mean a morphism of degree zero.

Let $\mathcal{A}$ and $\mathcal{A}'$ be two additive categories graded by the functors T and T', and let F be an additive functor from $\mathcal{A}$ to $\mathcal{A}'$. We say that F *is graded* if an isomorphism of functors* has been specified:

$$F \circ T = T' \circ F$$

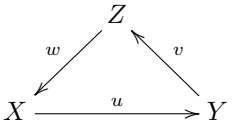
The definition of morphisms of graded functors is given in 4.1.1.

Let $\mathcal{A}$ be an additive category graded by the functor T. A triangle is a collection of three objects X, Y, Z and three morphisms $u : X \to Y$, $v : Y \to Z$, $w : Z \to T(X)$ ($\deg w = 1$). We use either the

by Jean-Louis Verdier

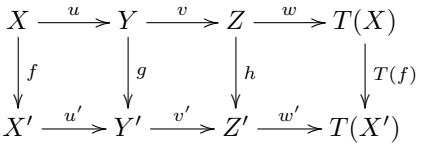
*Text amended in 1976: the original text required an equality $F \circ T = T' \circ F$. The case of functors in several variables is discussed in [2, XVIII 0.2]. It raises a problem of signs.

notation (X, Y, Z, u, v, w) or the following diagram for a triangle:



$\deg(w) = 1$

Let (X, Y, Z, u, v, w) and (X', Y', Z', u', v', w') be two triangles of $\mathcal{A}$. A morphism from the first triangle to the second is a collection of three morphisms $f : X \to X'$, $g : Y \to Y'$, $h : Z \to Z'$ such that the following diagram is commutative:



1.1.1

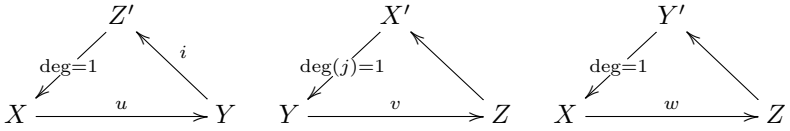
A *triangulated category* is an additive category graded by a translation functor and equipped with a family of triangles, called the family of distinguished triangles. This family must also satisfy the following axioms.

TR1 Every triangle isomorphic to a distinguished triangle is distinguished. Every morphism $u : X \to Y$ is contained in a distinguished triangle (X, Y, Z, u, v, w). The triangle $(X, X, 0, \mathrm{id}_X, 0, 0)$ is distinguished.

TR2 A triangle (X, Y, Z, u, v, w) is distinguished if and only if $(Y, Z, T(X), v, w, -T(u))$ is distinguished.

TR3 If (X, Y, Z, u, v, w) and (X', Y', Z', u', v', w') are distinguished, then for every morphism $(f, g) : u \to u'$, there exists a morphism $h : Z \to Z'$ such that (f, g, h) is a morphism of triangles.

TR4



If these are three distinguished triangles such that $w = v \circ u$, then there exist two morphisms $f : Z' \to Y'$, $g : Y' \to X'$ such that

1. (id_X, v, f) is a morphism of triangles;
2. (u, id_Z, g) is a morphism of triangles;
3. $(Z', Y', X', f, g, T(i)j)$ is a distinguished triangle.

If $\mathcal{C}$ and $\mathcal{C}'$ are two triangulated categories, an *exact functor* from $\mathcal{C}$ to $\mathcal{C}'$ is an additive graded functor that takes distinguished triangles to distinguished triangles. Morphisms of exact functors are morphisms of graded functors.

The following property follows immediately from axioms TR1, TR2, and TR3.

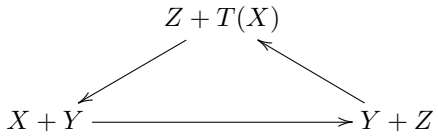
Proposition 1.1.2. *Let (X, Y, Z, u, v, w) be a distinguished triangle and let M be an object of a triangulated category. Then the following sequences are exact:*

$$\cdots \longrightarrow \hom^i(M,X) \xrightarrow{\hom(M,u)} \hom^i(M,Y) \xrightarrow{\hom(M,v)} \hom^i(M,Z) \xrightarrow{\hom(M,w)} \hom^{i+1}(M,X) \longrightarrow \cdots$$

$$\cdots \longrightarrow \hom^i(Z,M) \xrightarrow{\hom(v,M)} \hom^i(Y,M) \xrightarrow{\hom(u,M)} \hom^i(X,M) \xrightarrow{\hom(w,M)} \hom^{i+1}(Z,M) \longrightarrow \cdots$$

From this proposition one obtains statements of the following kind:

- If (f, g, h) is a morphism of distinguished triangles and f, g are isomorphisms, then h is an isomorphism.
- The distinguished triangles constructed on a morphism (TR1) are all isomorphic (*but the isomorphisms are not, in general, uniquely determined*).
- When one of the morphisms in a distinguished triangle has a kernel or a cokernel, the latter splits; that is, the triangle is of the form



- The sum of two distinguished triangles is distinguished. (One uses axiom TR4.)

1.2 Examples

1.2.1

We first fix some notation. Let $\mathcal{A}$ be an additive category. The following category will be denoted by $\mathsf{C}(\mathcal{A})$:

- The objects of $\mathsf{C}(\mathcal{A})$ are the complexes of $\mathcal{A}$, with no restriction on degree and with differential of degree $+1$.
- The morphisms of $\mathsf{C}(\mathcal{A})$ are the degree-preserving morphisms of complexes (those commuting with the differential).

Let T be the following functor. For every object $X^\bullet$ of $\mathsf{C}(\mathcal{A})$,

$$T(X^\bullet)_i = (X^\bullet)_{i+1}$$
$$d_i(T(X^\bullet)) = -d_{i+1}(X^\bullet).$$

On morphisms the functor T acts as follows: for every morphism $f : X^\bullet \to Y^\bullet$, the morphism $T(f) : T(X^\bullet) \to T(Y^\bullet)$ is defined by

$$T(f)_i = f_{i+1}.$$

It is immediate that T is an additive automorphism of $\mathsf{C}(\mathcal{A})$. The morphisms of degree n defined using this translation functor are then the morphisms that raise degree by n and commute or anticommute with the differential according to the parity of n. Thus one recovers the usual definition.

Let $\mathscr{C}$ denote the following family of triangles. A triangle $(X^\bullet, Y^\bullet, Z^\bullet, u, v, w)$ belongs to the family if[†]

[†]Text amended in 1976 to ensure the validity of 1.2.4.

- $Z^\bullet$ is the simple complex associated with the double complex

$$\cdots \longrightarrow 0 \longrightarrow X^\bullet \xrightarrow{u} Y^\bullet \longrightarrow 0 \longrightarrow \cdots$$

in which the objects of $Y^\bullet$ have first degree zero and the objects of $X^\bullet$ have first degree -1.

- v is the image, under the functor: double complex $\mapsto$ simple complex, of the following morphism of double complexes:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \longrightarrow & 0^\bullet & \longrightarrow & Y^\bullet & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \cdots & \longrightarrow & 0 & \longrightarrow & X^\bullet & \longrightarrow & Y^\bullet & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

- w is the negative of the image, under the same functor, of the following morphism of double complexes:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \longrightarrow & X^\bullet & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \\ \cdots & \longrightarrow & 0 & \longrightarrow & X^\bullet & \longrightarrow & Y^\bullet & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

1.2.2

The set of null-homotopic morphisms is a two-sided ideal (if f and g are composable and either f or g is null-homotopic, then the composite is null-homotopic; if f and g are null-homotopic, then $f \oplus g$ is null-homotopic). For every pair of objects $X^\bullet$, $Y^\bullet$, let $\mathrm{Htp}(X^\bullet, Y^\bullet)$ denote the subgroup of $\hom_{\mathsf{C}(\mathcal{A})}(X^\bullet, Y^\bullet)$ consisting of the null-homotopic morphisms. The objects of $\mathsf{K}(\mathcal{A})$ are then the objects of $\mathsf{C}(\mathcal{A})$, and its group of morphisms with source $X^\bullet$ and target $Y^\bullet$ is

$$\hom_{\mathsf{C}(\mathcal{A})}(X^\bullet, Y^\bullet) / \mathrm{Htp}(X^\bullet, Y^\bullet).$$

The composition law on the morphisms of $\mathsf{K}(\mathcal{A})$ is defined by passing the composition law on the morphisms of $\mathsf{C}(\mathcal{A})$ to the quotient. The category $\mathsf{K}(\mathcal{A})$ is additive. The translation functor T on $\mathsf{C}(\mathcal{A})$ passes to the quotient (the translate of a null-homotopic morphism is null-homotopic). The resulting functor is an additive automorphism of $\mathsf{K}(\mathcal{A})$, again denoted by T. By an abuse of notation, $\mathsf{K}(\mathcal{A})$ will henceforth denote the category $\mathsf{K}(\mathcal{A})$ graded by the functor T.

A distinguished triangle in $\mathsf{K}(\mathcal{A})$ is any triangle *isomorphic* to a triangle arising from $\mathscr{C}$. The family of distinguished triangles satisfies axioms TR1, TR2, TR3, and TR4. By an abuse of notation, $\mathsf{K}(\mathcal{A})$ will denote the category $\mathsf{K}(\mathcal{A})$ triangulated by the family of triangles defined above.

1.2.3

There is also a whole arsenal of subcategories:

- A complex is said to be bounded below if all but finitely many of its objects of negative degree are zero. Bounded-above and bounded complexes are defined similarly. Let $\mathsf{K}^+(\mathcal{A})$, $\mathsf{K}^-(\mathcal{A})$, and $\mathsf{K}^b(\mathcal{A})$ denote the full subcategories of $\mathsf{K}(\mathcal{A})$ generated by these families of objects. These subcategories are "*triangulated subcategories*" of $\mathsf{K}(\mathcal{A})$: every distinguished triangle two of whose objects belong to the subcategory is isomorphic to a triangle all three of whose objects belong to the subcategory.

- Suppose that $\mathcal{A}$ is an abelian category. A complex is said to be cohomologically bounded below if all but finitely many of its cohomology objects of negative degree are zero. Cohomologically bounded-above complexes, cohomologically bounded complexes, and acyclic complexes are defined similarly. Let $\mathsf{K}^{\infty,+}(\mathcal{A})$, $\mathsf{K}^{\infty,-}(\mathcal{A})$, $\mathsf{K}^{\infty,b}(\mathcal{A})$, and $\mathsf{K}^{\infty,\varnothing}(\mathcal{A})$ denote the full subcategories generated by these families of objects. They are triangulated subcategories.

  We shall also use the triangulated subcategories $\mathsf{K}^{+,b}(\mathcal{A})$, $\mathsf{K}^{+,\varnothing}(\mathcal{A})$, ... etc ... (the first superscript gives information about the objects of the complexes, and the second about the cohomology objects).

- Without assuming that $\mathcal{A}$ is abelian, let O be a set of objects of $\mathcal{A}$ stable under isomorphisms and direct sums. Let $O(\mathcal{A})$ denote the full subcategory of $\mathsf{C}(\mathcal{A})$ generated by the complexes whose objects in every degree belong to O. Let $\underline{O}(\mathcal{A})$ denote the corresponding triangulated subcategory of $\mathsf{K}(\mathcal{A})$. When $\mathcal{A}$ is abelian, one may take for O either the set I of injectives or the set P of projectives.

1.2.4

The category $\mathsf{K}(\mathcal{A})$ is uniquely determined by the data of $\mathsf{C}(\mathcal{A})$ and the family $\mathscr{C}$. More precisely, every graded additive functor from $\mathsf{C}(\mathcal{A})$ to a triangulated category that takes every triangle of $\mathscr{C}$ to a distinguished triangle factors uniquely through $\mathsf{K}(\mathcal{A})$, and the resulting functor is exact.

For every complex $X^\bullet$ of $\mathsf{C}(\mathcal{A})$, let $\bar{X}^\bullet$ denote the complex obtained from $X^\bullet$ by setting the differential equal to zero. A three-term sequence

$$0 \longrightarrow X^\bullet \longrightarrow Y^\bullet \longrightarrow Z^\bullet \longrightarrow 0$$

is called a semi-split sequence if the sequence

$$0 \longrightarrow \bar{X}^\bullet \longrightarrow \bar{Y}^\bullet \longrightarrow \bar{Z}^\bullet \longrightarrow 0$$

is split; that is, if it is isomorphic to the sequence

$$0 \longrightarrow \bar{X}^\bullet \longrightarrow \bar{X}^\bullet + \bar{Z}^\bullet \longrightarrow \bar{Z}^\bullet \longrightarrow 0.$$

Let $0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$ be a semi-split sequence. Choose a splitting, that is, for every n an isomorphism

$$(Y^\bullet)^n \simeq (X^\bullet)^n + (Z^\bullet)^n.$$

The matrix of the differential of $Y^\bullet$ in the above direct-sum decomposition has the form

$$\begin{pmatrix} d_{X^\bullet} & \varphi_n \\ 0 & d_{Z^\bullet} \end{pmatrix}$$

where φ_n is a morphism from $(Z^\bullet)^n$ to $(X^\bullet)^{n+1}$. The φ_n determine a morphism of complexes

$$\varphi : Z^\bullet \to T(X^\bullet).$$

When the splitting is changed, the homotopy class of φ does not change. Moreover, if $\underline{u}$, $\underline{v}$, and $\underline{\varphi}$ denote the images in $\mathsf{K}(\mathcal{A})$ of the morphisms u, v, and φ, then the triangle $(X^\bullet, Y^\bullet, Z^\bullet, \underline{u}, \underline{v}, \underline{\varphi})$ is distinguished. Let $\mathsf{SSS}(\mathcal{A})$ denote the category of semi-split sequences of $\mathsf{C}(\mathcal{A})$ and let $\mathsf{TrK}(\mathcal{A})$ denote the category of distinguished triangles of $\mathsf{K}(\mathcal{A})$. The preceding construction defines a functor

$$\rho : \mathsf{SSS}(\mathcal{A}) \to \mathsf{TrK}(\mathcal{A}).$$

This functor is essentially surjective.

1.3 Examples of cohomological and exact functors

Definition 1.3.1. *Let $\mathcal{A}$ be a triangulated category and let $\mathcal{B}$ be an abelian category. An additive functor $F : \mathcal{A} \to \mathcal{B}$ is called a* cohomological functor *if, for every distinguished triangle (X, Y, Z, u, v, w), the sequence*

$$F(X) \xrightarrow{F(u)} F(Y) \xrightarrow{F(v)} F(Z)$$

is exact.

The functor $F \circ T^i$ will often be denoted by F^i. By axiom TR2 for triangulated categories, there is an unbounded exact sequence

$$\cdots \longrightarrow F^i(X) \longrightarrow F^i(Y) \longrightarrow F^i(Z) \longrightarrow F^{i+1}(X) \longrightarrow \cdots$$

1.3.2 Examples

1. Let $\mathcal{A}$ be a triangulated category. For every object X of $\mathcal{A}$, the functor $\hom_{\mathcal{A}}(X, -)$ is cohomological. The functor $\hom_{\mathcal{A}}(-, X)$ is a cohomological functor $\mathcal{A}^\circ \to \mathsf{Ab}$.

2. Let $\mathcal{A}$ be an abelian category and let $\mathsf{C}(\mathcal{A})$ be the category of complexes of $\mathcal{A}$. Define the functor $\mathrm{H}^0 : \mathsf{C}(\mathcal{A}) \to \mathcal{A}$ as follows. For an object $X^\bullet$ of $\mathsf{C}(\mathcal{A})$, set

   $$\mathrm{H}^0(X^\bullet) = \ker d^0 / \operatorname{im} d^{-1}.$$

   The functor H^0 annihilates null-homotopic morphisms and therefore factors uniquely through $\mathsf{K}(\mathcal{A})$. The resulting functor will again be denoted by $\mathrm{H}^0 : \mathsf{K}(\mathcal{A}) \to \mathcal{A}$.

1.3.3 Example of an exact bifunctor

Let $\mathcal{A}$, $\mathcal{A}'$, and $\mathcal{A}''$ be three additive categories, and let

$$P : \mathcal{A} \times \mathcal{A}' \to \mathcal{A}''$$

be a bilinear functor (that is, additive in each argument separately). It induces the bilinear functor

$$P^* : \mathsf{C}(\mathcal{A}) \times \mathsf{C}(\mathcal{A}') \to \mathsf{C}(\mathcal{A}'')$$

as follows.

Let $X^\bullet$ be an object of $\mathsf{C}(\mathcal{A})$ and let $Y^\bullet$ be an object of $\mathsf{C}(\mathcal{A}')$. Then $P(X^\bullet, Y^\bullet)$ is a double complex of $\mathcal{A}''$. Set $P^*(X^\bullet, Y^\bullet) =$ the simple complex associated with $P(X^\bullet, Y^\bullet)$.

Let f be a null-homotopic morphism of $\mathsf{C}(\mathcal{A})$ (respectively $\mathsf{C}(\mathcal{A}')$), and let $Z^\bullet$ be an object of $\mathsf{C}(\mathcal{A}')$ (respectively $\mathsf{C}(\mathcal{A})$). The morphism $P^*(f, Z^\bullet)$ (respectively $P^*(Z^\bullet, f)$) is then null-homotopic. It follows that P^* uniquely defines a functor

$$P^\bullet : \mathsf{K}(\mathcal{A}) \times \mathsf{K}(\mathcal{A}') \to \mathsf{K}(\mathcal{A}'').$$

The bifunctor $P^\bullet$ is exact.

In particular, let $\mathcal{A}$ be an additive category. For P one may take the functor

$$\begin{aligned} \mathcal{A}^\circ \times \mathcal{A} &\to \mathsf{Ab} \\ (X, Y) &\mapsto \hom(X, Y) \end{aligned}$$

The preceding construction then gives a functor

$$\hom^\bullet : \mathsf{K}(\mathcal{A})^\circ \times \mathsf{K}(\mathcal{A}) \to \mathsf{K}(\mathsf{Ab})$$

which, when composed with the functor $\mathrm{H}^0 : \mathsf{K}(\mathsf{Ab}) \to \mathsf{Ab}$, plainly gives back the functor $\hom_{\mathsf{K}(\mathcal{A})}$.

2 Quotient categories

2.1 Thick subcategories. Multiplicative systems of morphisms

We begin with two definitions.

Definition 2.1.1. *A subcategory $\mathcal{B}$ of a triangulated category $\mathcal{A}$ is said to be* thick *if $\mathcal{B}$ is a full triangulated subcategory of $\mathcal{A}$ and, in addition, $\mathcal{B}$ has the following property:*

For every morphism $f : X \to Y$ that factors through an object of $\mathcal{B}$ and is contained in a distinguished triangle (X, Y, Z, f, g, h) in which Z is an object of $\mathcal{B}$, the source of f and the target of f are objects of $\mathcal{B}$.

The set of thick subcategories of $\mathcal{A}$ will be denoted by $\mathscr{N}$.

The intersection of an arbitrary family of thick subcategories is a thick subcategory.

Definition 2.1.2. *Let $\mathcal{A}$ be a triangulated category. A set S of morphisms of $\mathcal{A}$ is called a* multiplicative system of morphisms *if it has the following five properties.*

FR1 *If $f, g \in S$ and f and g are composable, then $f \circ g \in S$. For every object X of $\mathcal{A}$, the identity morphism of X belongs to S.*

FR2 *In the category $\mathcal{A}$, every diagram*

$$\begin{array}{ccc} & & Y \\ & & \downarrow{\scriptstyle s\in S} \\ Z & \xrightarrow{f} & X \end{array}$$

can be completed to a commutative diagram

$$\begin{array}{ccc} P & \xrightarrow{g} & Y \\ \downarrow{\scriptstyle t\in S} & & \downarrow{\scriptstyle s\in S} \\ Z & \xrightarrow{f} & X \end{array}$$

Moreover, the symmetric property holds.

FR3 *If f and g are morphisms, the following properties are equivalent:*

i) there exists $s \in S$ such that $s \circ f = s \circ g$;

ii) there exists $t \in S$ such that $f \circ t = g \circ t$.

FR4 *Let T be the translation functor of $\mathcal{A}$. For every element s of S, one has $T(s) \in S$.*

FR5 *Let (X, Y, Z, u, v, w) and (X', Y', Z', u', v', w') be two distinguished triangles, and let f and g be two elements of S such that the pair (f, g) is a morphism from u to u'. There exists a morphism $h \in S$ such that (f, g, h) is a morphism of triangles.*

A multiplicative system of morphisms is said to be *saturated* if it has the following property:

A morphism f belongs to S if and only if there exist two morphisms g and g' such that $g \circ f \in S$ and $f \circ g' \in S$.

The intersection of an arbitrary family of saturated multiplicative systems is a saturated multiplicative system.

The set of saturated multiplicative systems will be denoted by $\mathscr{S}$.

Let $\mathcal{B}$ be a thick subcategory. Denote by $\varphi(\mathcal{B})$ the set of morphisms f contained in a distinguished triangle (X, Y, Z, f, g, h) in which Z is an object of $\mathcal{B}$. Then $\varphi(\mathcal{B})$ is a saturated multiplicative system.

Let S be a saturated multiplicative system. Denote by $\psi(S)$ the full subcategory generated by the objects Z contained in a distinguished triangle (X, Y, Z, f, g, h) in which f is an element of S. The subcategory $\psi(S)$ is thick.

The main result of this subsection is the following: φ is an order-preserving isomorphism from $\mathscr{N}$ onto $\mathscr{S}$, where the order relation is defined by inclusion. Its inverse is precisely ψ.

2.2 Universal problems

Let $\mathcal{A}$ and $\mathcal{A}'$ be two triangulated categories and let F be an exact functor from $\mathcal{A}$ to $\mathcal{A}'$. Let $S(F)$ be the set of morphisms of $\mathcal{A}$ that F takes to isomorphisms, and let $\mathcal{B}(F)$ be the full subcategory generated by the objects of $\mathcal{A}$ that F takes to zero objects of $\mathcal{A}'$. Then $S(F)$ is a saturated multiplicative system and $\mathcal{B}(F)$ is a thick subcategory. Moreover, $S(F) = \varphi(\mathcal{B}(F))$. Now let $\mathcal{A}$ be a triangulated category, let $\mathcal{B}$ be a thick subcategory, and let $S = \varphi(\mathcal{B})$ be the corresponding saturated multiplicative system. The following two universal problems are equivalent.

Problem 1 Find a triangulated category $\mathcal{A}/\mathcal{B}$ and an exact functor $Q : \mathcal{A} \to \mathcal{A}/\mathcal{B}$ such that every exact functor from $\mathcal{A}$ to a triangulated category $\mathcal{A}'$ that takes the objects of $\mathcal{B}$ to zero objects of $\mathcal{A}'$ admits a unique factorization of the form $G \circ Q$, where G is exact.

Problem 2 Find a triangulated category $\mathcal{A}_S$ and an exact functor $Q : \mathcal{A} \to \mathcal{A}_S$ such that every exact functor from $\mathcal{A}$ to a triangulated category $\mathcal{A}'$ that takes the morphisms of S to isomorphisms of $\mathcal{A}'$ admits a unique factorization of the form $G \circ Q$, where G is exact.

(The author asks the reader to excuse this use of obsolete and reactionary language.) In the next subsection we shall show that Problem 2 has a solution. Consequently, the corresponding Problem 1 has the same solution.

2.3 Calculus of fractions

2.3.1

Let $\mathcal{A}$ be a triangulated category and let S be a saturated multiplicative system. Denote by $\mathcal{A}[S^{-1}]$ the category of fractions of $\mathcal{A}$ with respect to the set of morphisms S, and by Q the canonical functor from $\mathcal{A}$ to $\mathcal{A}[S^{-1}]$. The pair $(\mathcal{A}[S^{-1}], Q)$ solves the following problem: every functor from $\mathcal{A}$ to an arbitrary category (not necessarily additive) that takes morphisms of S to isomorphisms factors uniquely through $(\mathcal{A}[S^{-1}], Q)$. Such a pair always exists, without any hypothesis on the set of morphisms S [18, 1.1][‡]. However, because the set of morphisms S is a saturated multiplicative system, it has properties FR1, FR2, and FR3. Consequently, by [18], the category $\mathcal{A}[S^{-1}]$ can be obtained by a right or left calculus of

[‡]The original text references [C.G.G] here. Most likely, this is the lost work *Catégories et foncteurs* by Chevalley, Gabriel and Grothendieck, referenced in[31].

fractions. We recall how $\mathcal{A}[S^{-1}]$ is obtained by a right calculus of fractions. (The left calculus of fractions is obtained by reversing the arrows.)

2.3.2

The objects of $\mathcal{A}[S^{-1}]$ are the objects of $\mathcal{A}$.

- For every object X of $\mathcal{A}$, let S_X denote the category of arrows belonging to S and having target X. To every object Y of $\mathcal{A}$ one associates, functorially in Y, the following set-valued functor on S_X° (S_X° denotes the opposite category):

$$H_Y : s \mapsto \hom(\text{source}(s), Y).$$

  Set

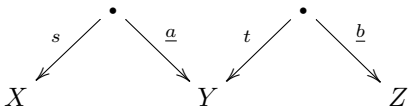
$$\hom_{\mathcal{A}[S^{-1}]}(X, Y) = \varinjlim_{S_X^\circ} H_Y.$$

  For all matters concerning direct limits, we refer to [1].

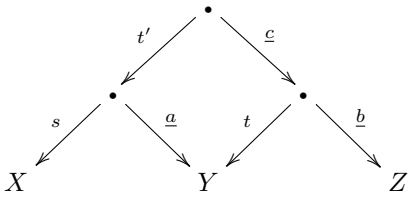
  One then observes with pleasure, using FR1, FR2, and FR3, that the category S_X° has properties L1, L2, and L3. Direct limits therefore have all the good properties of direct limits over filtered ordered sets.

- Let X, Y, and Z be three objects of $\mathcal{A}$, let a be an element of $\hom_{\mathcal{A}[S^{-1}]}(X, Y)$, and let b be an element of $\hom_{\mathcal{A}[S^{-1}]}(Y, Z)$. Let s be an object of S_X, let $\underline{a}$ be an element of $\hom(\text{source}(s), Y)$ whose image is a, and let an object of S_Y, $\underline{b}$ be an element of $\hom(\text{source}(t), Z)$ whose image is b. There is then a diagram

(1)



  which, by FR2, can be completed to a diagram



  Let $b \circ a$ denote the image in $\hom_{\mathcal{A}[S^{-1}]}(X, Z)$ of the morphism $\underline{b} \circ \underline{c}$. Using properties FR1, FR2, and FR3, one checks that $b \circ a$ is independent of the chosen representatives $\underline{a}$ and $\underline{b}$, and also of the way in which diagram (1) is completed. One further checks that this defines a category $\mathcal{A}[S^{-1}]$.

We denote by Q the evident functor $\mathcal{A} \to \mathcal{A}[S^{-1}]$.

Since $\mathcal{A}$ is additive, $\mathcal{A}[S^{-1}]$ is an additive category. Moreover, using FR4, one proves that there exists a unique translation functor on $\mathcal{A}[S^{-1}]$, denoted by T, satisfying

$$Q \circ T = T \circ Q.$$

Finally, using FR5, one proves that there exists a unique triangulated structure on $\mathcal{A}[S^{-1}]$ for which Q is exact. The distinguished triangles of $\mathcal{A}[S^{-1}]$ are precisely the triangles isomorphic to the images under Q of the distinguished triangles of $\mathcal{A}$. Denoting by $\mathcal{A}_S$ the category $\mathcal{A}[S^{-1}]$ equipped with this triangulated structure, one proves without difficulty that the pair $(\mathcal{A}_S, Q)$ is a solution to Problem 2.

Definition 2.3.3. *Let $\mathcal{A}$ be a triangulated category, let $\mathcal{B}$ be a thick subcategory of $\mathcal{A}$, and let $(Q, \mathcal{A}/\mathcal{B})$ be the solution to Problem 1 (§2.2). The category $\mathcal{A}/\mathcal{B}$ is called the* quotient category *of $\mathcal{A}$ by $\mathcal{B}$, and Q is called the canonical quotient functor. More generally, let $\mathcal{N}$ be a triangulated subcategory of $\mathcal{A}$, and let $\underline{\mathcal{N}}$ be the smallest thick subcategory containing $\mathcal{N}$. By definition, the quotient category of $\mathcal{A}$ by $\mathcal{N}$, denoted $\mathcal{A}/\mathcal{N}$, is $\mathcal{A}/\underline{\mathcal{N}}$.*

2.4 Properties of quotient categories

2.4.1

Let $\mathcal{A}$ be a triangulated category and let H be a cohomological functor with values in an abelian category $\mathcal{G}$. Let $\mathcal{B}_H$ be the full subcategory generated by the objects all of whose translates are taken by H to zero objects of $\mathcal{G}$, and let S_H be the set of morphisms all of whose translates are taken by H to isomorphisms of $\mathcal{G}$. Then $\mathcal{B}_H$ is a thick subcategory of $\mathcal{A}$, and S_H is a saturated multiplicative system. Moreover, $S_H = \varphi(\mathcal{B}_H)$ (§2.1). The functor H factors uniquely through $(Q, \mathcal{A}/\mathcal{B}_H)$.

Theorem 2.4.2. *Let $\mathcal{A}$ be a triangulated category, let $\mathcal{B}$ be a triangulated subcategory, let $\mathcal{N}$ be a thick subcategory of $\mathcal{A}$, and let $S = \varphi(\mathcal{N})$ be the corresponding saturated multiplicative system.*

(a) The category $\mathcal{N} \cap \mathcal{B}$ is a thick subcategory of $\mathcal{B}$. The corresponding multiplicative system is $S \cap \mathcal{B}$.

(b) The following two properties are equivalent:

(i) For every object X of $\mathcal{B}$ and every morphism $s : R \to X$, where s belongs to S, there exists a morphism $s' : R' \to R$ such that $s \circ s'$ belongs to $S \cap \mathcal{B}$; and $R' \in \operatorname{Ob} \mathcal{B}$.

(ii) Every morphism from an object X of $\mathcal{B}$ to an object Y of $\mathcal{N}$ factors through an object of $\mathcal{N} \cap \mathcal{B}$.

(c) The properties (i)' and (ii)' obtained from (i) and (ii) by reversing the arrows are equivalent.

(d) If properties (i) and (ii) hold, or if properties (i)' and (ii)' hold, the canonical functor

$$\mathcal{B}/\mathcal{N} \cap \mathcal{B} \longrightarrow \mathcal{A}/\mathcal{N}$$

is faithful.

Since this functor is injective on objects, it realizes $\mathcal{B}/\mathcal{N} \cap \mathcal{B}$ as a subcategory of $\mathcal{A}/\mathcal{N}$.

(e) If, in addition, $\mathcal{B}$ is a full subcategory of $\mathcal{A}$, then $\mathcal{B}/\mathcal{N} \cap \mathcal{B}$ is a full subcategory of $\mathcal{A}/\mathcal{N}$.

Corollary 2.4.3. *Let $\mathcal{A} \subset \mathcal{B} \subset \mathcal{C}$ be three triangulated categories, with $\mathcal{A}$ a thick subcategory of $\mathcal{B}$ and $\mathcal{B}$ a thick subcategory of $\mathcal{C}$. Then $\mathcal{A}$ is a thick subcategory of $\mathcal{C}$, and $\mathcal{B}/\mathcal{A}$ is a thick subcategory of $\mathcal{C}/\mathcal{A}$. The canonical functor $\mathcal{C}/\mathcal{B} \to (\mathcal{C}/\mathcal{A})/(\mathcal{B}/\mathcal{A})$ is an isomorphism.*

2.5 Properties of the quotient functor

Let $\mathcal{A}$ be a triangulated category, let $\mathcal{B}$ be a thick subcategory, let S be the corresponding multiplicative system, and let $Q : \mathcal{A} \to \mathcal{A}/\mathcal{B}$ be the canonical quotient functor.

Proposition 2.5.1. *For objects X and Y of $\mathcal{A}$, the following assertions are equivalent:*

(a) Every morphism $f : X \to Y$ for which there exists a morphism $s : Z \to X$ in S satisfying $f \circ s = 0$ is zero.

(b) Every morphism $f : X \to Y$ for which there exists a morphism $s : Y \to Z$, $s \in S$, satisfying $s \circ f = 0$ is zero.

(c) Every morphism $f : X \to Y$ that factors through an object of $\mathcal{B}$ is zero.

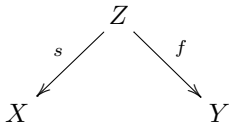
(d) The canonical map

$$\hom_{\mathcal{A}}(X, Y) \to \hom_{\mathcal{A}/\mathcal{B}}(Q(X), Q(Y))$$

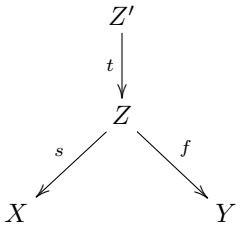
is injective.

Proposition 2.5.2. *Let X and Y be two objects of $\mathcal{A}$. The following assertions are equivalent:*

(a) Every diagram ($s \in S$)



can be completed to a diagram: ($s, t \in S$):



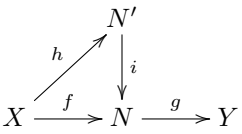
in which $f \circ t$ factors through $(X, s \circ t)$.

(b) The assertion obtained by reversing the direction of the arrows in (a) and interchanging X and Y.

(c) Every diagram

$$X \xrightarrow{f} N \xrightarrow{g} Y$$

in which $g \circ f = 0$ and N is an object of $\mathcal{B}$ can be completed to a diagram



in which $f = i \circ h$ and $g \circ i = 0$.

(d) The assertion obtained by reversing the arrows in (c) and interchanging X and Y.

(e) The canonical map

$$\hom_{\mathcal{A}}(X, Y) \xrightarrow{Q} \hom_{\mathcal{A}/\mathcal{B}}(Q(X), Q(Y))$$

is surjective.

Proposition 2.5.3. *Let X be an object of $\mathcal{A}$. The following assertions are equivalent:*

(a) For every object Y of $\mathcal{A}$, the canonical map

$$\hom_{\mathcal{A}}(X,Y) \xrightarrow{Q} \hom_{\mathcal{A}/\mathcal{B}}(Q(X),Q(Y))$$

is an isomorphism.

(b) Every morphism from X to an object of $\mathcal{B}$ is zero.

Similarly, the following two assertions are equivalent:

(a)' For every object Y of $\mathcal{A}$, the canonical map

$$\hom_{\mathcal{A}}(Y,X) \xrightarrow{Q} \hom_{\mathcal{A}/\mathcal{B}}(Q(Y),Q(X))$$

is an isomorphism.

(b)' Every morphism from an object of $\mathcal{B}$ to X is zero.

Definition 2.5.4. *Every object having the equivalent properties (a) and (b) of the preceding proposition is called an* object free on the left over $\mathcal{A}/\mathcal{B}$, *or a* left Q-free object. *Similarly, every object having the equivalent properties (a)' and (b)' is called an* object free on the right over $\mathcal{A}/\mathcal{B}$, *or a* right Q-free object. *It is determined up to isomorphism by the knowledge of $Q(X) \in \operatorname{Ob} \mathcal{A}/\mathcal{B}$.*

2.6 Functors adjoint to the quotient functor

Definition 2.6.1. *Two triangulated subcategories $\mathcal{N}$ and $\mathcal{N}'$ of a triangulated category $\mathcal{A}$ are said to be* orthogonal *if, for every object X of $\mathcal{N}$ and every object Y of $\mathcal{N}'$, one has*

$$\hom_{\mathcal{A}}(X,Y) = 0$$

The category $\mathcal{N}'$ is then said to be right orthogonal to $\mathcal{N}$, and $\mathcal{N}$ left orthogonal to $\mathcal{N}'$.

Proposition 2.6.2. *Let $\mathcal{N}$ be a triangulated subcategory of a triangulated category $\mathcal{A}$. The full category $\mathcal{N}^{\perp}$ (respectively ${}^{\perp}\mathcal{N}$) generated by the objects X of $\mathcal{A}$ such that, for every object Y of $\mathcal{N}$, $\hom_{\mathcal{A}}(Y,X) = 0$ (respectively $\hom_{\mathcal{A}}(X,Y) = 0$), is a thick subcategory of $\mathcal{A}$. By an abuse of language, the category $\mathcal{N}^{\perp}$ is called the right orthogonal of $\mathcal{N}$.*

This proposition allows us to restate Proposition 2.5.3.

Proposition 2.6.3. *Let $\mathcal{B} \subset \mathcal{A}$ be a thick subcategory of a triangulated category $\mathcal{A}$. The full category generated by the objects free on the right over $\mathcal{A}/\mathcal{B}$ (respectively on the left) is precisely the right (respectively left) orthogonal of $\mathcal{B}$.*

Proposition 2.6.4. *Let $\mathcal{A}$ be a triangulated category, let $\mathcal{B}$ be a thick subcategory of $\mathcal{A}$, let $\mathcal{B}^{\perp}$ be the right orthogonal of $\mathcal{B}$, let $S(\mathcal{B})$ and $S(\mathcal{B}^{\perp})$ be the corresponding multiplicative systems, and let Q and $Q^{\perp}$ be the canonical quotient functors. For an object X of $\mathcal{A}$, consider the following properties:*

(i) X maps by a morphism of $S(\mathcal{B})$ to an object of $\mathcal{B}^{\perp}$.

(ii) X receives, by a morphism of $S(\mathcal{B}^{\perp})$, an object of $\mathcal{B}$.

(iii) The category $S_X(\mathcal{B})$ of arrows of $S(\mathcal{B})$ with source X has a final object.

(iv) The category $S^X(\mathcal{B}^\perp)$ of arrows of $S(\mathcal{B}^\perp)$ with target X has an initial object.

(v) The category $\mathcal{B}/X$ of objects of $\mathcal{B}$ over X has a final object.

(vi) The category $\mathcal{B}^\perp/X$ of objects of $\mathcal{B}^\perp$ under X has an initial object.

Then

$$(i) \Longleftrightarrow (ii) \Longleftrightarrow (iii) \Longleftrightarrow (v) \Rightarrow (iv), \quad (v) \Rightarrow (vi)$$

If, in addition, ${}^\perp(\mathcal{B}^\perp) = \mathcal{B}$, then all the properties are equivalent.

Let $\mathcal{B}$ be a thick subcategory of a triangulated category $\mathcal{A}$, let i be the inclusion functor from $\mathcal{B}$ into $\mathcal{A}$, let Q be the quotient functor from $\mathcal{A}$ to $\mathcal{A}/\mathcal{B}$, let $\mathcal{B}^\perp$ be the right orthogonal of $\mathcal{B}$, and let $i^\perp$ and $Q^\perp$ be the corresponding functors. Proposition 2.6.4 immediately gives the following propositions.

Proposition 2.6.5. *The following two properties are equivalent:*

(i) The functor i has a right adjoint.

(ii) The functor Q has a right adjoint.

The proposition remains true when "right" is replaced by "left."

Proposition 2.6.6. *The following two properties are equivalent:*

(i) The functor i has a right adjoint.

(ii) The functor $i^\perp$ has a left adjoint. The left orthogonal of the category $\mathcal{B}^\perp$ is equal to $\mathcal{B}$.

Proposition 2.6.7. *Suppose the properties of Propositions 2.6.5 and 2.6.6 hold. Let i^* and Q^* be the right adjoints of i and Q. Similarly, let ${}^*i^\perp$ and ${}^*Q^\perp$ be the left adjoints of $i^\perp$ and $Q^\perp$. All these functors are exact. Moreover:*

There is a functorial isomorphism $Q^* \circ Q \xrightarrow{\sim} i^\perp \circ^* i^\perp$ such that the following diagram is commutative:

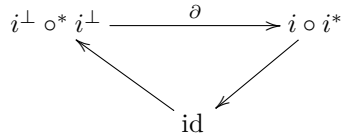
$$\begin{array}{ccc} & \mathrm{id} & \\ \swarrow & & \searrow \\ Q^* \circ Q & \xrightarrow{\sim} & i^\perp \circ^* i^\perp \end{array}$$

(the oblique arrows are defined by the adjunction properties).

Similarly, there is an isomorphism ${}^*Q^\perp \circ Q^\perp \xrightarrow{\sim} i \circ i^*$ such that the following diagram is commutative:

$$\begin{array}{ccc} {}^*Q^\perp \circ Q^\perp & \xrightarrow{\sim} & i \circ i^* \\ & \searrow \quad \swarrow & \\ & \mathrm{id} & \end{array}$$

Finally, there is a functorial morphism of degree 1, $\partial : i^{\perp} \circ^* i^{\perp} \to i \circ i^*$, *such that the following triangle is distinguished* ($\deg \partial = 1$)*:*



Such a morphism ∂ *is unique and satisfies* $\partial(TX) = -T\partial(X)$ ($X \in \operatorname{Ob} \mathcal{A}$, T *the translation).*

3 Derived categories of an abelian category

3.1 Derived categories

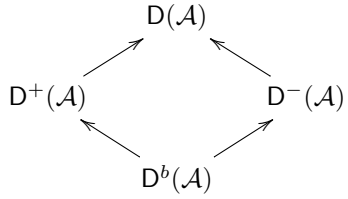
We use the notation of 1.2.3. We shall also use the following notation.

3.1.1 Notation

Let $\mathcal{A}$ be an abelian category. Set

$$\begin{aligned} \mathsf{D}(\mathcal{A}) &= \mathsf{K}^{\infty,\infty}(\mathcal{A})/\mathsf{K}^{\infty,\varnothing}(\mathcal{A}) \\ \mathsf{D}^b(\mathcal{A}) &= \mathsf{K}^{b,b}(\mathcal{A})/\mathsf{K}^{b,\varnothing}(\mathcal{A}) \\ \mathsf{D}^+(\mathcal{A}) &= \mathsf{K}^{+,+}(\mathcal{A})/\mathsf{K}^{+,\varnothing}(\mathcal{A}) \\ \mathsf{D}^-(\mathcal{A}) &= \mathsf{K}^{-,-}(\mathcal{A})/\mathsf{K}^{-,\varnothing}(\mathcal{A}). \end{aligned}$$

Proposition 3.1.2. *The natural functors appearing in the following diagram*



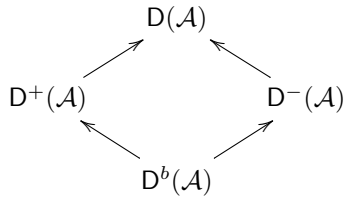
are fully faithful.

The natural functors

$$\begin{aligned} \mathsf{D}^b(\mathcal{A}) = \mathsf{K}^{b,b}(\mathcal{A})/\mathsf{K}^{b,\varnothing}(\mathcal{A}) &\to \mathsf{K}^{+,b}(\mathcal{A})/\mathsf{K}^{+,\varnothing}(\mathcal{A}) \\ \mathsf{D}^b(\mathcal{A}) = \mathsf{K}^{b,b}(\mathcal{A})/\mathsf{K}^{b,\varnothing}(\mathcal{A}) &\to \mathsf{K}^{-,b}(\mathcal{A})/\mathsf{K}^{-,\varnothing}(\mathcal{A}) \end{aligned}$$

are equivalences of categories.

Since the functors in the diagram



are injective on objects, they realize the categories $\mathsf{D}^b(\mathcal{A})$, $\mathsf{D}^+(\mathcal{A})$, *and* $\mathsf{D}^-(\mathcal{A})$ *as full subcategories of* $\mathsf{D}(\mathcal{A})$.

The proof of this proposition is contained in Theorem 2.4.2.

Definition 3.1.3. *The category* $\mathsf{D}(\mathcal{A})$ *is called the* derived category *of* $\mathcal{A}$. *The objects of* $\mathsf{D}(\mathcal{A})$ *isomorphic to objects of* $\mathsf{D}^b(\mathcal{A})$ *are called the* bounded *objects of* $\mathsf{D}(\mathcal{A})$. *The objects of* $\mathsf{D}(\mathcal{A})$ *isomorphic to objects of* $\mathsf{D}^+(\mathcal{A})$ *are called the* left-bounded *objects, and the objects of* $\mathsf{D}(\mathcal{A})$ *isomorphic to objects of* $\mathsf{D}^-(\mathcal{A})$ *are called the* right-bounded *objects.*

Let D denote the canonical functor

$$D : \mathsf{C}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}).$$

The "restriction" of D to the full subcategory $\mathcal{A}$ of $\mathsf{C}(\mathcal{A})$ will again be denoted by D. The functor D restricted to $\mathcal{A}$ is fully faithful. The functor D factors uniquely through $\mathsf{K}(\mathcal{A})$. Let X and Y be two objects of $\mathcal{A}$.

One readily checks that, for every $m > 0$, the groups

$$\hom_{\mathsf{D}(\mathcal{A})}(D(X), T^{-m}D(Y)) \qquad (T \text{ is the translation functor})$$

are zero. The cohomological dimension of $\mathcal{A}$ is the least integer n such that, for every $m > n$,

$$\hom_{\mathsf{D}(\mathcal{A})}(D(X), T^{m}D(Y)) = 0 \qquad \text{for every pair } X, Y \text{ of objects of } \mathcal{A}.$$

If there is no such integer, the cohomological dimension of $\mathcal{A}$ is said to be infinite.

Let $\mathsf{I}^+(\mathcal{A})$ (respectively $\mathsf{I}(\mathcal{A})$) denote the full triangulated subcategory of $\mathsf{K}^+(\mathcal{A})$ (respectively of $\mathsf{K}(\mathcal{A})$) defined by the complexes whose objects are injective in every degree.

Proposition 3.1.4. *Suppose that the category* $\mathcal{A}$ *has enough injectives. The natural functor*

$$Q^+ : \mathsf{K}^+(\mathcal{A}) \to \mathsf{D}^+(\mathcal{A})$$

induces an equivalence of categories

$$\mathsf{I}^+(\mathcal{A}) \to \mathsf{D}^+(\mathcal{A}).$$

The quasi-inverse functor is a right adjoint of Q^+.

If, in addition, $\mathcal{A}$ has finite cohomological dimension, the preceding assertion remains true after the superscripts + are omitted.

There is an analogous statement concerning projectives.

The proof of Proposition 3.1.4 relies on 2.6.4.

The triangulated structure of $\mathsf{D}(\mathcal{A})$ is made more explicit by the following proposition.

Let $\mathsf{E}(\mathcal{A})$ be the category of three-term exact sequences in $\mathsf{C}(\mathcal{A})$. The category $\mathsf{SSS}(\mathcal{A})$ of semi-split exact sequences (§1.2.4) is a full subcategory of $\mathsf{E}(\mathcal{A})$. We defined (loc. cit.) a functor

$$\rho : \mathsf{SSS}(\mathcal{A}) \to \mathsf{TrK}(\mathcal{A})$$

where $\mathsf{TrK}(\mathcal{A})$ denotes the category of distinguished triangles of $\mathsf{K}(\mathcal{A})$. This gives a functor

$$\sigma : \mathsf{SSS}(\mathcal{A}) \to \mathsf{TrD}(\mathcal{A})$$

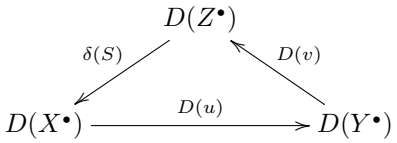
where $\mathsf{TrD}(\mathcal{A})$ denotes the category of distinguished triangles of $\mathsf{D}(\mathcal{A})$.

Proposition 3.1.5. *There exists a unique functor*

$$\Pi : \mathsf{E}(\mathcal{A}) \to \mathsf{TrD}(\mathcal{A})$$

of the following form ($\deg \delta(S) = 1$)*:*

$$(S = 0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0) \mapsto$$



whose restriction to $\mathsf{SSS}(\mathcal{A})$ *is* σ*.* This functor is essentially surjective.

Proposition 3.1.6. *The canonical cohomological functor*

$$\mathrm{H}^0 : \mathsf{K}(\mathcal{A}) \to \mathcal{A}$$

factors uniquely through a functor that we again denote by H^0*,*

$$\mathrm{H}^0 : \mathsf{D}(\mathcal{A}) \to \mathcal{A}.$$

On $\mathsf{D}(\mathcal{A})$*, the functor* H^0 *and its translates* H^i *form a conservative system; that is, a morphism* $f : X^\bullet \to Y^\bullet$ *is an isomorphism if and only if, for every integer* i*,*

$$\mathrm{H}^i(f) : \mathrm{H}^i(X) \to \mathrm{H}^i(Y)$$

is an isomorphism.

Definition 3.1.7. *A morphism* $f : X^\bullet \to Y^\bullet$ *of* $\mathsf{C}(\mathcal{A})$ *(respectively of* $\mathsf{K}(\mathcal{A})$*) is called a* quasi-isomorphism *if, for every integer* i*,*

$$\mathrm{H}^i(f) : \mathrm{H}^i(X) \to \mathrm{H}^i(Y)$$

is an isomorphism.

By the preceding proposition, the quasi-isomorphisms are the morphisms that become isomorphisms in $\mathsf{D}(\mathcal{A})$.

Definition 3.1.8. *Let* σ *be a set of objects of* $\mathcal{A}$ *and let* $X^\bullet$ *be an object of* $\mathsf{C}(\mathcal{A})$ *(respectively* $\mathsf{K}(\mathcal{A})$*). A right (respectively left) resolution of type* σ *of* $X^\bullet$ *is a quasi-isomorphism* $X^\bullet \to V^\bullet$ *(respectively* $V^\bullet \to X^\bullet$*), where* $V^\bullet$ *is a complex all of whose objects belong to* σ*. We say* injective resolution *(respectively projective resolution) instead of right (respectively left) resolution of injective (respectively projective) type.*

In general, we shall omit the words "right" or "left" when no ambiguity can result.

Proposition 3.1.9. *1. Suppose that every object of* $\mathcal{A}$ *embeds in an object of* σ *(respectively is a quotient of an object of* σ*). Then every object of* $\mathsf{C}^+(\mathcal{A})$ *(respectively* $\mathsf{C}^-(\mathcal{A})$*) has a right (respectively left) resolution of type* σ *in* $\mathsf{C}^+(\mathcal{A})$ *(respectively in* $\mathsf{C}^-(\mathcal{A})$*).*

2. Suppose, in addition, that there exists an integer n *such that for every exact sequence*

$$Y^0 \to Y^1 \to \cdots \to Y^{n-1} \to Y^n \to 0$$

$$\text{(respectively } 0 \to Y^n \to Y^{n-1} \to \cdots \to Y^1 \to Y^0\text{)}$$

of objects of $\mathcal{A}$ *in which, for every* i*,* $0 \leqslant i \leqslant n-1$*,* Y^i *is an object of* σ*, the object* Y^n *is also an object of* σ*. Then every object of* $\mathsf{C}(\mathcal{A})$ *has a right (respectively left) resolution of type* σ*.*

3.2 Study of Ext

Let $X^\bullet$ and $Y^\bullet$ be two objects of $\mathsf{C}(\mathcal{A})$ (respectively $\mathsf{K}(\mathcal{A})$), and let $D : \mathsf{C}(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$ (respectively $Q : \mathsf{K}(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$) be the canonical functor.

Definition 3.2.1. *We call the* ith hyper-Ext*, and denote by*

$$(X^\bullet, Y^\bullet) \mapsto \mathrm{Ext}^i(X^\bullet, Y^\bullet)$$

the functor $\hom_{\mathsf{D}(\mathcal{A})}(D(X^\bullet), T^i D(Y^\bullet))$ *(respectively* $\hom_{\mathsf{D}(\mathcal{A})}(Q(X^\bullet), T^i Q(Y^\bullet))$*).*

Proposition 3.2.2. *Let* $0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$ *be an exact sequence in* $\mathsf{C}(\mathcal{A})$*, and let* $V^\bullet$ *be an object of* $\mathsf{C}(\mathcal{A})$*. There are unbounded exact sequences*

$$\cdots \longrightarrow \mathrm{Ext}^i(V^\bullet, X^\bullet) \longrightarrow \mathrm{Ext}^i(V^\bullet, Y^\bullet) \longrightarrow \mathrm{Ext}^i(V^\bullet, Z^\bullet) \xrightarrow{\delta} \mathrm{Ext}^{i+1}(V^\bullet, X^\bullet) \longrightarrow \cdots$$

$$\cdots \longrightarrow \mathrm{Ext}^i(Z^\bullet, V^\bullet) \longrightarrow \mathrm{Ext}^i(Y^\bullet, V^\bullet) \longrightarrow \mathrm{Ext}^i(X^\bullet, V^\bullet) \xrightarrow{\delta} \mathrm{Ext}^{i+1}(Z^\bullet, V^\bullet) \longrightarrow \cdots$$

Let $X^\bullet$ and $Y^\bullet$ be two objects of $\mathsf{K}(\mathcal{A})$. Denote by $\mathsf{Qis}/X^\bullet$ (respectively $\mathsf{Qis}^+/X^\bullet$, $\mathsf{Qis}^-/X^\bullet$, $\mathsf{Qis}^b/X^\bullet$) the category of quasi-isomorphisms with target $X^\bullet$ and source in $\mathsf{K}(\mathcal{A})$ (respectively $\mathsf{K}^+(\mathcal{A})$, $\mathsf{K}^-(\mathcal{A})$, $\mathsf{K}^b(\mathcal{A})$). Similarly, define the categories $Y^\bullet/\mathsf{Qis}$, $Y^\bullet/\mathsf{Qis}^+$,... (quasi-isomorphisms with source $Y^\bullet$).

The following proposition summarizes the various equivalent definitions of Ext^i.

Proposition 3.2.3. *1. There are isomorphisms of functors*

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) = \hom_{\mathsf{D}(\mathcal{A})}(Q(X^\bullet), Q(Y^\bullet)) \xrightarrow{\sim} \varinjlim_{(\mathsf{Qis}/X^\bullet)^\circ} \hom_{\mathsf{K}(\mathcal{A})}(-, Y^\bullet) \xrightarrow{\sim}$$

$$\varinjlim_{Y^\bullet/\mathsf{Qis}} \hom_{\mathsf{K}(\mathcal{A})}(X^\bullet, -) \xrightarrow{\sim} \varinjlim_{(\mathsf{Qis}/X^\bullet)^\circ \times Y^\bullet/\mathsf{Qis}} \hom_{\mathsf{K}(\mathcal{A})}(-, -).$$

2. If $X^\bullet$ *is an object of* $\mathsf{K}^-(\mathcal{A})$*,*

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \varinjlim_{(\mathsf{Qis}^-/X^\bullet)^\circ} \hom_{\mathsf{K}(\mathcal{A})}(-, Y^\bullet)$$

If $Y^\bullet$ *is an object of* $\mathsf{K}^+(\mathcal{A})$*,*

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \varinjlim_{Y^\bullet/\mathsf{Qis}} \hom_{\mathsf{K}(\mathcal{A})}(X^\bullet, -)$$

3. If $X^\bullet$ *is an object of* $\mathsf{K}^-(\mathcal{A})$ *and* $Y^\bullet$ *is an object of* $\mathsf{K}^b(\mathcal{A})$*,*

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \varinjlim_{Y^\bullet/\mathsf{Qis}^b} \hom_{\mathsf{K}(\mathcal{A})}(X^\bullet, -)$$

If $X^\bullet$ *is an object of* $\mathsf{K}^b(\mathcal{A})$ *and* $Y^\bullet$ *is an object of* $\mathsf{K}^+(\mathcal{A})$*,*

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \varinjlim_{(\mathsf{Qis}^b/X^\bullet)^\circ} \hom_{\mathsf{K}(\mathcal{A})}(-, Y^\bullet)$$

4. If the category $\mathcal{A}$ has enough injectives and $Y^\bullet$ is an object of $\mathsf{K}^+(\mathcal{A})$, then $Y^\bullet$ has an injective resolution, unique up to isomorphism in $\mathsf{K}^+(\mathcal{A})$,

$$Y^\bullet \to I(Y^\bullet)$$

and

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \mathrm{hom}_{\mathsf{K}(\mathcal{A})}(X^\bullet, I(Y^\bullet)).$$

There is an analogous statement for projectives. 5. If the category $\mathcal{A}$ has enough injectives and if $\mathcal{A}$ has finite cohomological dimension*, then every object $Y^\bullet$ of $\mathsf{K}(\mathcal{A})$ has a unique injective resolution*

$$Y^\bullet \to I(Y^\bullet)$$

and

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \xrightarrow{\sim} \mathrm{hom}_{\mathsf{K}(\mathcal{A})}(X^\bullet, I(Y^\bullet))$$

There is an analogous statement for projectives.

Remark: On making explicit the isomorphism of Proposition 3.2.3(3), one recovers the Yoneda definition.

4 Derived functors

4.1 Definition of derived functors

Definition 4.1.1. *Let $\mathcal{C}$ and $\mathcal{C}'$ be two graded categories (we denote by T the translation functor of $\mathcal{C}$ and $\mathcal{C}'$), and let F and G be two graded functors from $\mathcal{C}$ to $\mathcal{C}'$. A* morphism of graded functors *is a morphism of functors*

$$u : F \to G$$

having the following property:

For every object X of $\mathcal{C}$, the following diagram is commutative:

$$\begin{array}{ccc} F(TX) & \xrightarrow{u(TX)} & G(TX) \\ \uparrow \wr & & \uparrow \wr \\ TF(X) & \xrightarrow{Tu(X)} & TG(X) \end{array}$$

Let $\mathcal{C}$ and $\mathcal{C}'$ be two triangulated categories. Denote by $\mathsf{Fex}(\mathcal{C}, \mathcal{C}')$ the category of exact functors from $\mathcal{C}$ to $\mathcal{C}'$, the morphisms between two functors being morphisms of graded functors.

Let $\mathcal{A}$ and $\mathcal{B}$ be two abelian categories and let $\Phi : \mathsf{K}^*(\mathcal{A}) \to \mathsf{K}^{*'}(\mathcal{B})$ be an exact functor ($*$ and $*'$ denote one of the signs $+$, $-$, b, or v, "empty"). By composition, the canonical functor $Q : \mathsf{K}^*(\mathcal{A}) \to \mathsf{D}^*(\mathcal{A})$ gives a functor

$$\mathsf{Fex}(\mathsf{D}^*(\mathcal{A}), \mathsf{D}^{*'}(\mathcal{B})) \longrightarrow \mathsf{Fex}(\mathsf{K}^*(\mathcal{A}), \mathsf{D}^{*'}(\mathcal{B}))$$

and hence (also denoting by Q' the canonical functor $\mathsf{K}^{*'}(\mathcal{B}) \to \mathsf{D}^{*'}(\mathcal{B})$) a functor $\%$ (respectively $\%'$) $\mathsf{Fex}(\mathsf{D}^*(\mathcal{A}), \mathsf{D}^{*'}(\mathcal{B})) \to \mathsf{Ab}$:

$$\Psi \mapsto \mathrm{hom}(Q' \circ \Phi, \Psi \circ Q)$$
$$(\text{respectively } \Psi \mapsto \mathrm{hom}(\Psi \circ Q, Q' \circ \Phi) \ ).$$

Definition 4.1.2. *We say that* Φ *admits a* total right derived functor *(respectively total left derived functor) if the functor* $\%$ *(respectively* $\%'$*) is representable. An object representing* $\%$ *(respectively* $\%'$*) is called the total right (respectively left) derived functor of* Φ *and is denoted by* $\mathsf{R}\Phi$ *(respectively* $\mathsf{L}\Phi$*).*[§]

4.1.3 Notation

Let $\mathcal{A}$ and $\mathcal{B}$ be two abelian categories and let $f : \mathcal{A} \to \mathcal{B}$ be an additive functor. The *total right derived functor* of $\mathsf{K}^+(f) : \mathsf{K}^+(\mathcal{A}) \to \mathsf{K}^+(\mathcal{B})$ is denoted by R^+f. The functors R^-f, $\mathsf{R}f$, L^+f, L^-f, and $\mathsf{L}f$ are defined similarly.

Likewise, let $F : \mathsf{K}^*(\mathcal{A}) \to \mathcal{B}$ be a cohomological functor. By composition, the canonical quotient functor $Q : \mathsf{K}^*(\mathcal{A}) \to \mathsf{D}^*(\mathcal{A})$ defines a functor

$$\mathsf{Foco}(\mathsf{D}^*(\mathcal{A}), \mathcal{B}) \longrightarrow \mathsf{Foco}(\mathsf{K}^*(\mathcal{A}), \mathcal{B})$$

(where $\mathsf{Foco}(-,-)$ denotes the category of cohomological functors), and hence a functor $\%$ (respectively $\%'$) $\mathsf{Foco}(\mathsf{D}^*(\mathcal{A}), \mathcal{B}) \to \mathsf{Ab}$ given by $G \mapsto \hom(F, G \circ Q)$ (respectively $G \mapsto \hom(G \circ Q, F)$).

Definition 4.1.4. *We say that the functor* F *admits a* right cohomological derived functor *(respectively left cohomological derived functor) if the functor* $\%$ *(respectively* $\%'$*) is representable. An object representing* $\%$ *(respectively* $\%'$*) is called the right (respectively left) cohomological derived functor and is denoted by* $\mathsf{R}F$ *(respectively* $\mathsf{L}F$*).*

4.1.5 Notation

Let $\mathcal{A}$ and $\mathcal{B}$ be two abelian categories and let $f : \mathcal{A} \to \mathcal{B}$ be an additive functor. The right cohomological derived functor of

$$\mathrm{H}^0 \circ \mathsf{K}^+f : \mathsf{K}^+(\mathcal{A}) \to \mathcal{B}$$

is denoted by R^+f. When no confusion can result, the functor $\mathsf{R}^+f \circ T^i$ (T is the translation functor) is denoted by R^if. The right cohomological derived functor of $\mathrm{H} \circ \mathsf{K}f$ is denoted by $\mathsf{R}f$. When no confusion can result, the functor $\mathsf{R}f \circ T^i$ is denoted by R^if.

The functors L^-f, $\mathsf{L}f$, and L^if are defined similarly.

4.1.6 Remarks

As we shall see in the next subsection, these definitions include the classical definition of derived functors when the categories have enough injectives or projectives and the complexes are suitably bounded. They are nevertheless more general, but without further hypotheses they are not very manageable. For example, let $f : \mathcal{A} \to \mathcal{B}$ be an additive functor between two abelian categories. Suppose that the functors R^+f and $\mathsf{R}f$ exist. I do not know how to prove that $\mathsf{R}f$ induces on $\mathsf{D}^+(\mathcal{A})$ the functor R^+f. Suppose, in addition, that the functor R^+f (cohomological derived) exists. I do not know how to prove that the functor $\mathrm{H}^0 \circ \mathsf{R}^+f$ is isomorphic to R^+f. In practice, however, these properties will hold.

4.2 Existence of derived functors

Proposition 4.2.1. *The hypotheses are those of Definition 4.1.4. Suppose, in addition, that* $\mathcal{A}$ *is a* $\mathscr{U}$*-category, where* $\mathscr{U}$ *is a universe such that the set* $\mathbb{Z}$ *of rational integers belongs to* $\mathscr{U}$*, and such that the set of objects of* $\mathcal{A}$ *belongs to* $\mathscr{U}$*. Suppose also that* $\mathcal{B}$ *is the category of abelian* $\mathscr{U}$*-groups, or more*

[§]A more manageable definition (and one equivalent in practice) is given in [2, XVII 1.2].

generally the category of sheaves of abelian $\mathscr{U}$-groups on an arbitrary $\mathscr{U}$-site. (A $\mathscr{U}$-site is a $\mathscr{U}$-category whose set of objects belongs to $\mathscr{U}$, equipped with a topology.) The functor F admits a right cohomological derived functor. This functor is calculated as follows:

Let X be an object of $\mathsf{D}^*(\mathcal{A})$, and let $\underline{X}$ be the object of $\mathsf{K}^*(\mathcal{A})$ above X. There is a functorial isomorphism

$$\mathsf{R}F(X) = \varinjlim_{\underline{X}/\mathsf{Qis}^*} F(-).$$

In particular, let $f : \mathcal{A} \to \mathcal{B}$ be an additive functor. The functors $\mathsf{R}f$ and R^+f exist and are calculated as above. The functor $\mathsf{R}f$ induces on $\mathsf{D}^+(\mathcal{A})$ the functor R^+f. Since no confusion can then result, we shall use the notation R^if.

4.2.2 Remarks

Proposition 4.2.1 can be expressed in a more general form in the setting of triangulated categories. The hypotheses on the category $\mathcal{B}$ needed to ensure the validity of Proposition 4.2.1 are hypotheses asserting the existence and exactness of direct limits over pseudo-filtered $\mathscr{U}$-categories whose sets of objects belong to $\mathscr{U}$. Proposition 4.2.1 introduces an asymmetry between right and left derived functors, at least in practice.

Theorem 4.2.3. *The data are those of Definition 4.1.2. Suppose, in addition, that a full triangulated subcategory of $\mathsf{K}^*(\mathcal{A})$, denoted $\mathcal{C}$, is given, and denote by $\mathcal{N}$ the full triangulated subcategory of the acyclic objects of $\mathcal{C}$. Suppose that*

(1) Every object of $\mathcal{N}$ is taken by Φ to an acyclic object of $\mathsf{K}^{*'}(\mathcal{B})$.

(2) Every object X of $\mathsf{K}^*(\mathcal{A})$ is the source (respectively target) of a quasi-isomorphism whose target (respectively source) is an object of $\mathcal{C}$.

a) Then Φ admits a total right (respectively left) derived functor.

b) The functor $\mathsf{R}\Phi$ (respectively $\mathsf{L}\Phi$) is obtained as follows:

The restriction of the functor $Q' \circ \Phi : \mathsf{K}^*(\mathcal{A}) \to \mathsf{D}^{*'}(\mathcal{B})$ to $\mathcal{C}$ vanishes on the objects of $\mathcal{N}$ and therefore factors through $\mathcal{C}/\mathcal{N}$. This gives a functor $\Phi' : \mathcal{C}/\mathcal{N} \to \mathsf{D}^{*'}(\mathcal{B})$. But the natural functor $\mathcal{C}/\mathcal{N} \to \mathsf{D}^*(\mathcal{A})$ is an equivalence of categories (Theorem 2.4.2). Composing with its quasi-inverse gives the functor $\mathsf{R}\Phi$ (respectively $\mathsf{L}\Phi$).

c) Let X be an object of $\mathsf{K}^*(\mathcal{A})$, let Y be an object of $\mathcal{C}$, and let $X \to Y$ (respectively $Y \to X$) be a quasi-isomorphism. The object $\mathsf{R}\Phi \circ Q(X)$ (respectively $\mathsf{L}\Phi \circ Q(X)$) is isomorphic to $Q' \circ \Phi(Y)$.

d) The functor $X \mapsto \mathsf{R}\Phi \circ Q(X)$ (respectively $X \mapsto \mathsf{L}\Phi \circ Q(X)$) is isomorphic to the functor $X \mapsto \varinjlim_{X/\mathsf{Qis}^*} Q' \circ \Phi(-)$ (respectively $X \mapsto \varinjlim_{\mathsf{Qis}^*/X} Q' \circ \Phi(-)$).

e) The functor $\mathrm{H}^0 \circ \Phi : \mathsf{K}^*(\mathcal{A}) \to \mathcal{B}$ admits a right (respectively left) cohomological derived functor, which is precisely $\mathrm{H}^0 \circ \mathsf{R}\Phi$ (respectively $\mathrm{H}^0 \circ \mathsf{L}\Phi$).

Corollary 4.2.4. *Suppose that $* = +$ (respectively $* = -$) in the hypotheses of the preceding theorem and that $\mathcal{A}$ has enough injectives (respectively projectives). Then Theorem 4.2.3 applies with $\mathcal{C}$ the category of complexes of injectives (respectively projectives). Suppose in addition that $\mathcal{A}$ has finite cohomological dimension. Then the theorem applies without restriction on $*$, with $\mathcal{C}$ the category of complexes of injectives (respectively projectives).*

Definition 4.2.5. *Let $\mathcal{A}$ and $\mathcal{B}$ be two abelian categories and let $f : \mathcal{A} \to \mathcal{B}$ be an additive functor.*

We say that f has finite cohomological dimension *on the right (respectively left) if R^+f (respectively L^-f) exists and there is an integer $m \geqslant 0$ such that, for every integer $n > m$, R^nf (respectively $\mathsf{L}^{-n}f$) is zero on the objects arising from $\mathcal{A}$ under the functor D. The* cohomological dimension *on the right (respectively left) of f is then the least integer m having this property. If R^+f (respectively L^-f) exists and f does not have finite cohomological dimension, we say that f has infinite cohomological dimension.*

An object X of $\mathcal{A}$ is said to be f-acyclic *on the right (respectively on the left) if, for every integer $n > 0$, $\mathsf{R}^nf(D(X)) = 0$ (respectively $\mathsf{L}^{-n}f(D(X)) = 0$).*

Corollary 4.2.6. *The data are those of Definition 4.2.5. Suppose there exists a set σ of objects of $\mathcal{A}$, stable under direct sums, having the following properties:*

1. *Every acyclic complex $\in \mathrm{Ob}\,\mathsf{K}^+(\mathcal{A})$ (respectively $\in \mathrm{Ob}\,\mathsf{K}^-(\mathcal{A})$) of objects of σ is taken by f to an acyclic complex.*
2. *Every object X of $\mathcal{A}$ embeds in (respectively is a quotient of) an object of σ. The hypotheses of Theorem 4.2.3 hold with $* = +$ (respectively $* = -$) and with $\mathcal{C}$ the category of complexes whose objects in every degree belong to σ. The elements of σ are right (respectively left) f-acyclic objects.*

If, in addition, f has finite right (respectively left) cohomological dimension, the set of right (respectively left) f-acyclic objects has, in addition to properties (1) and (2) above, property (2) of Proposition 3.1.9. Theorem 4.2.3 then applies to the functor $\mathsf{K}f$, with $\mathcal{C}$ the category of complexes whose objects are right (respectively left) f-acyclic in every degree. The functor $\mathsf{R}f$ (respectively $\mathsf{L}f$) exists and induces on $\mathsf{D}^+(\mathcal{A})$ the functor R^+f (respectively L^+f), and on $\mathsf{D}^-(\mathcal{A})$ the functor R^-f (respectively L^-f).

4.3 Product. Composition

Let $\mathcal{A}$ be an abelian category, let $F : \mathsf{D}(\mathcal{A}) \to \mathsf{Ab}$ be a cohomological functor, and let $X^\bullet$ and $Y^\bullet$ be two objects of $\mathsf{C}(\mathcal{A})$. The functor F defines a map

$$\mathrm{Ext}^0(X^\bullet, Y^\bullet) \to \mathrm{hom}_{\mathsf{Ab}}(F(D(X^\bullet)), F(D(Y^\bullet)))$$

and hence a pairing $F(D(X^\bullet)) \times \mathrm{Ext}^0(X^\bullet, Y^\bullet) \to F(D(Y^\bullet))$.

Applying translations to this pairing gives pairings

$$F^i(D(X^\bullet)) \times \mathrm{Ext}^j(X^\bullet, Y^\bullet) \to F^{i+j}(D(Y^\bullet)).$$

By definition, when these pairings are applied to cohomological derived functors, they are precisely the Yoneda products. The properties of these products follow immediately from this definition. We shall not develop them here.

Let $\mathcal{A}$, $\mathcal{A}'$, and $\mathcal{A}''$ be three abelian categories, and let $f : \mathcal{A} \to \mathcal{A}'$ and $g : \mathcal{A}' \to \mathcal{A}''$ be two additive functors. Suppose that the functors R^+f and R^+g exist. I do not know how to deduce that the functor $\mathsf{R}^+g \circ f$ exists, and even if the latter functor exists, the formula

$$\mathsf{R}^+g \circ f \xrightarrow{\sim} \mathsf{R}^+g \circ \mathsf{R}^+f$$

is probably not true in general. Nevertheless, one has the following proposition.

Proposition 4.3.1. *Suppose that the category $\mathcal{A}'$ has enough right g-acyclic objects and that $\mathcal{A}$ has enough right f-acyclic objects that f takes to right g-acyclic objects. Then the functor $\mathsf{R}^+g \circ f$ exists and there is an isomorphism*

$$\mathsf{R}^+g \circ f \xrightarrow{\sim} \mathsf{R}^+g \circ \mathsf{R}^+f.$$

There are analogous statements for left derived functors.

5 Examples

5.1 The functor $\mathsf{Rhom}^\bullet$

Let $\mathcal{A}$ be an abelian category with enough injectives.

The functor $Y \mapsto \hom^\bullet(X^\bullet, Y^\bullet)$ (1.3), where $X^\bullet$ is an object of $\mathsf{K}(\mathcal{A})$ and $Y^\bullet$ an object of $\mathsf{K}^+(\mathcal{A})$, has a total right derived functor (Corollary 4.2.4), denoted by $\mathsf{R}\hom^\bullet(X^\bullet, -)$. Let Y be an object of $\mathsf{D}^+(\mathcal{A})$. The functor

$$\begin{aligned} \mathsf{K}(\mathcal{A}) &\to \mathsf{D}(\mathsf{Ab}) \\ X^\bullet &\mapsto \mathsf{R}\hom^\bullet(X^\bullet, Y) \end{aligned}$$

is exact and vanishes on acyclic complexes, hence gives an exact bifunctor

$$\mathsf{D}(\mathcal{A})^\circ \times \mathsf{D}^+(\mathcal{A}) \longrightarrow \mathsf{D}(\mathsf{Ab})$$

which we denote by $\mathsf{Rhom}^\bullet$.

When $\mathcal{A}$ has finite cohomological dimension, the functor $\mathsf{Rhom}^\bullet$ extends to $\mathsf{D}(\mathcal{A})^\circ \times \mathsf{D}(\mathcal{A})$ (loc. cit.).

If, in addition, $\mathcal{A}$ has enough projectives, the functor hom can be derived with respect to the first argument. The two definitions coincide on their common domain of validity.

5.2 $\mathscr{T}$or of sheaves

Let E be a ringed site (in particular, a ringed topological space), let $\mathscr{A}$ be its sheaf of rings, and let $\mathsf{Mod}(E)$ be the category of $\mathscr{A}$-modules. An object X of $\mathsf{Mod}(E)$ is said to be *$\mathscr{A}$-flat* if, for every exact sequence

$$0 \longrightarrow Y' \longrightarrow Y \longrightarrow Y'' \longrightarrow 0$$

the sequence $0 \to Y' \otimes_{\mathscr{A}} X \to Y \otimes_{\mathscr{A}} X \to Y'' \otimes_{\mathscr{A}} X \to 0$ is exact. There are enough $\mathscr{A}$-flat objects: the direct sums of objects $\mathscr{A}_U$, where U is an object of E and $\mathscr{A}_U$ is the sheaf $\mathscr{A}$ "extended by zero outside U." Thus Theorem 4.2.3 applies, and one can define the functor

$$\begin{aligned} \mathsf{D}^-(E, \mathscr{A}) \times \mathsf{D}^-(E, \mathscr{A}) &\to \mathsf{D}^-(E, \mathscr{A}) \\ (X, Y) &\mapsto X \overset{\mathsf{L}}{\otimes} Y \end{aligned}$$

the total tensor product, which is the left derived functor of the tensor product. Passing to cohomology defines the $\mathscr{T}\mathrm{or}_i^{\mathscr{A}}(X, Y)$, the local hyper-Tor groups.

Since the category $\mathsf{Mod}(E)$ has enough injectives, one can derive the functor $\hom^\bullet(X, Y)$, the complex of local homomorphisms, and thus obtain $\mathsf{Rhom}(X, Y)$ ($X \in \mathsf{D}$, $Y \in \mathsf{D}^+$).

Let X and Y be two objects of $\mathsf{D}^-(E, \mathscr{A})$ and let Z be an object of $\mathsf{D}^+(E, \mathscr{A})$. There is a functorial isomorphism

$$\hom_{\mathsf{D}(E,\mathscr{A})}(X \overset{\mathsf{L}}{\otimes} Y, Z) \xrightarrow{\sim} \hom_{\mathsf{D}(E,\mathscr{A})}(X, \mathsf{Rhom}(Y, Z)).$$

5.3 Left derived functor of the inverse image

Let $f : E \to E'$ be a map of ringed topological spaces. The inverse-image functor

$$f^* : \mathsf{Mod}(E') \to \mathsf{Mod}(E)$$

is not necessarily exact. But one knows how to define f^*-flat objects, and there are enough such objects. Thus one can define the functor $\mathsf{L}^- f^*$. There is likewise a duality formula:

Let X be an object of $\mathsf{D}^-(E', \mathscr{A}')$ and let Y be an object of $\mathsf{D}^+(E, \mathscr{A})$. There is a functorial isomorphism

$$\hom_{\mathsf{D}(E,\mathscr{A})}(\mathsf{L}^- f^*(X), Y) \xrightarrow{\sim} \hom_{\mathsf{D}(E',\mathscr{A}')}(X, \mathsf{R}^+ f_*(Y)).$$

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